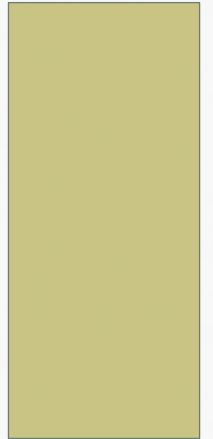
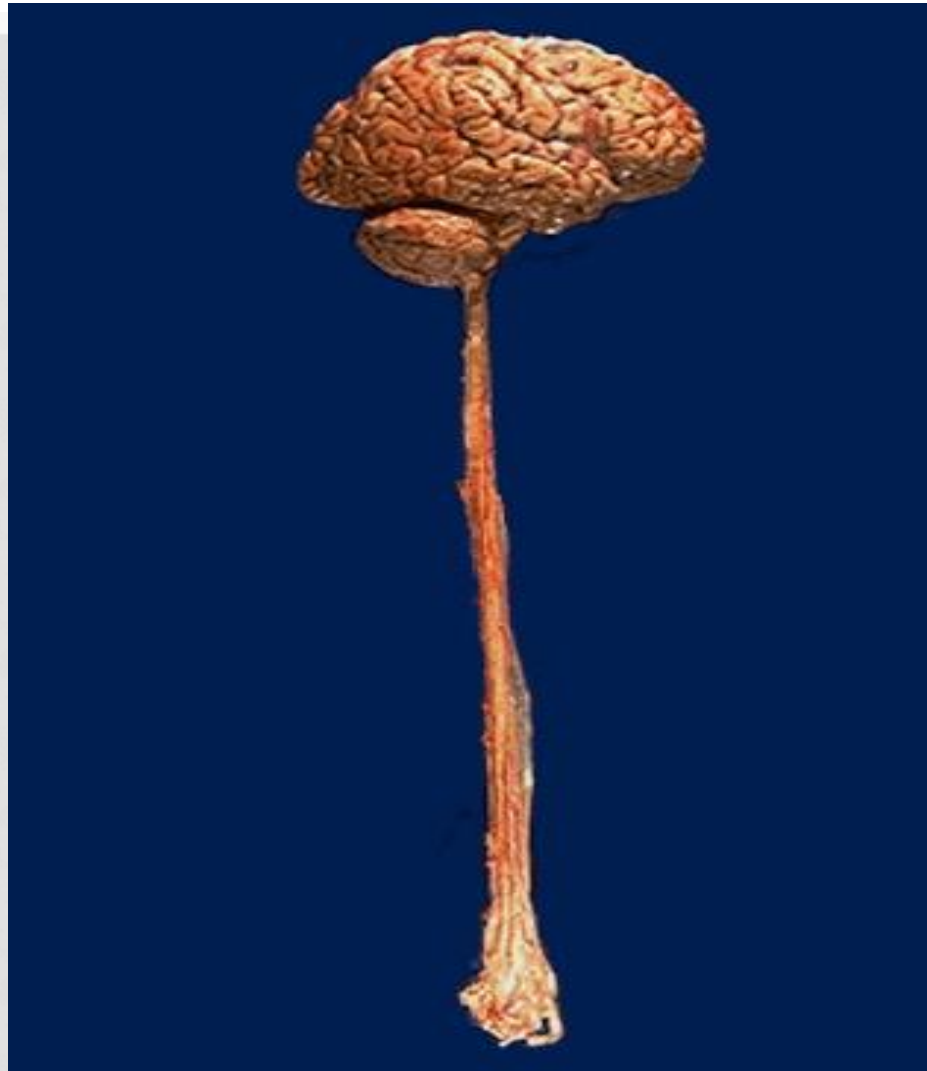


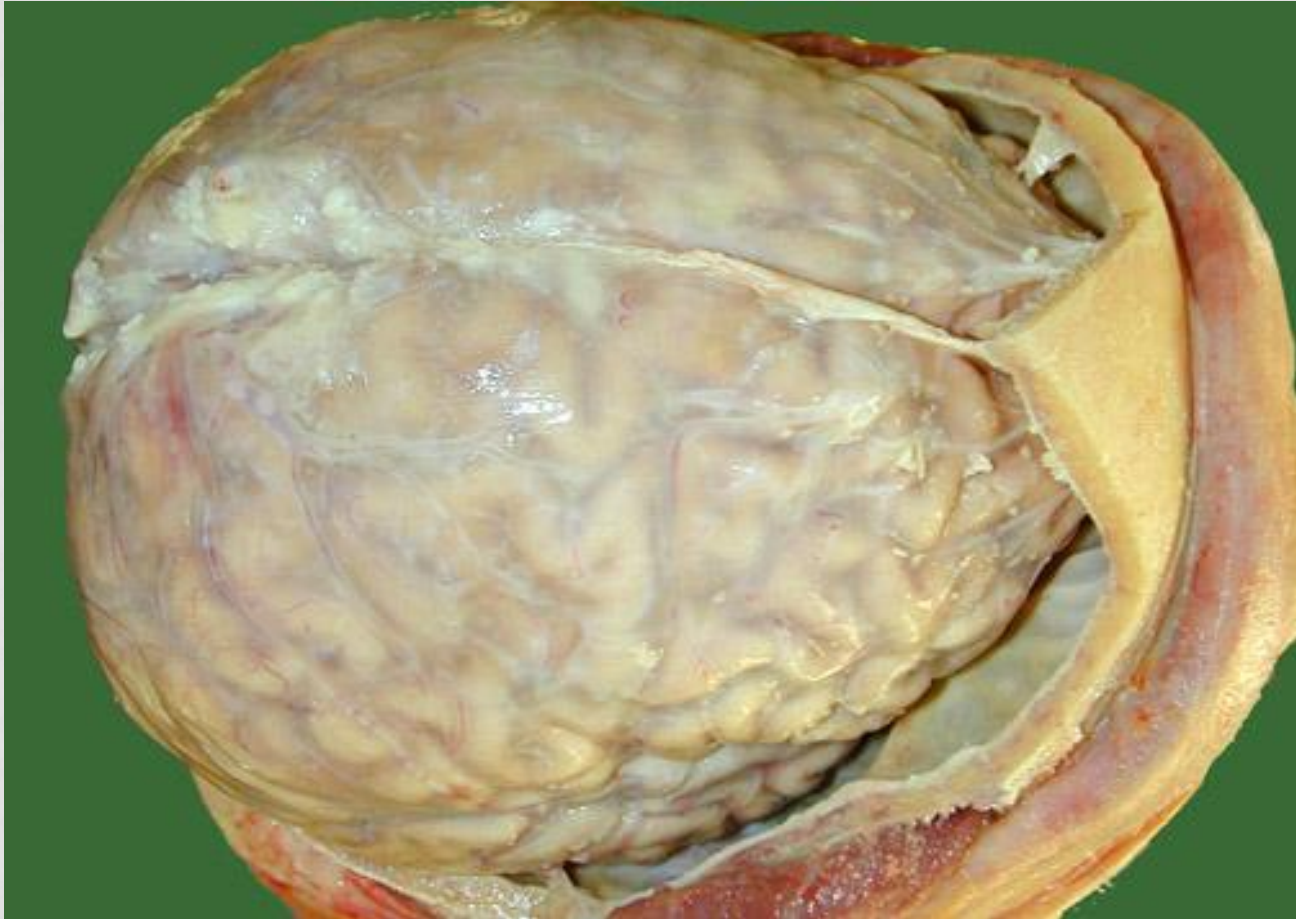
CENTRAL NERVOUS SYSTEM

CNS I

BASICS
CIRCULATION
TRAUMA











Considerations of anatomy and physiology have important applications to diseases of the central nervous system (CNS), particularly their effects and spread.

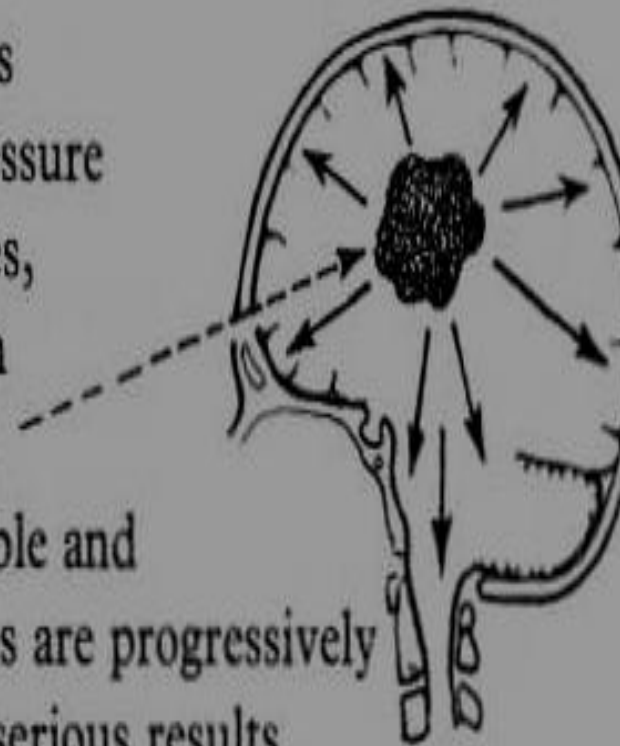
The anatomy of the various coverings is important.

The skull and vertebrae

form a rigid compartment protecting the delicate CNS tissues.

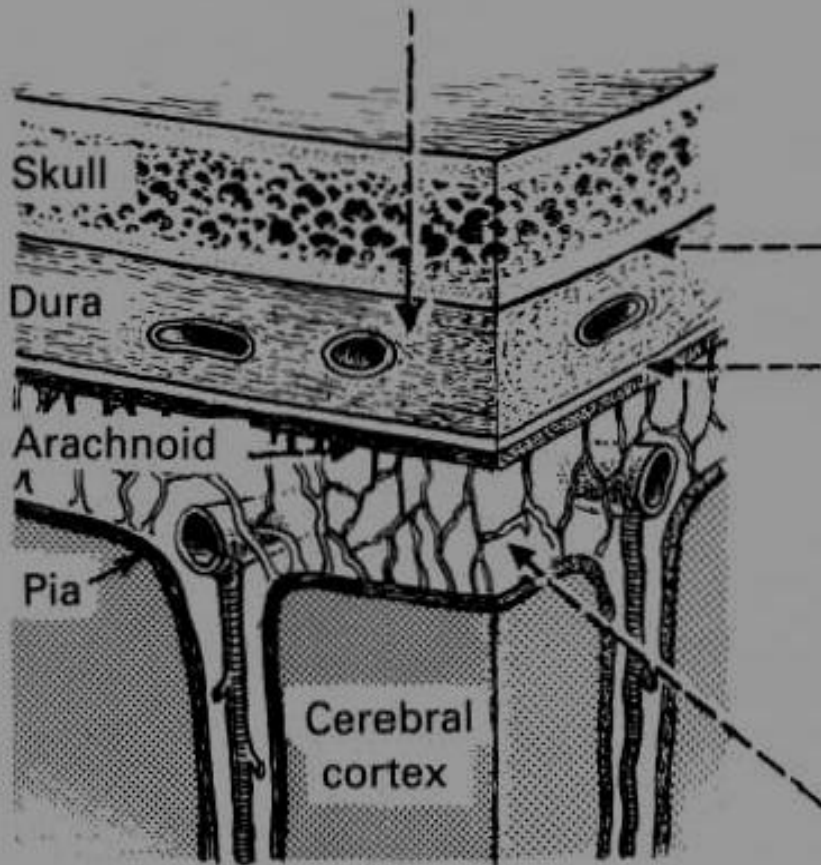


This rigidity has serious disadvantages when pressure inside the skull increases, e.g. an expanding lesion soon takes up the small reserves of space available and the delicate brain tissues are progressively compressed, with very serious results.



The detailed arrangement of the meninges is important.

The thick **dura**, closely applied to the skull, acts as the periosteum – its rigid reflections (falx cerebri and tentorium cerebelli) complicate the effects of increased intracranial pressure.



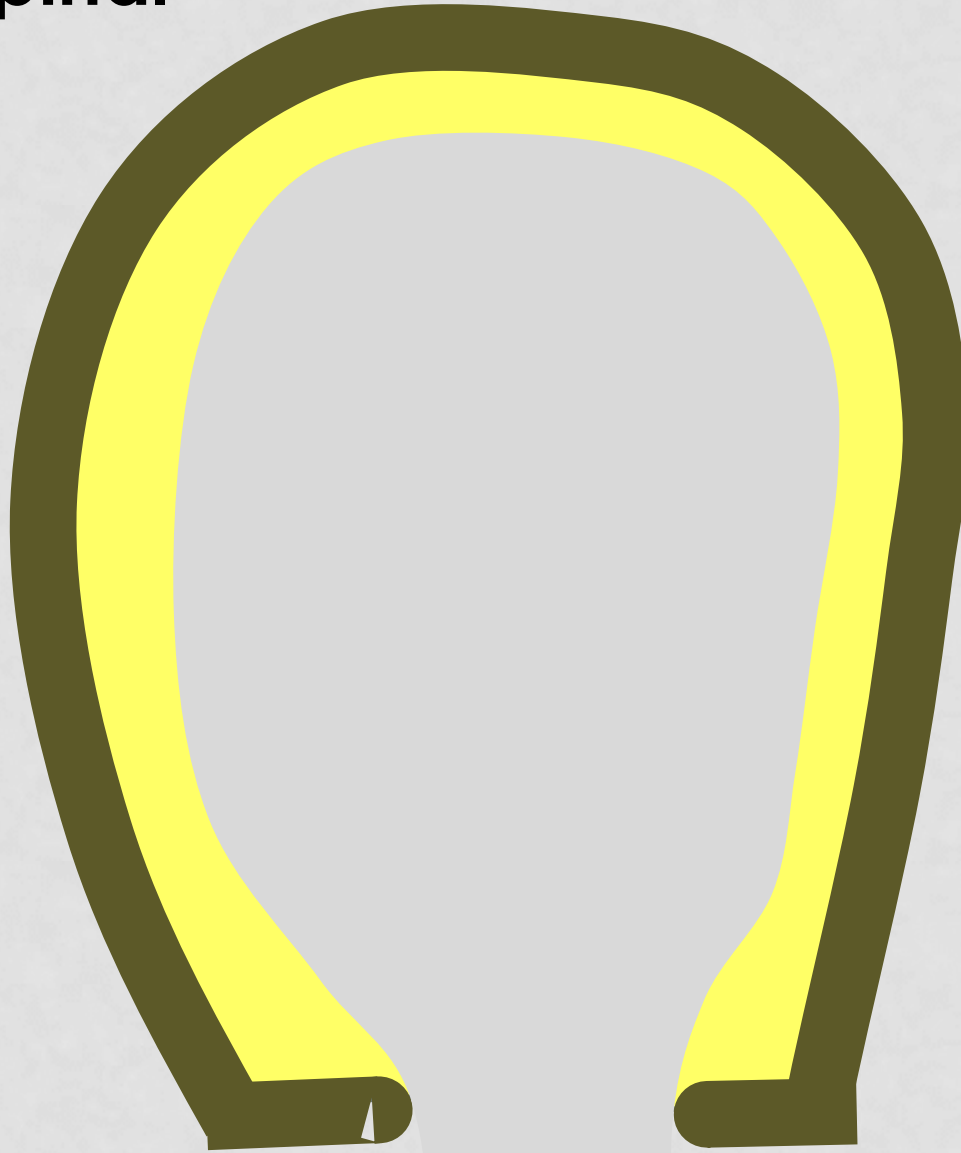
Extradural lesions tend to be localised; they have to strip the dura from the bone.

Subdural lesions remain local but can spread more widely since the arachnoid and dura are loosely attached.

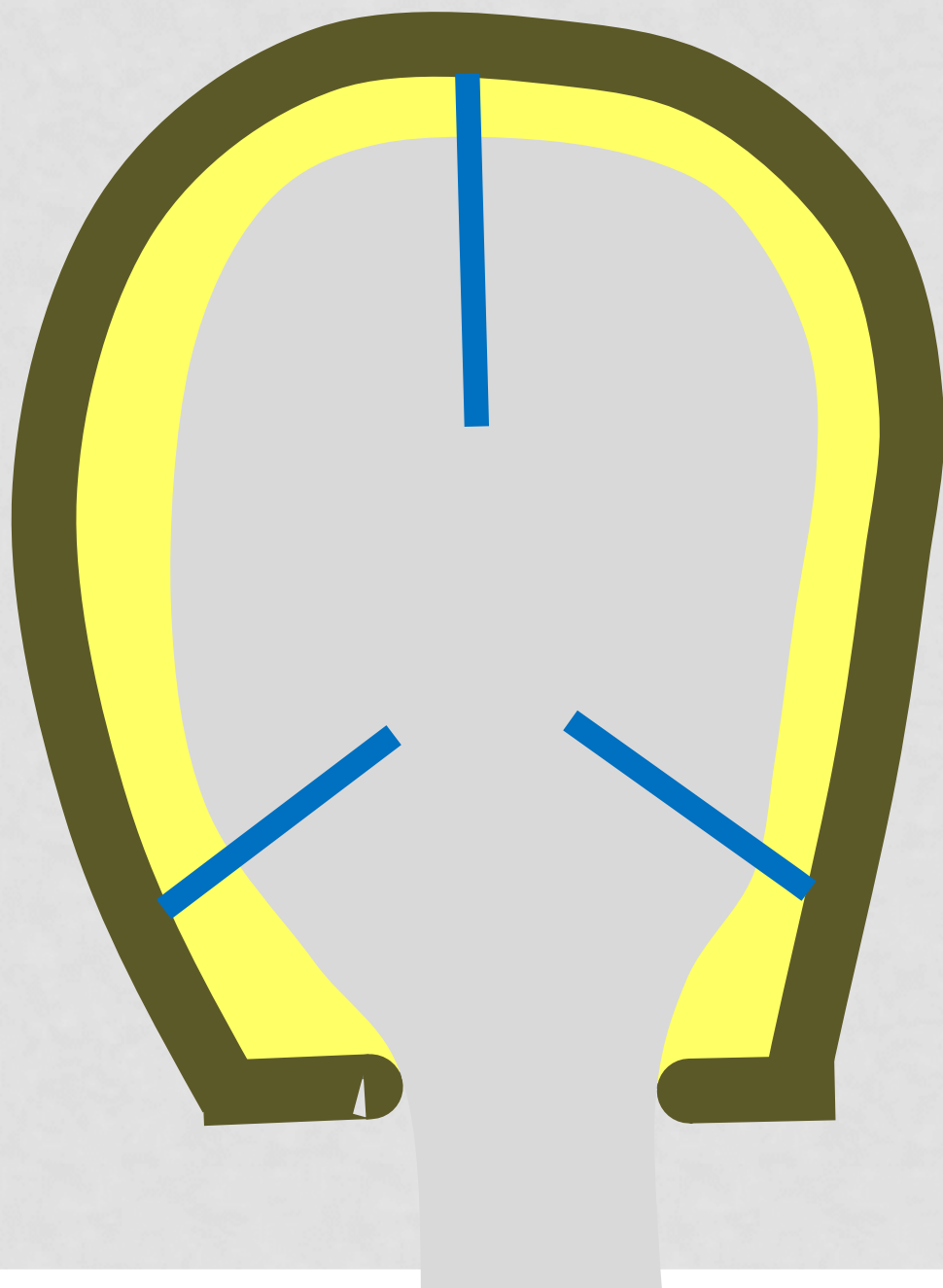
The **arachnoid**, a delicate membrane, is loosely attached to the dura and sends trabeculae across the subarachnoid space which contains the cerebral blood vessels and the CSF.

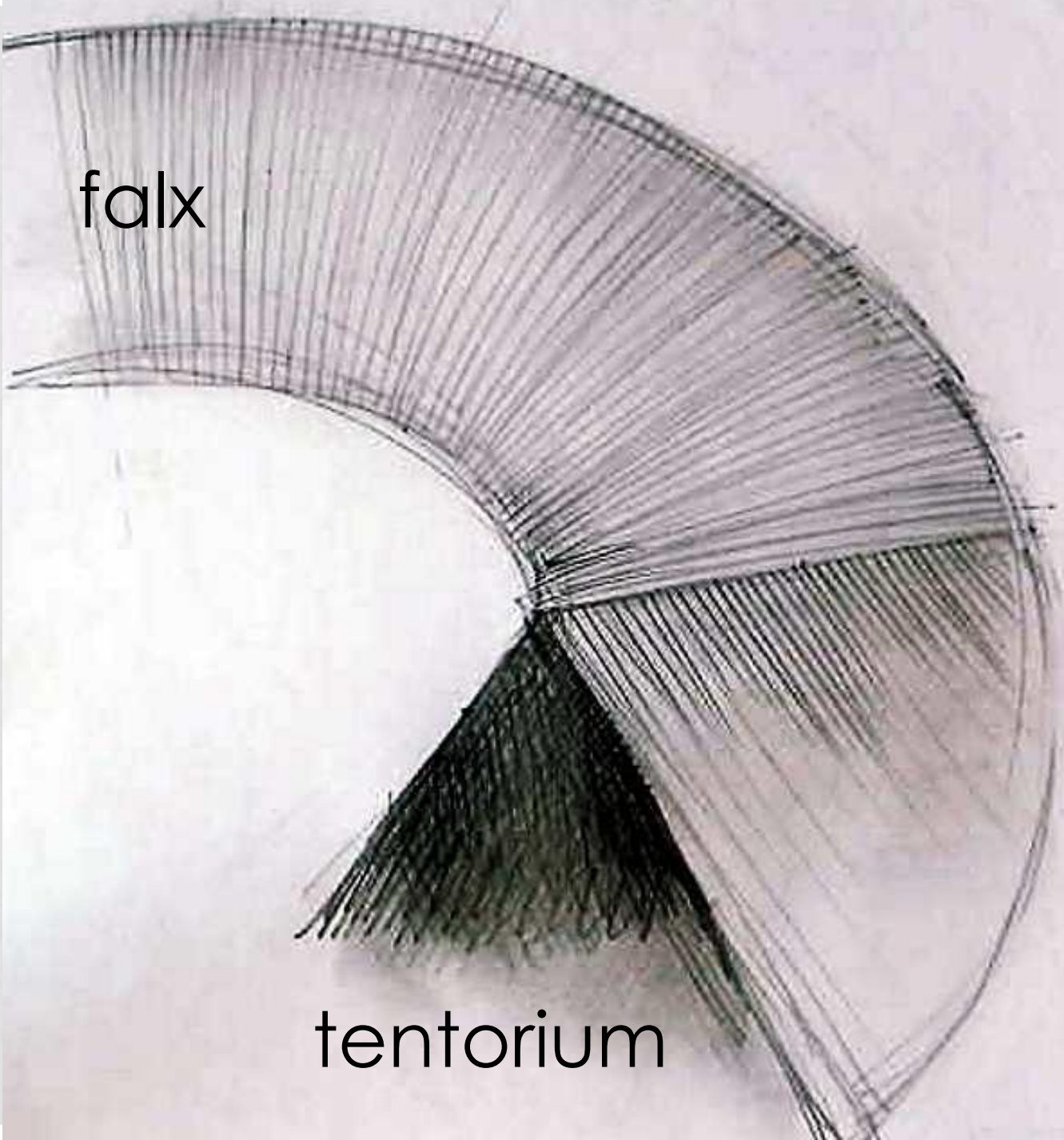
Disease in the **subarachnoid** space can spread widely over the whole surface of the brain and spinal cord but is prevented from penetrating into the brain tissue by the **pia**.

**Cerebrospinal
fluid**



CSF





A hand-drawn anatomical diagram of the falx cerebri and tentorium cerebelli. The falx is depicted as a large, crescent-shaped structure with numerous fine, parallel lines radiating from its free margin. The tentorium is shown as a smaller, triangular structure with similar fine, parallel lines, positioned below the falx. The drawing is rendered in a light, sketchy style on a white background.

falx

tentorium

shield

bath

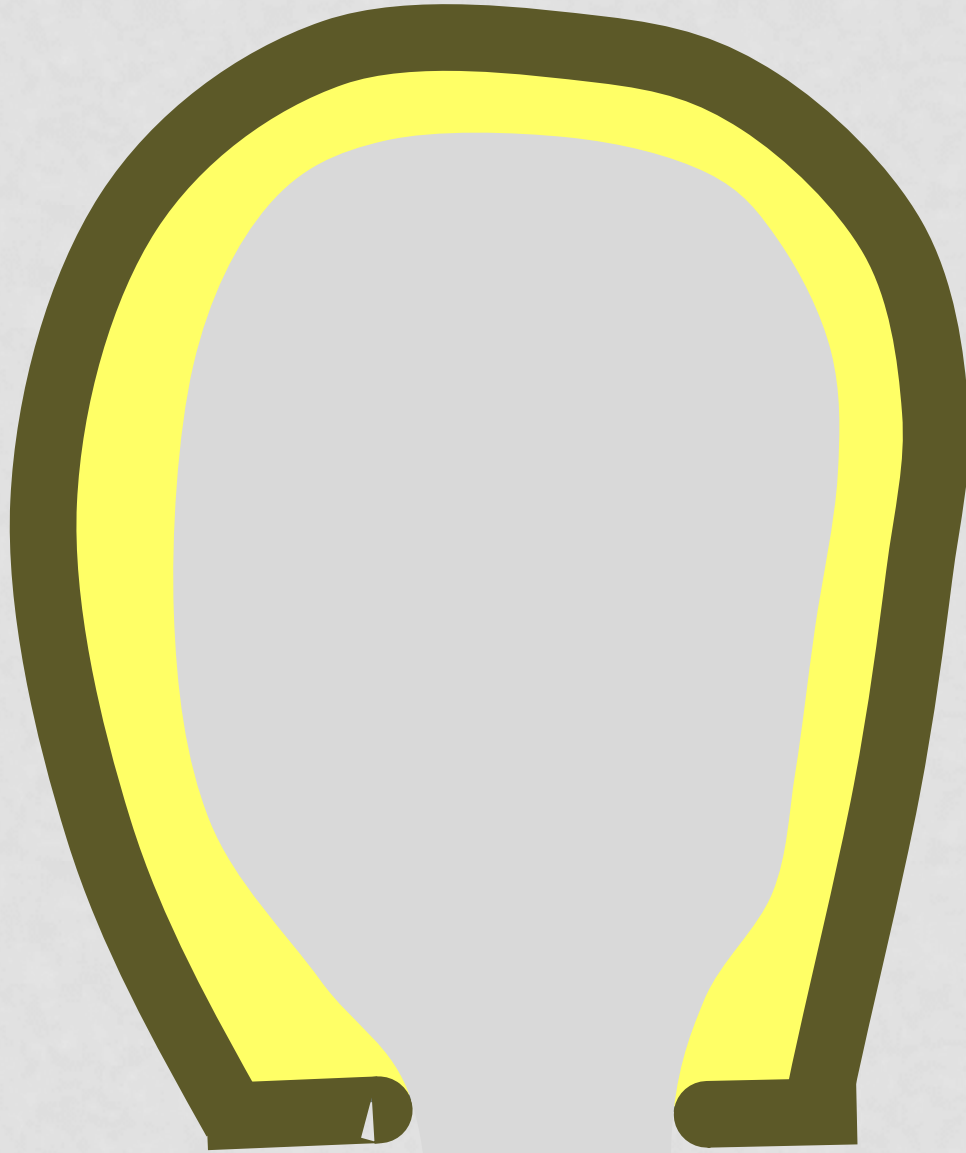
septae



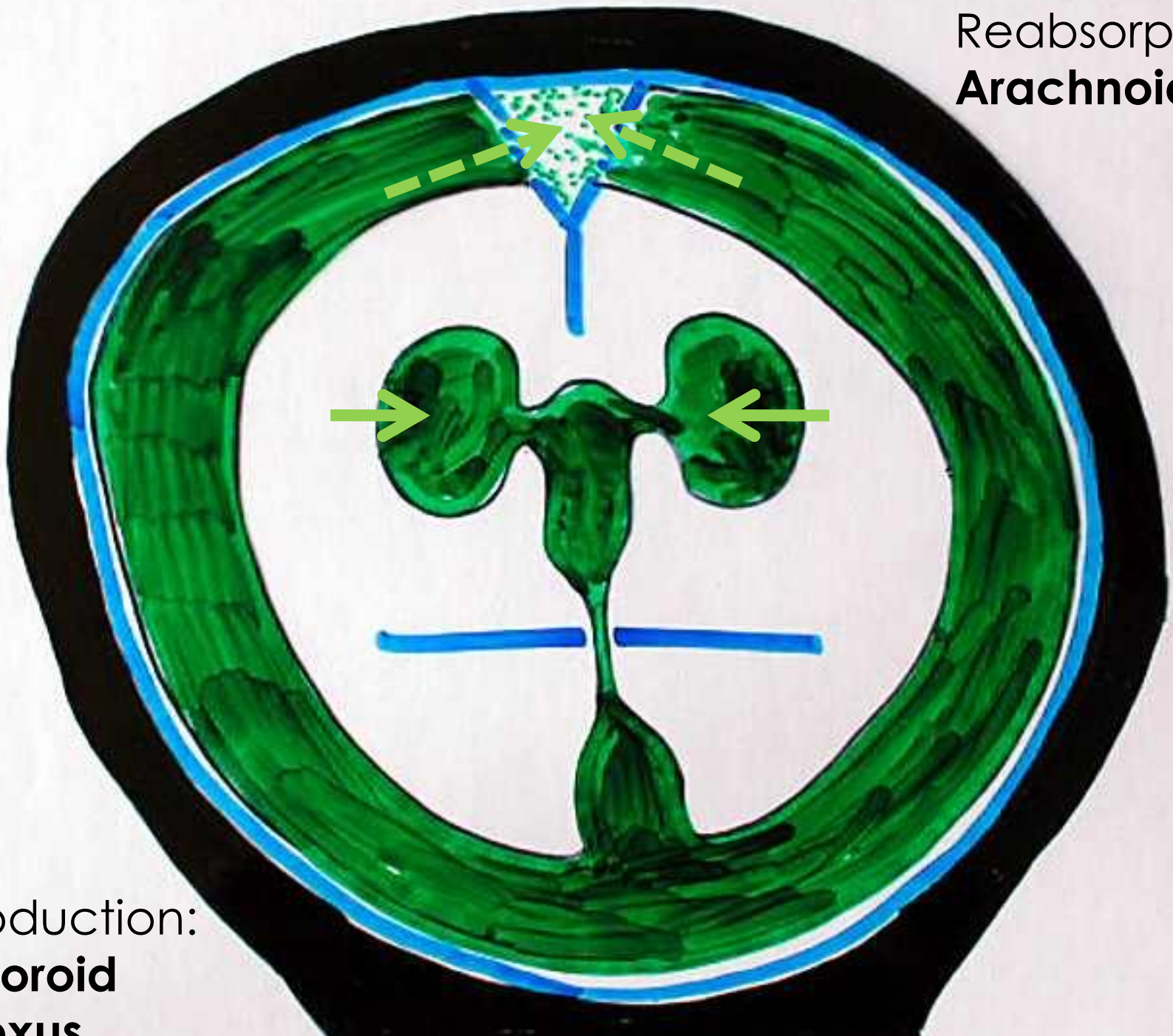
protection

fixation

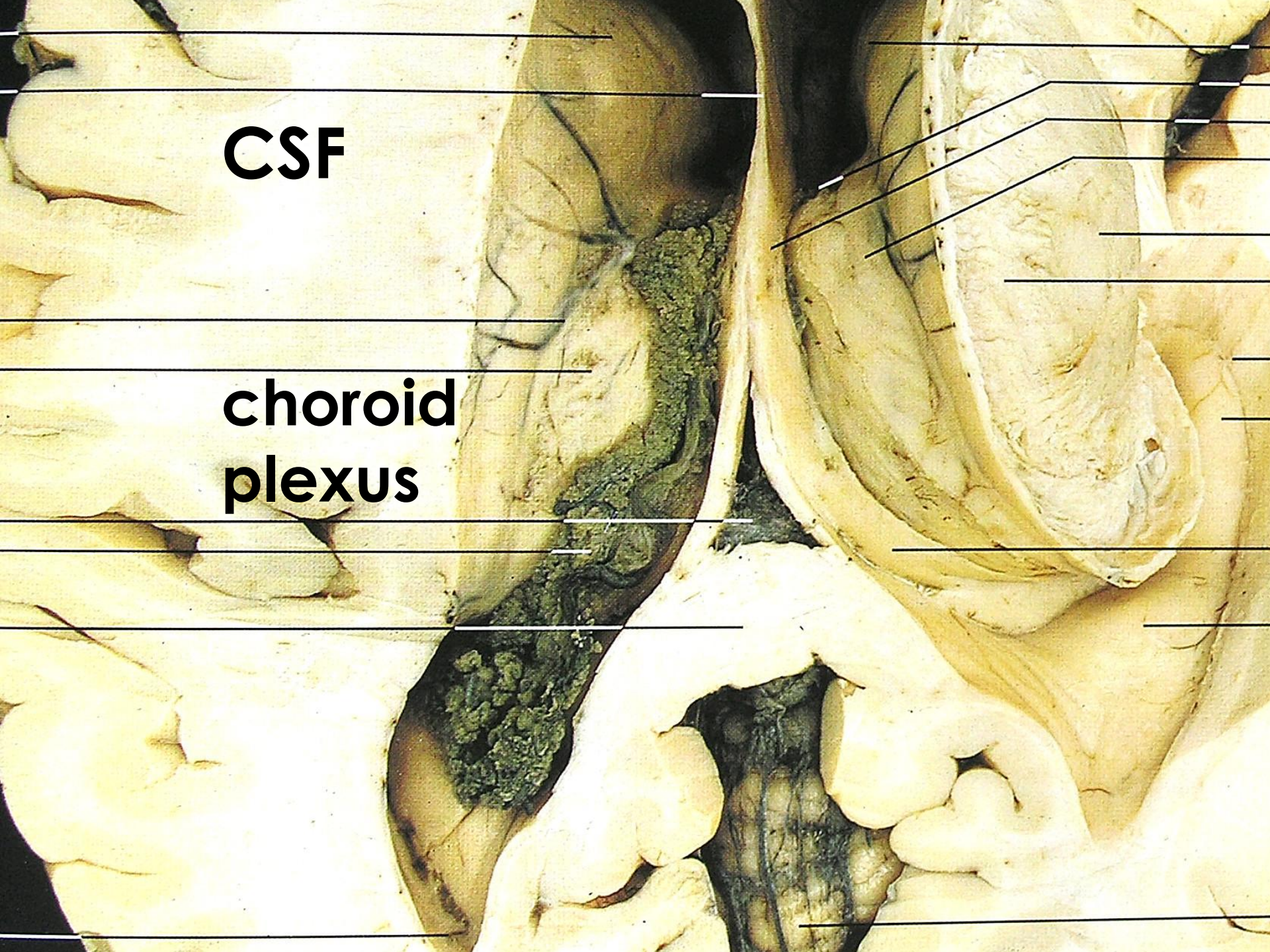
CSF



Reabsorption:
Arachnoid villi



Production:
**Choroid
plexus**



This anatomical dissection shows a cross-section of the choroid plexus, a cluster of capillaries in the brain that produces cerebrospinal fluid. The plexus is visible as a dark, granular mass. Surrounding it are the layers of the meninges (dura mater, arachnoid mater, and pia mater) and the underlying brain tissue. The cerebrospinal fluid (CSF) is seen in the ventricular space. Labels with leader lines identify the CSF, choroid plexus, and various meningeal and brain structures.

CSF

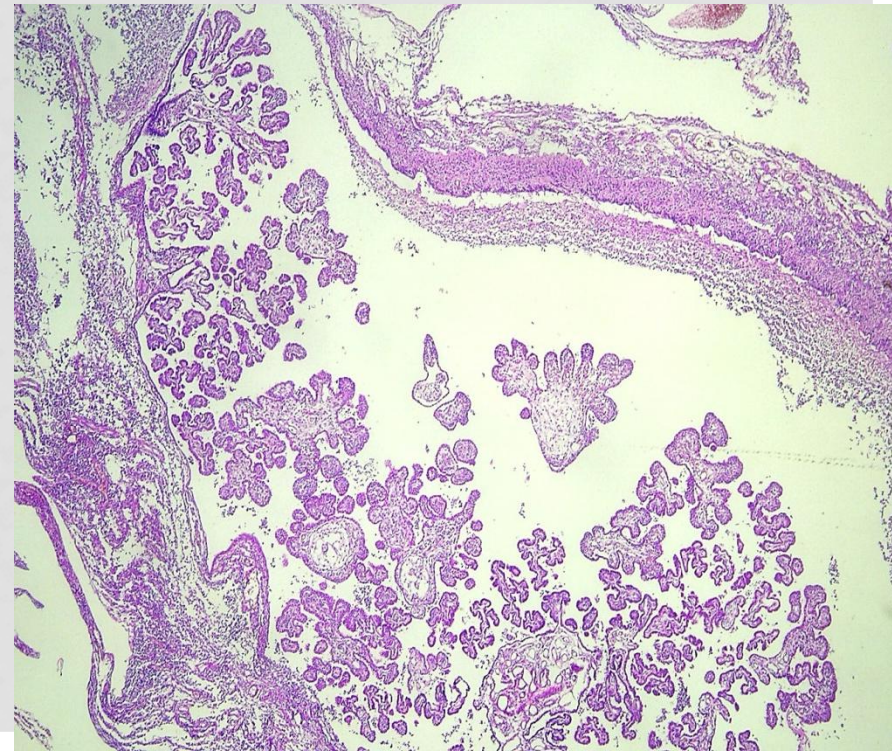
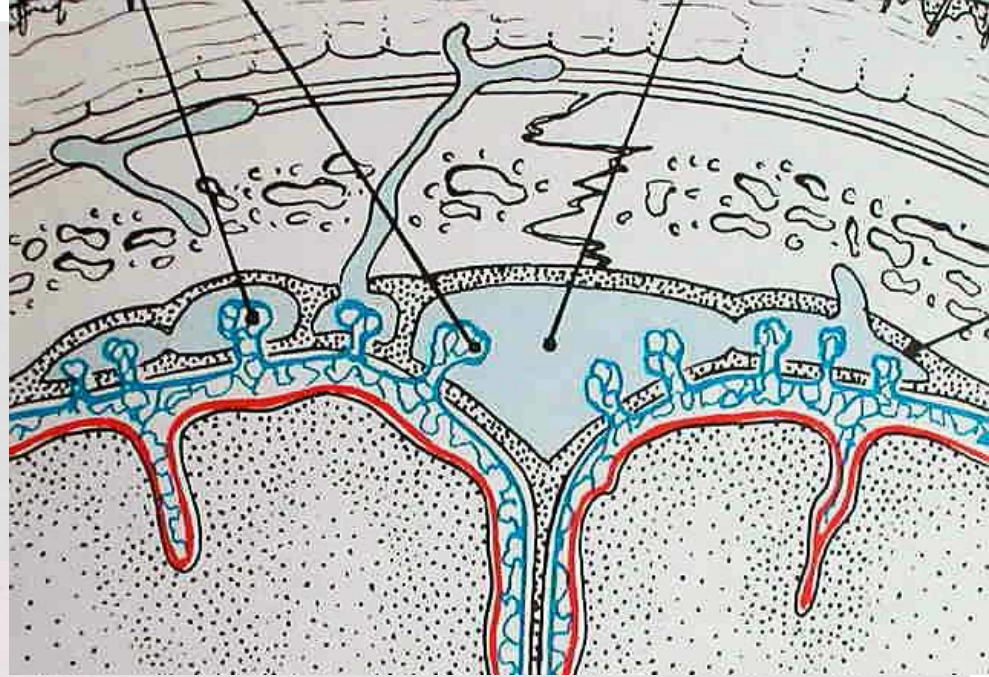
**choroid
plexus**

Cerebro spinal Fluid **CSF**

Volume: 100 – 150 ml

Replaced 3 - 4 x per day

Selective secretion



Substance	CSF	Plasma	Ratio CSF/plasma
Na+ [meq/kg H ₂ O]	147.0	150.0	0.89
K+ [meq/kg H ₂ O]	2.9	4.6	0.62
Mg++ [meq/kg H ₂ O]	2.2	1.6	1.39
Ca++ [meq/kg H ₂ O]	2.3	4.7	0.49
Osmolality [mosm/kg H ₂ O]	289.0	289.0	1.0
Protein [mg/dl]	20.0	6000.0	0.003
Glucose [mg/dl]	64.0	100.0	0.64
Cholesterol [mg/dl]	0.2	175.0	0.001

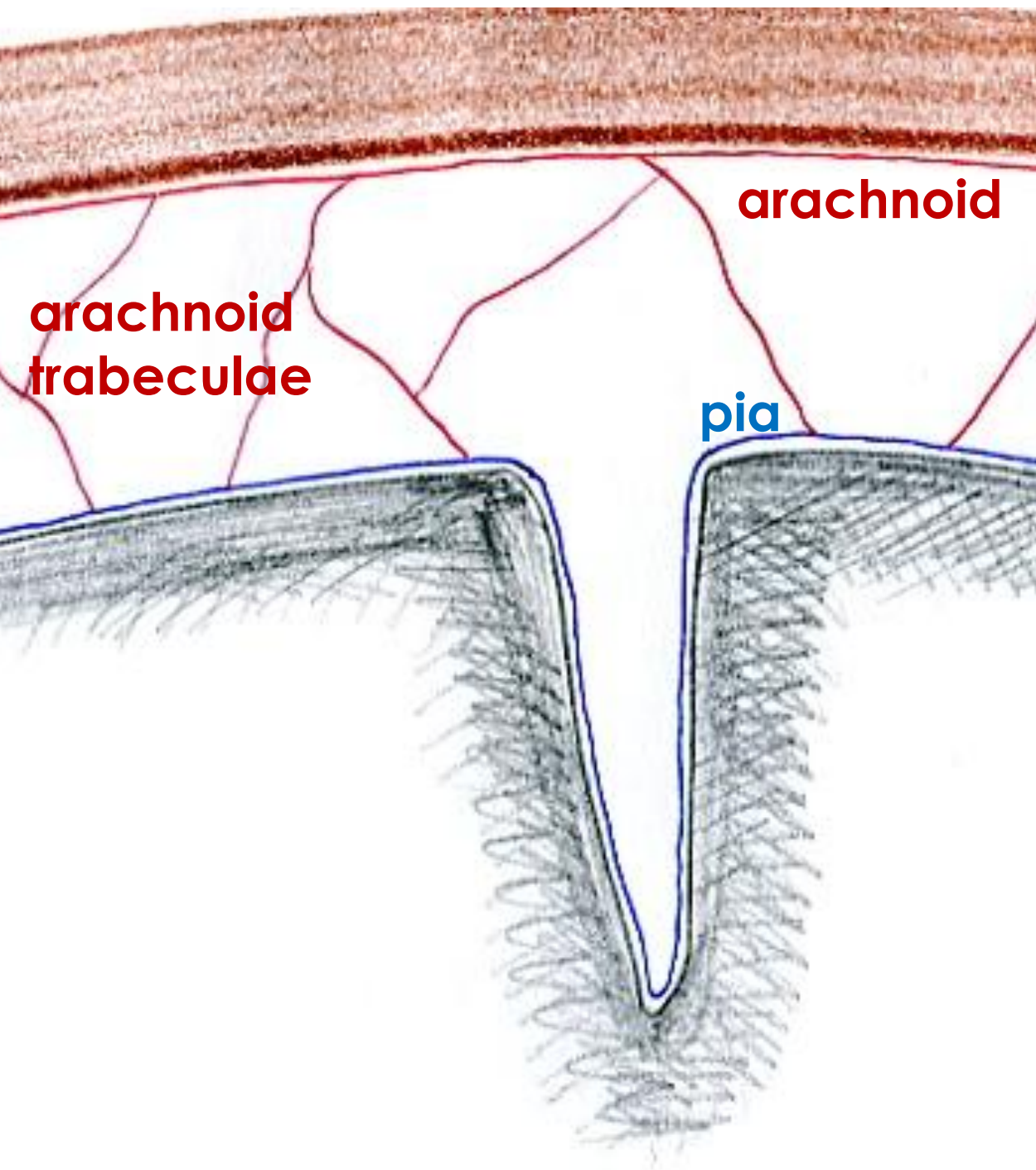
Concentration of various substances in CSF and plasma

Medical Physiology, Ganong, 18th edit.

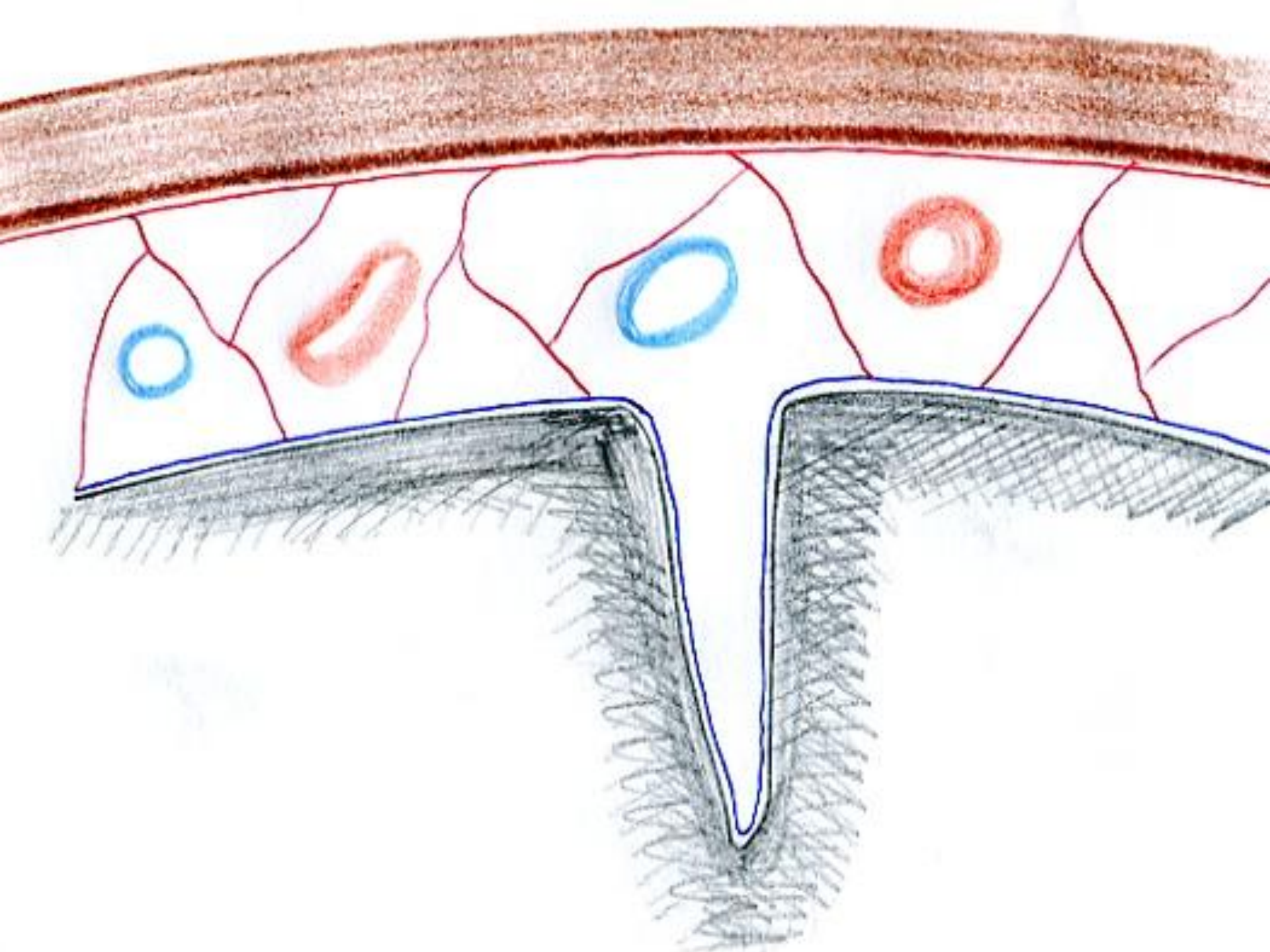
Foramina LUSCHKAE

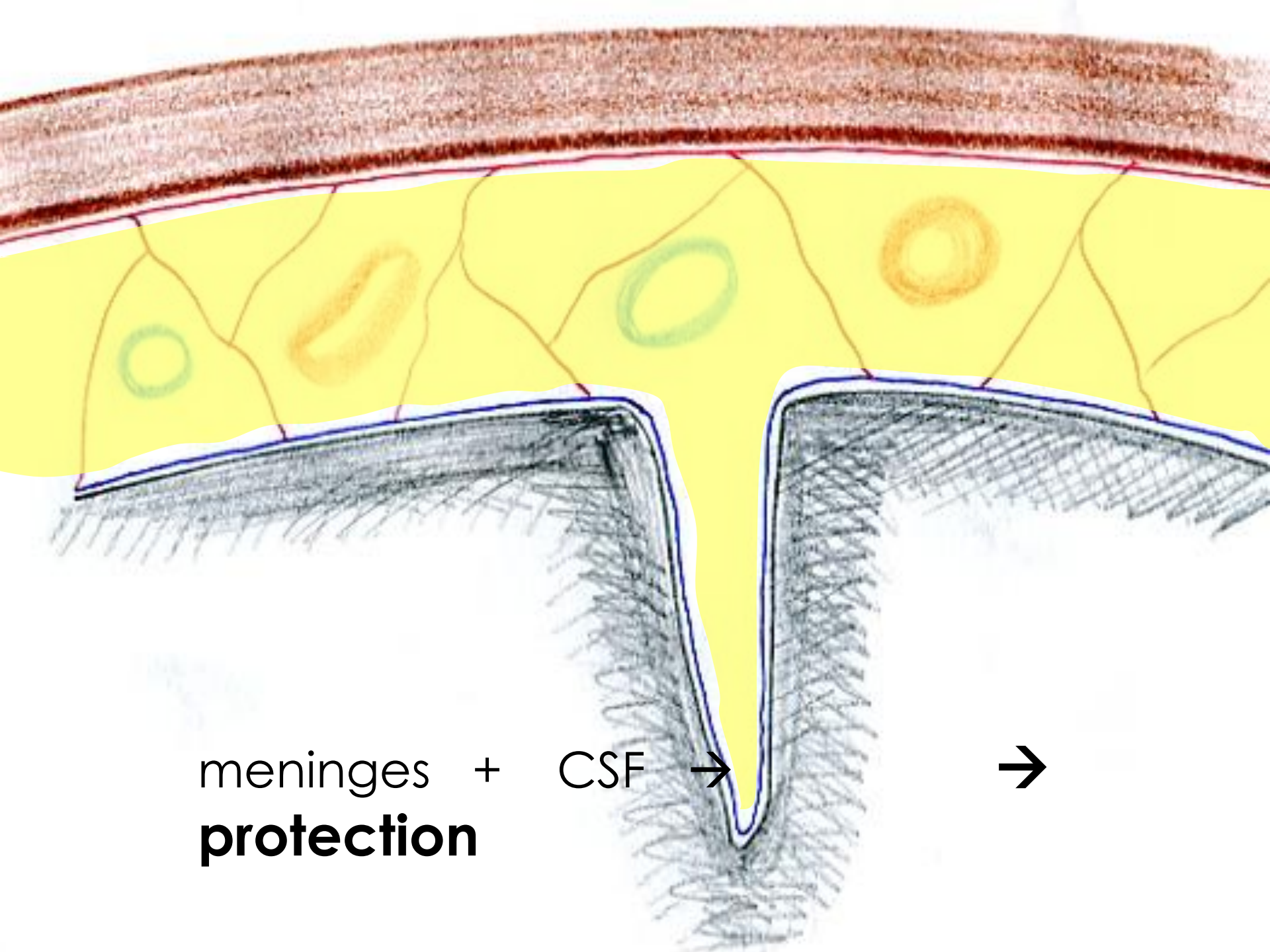
Foramen
MAGENDI





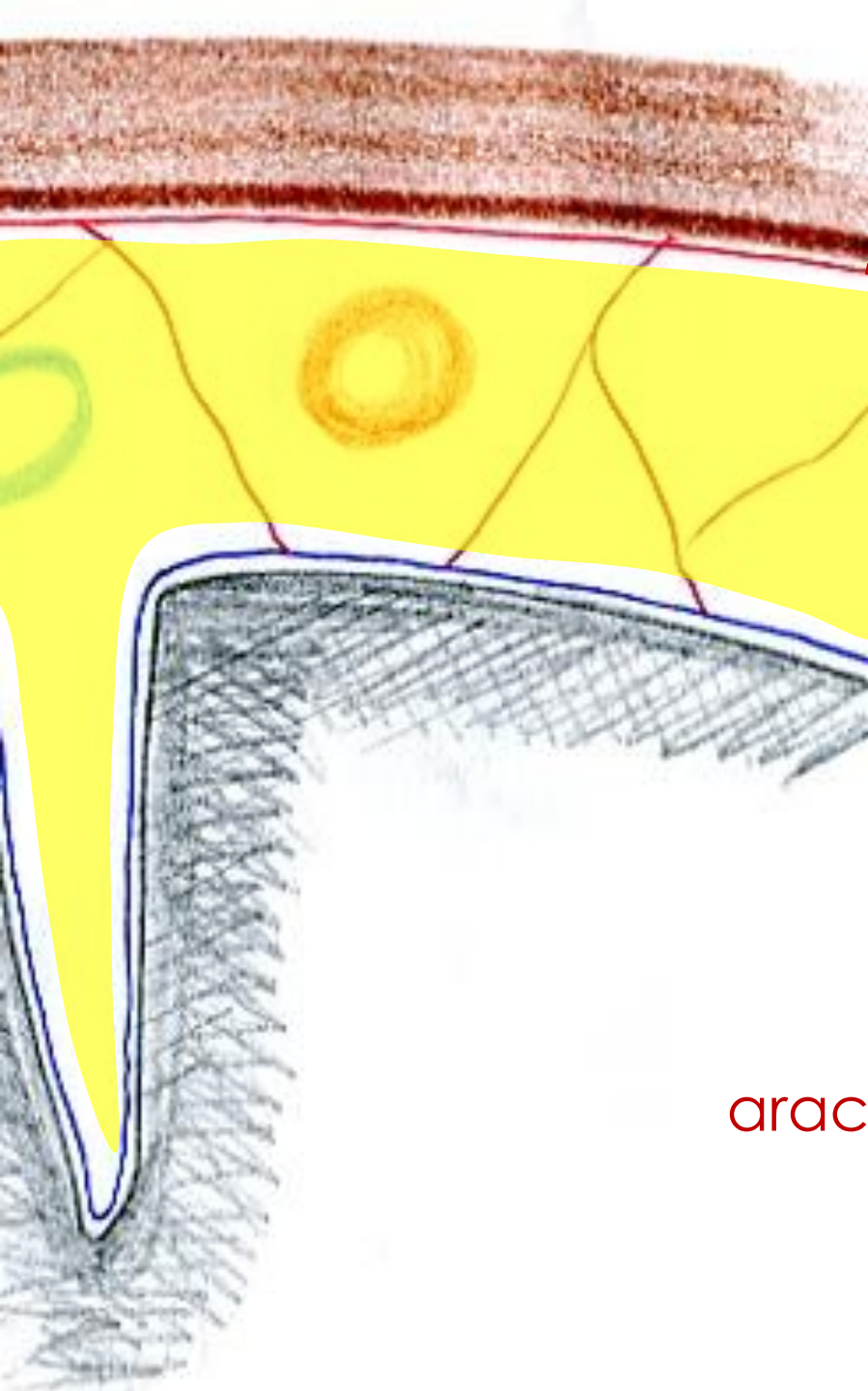
dura =
pachymeninx
leptomeninx





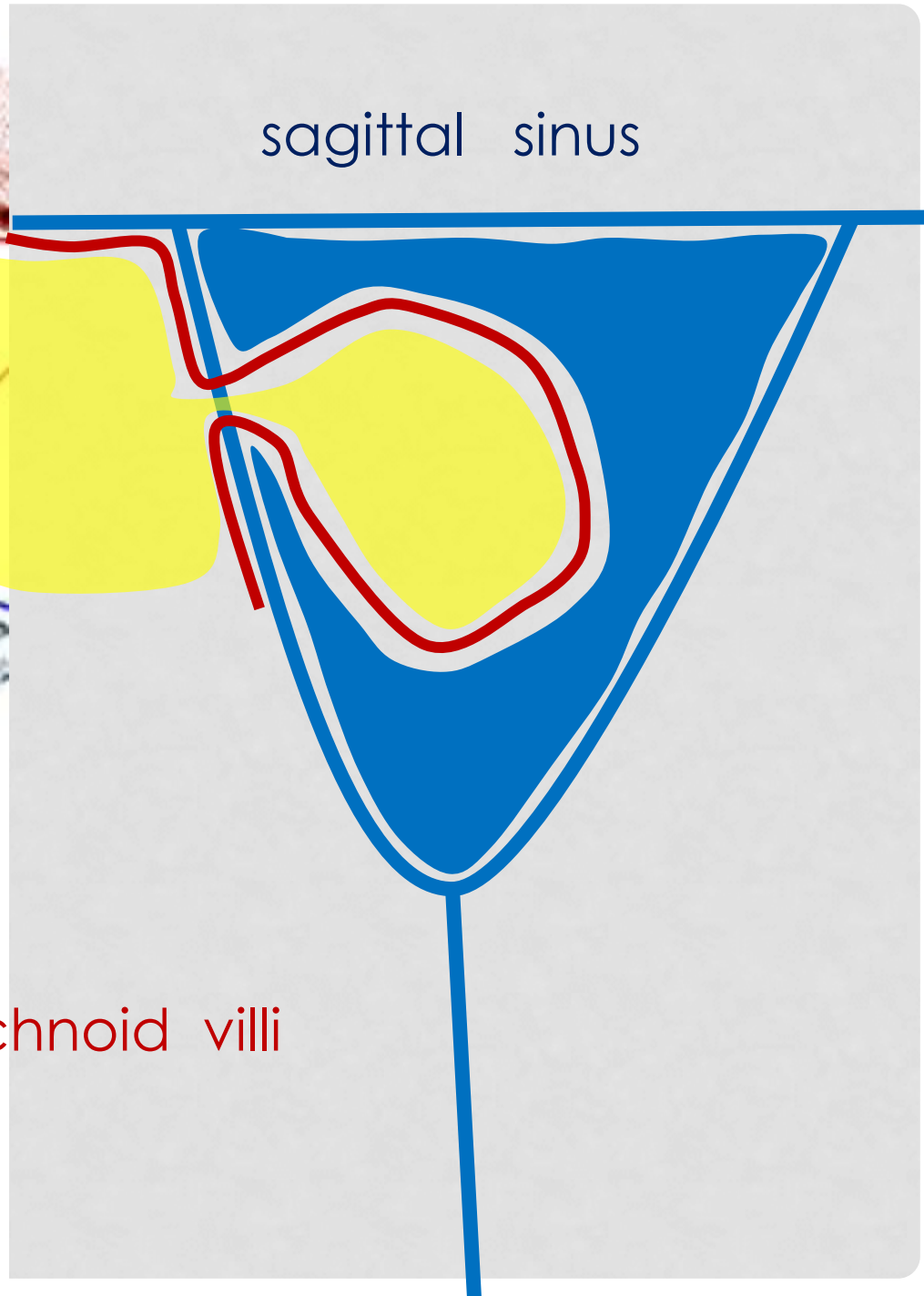
meninges + CSF →
protection



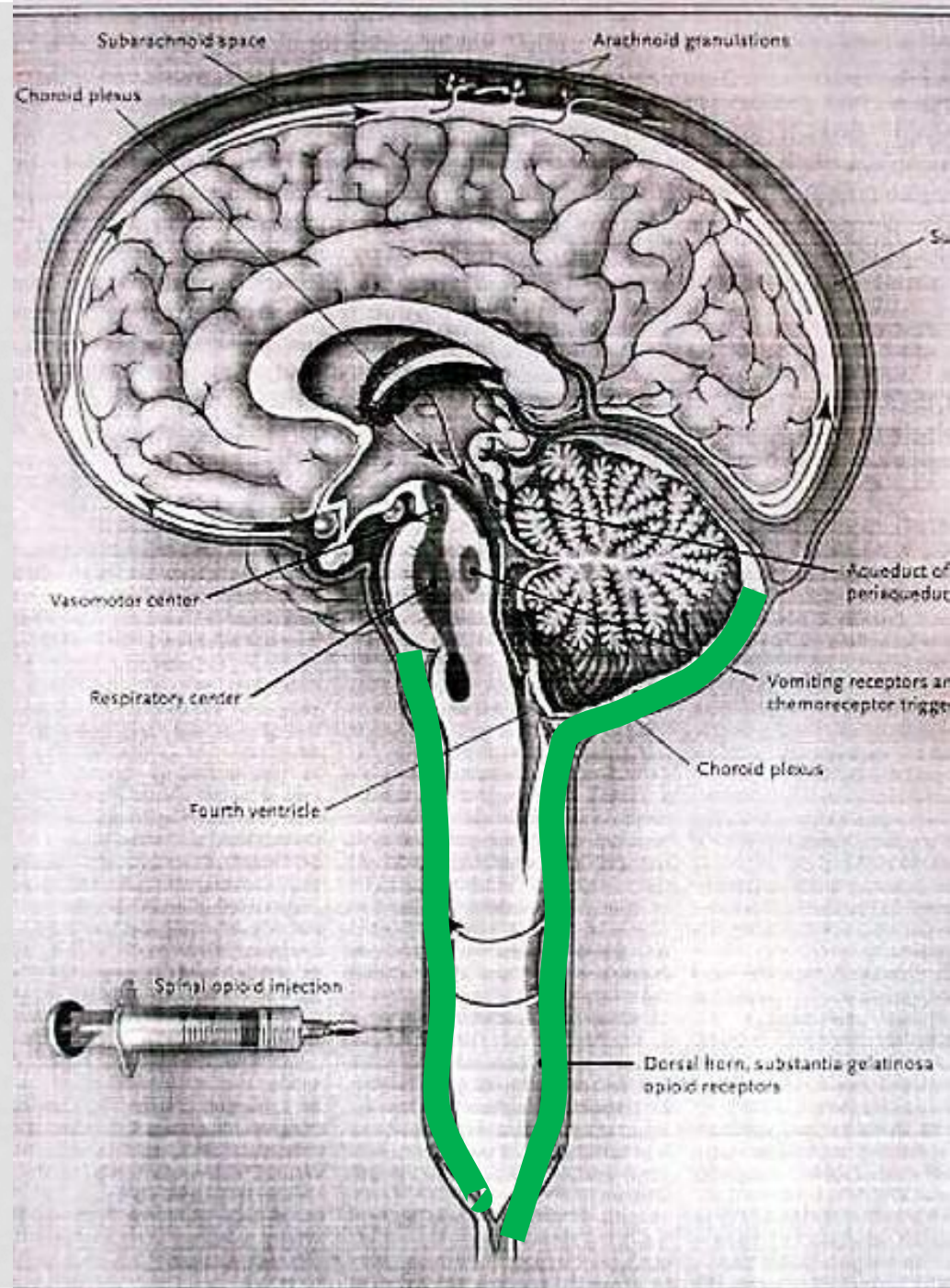


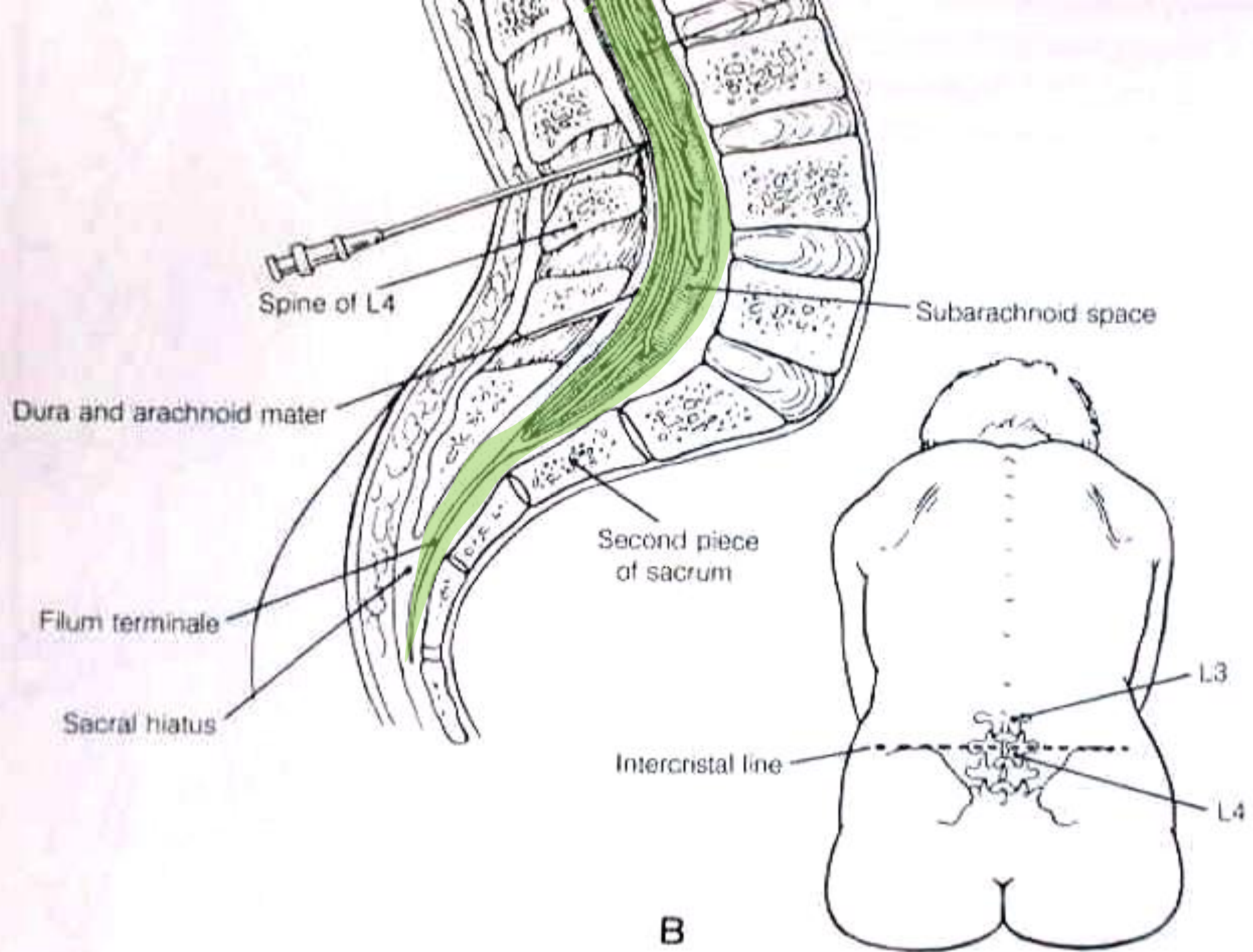
sagittal sinus

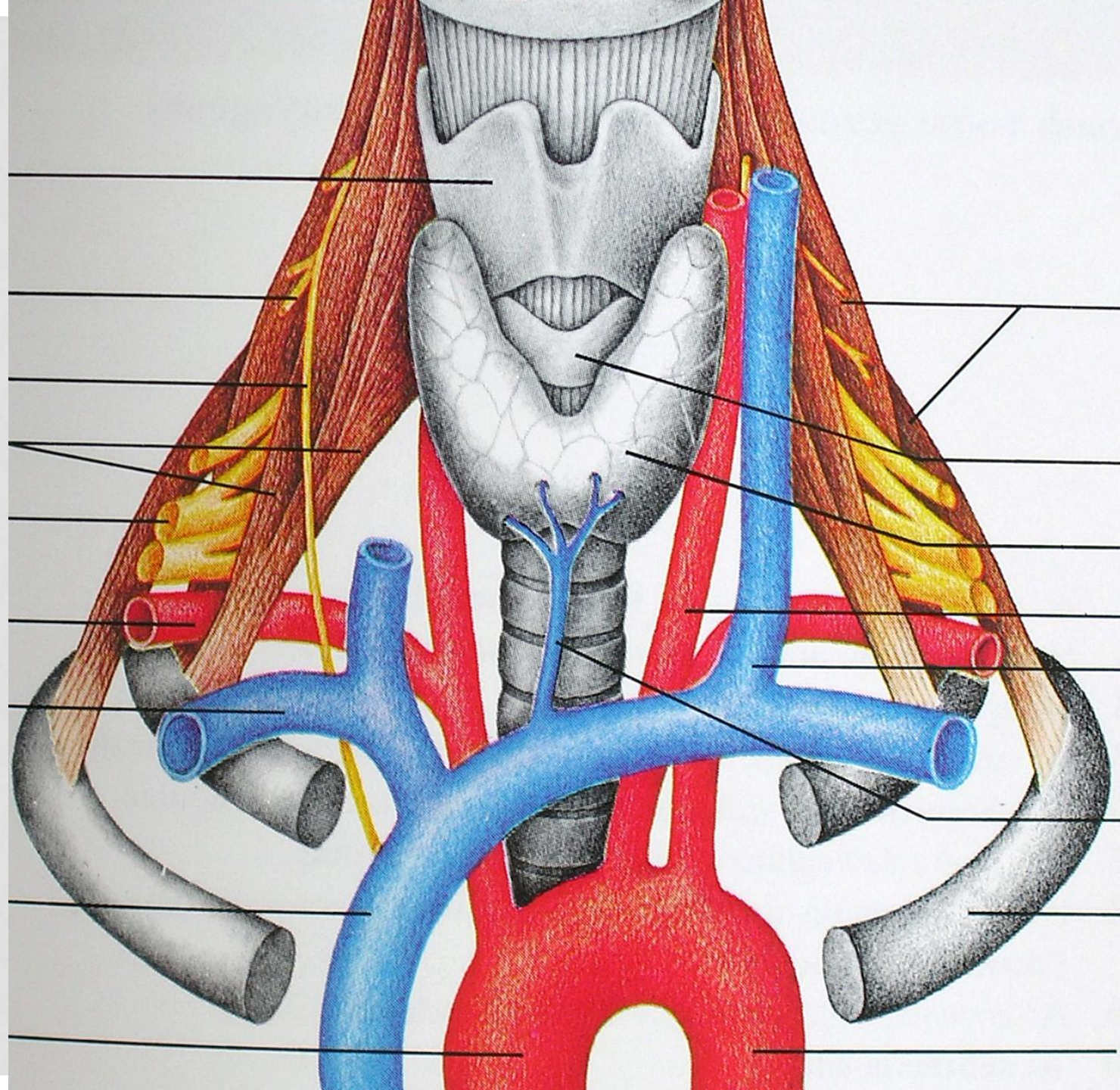
arachnoid villi

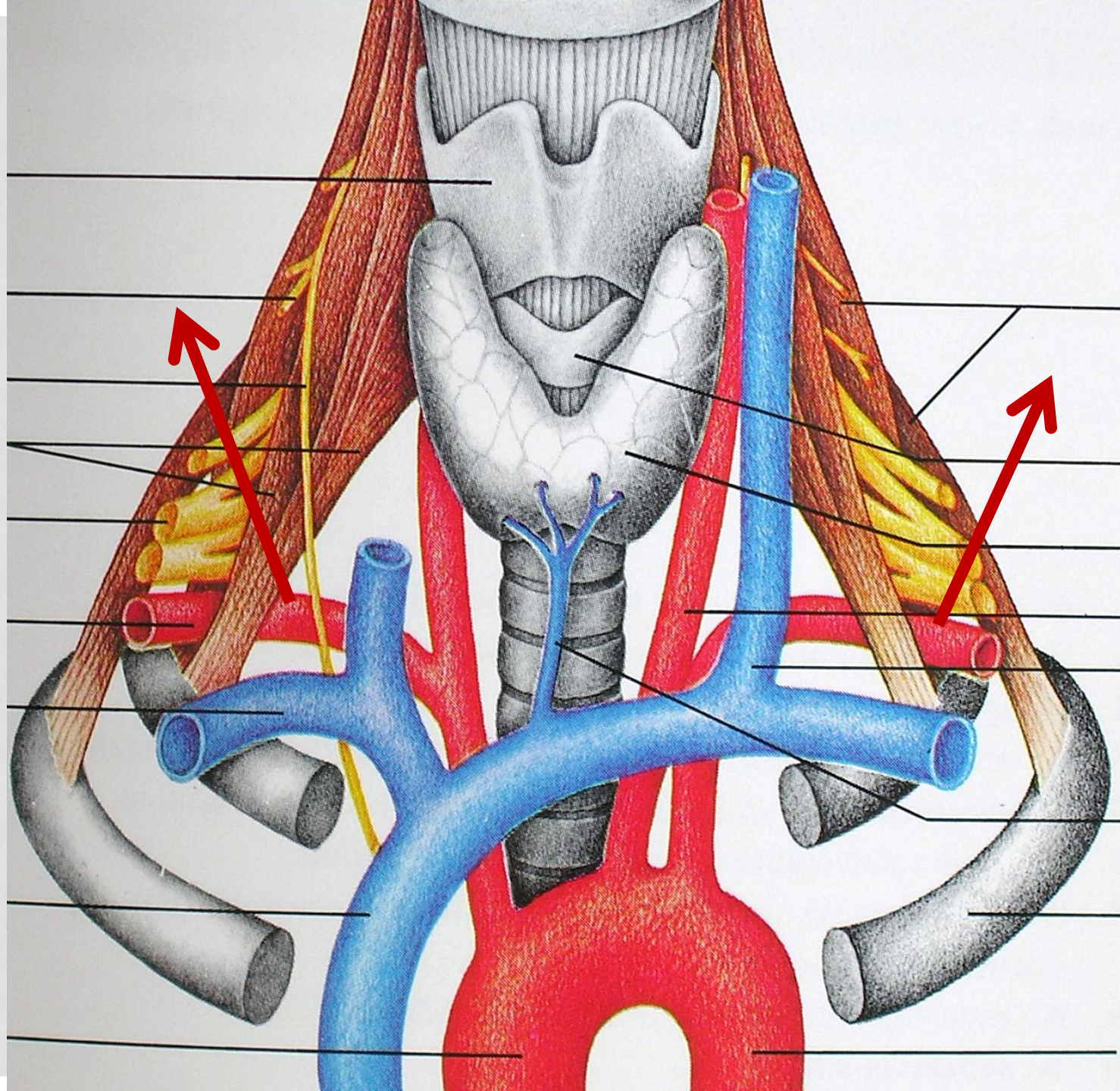


Lumbar puncture

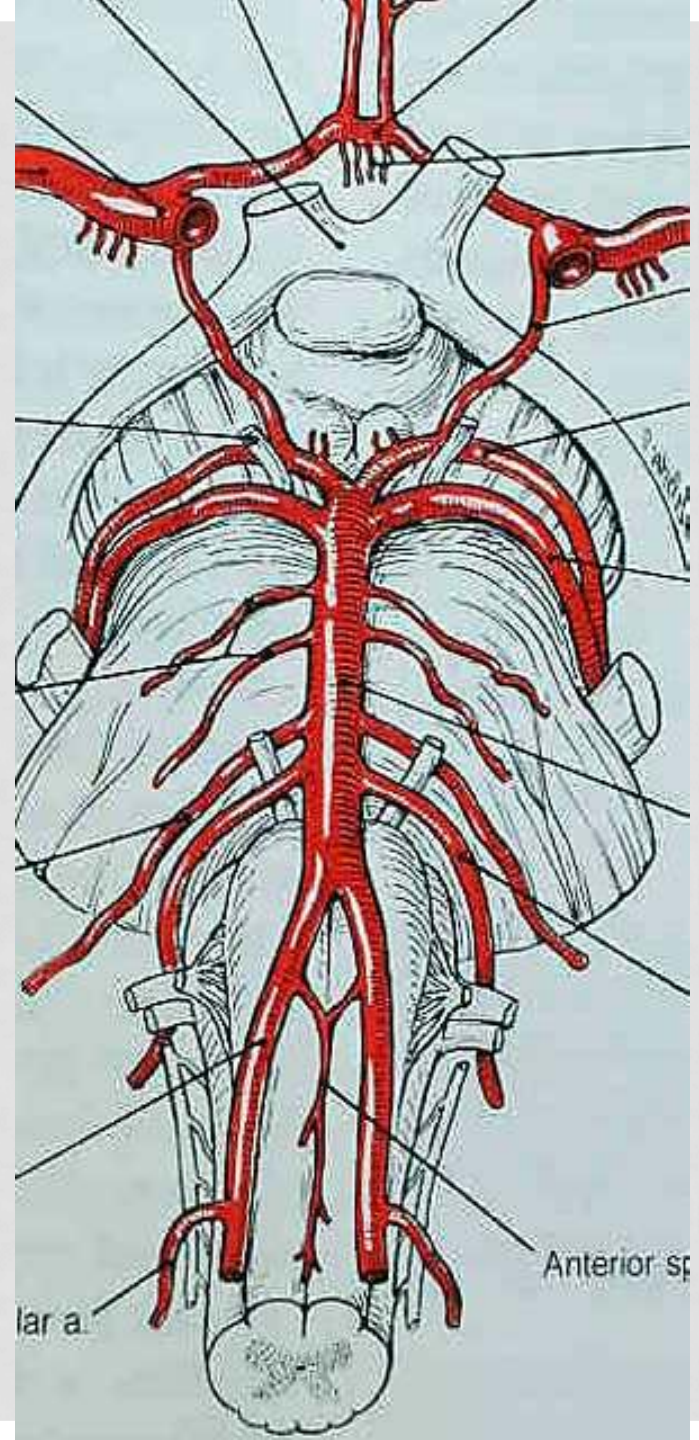
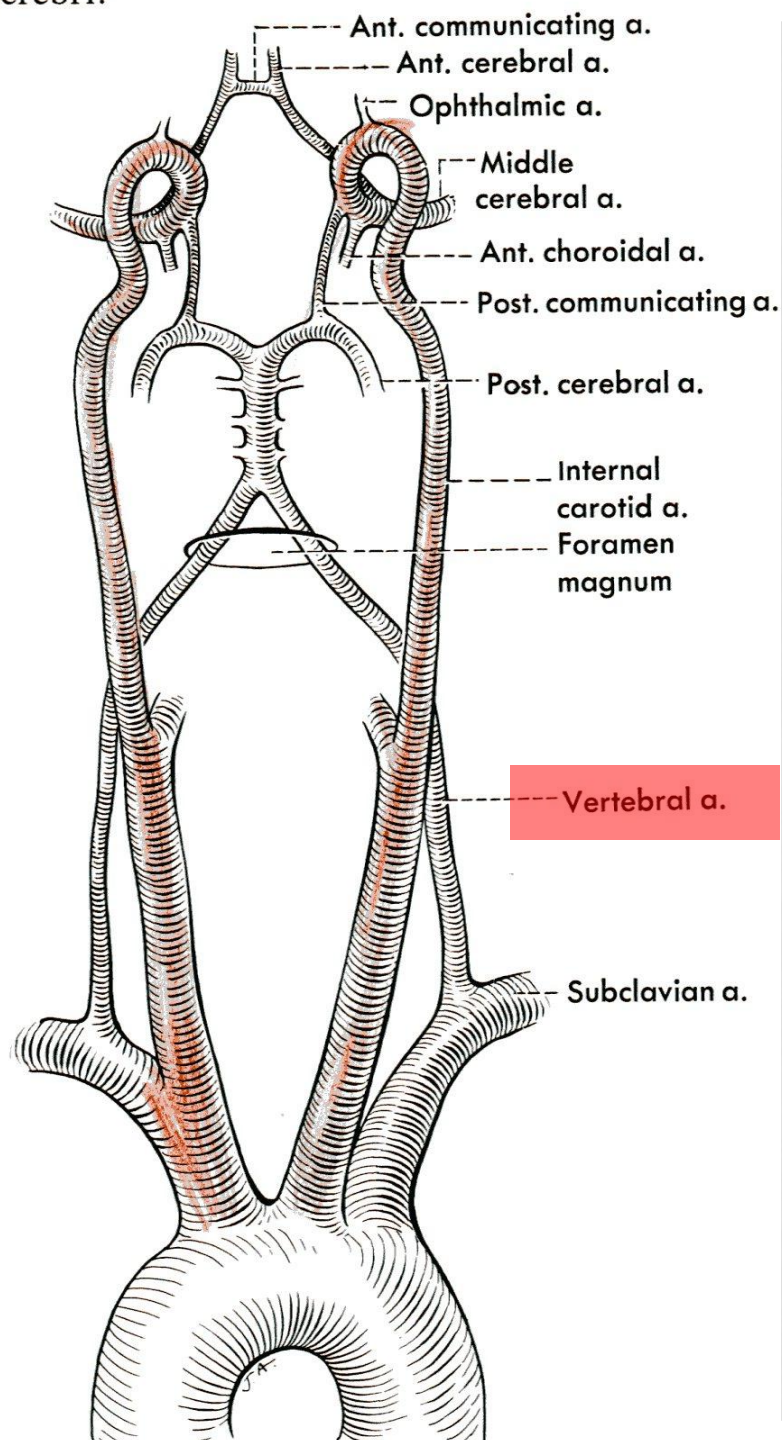


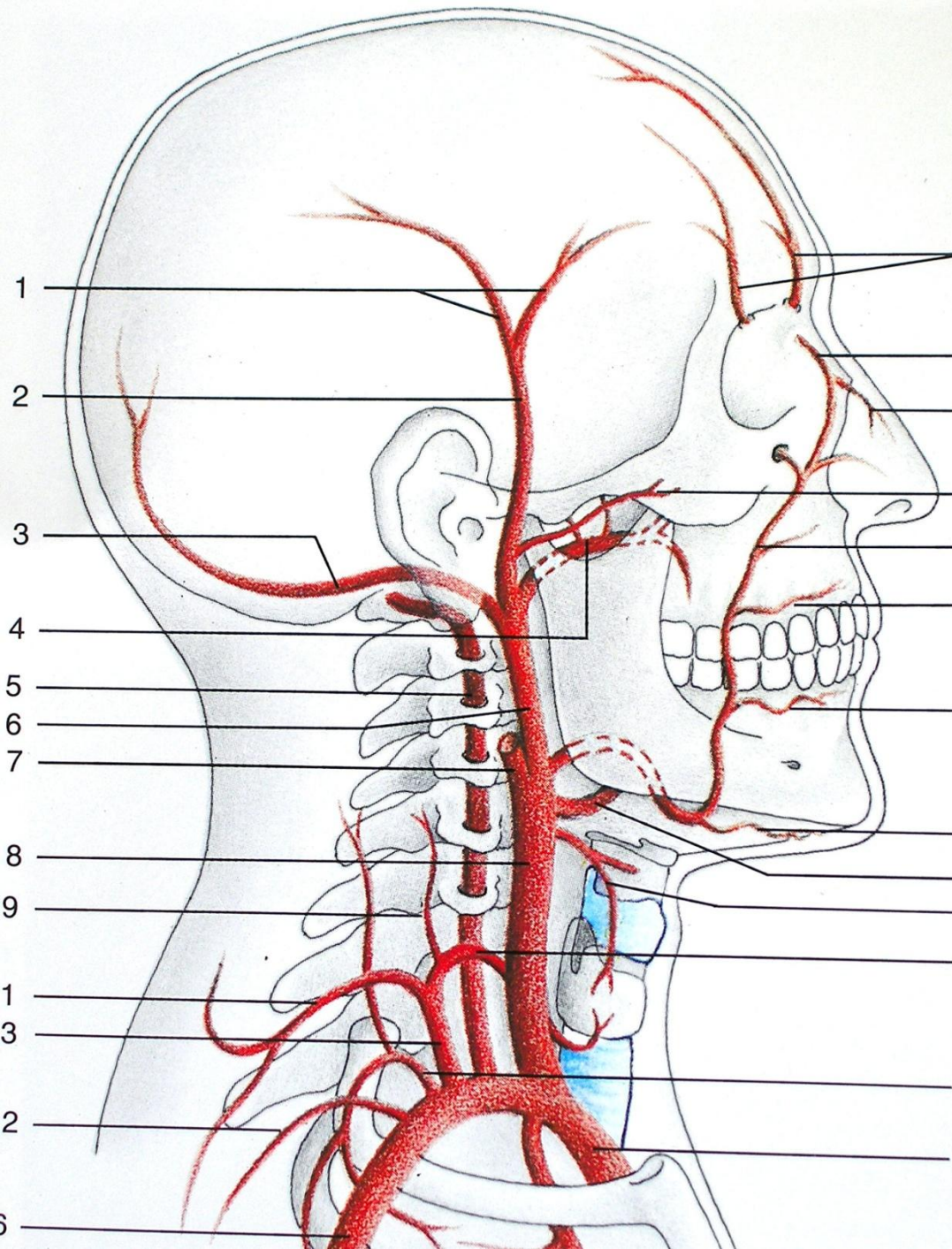






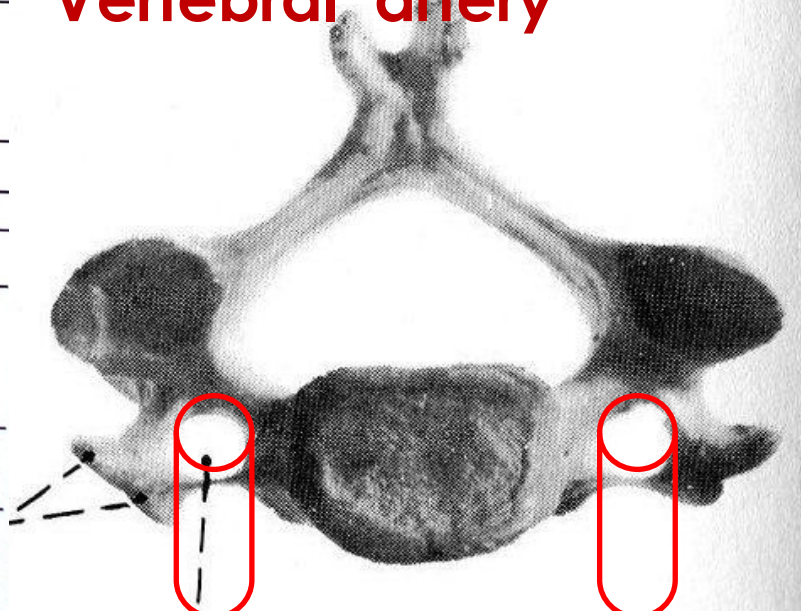
cerebrum.



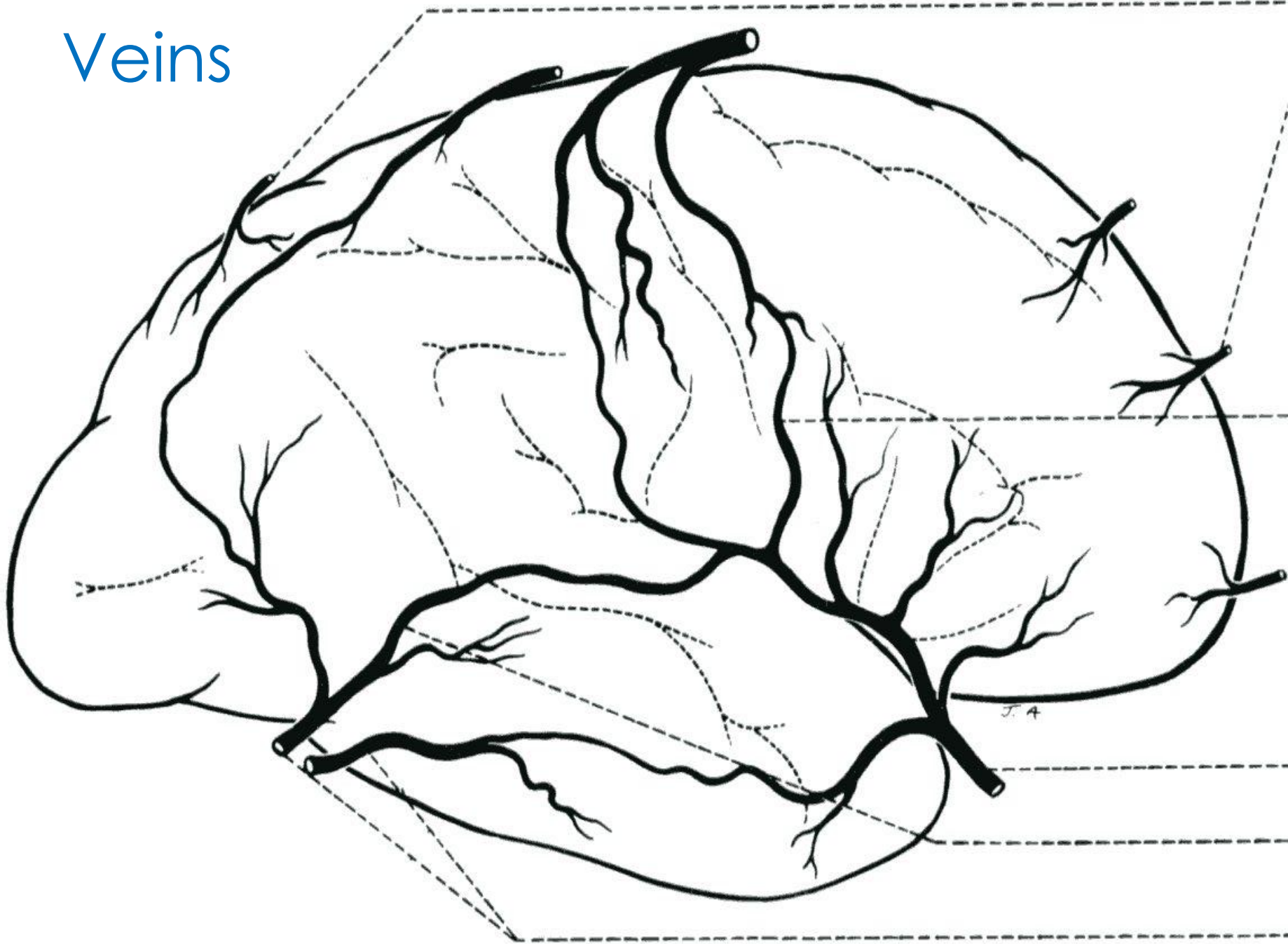


Middle meningeal artery

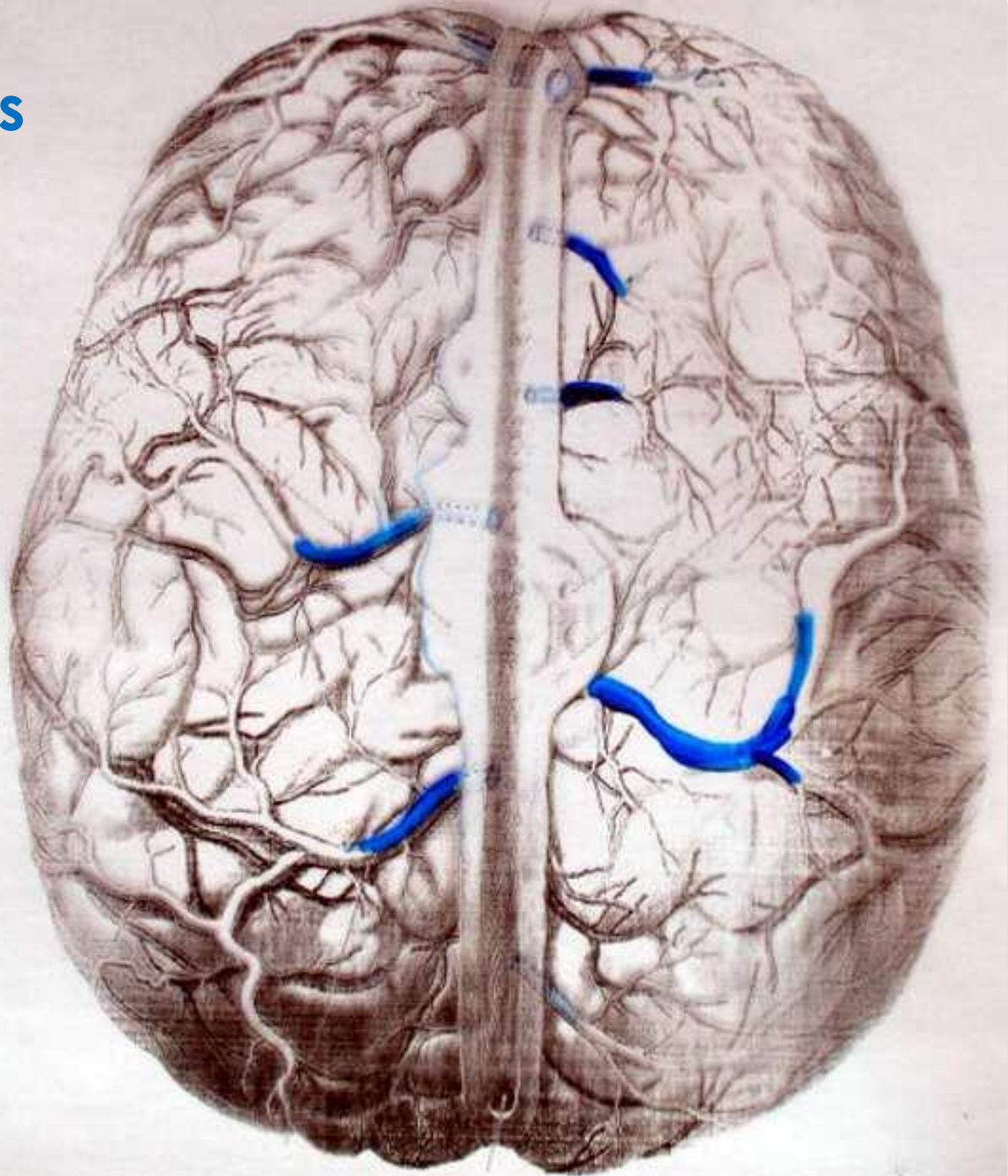
Vertebral artery



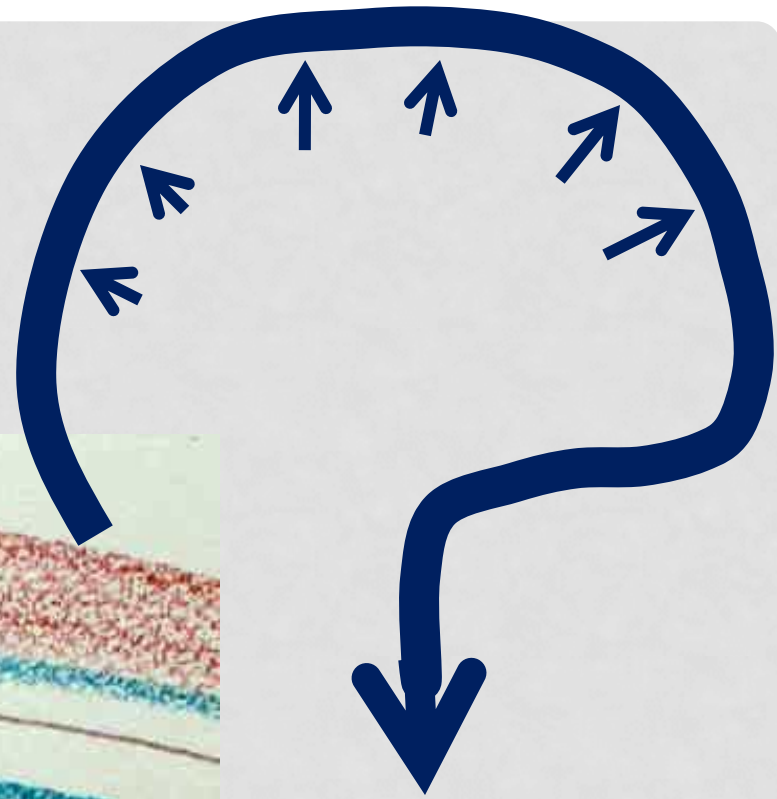
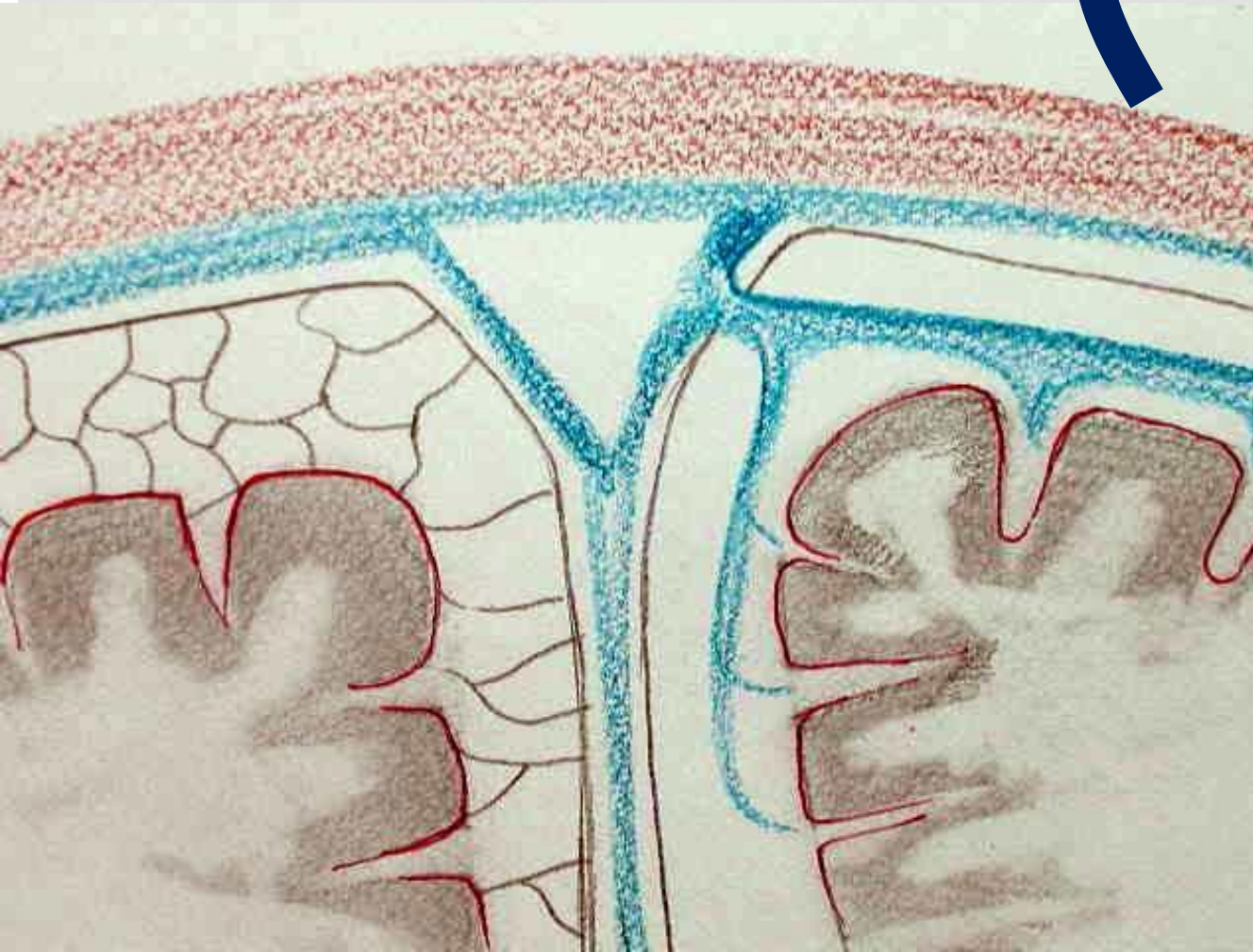
Veins

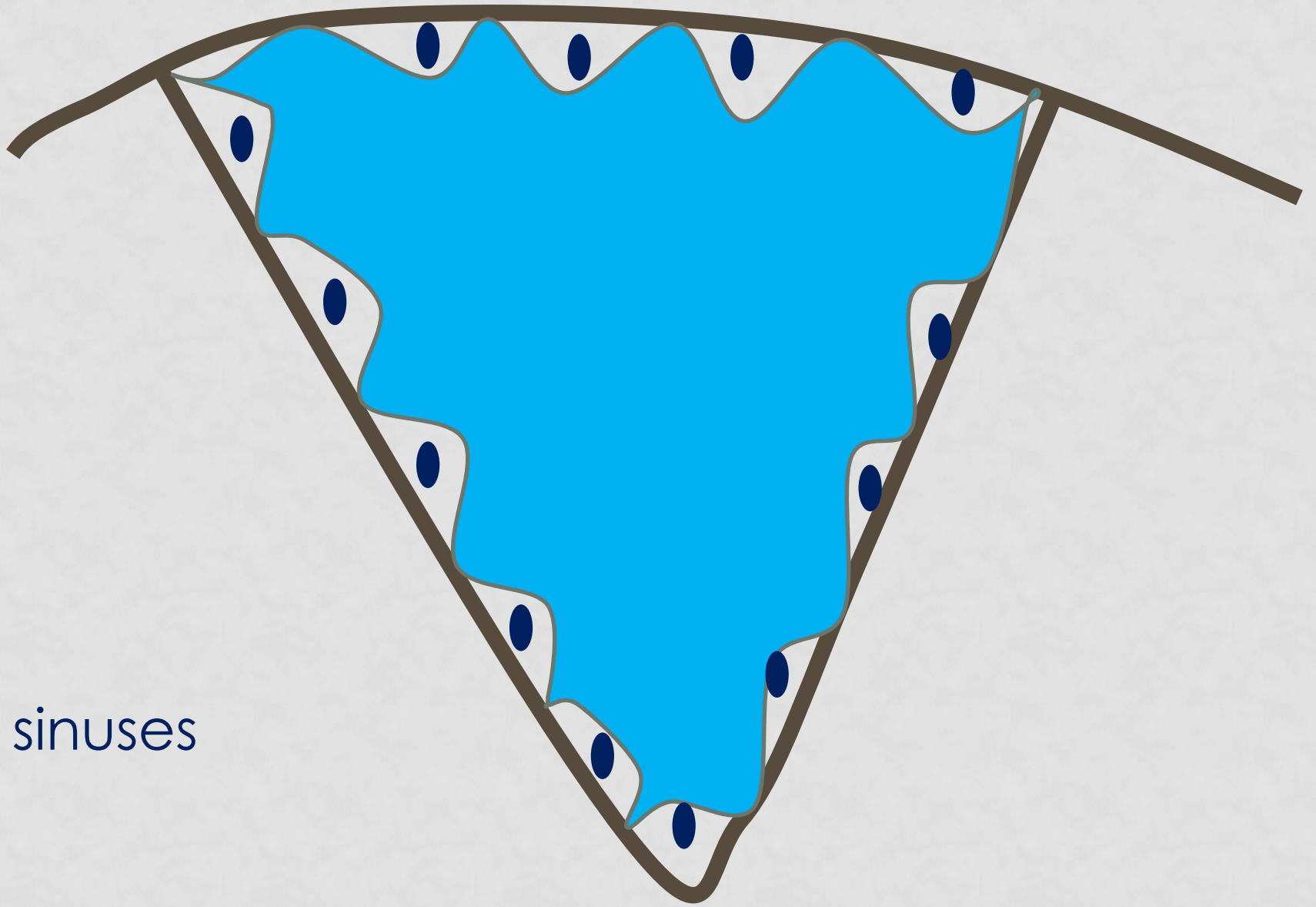


Bridging veins



superficial vv → bridging vv →
sinuses





sinuses

Superior sagittal sinus

Inferior sagittal sinus

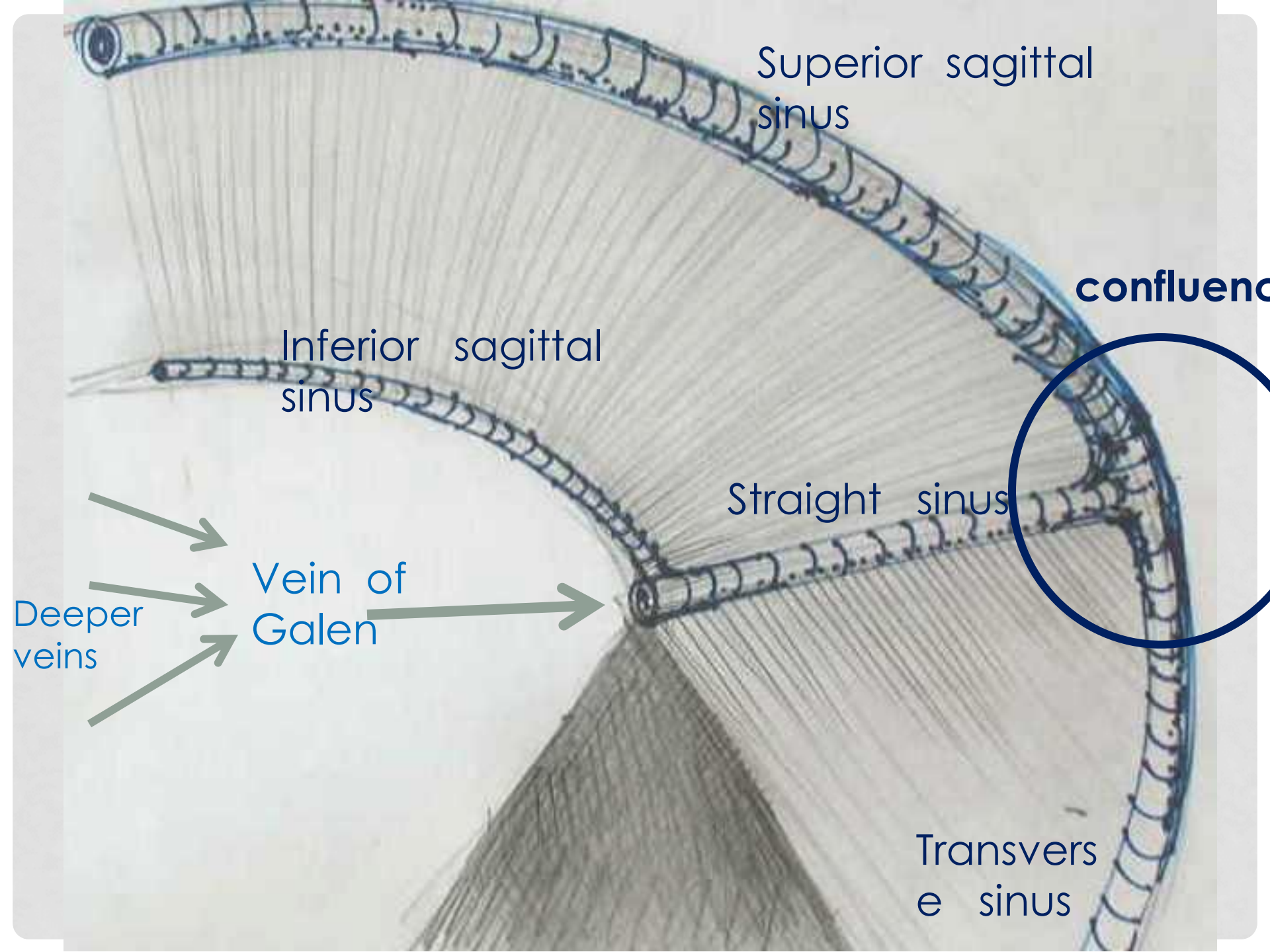
Straight sinus

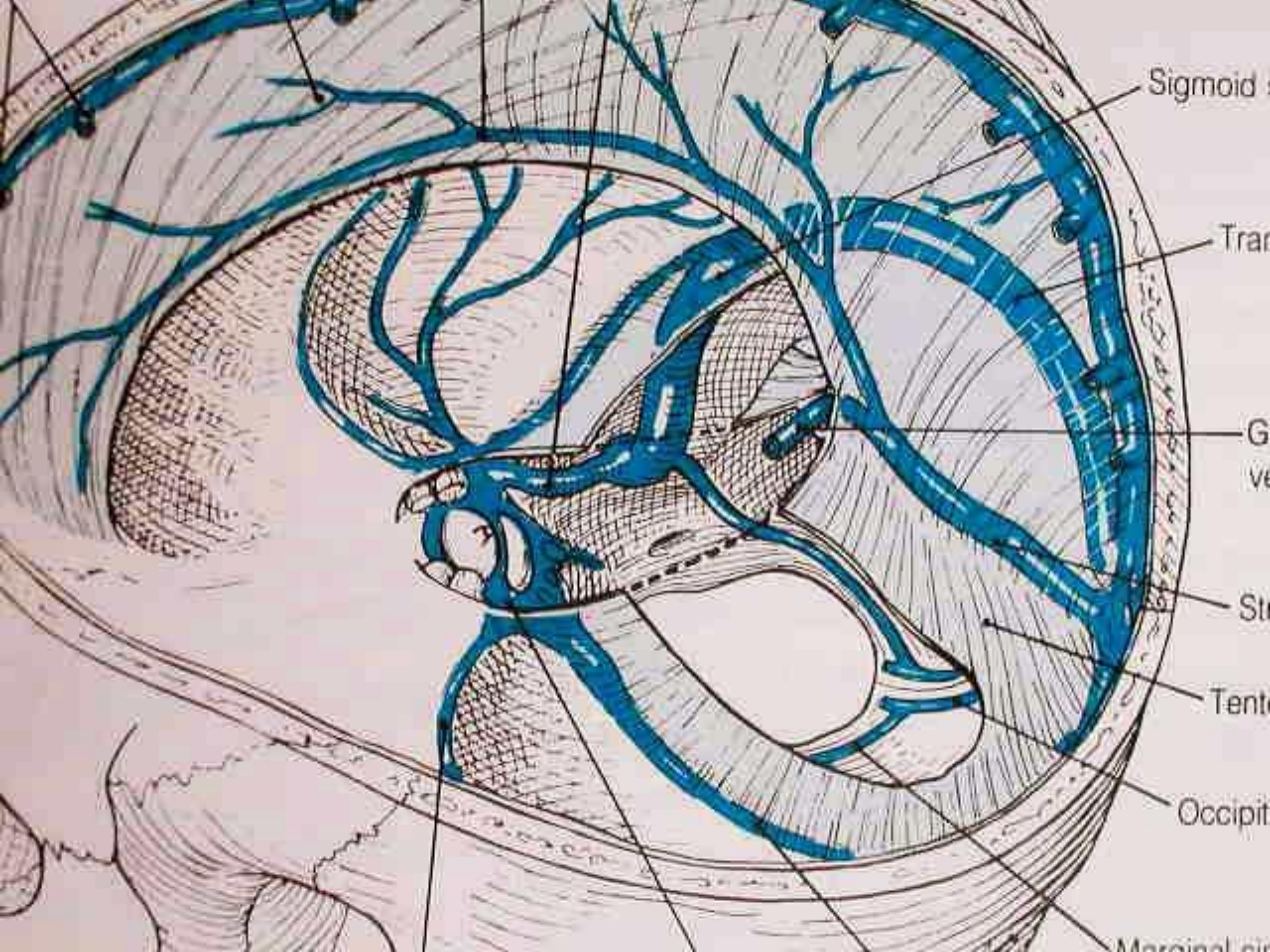
confluence

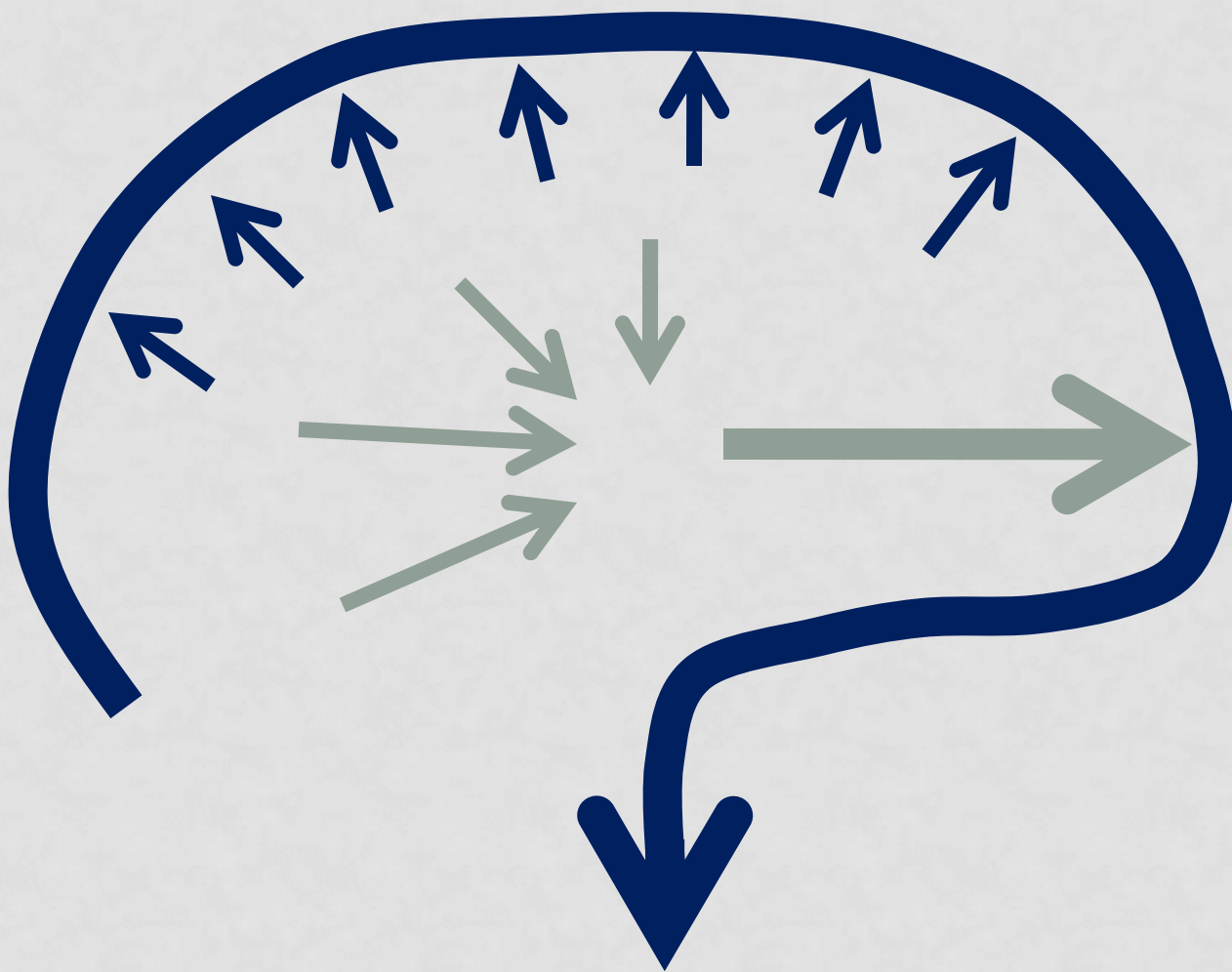
Vein of Galen

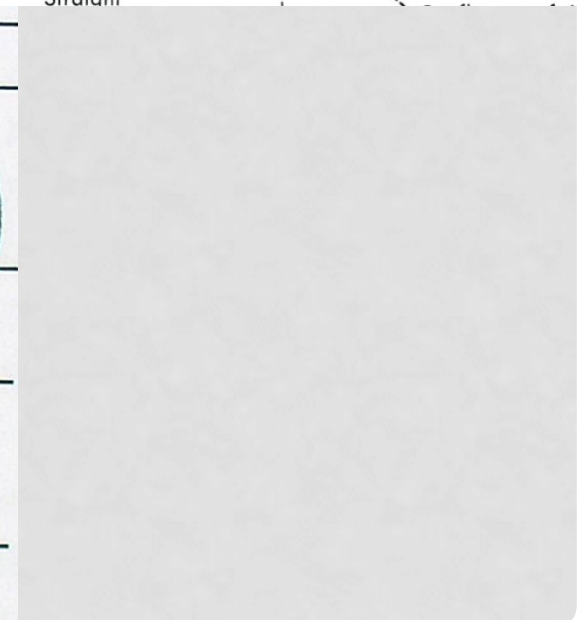
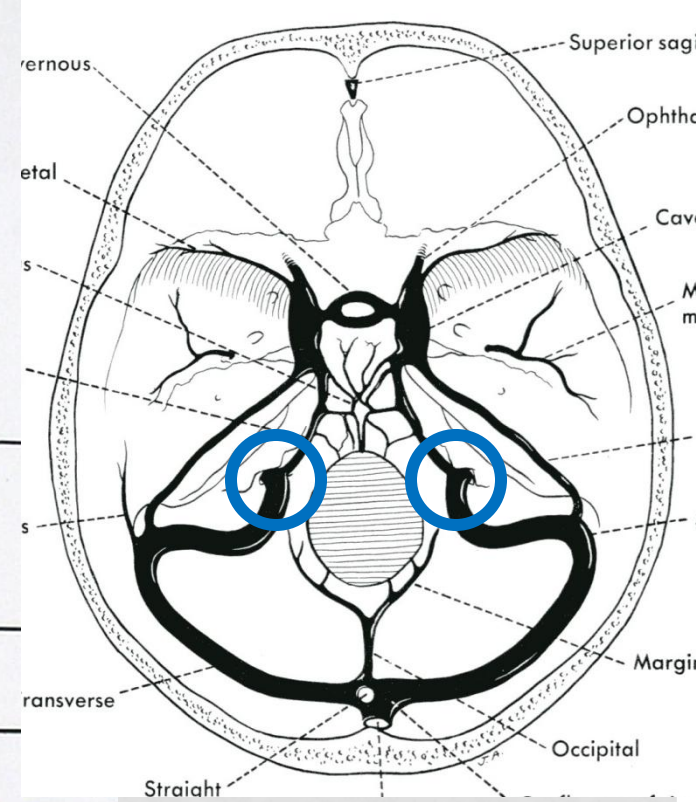
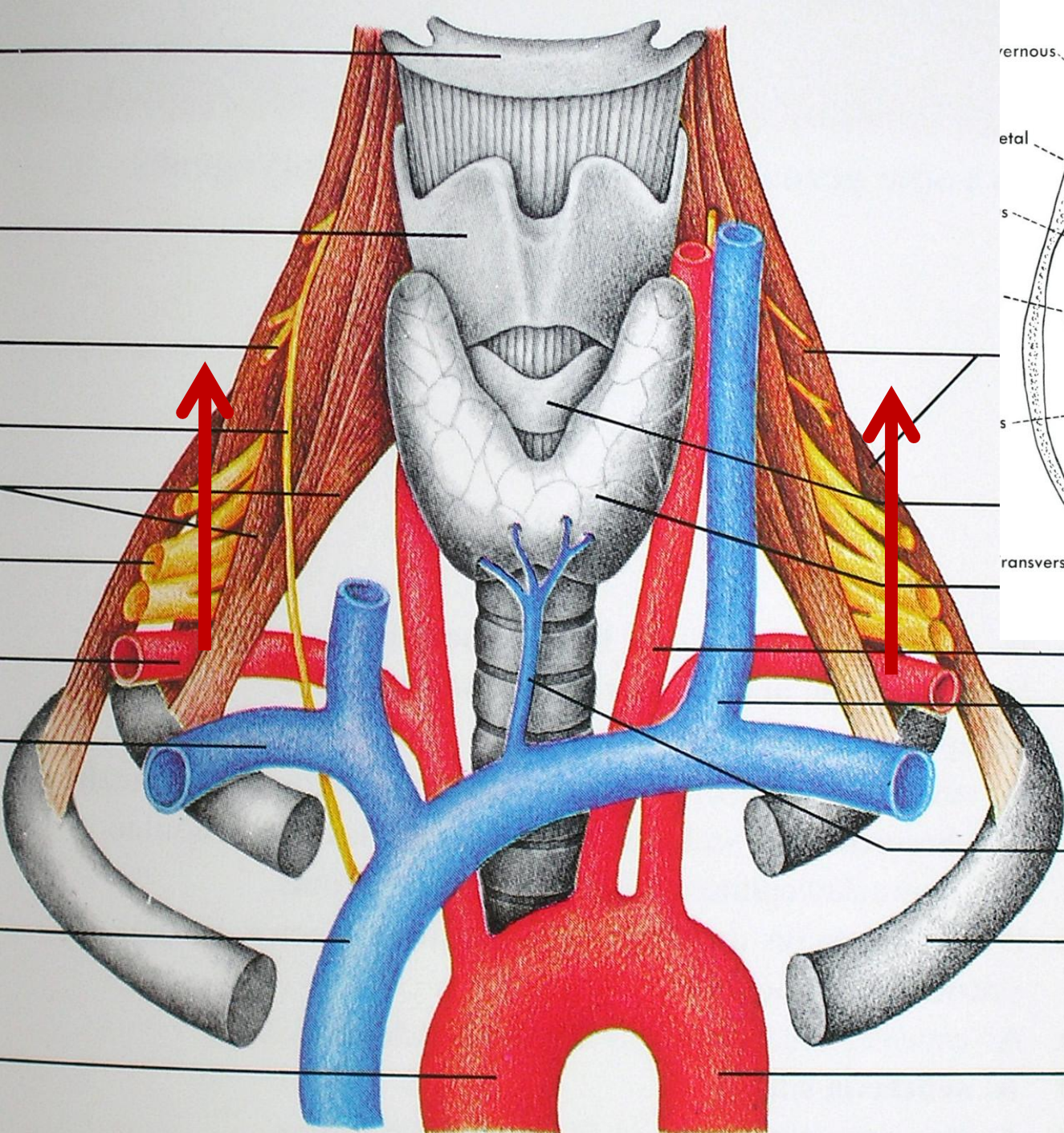
Deeper veins

Transverse sinus









CELLS

- The principal cells of the CNS are neurons, glia & the cells that compose the meninges & blood vessels

Neurons

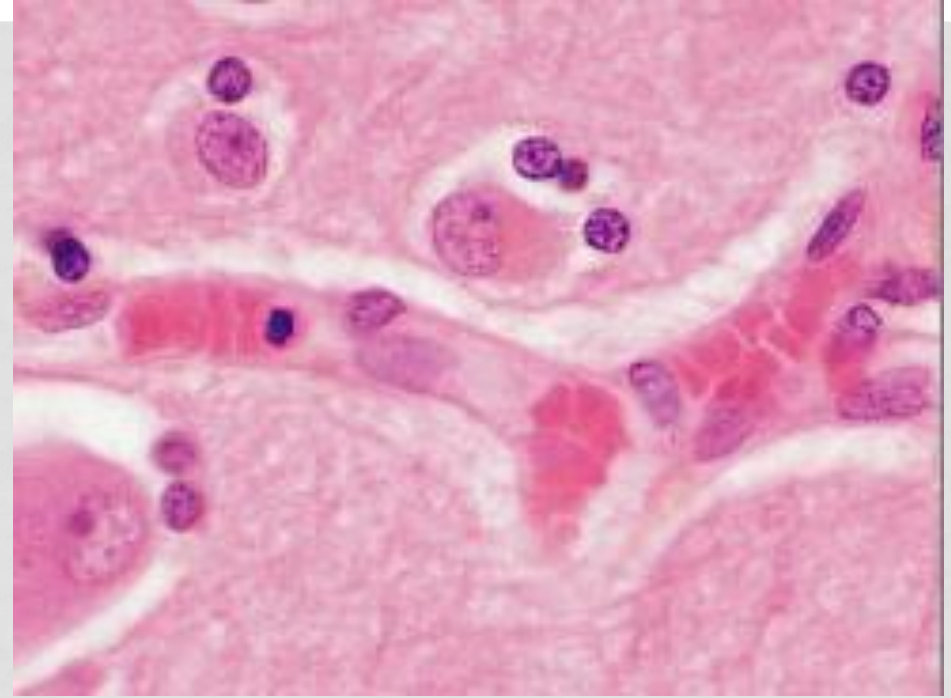
- Are the structural & functional unit of the CNS.
- Are highly specialized post-mitotic cells which can't be replaced after cell death.

Glia

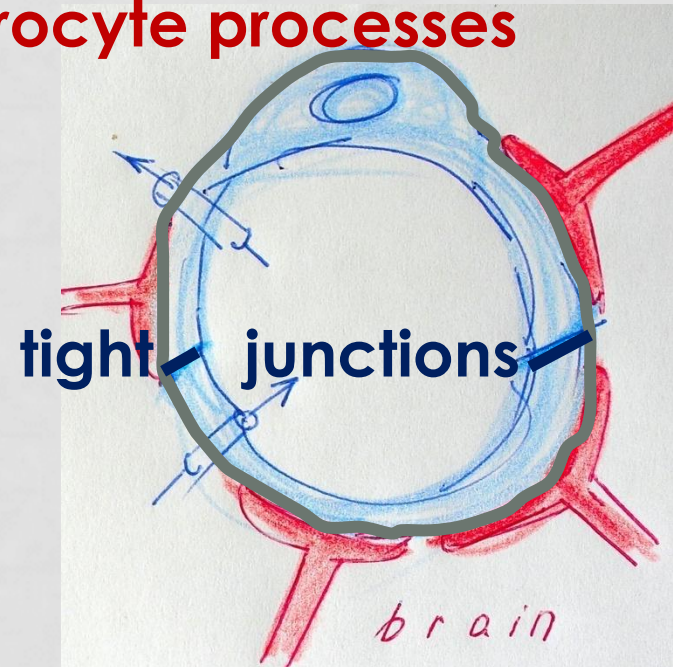
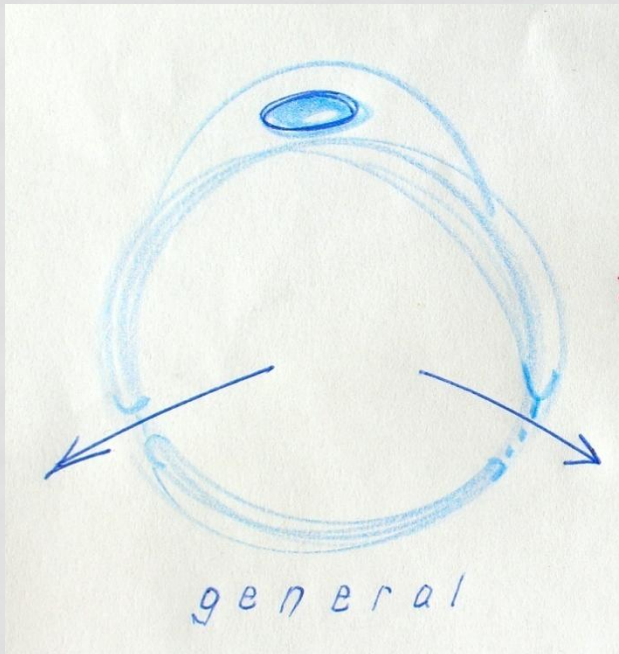
- Glial cells are derived from neuroectoderm & bone marrow. They have role in normal functions & reactions to injury. Include astrocyte, oligodendrocyte, microglia & ependymal cells

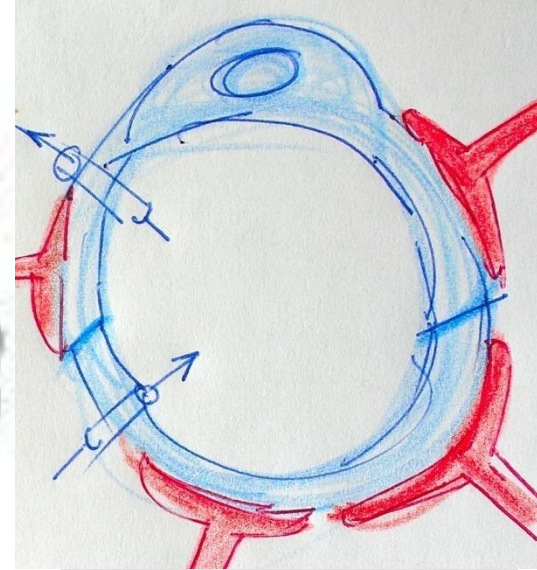
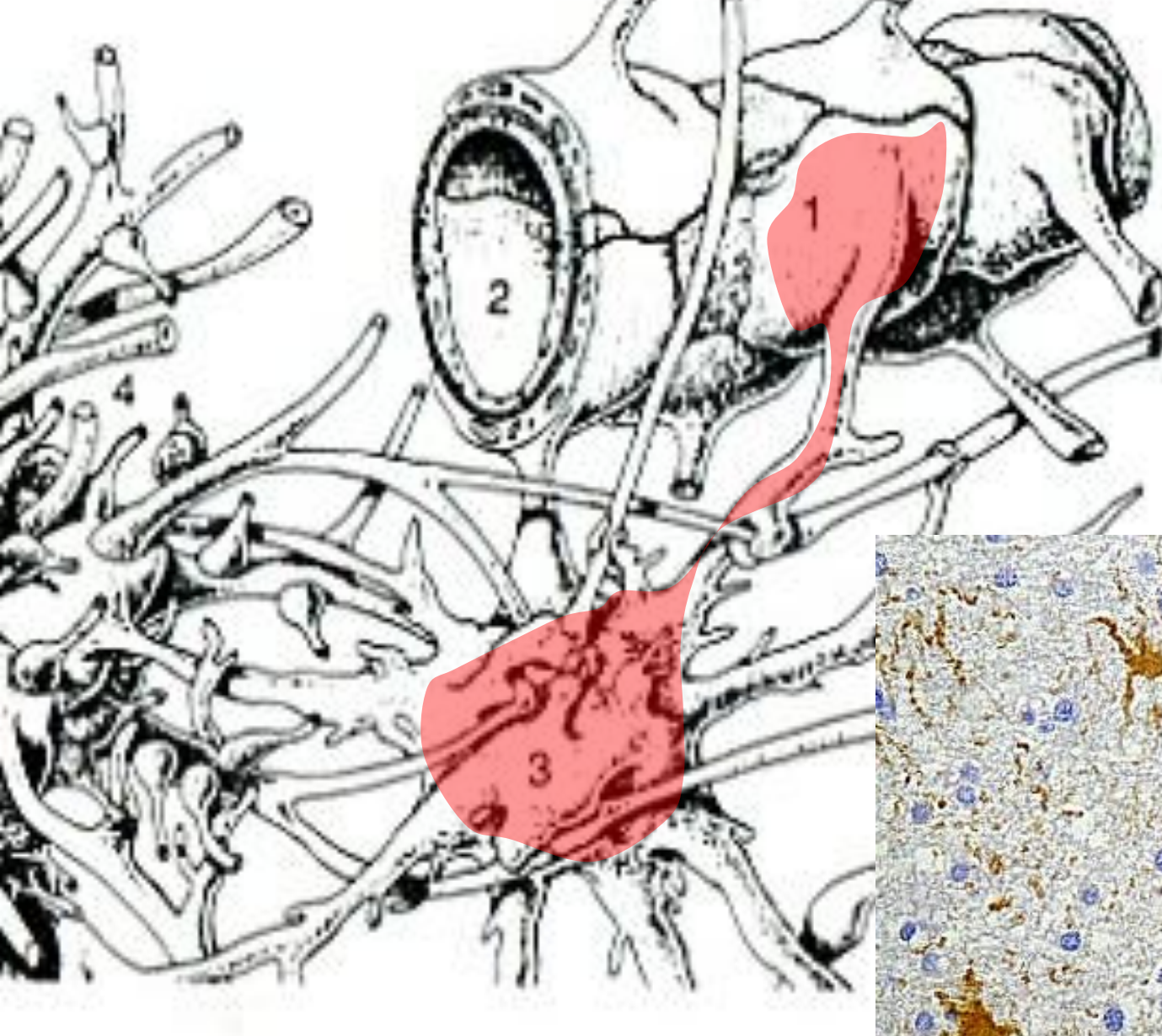
CELLS OF THE NERVOUS SYSTEM

1. Neuron: functions in neural transmission, most vulnerable cell, limited regeneration
2. Astrocyte: major supporting & reactive cell of CNS forms 'scar'
3. Oligodendrocyte: highly vulnerable, limited proliferation, forms myelin sheath
4. Ependymal cell: vulnerable, limited regeneration, lines ventricles (ependymal granulations)
5. Microglial cell: monocyte/macrophage (bone marrow) derived phagocytic cell, antigen presentation, producer of cytokines, inflammatory cell

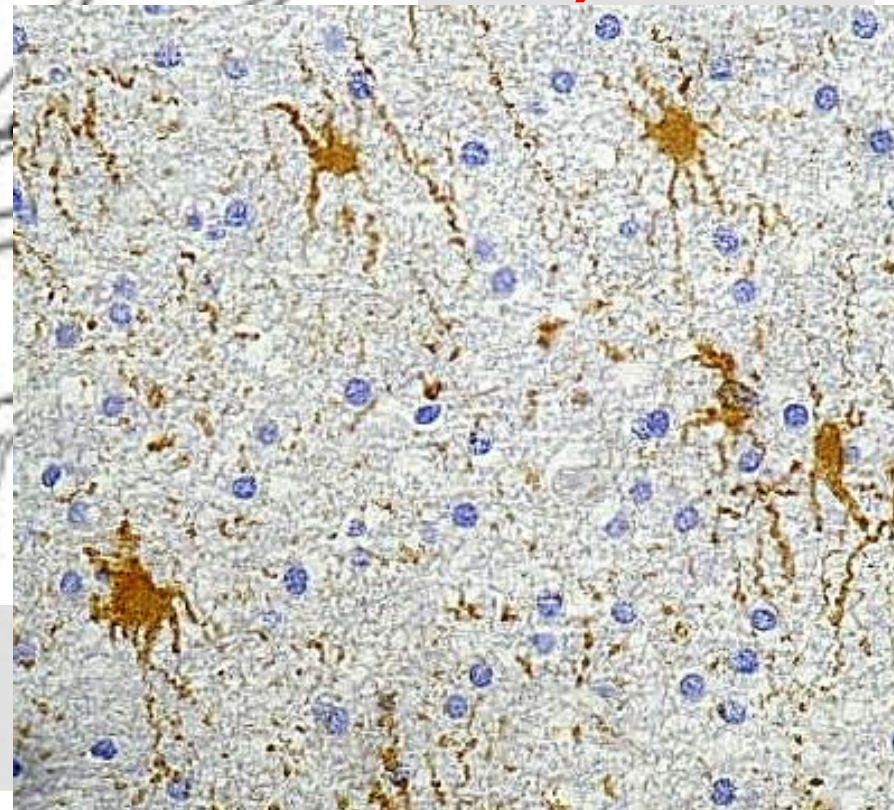


astrocyte processes

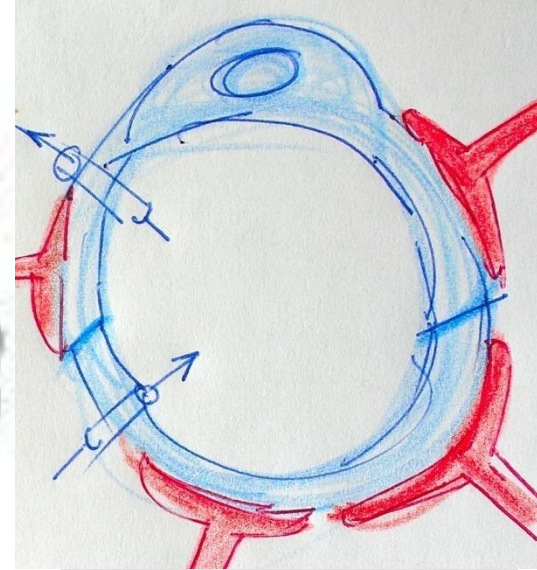
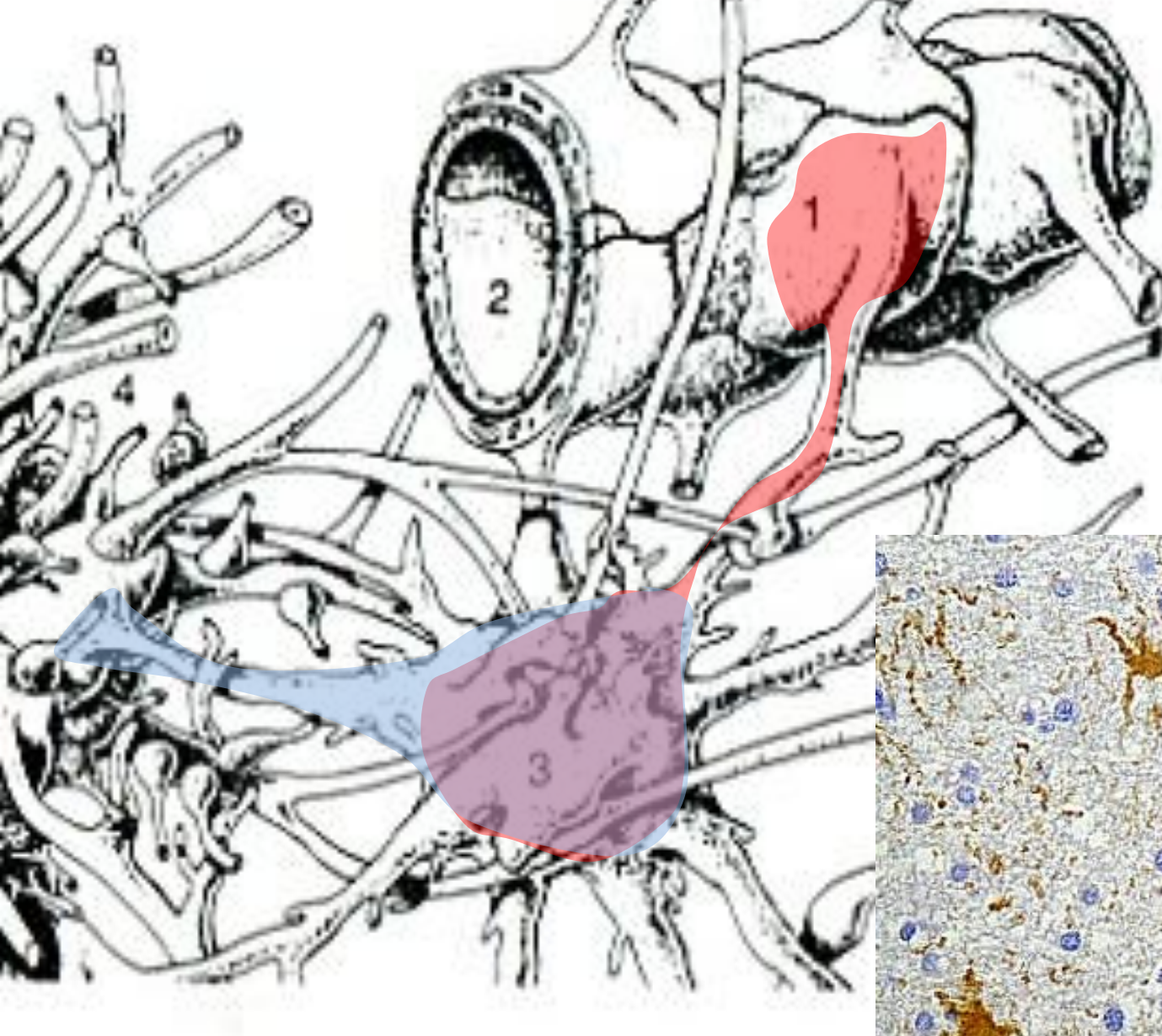




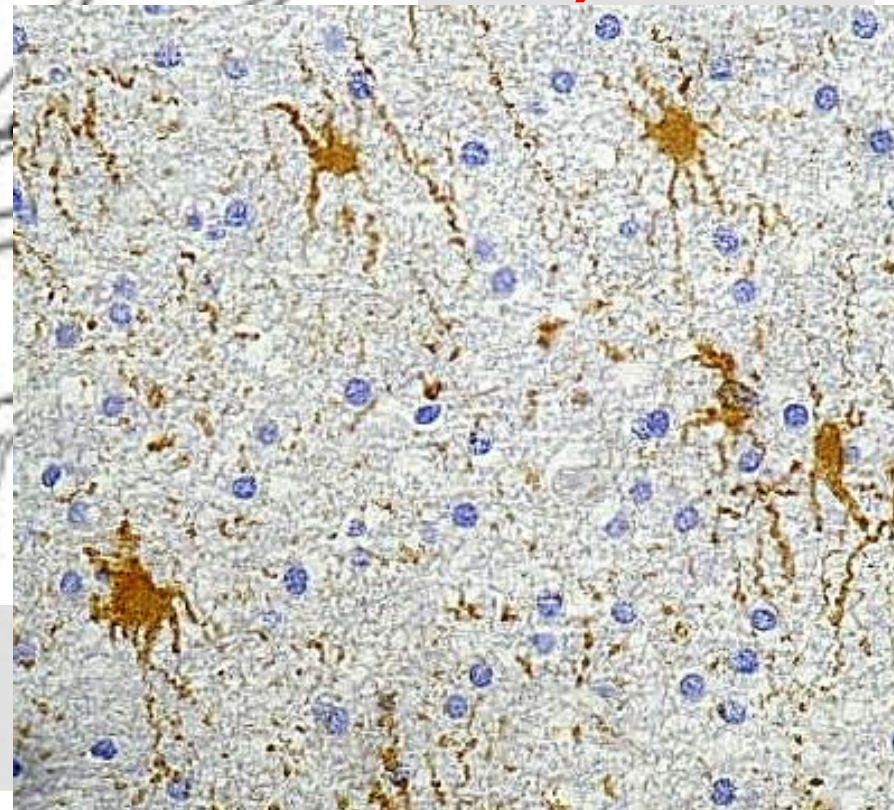
astrocyte



Blood-brain barrier BBB

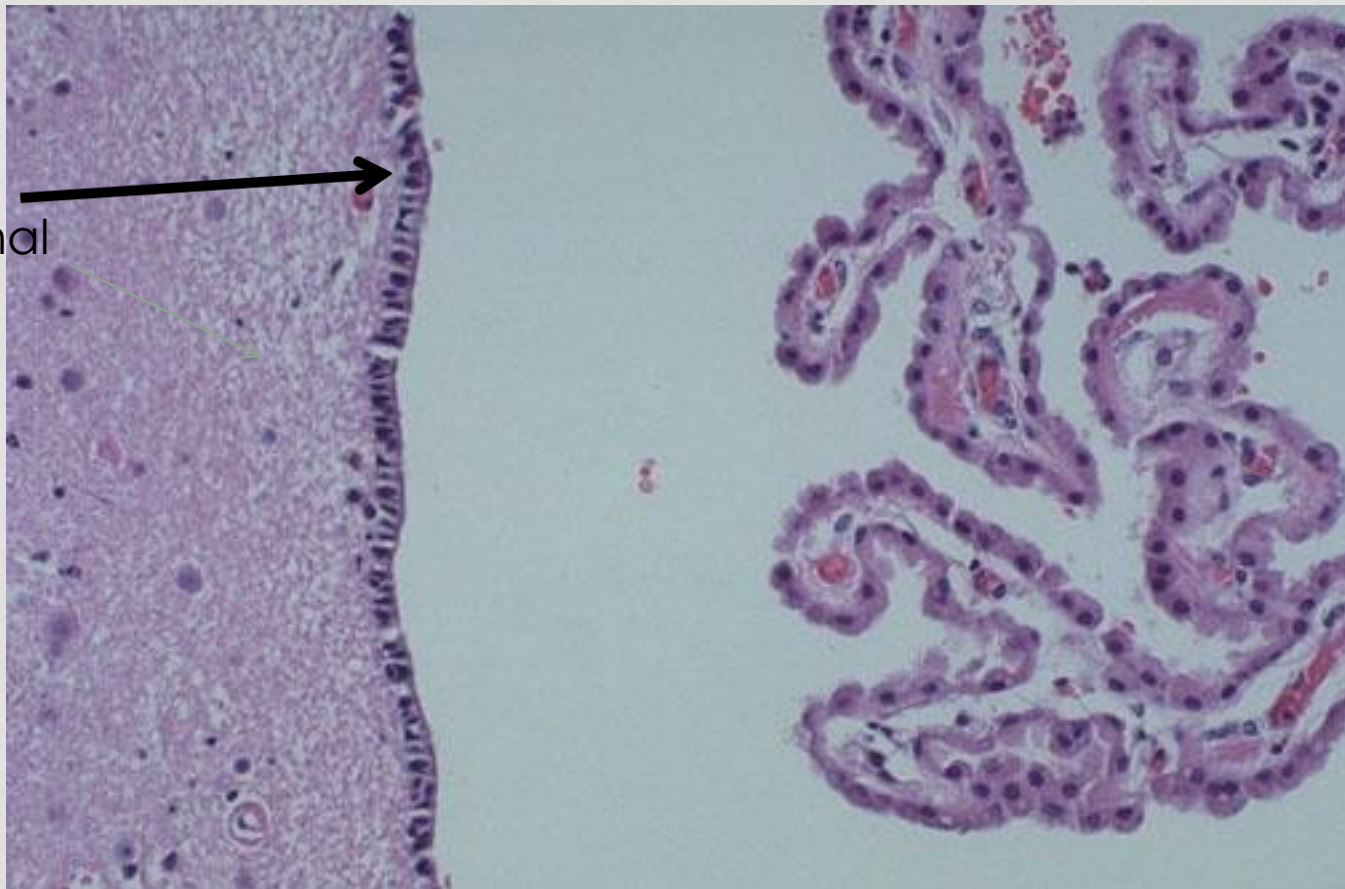


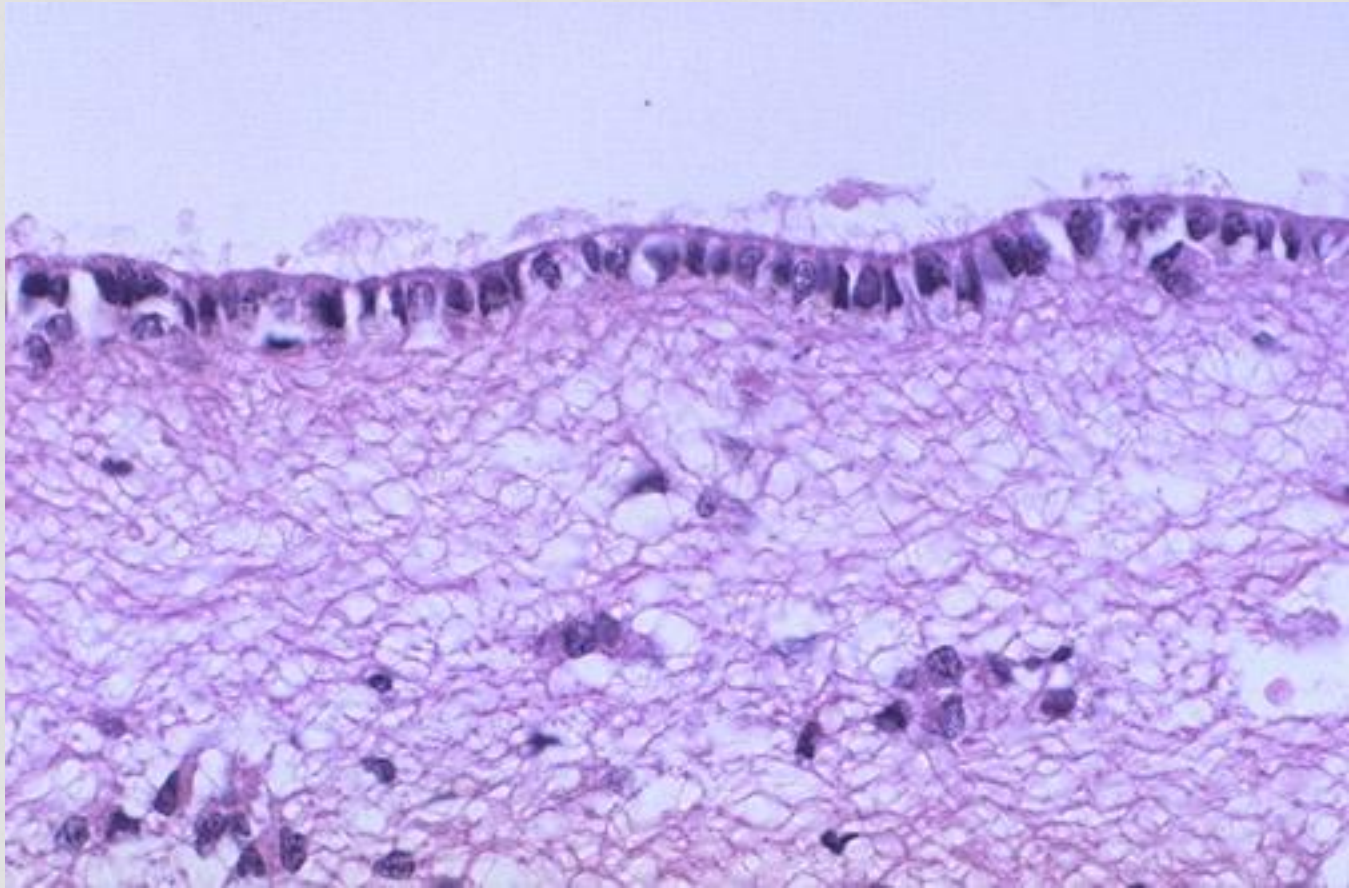
astrocyte

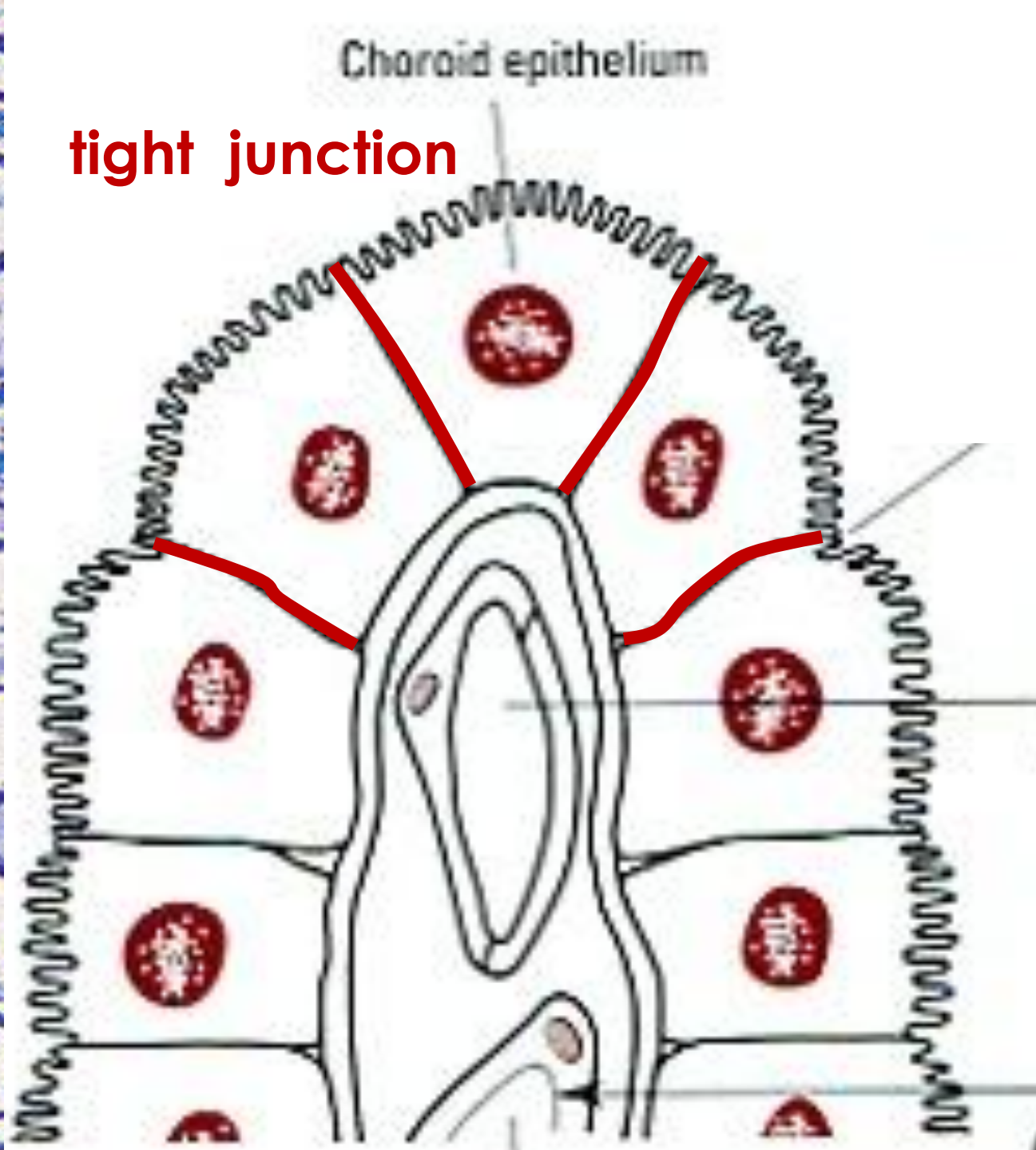
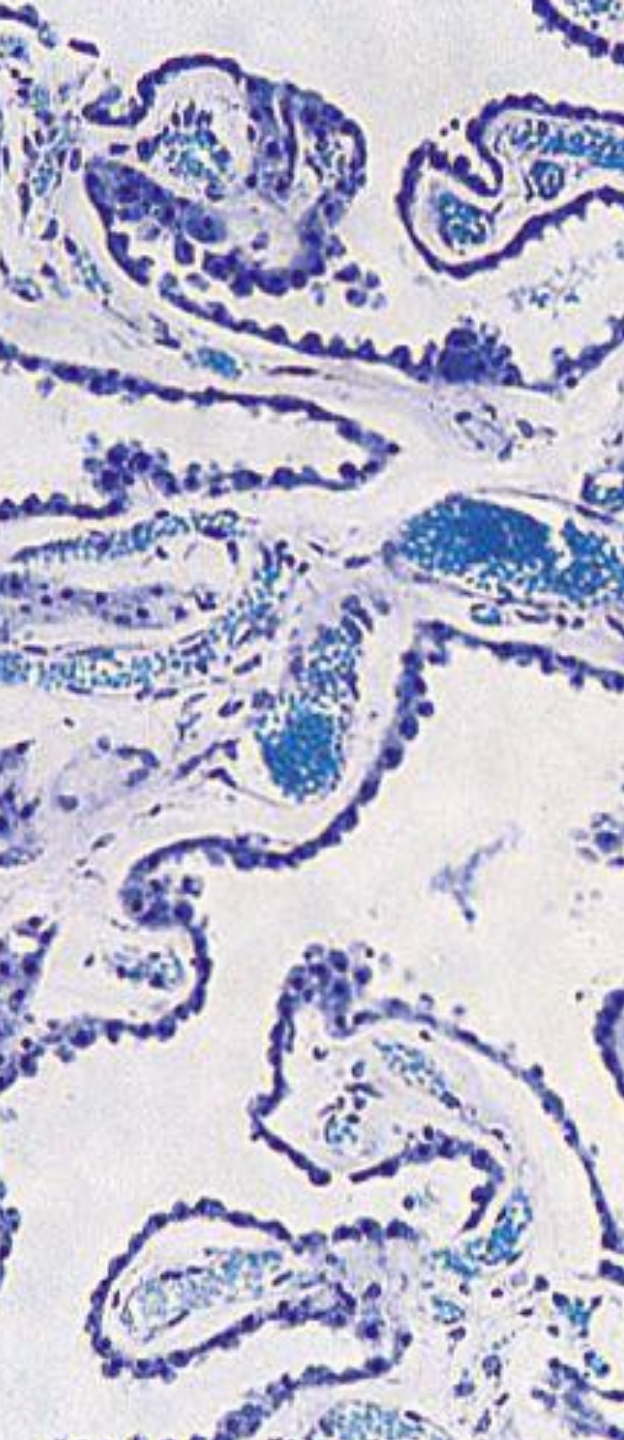


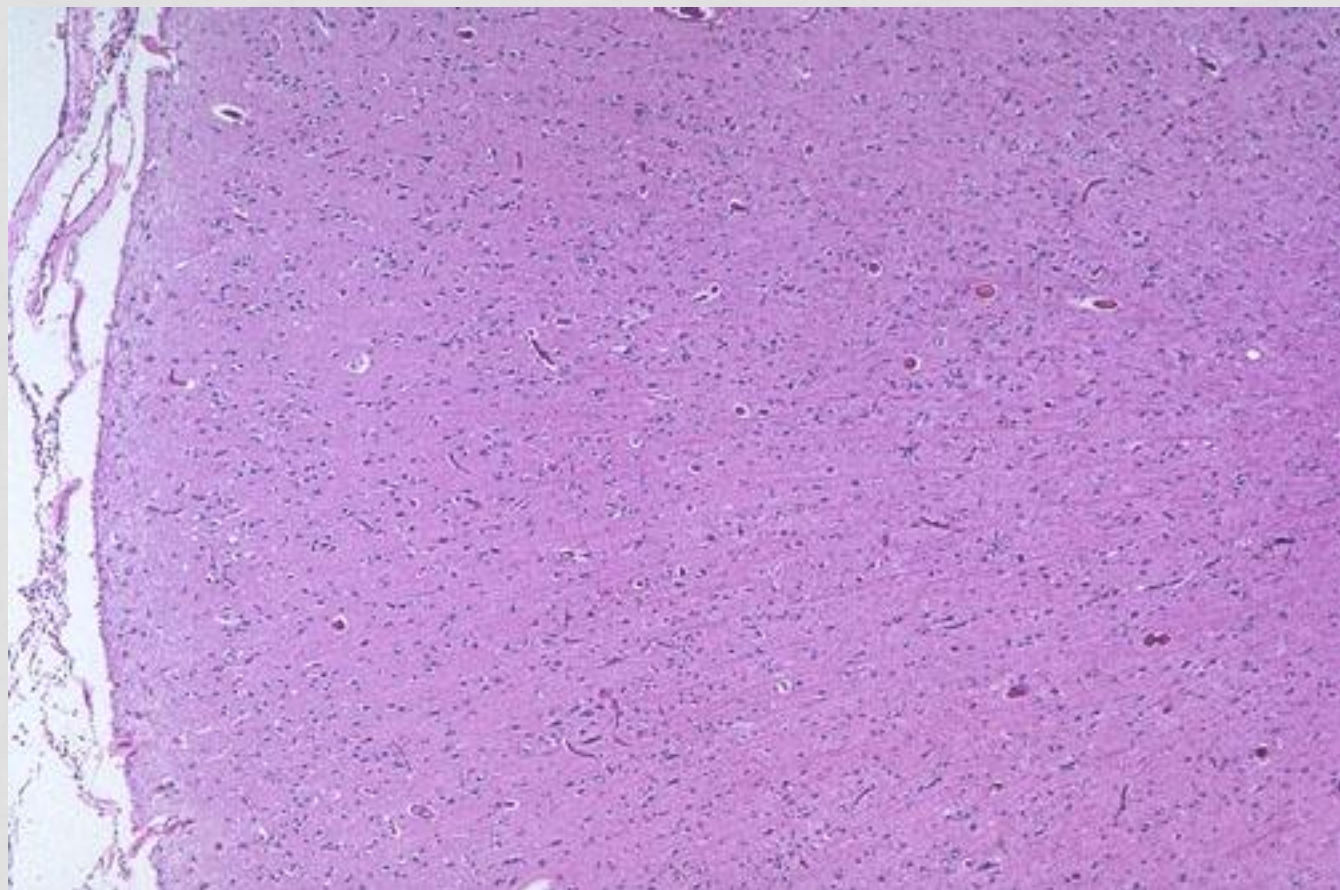
Blood-brain barrier BBB

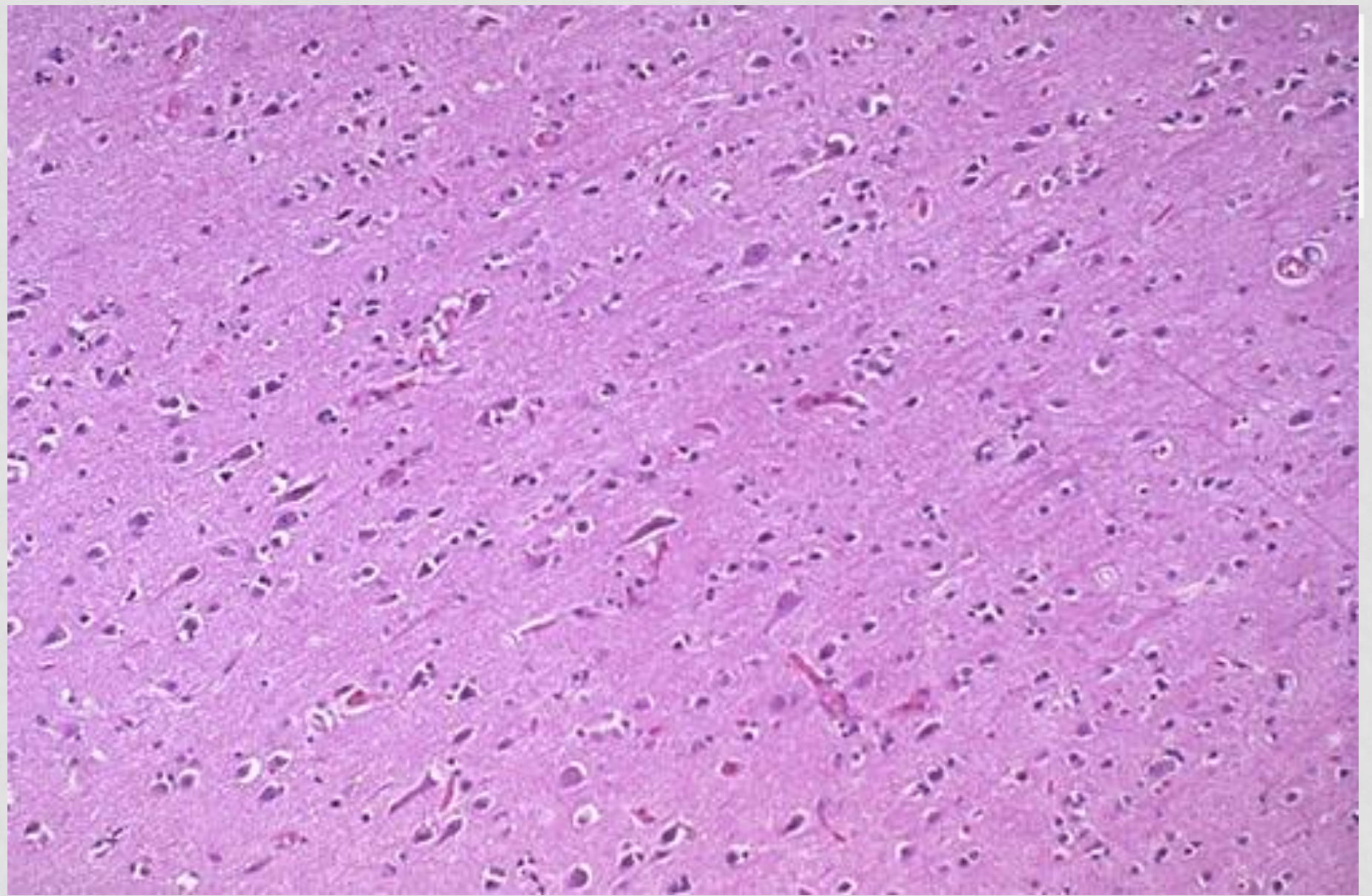
Ependymal
cells

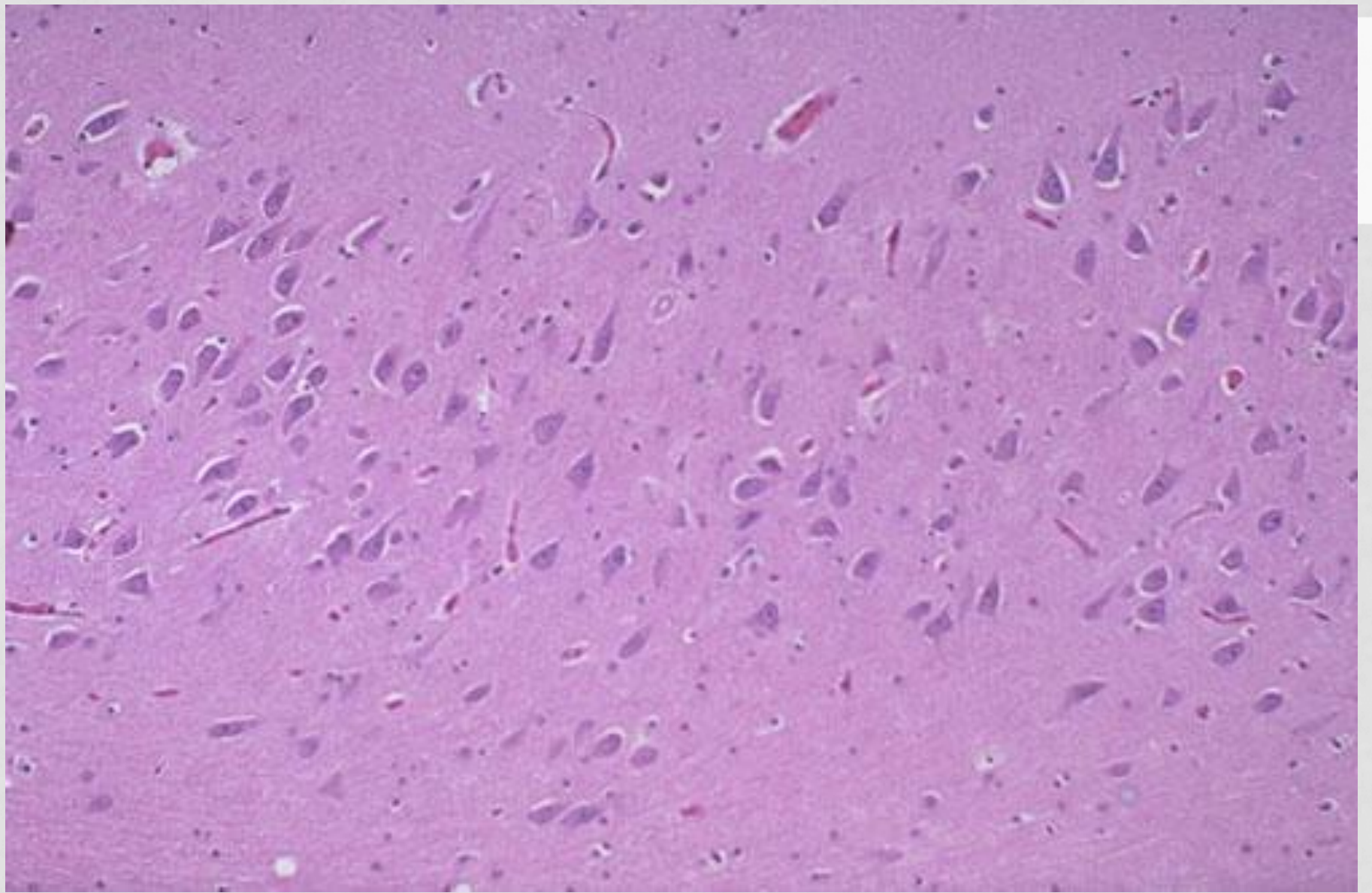












Cellular pathology of the central nervous system

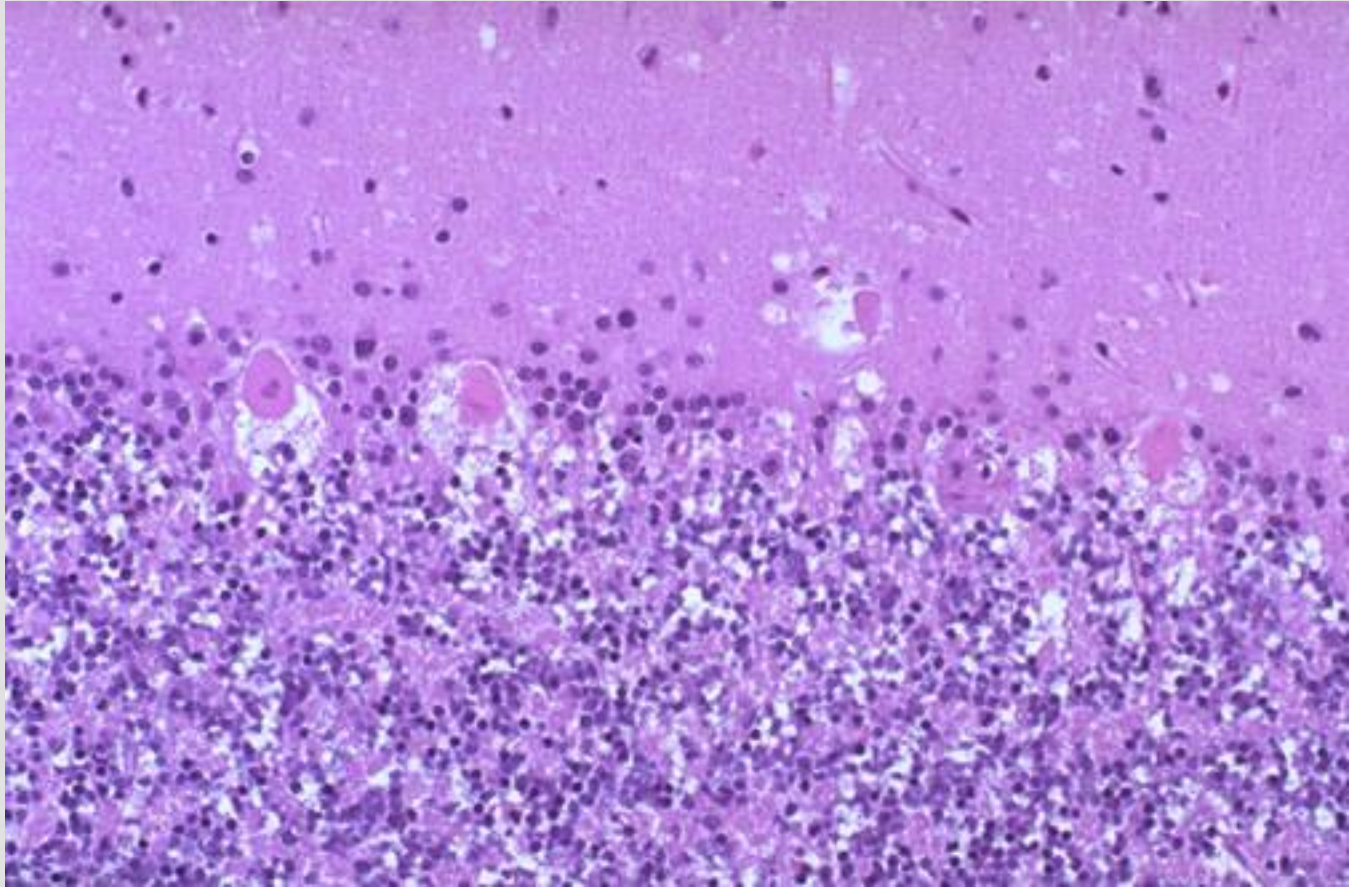
- The tissues of the CNS undergo fairly stereotyped changes in response to diverse types of injury

Reactions of neurons to injury

- **Acute neuronal injury (Red neuron)**

It refers to spectrum of changes that accompany acute CNS hypoxia /ischemia or other acute insults that ultimately lead to death of the cell

Morphologic features of red neurons consists of shrinkage of the cell body, pyknosis of the nucleus with intense eosinophilia of the cytoplasm

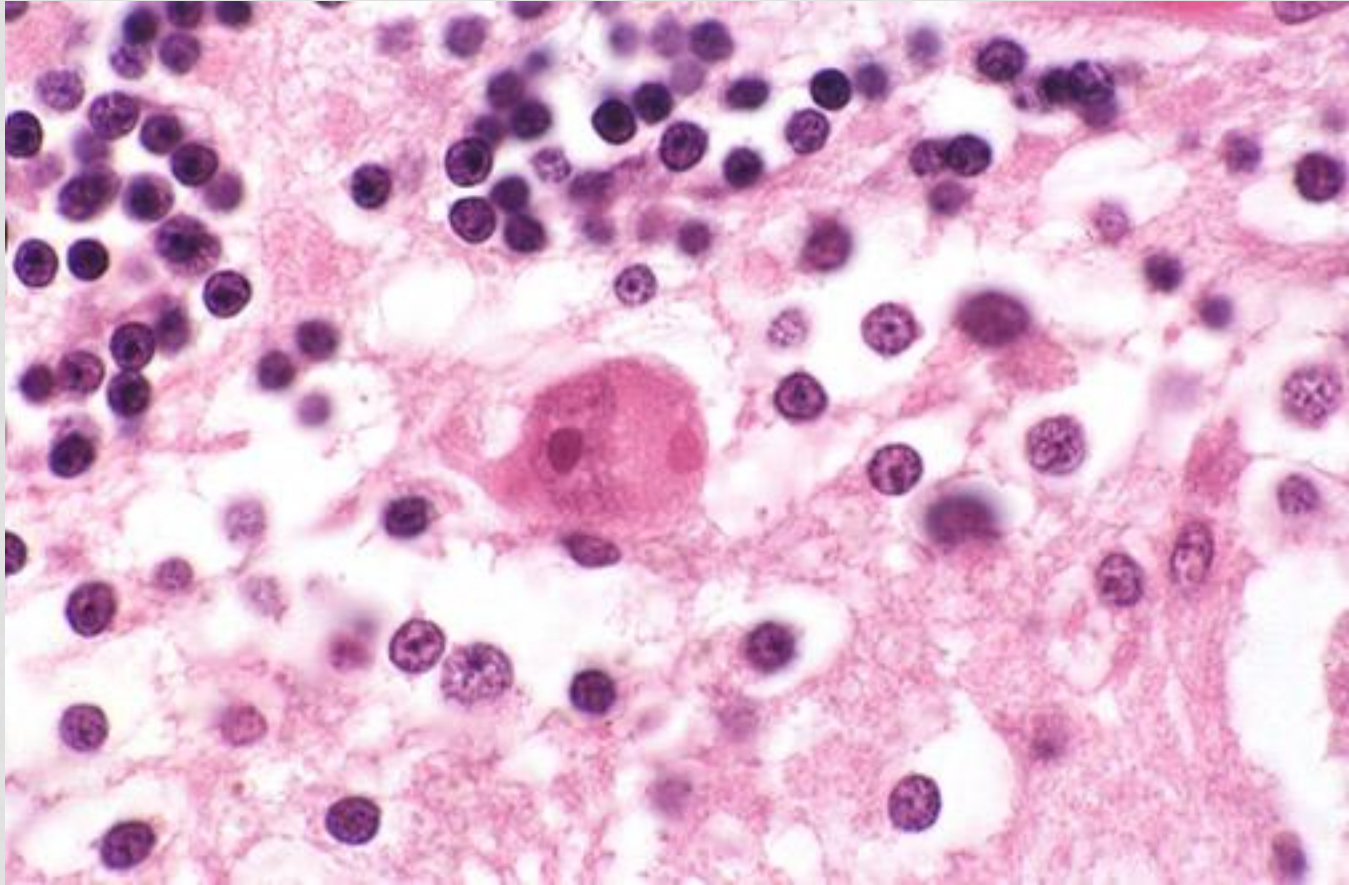


- **Subacute & chronic neuronal injury (degeneration)**
refers to conditions leading to neuronal death occurring as a result of a progressive disease process of some duration . The characteristic histologic picture is cell loss , often selectively involving functionally related system of neurons & reactive gliosis

- **Axonal reaction (central chromatolysis)** refers to the reaction within a cell body that attends regeneration of the axon
- It is accompanied by increased RNA & protein synthesis & its most important effect is axonal sprouting.

- Neuronal damage may be associated with wide range of subcellular alterations in the neuronal organelles & cytoskeleton. Neuronal inclusions may occur as a manifestation of aging , when there are intracytoplasmic accumulation of complex lipids(**lipofuscin**).

Viral infection can lead to abnormal intranuclear inclusion ,as seen in herpetic infection (cowdery body), cytoplasmic inclusions as seen in rabies (Negri body) or both nucleus & cytoplasm (cytomegalovirus)



Reactions of astrocytes to injury

- **Gliososis**

Astrocytes undergo both hyperplasia & hypertrophy . This is the most important histopathologic indicator of CNS injury

- **Cellular swelling** - occurs regularly in acute insults such as hypoxia

The **astrocytes** with their numerous fibrillary processes give structural support.
They are less susceptible to damage than neurones.

Normal



Damage not
sufficient to
kill the cell

Cells multiply,
and enlarge
→
Increase in
fibril
production



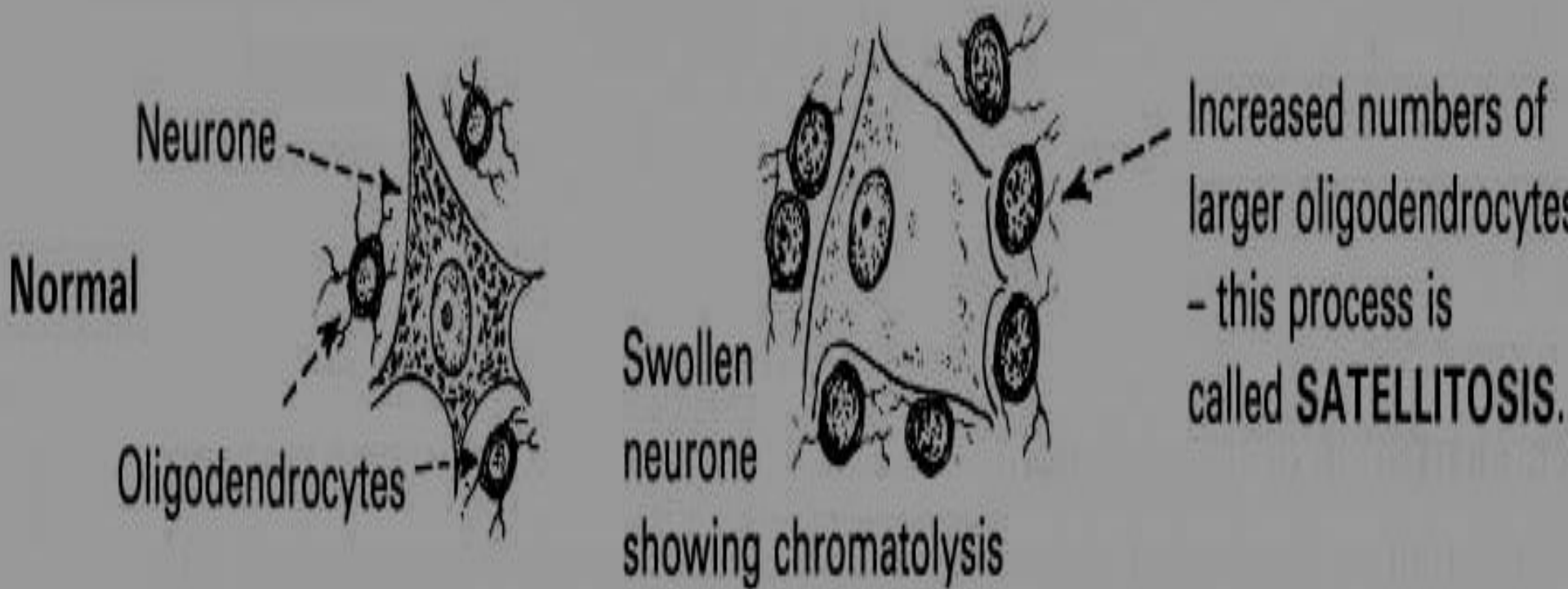
Later, cells
atrophy
→
leaving dense
meshwork
of fibres



This process is called GLIOSIS. It is a feature of many diseases and is analogous to scar tissue. These cells and fibres contain Glial Fibrillary Acidic Protein (GFAP), recognition of which, using antibodies, is useful in histological sections.

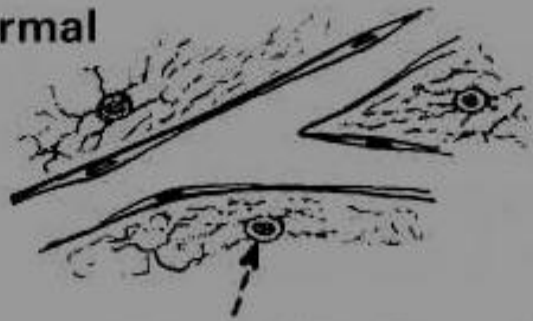
Note: Collagenous scar tissue is only formed in the CNS when mesodermal structures such as large blood vessels and their sheaths are damaged.

The **oligodendrocytes** – small cells with short processes – have a nutritive function in respect of neurones and especially myelin. This reaction is best seen when neurones are damaged.

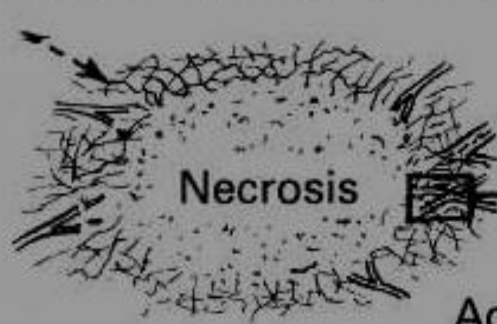


The **MICROGLIAL** cells are members of the mononuclear-phagocytic system. Reaction is best seen when there is necrosis of tissues.

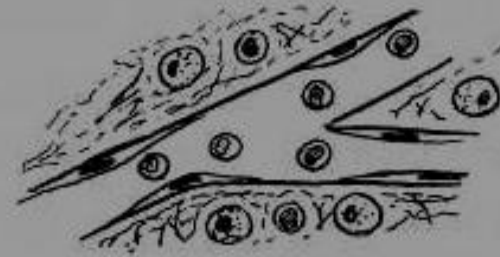
Normal



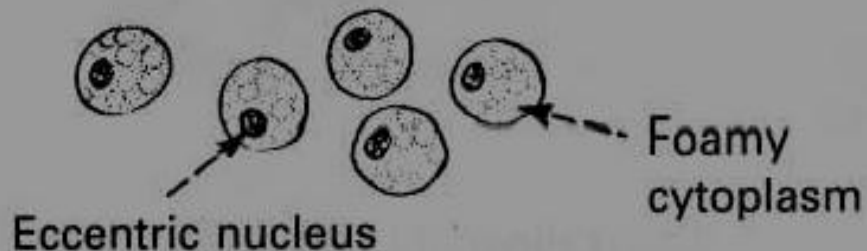
Microglial cells resident near blood vessels



Activated microglia phagocytose degradation products, e.g. lipids, haemosiderin. Blood monocytes augment resident cell



Microglial cells are well seen in and around infarcts; the activated cells which have ingested lipids are strikingly different from the small inactive microglia.



These cells have been given a variety of names: e.g. 'gitter' cells, or lipophages.

MECHANISMS OF DYSFUNCTION CAUSING DISEASE

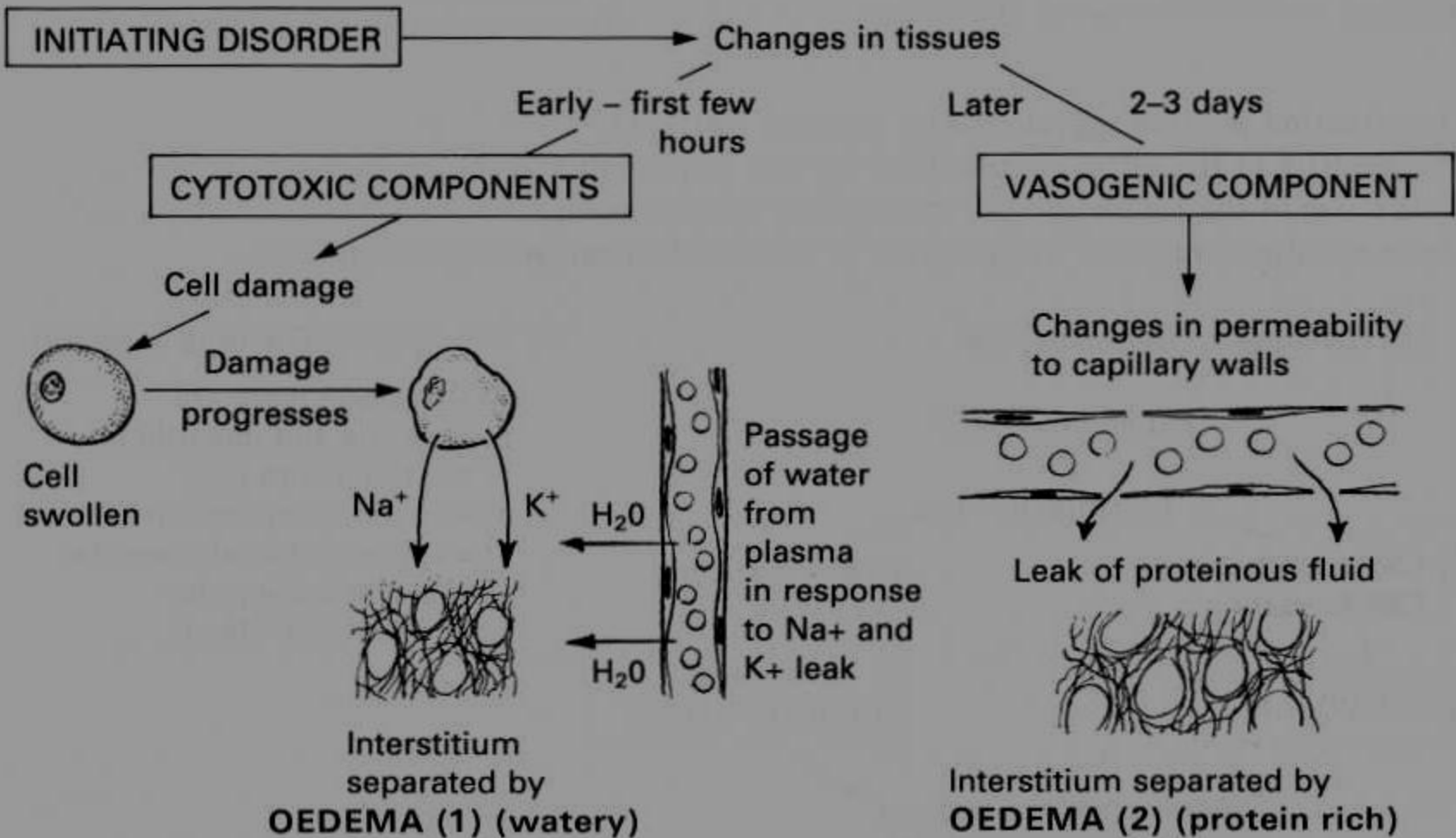
- Pathophysiological (toxic/metabolic)
- Structural
 - Focal lesions correlate with localizing symptoms
 - System degenerations correlate with functionally localizing symptoms (ie motor neuron disease)
- Increased intracranial pressure (generalized or focal), can cause global symptoms or brain herniation since the volume of the brain is fixed by the skull

Cerebral edema

There are two mechanisms

- **Vasogenic edema** – occurs when the integrity of the normal blood brain barrier is disrupted & increased vascular permeability occurs. Fluid escape from intravascular compartment into the intercellular space of brain
- **Cytotoxic edema** – an increase in intracellular fluid secondary to neuronal, glial or endothelial cell membrane injury

The pathological mechanism is as follows:



EDEMA, HERNIATION, AND HYDROCEPHALUS

Cerebral Edema :is the accumulation of excess fluid within the brain parenchyma

Localised conditions

Examples:

Infarcts, and local ischaemia

Haematomas (due to vessel rupture and injury)

Tumours

Generalised conditions

Examples: Intoxications

Metabolic disturbances, e.g. hypoglycaemia

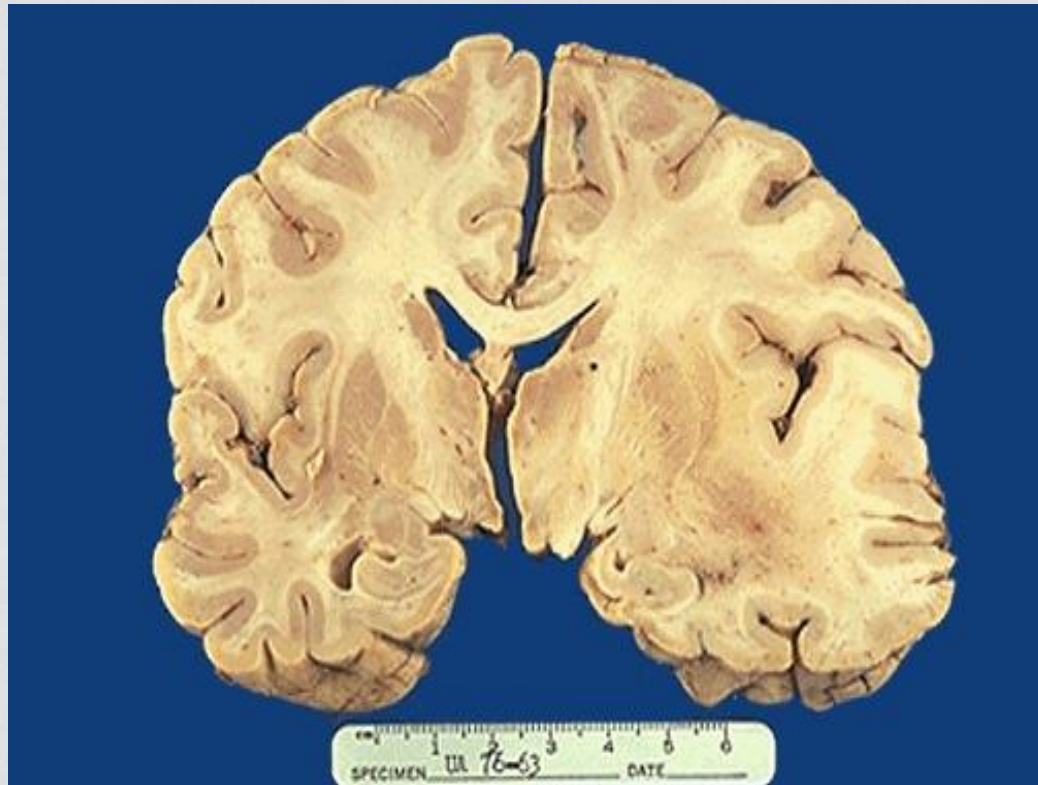
Generalised hypoxia

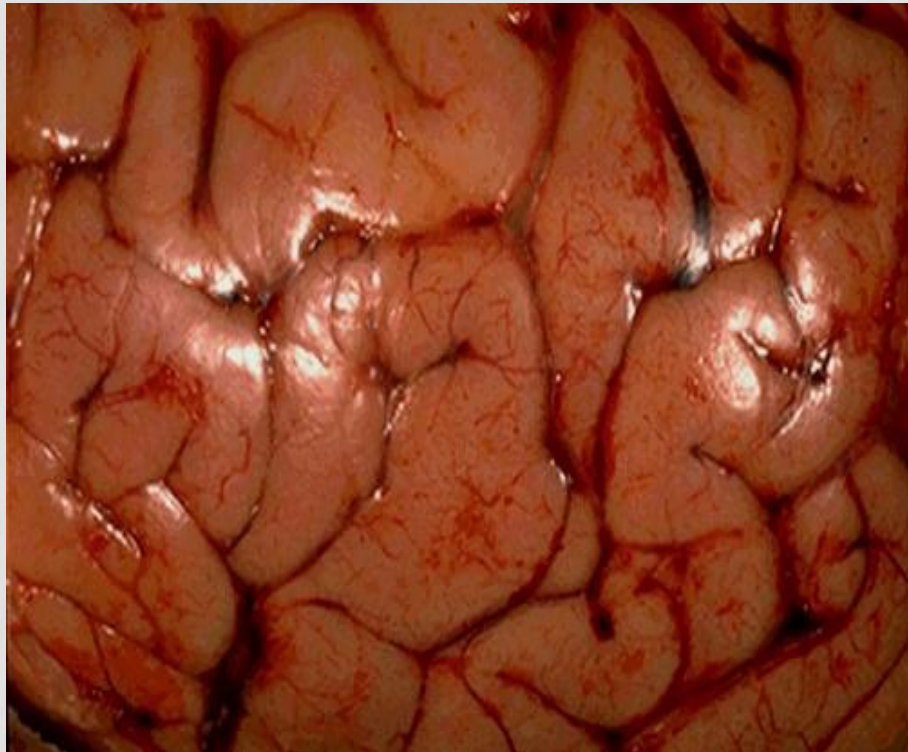
Severe head trauma

Malignant hypertension

Morphology

- The edematous brain overfill the cranial vault. Gyri is flattened & sulci is narrowed and the ventricular cavities are compressed . As the brain expands, herniation may occur





Hydrocephalus

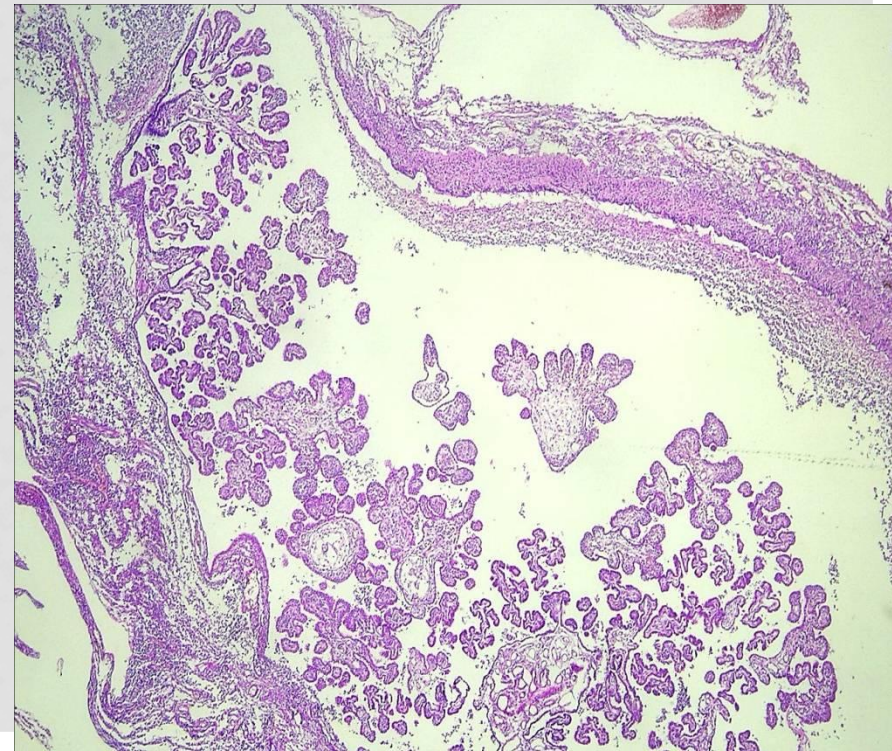
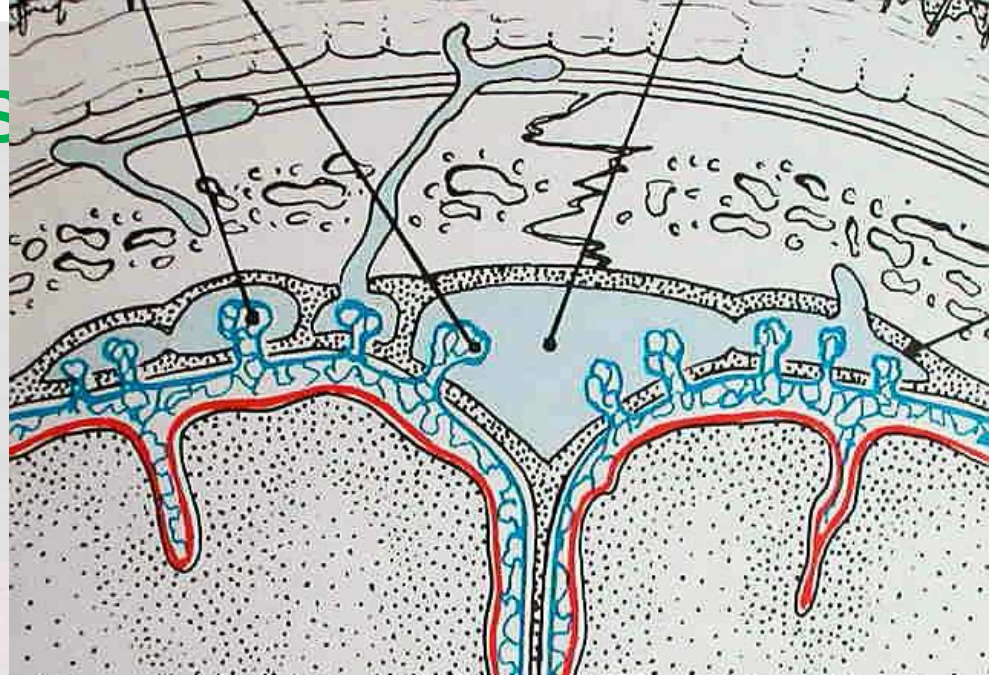
- Accumulation of excessive CSF within ventricular system , which results in increased intracranial pressure

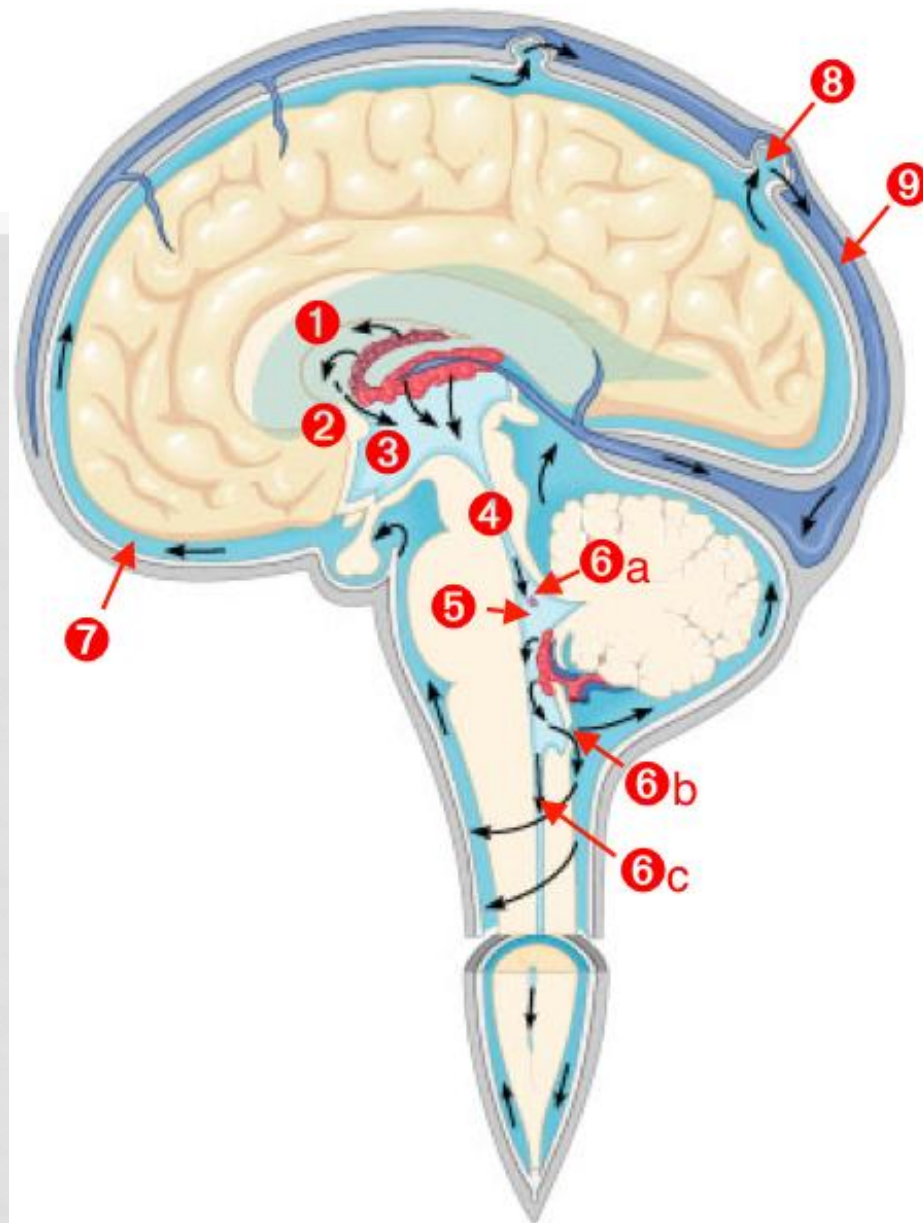
Cerebro spinal Fluid CSF

Volume: 100 – 150 ml

Replaced 3 - 4 x per day

Selective secretion





- Most cases occur due to impairments of flow & resorption of CSF. In rare cases over production of CSF may be responsible. (Tumors of the choroid plexus)

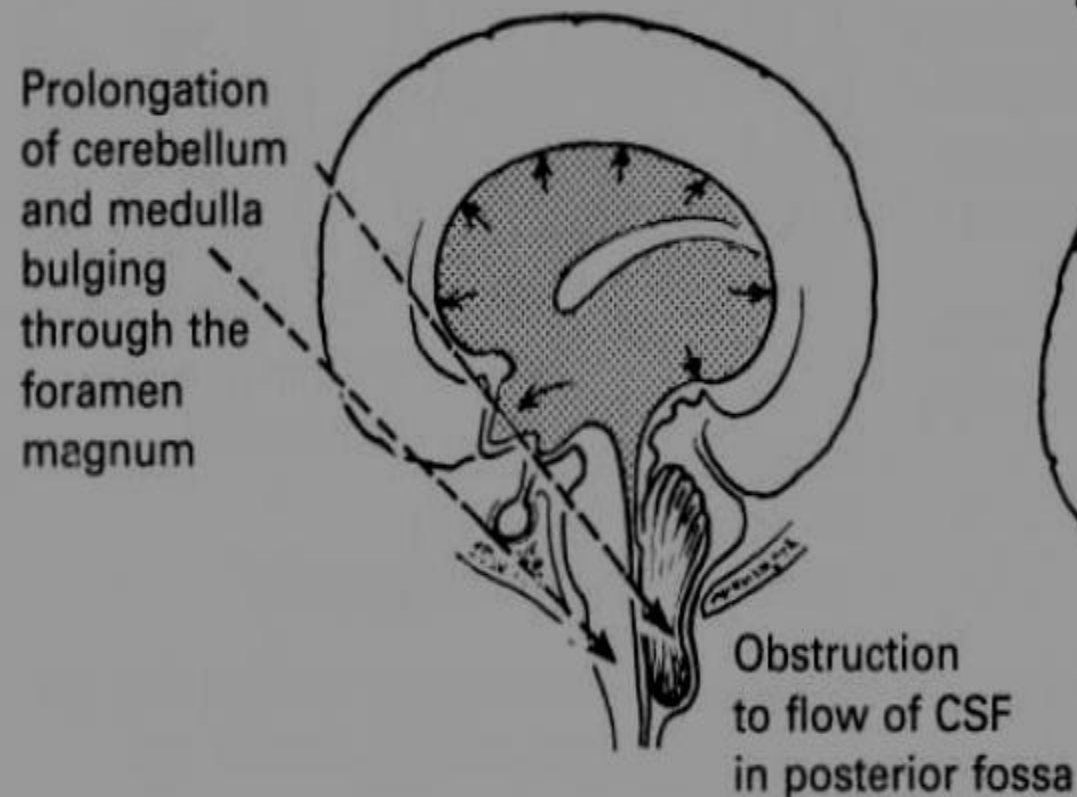
CAUSES

The diseases causing hydrocephalus fall into two groups:

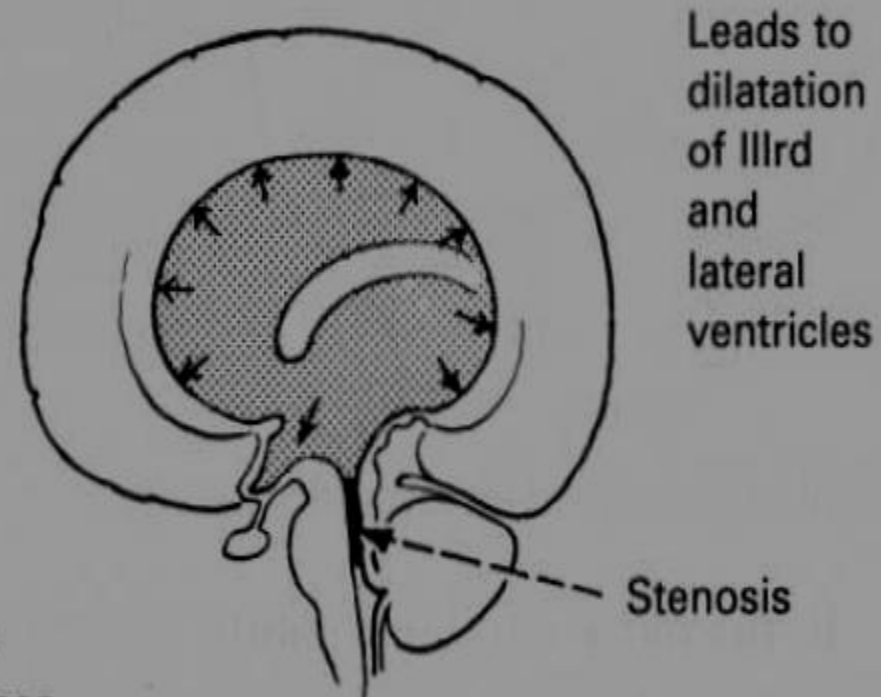
1. **Congenital (developmental) abnormalities**

The common conditions are:

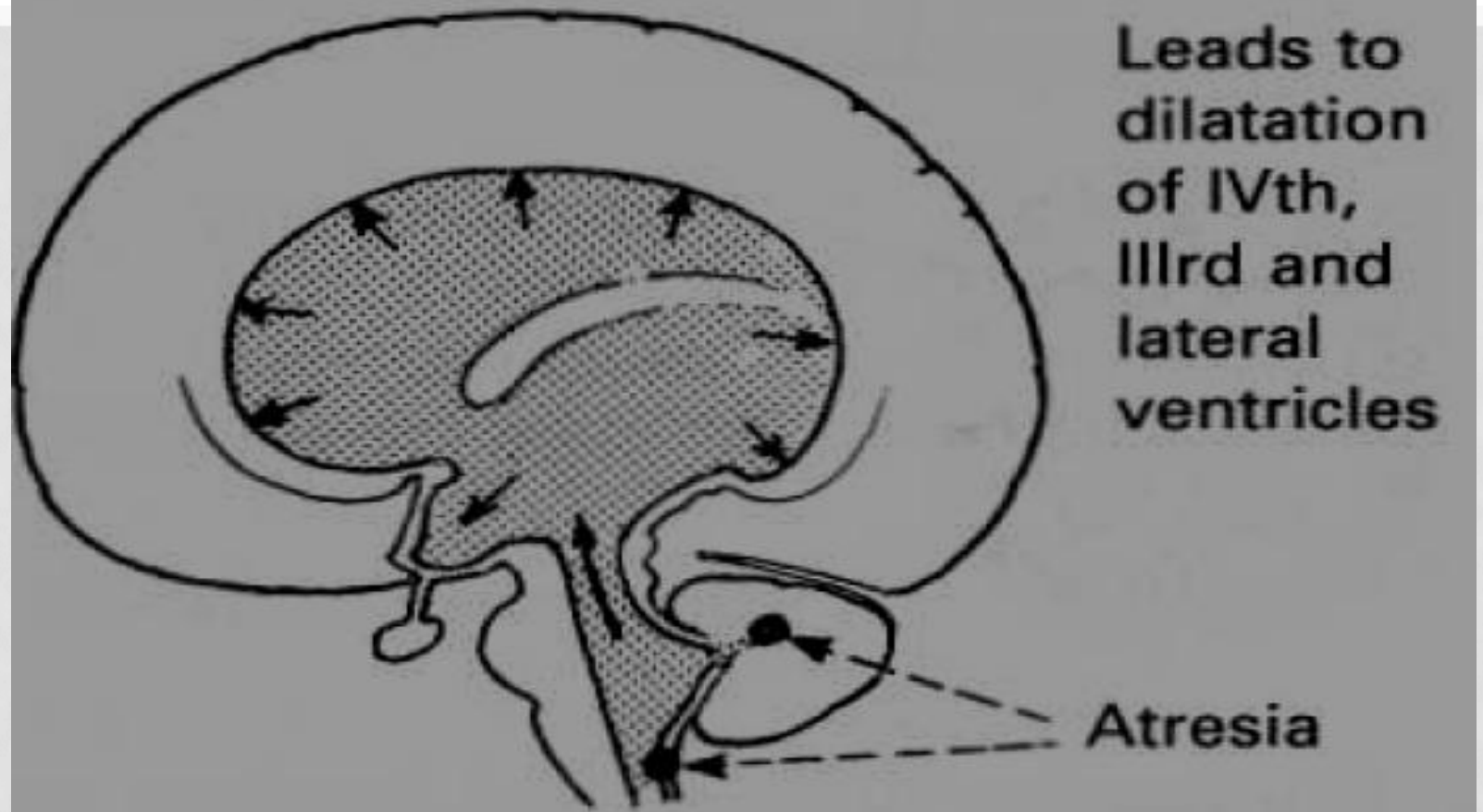
(a) *The Arnold-Chiari malformation*



(b) *Congenital stenosis or atresia of aqueduct of Sylvius*



**(c) *Atresia of foramina of
Magendie and Luschka***



2. Acquired hydrocephalus

Of the many possible conditions causing hydrocephalus, the following are most common:

(a) cerebral tumour (primary or secondary) and (b) scarring of the meninges following meningitis or subarachnoid haemorrhage.

As already explained, whether any particular disease produces hydrocephalus depends largely on the site affected.

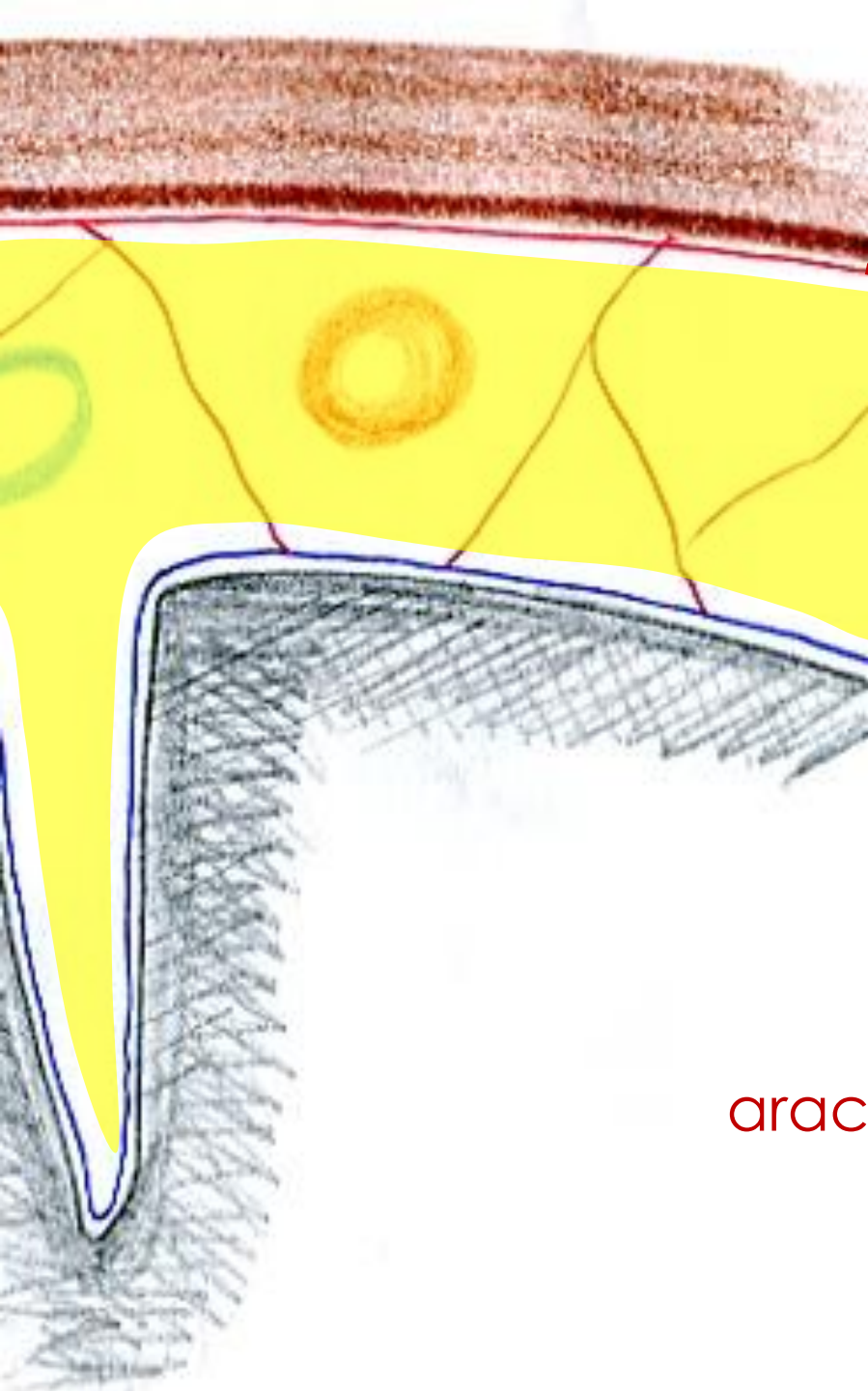
Depending the presence or absence of obstacle to the flow of CSF classified as

i.noncommunicating hydrocephalus

there is an obstacle to the flow of CSF within the ventricular system, then a portion of the ventricles enlarges while the remainder does not

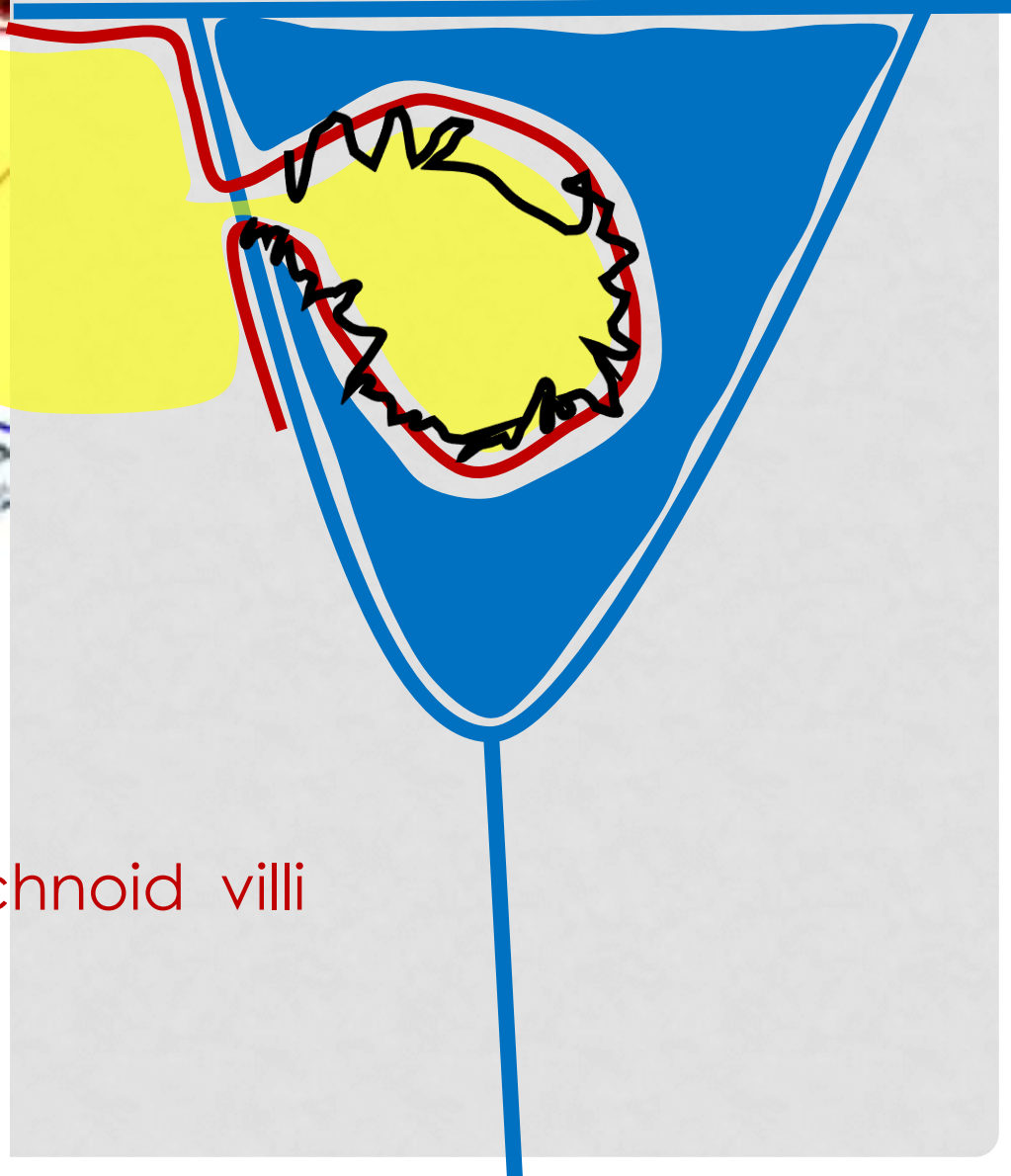
ii.communicating hydrocephalus

- all of the ventricular system is enlarged; here the cause is most often reduced resorption of CSF.



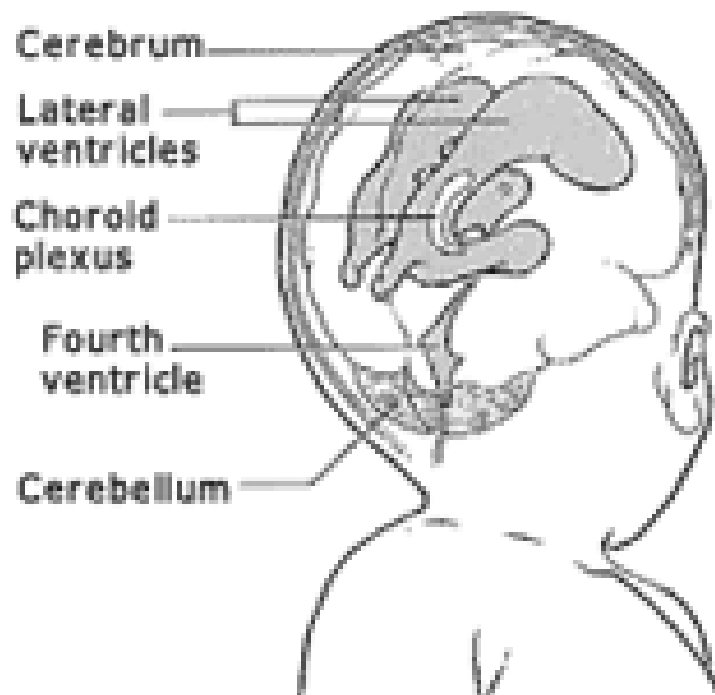
sagittal sinus

arachnoid villi

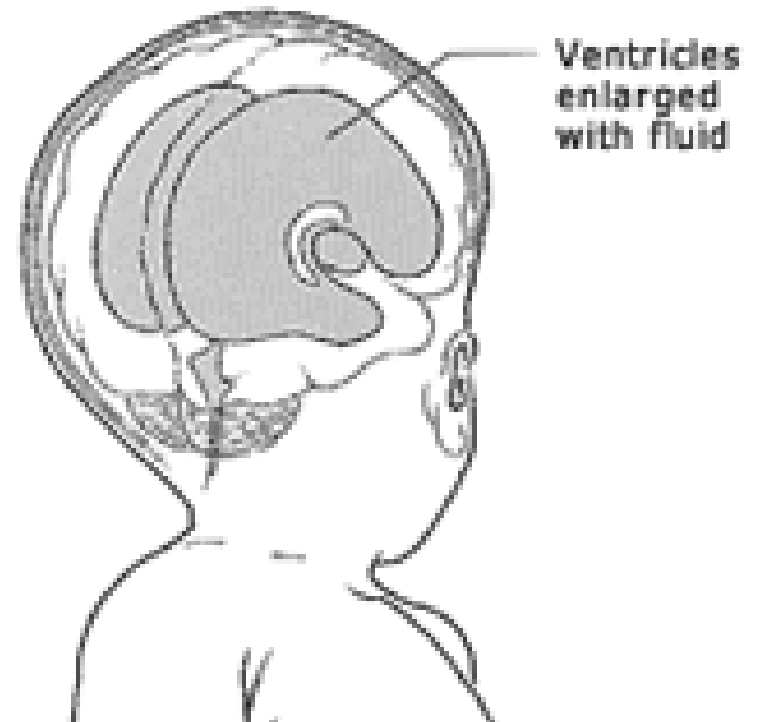


- Whatever is the cause , an increased volume of CSF within the ventricles expands them & can cause an elevation in intracranial pressure

Brain with no hydrocephalus



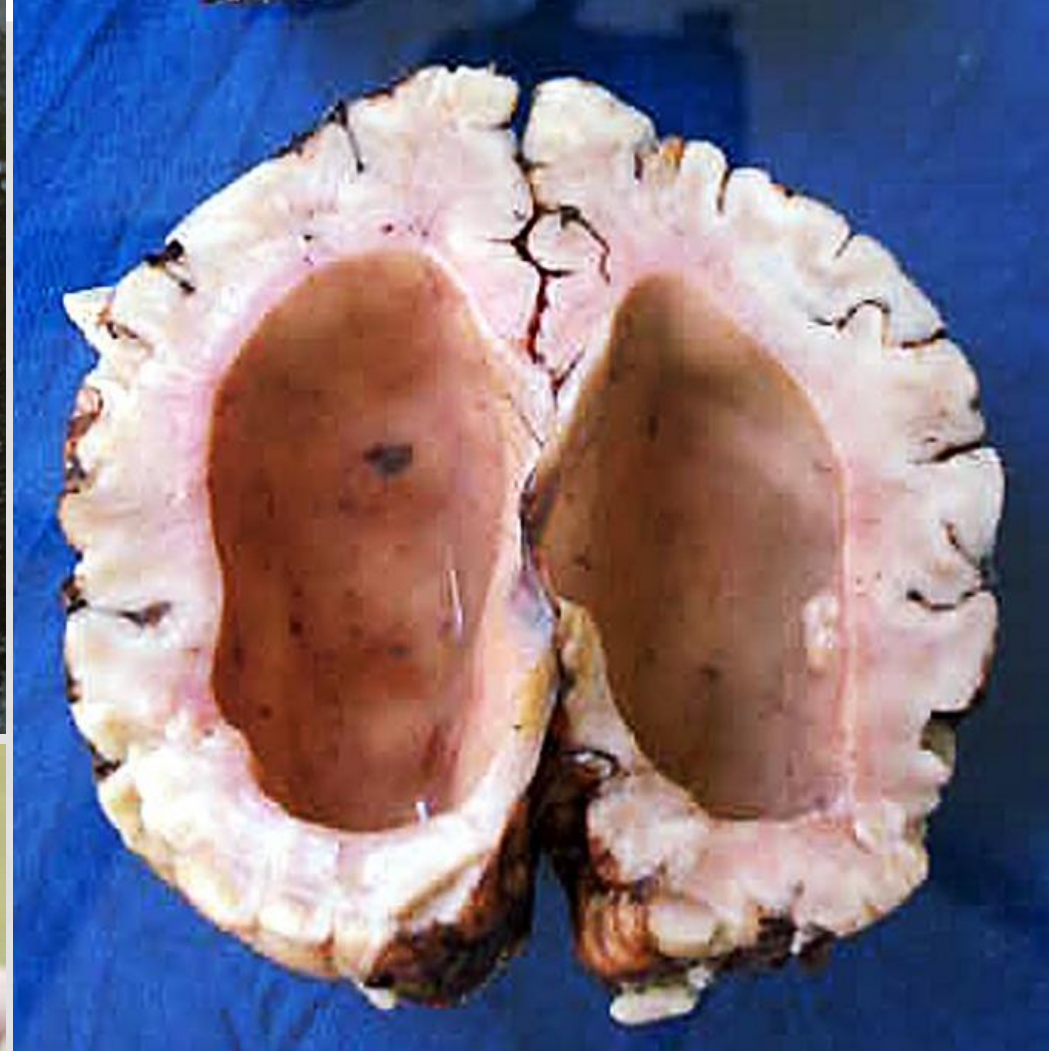
Brain with hydrocephalus



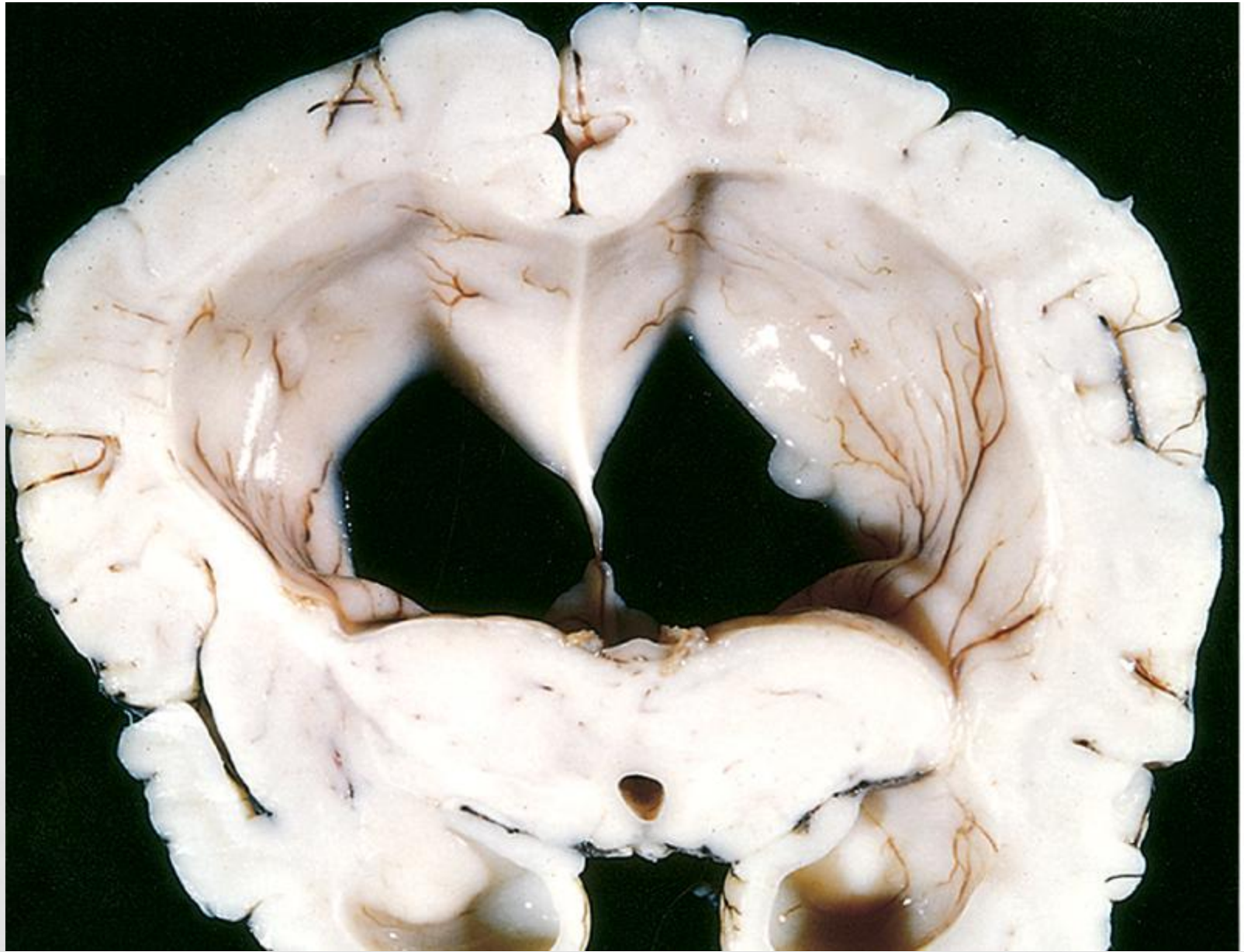
- If hydrocephalus occurs before closure of the sutures, there is enlargement of the head, manifested by an increase in head circumference .
- Hydrocephalus developing after fusion of the sutures in contrast , is associated with expansion of the ventricles & increased intracranial pressure, without a change in head circumference



- Communicating hydrocephalus – there is enlargement of the entire ventricular system
- Non communicating – if portion of the ventricular system is enlarged (eg tumor in the third ventricle)



Hydrocephalus internus
Non-communicating



MALFORMATION AND DEVELOPMENTAL DISEASES

Malformation and Developmental Diseases

- Malformations of the CNS account for approximately one third of congenital malformations and are an important cause of infant mortality and long-term disability.

- Some malformations have now been linked to clearly defined chromosomal and/or genetic abnormalities, and others have been associated with specific environmental factors. In the majority of cases, however, the cause of the CNS malformation is unknown.

Neural tube defect

- Failure of a portion of the neural tube to close or reopening of a region of the tube
- All are characterized by abnormalities involving some combination of neural tissue , meninges & overlying bone & soft tissue
- Neural tube defect account for most CNS malformation
- The etiology is unknown . Folate deficiency during initial weeks of gestation has been implicated
- Elevation of maternal blood α -fetoprotein occurs

Anencephaly is malformation of the anterior end the neural tube with absence of the brain & calvarium.

- Incidence is 1 to 5 per 1000 live births
- It is thought to develop at ~ 28 wks of gestation
- The forebrain development is disrupted . The posterior fossa structures may be spared.

Clinical Features

- Anencephaly is incompatible with prolonged extrauterine survival, and most fetuses die within minutes to hours after birth.
- The diagnosis of anencephaly can be made prenatally. Hydramnios is common in anencephalic gestations, and concentrations of α -fetoprotein and acetylcholinesterase are increased in the amniotic fluid.
- Ultrasonography can detect the presence of anencephaly in the later stages of the first trimester.



Encephalocele

- Diverticulum of malformed brain tissue through defect in cranium . It most often occurs in the occipital region or in the posterior fossa

- The most common forms of neural tube defect involve the spinal cord & are caused **by a failure of closure or reopening of the caudal portions of neural tube.**
- **Spina bifida or dysraphism** may be asymptomatic bony defect (spina bifida occulta) or severe malformation with disorganised spinal cord segment associated with an overlying meningeal outpouching.

- Meningomyelocele- extension of CNS tissue through the vertebral column. Meningocele applies when there is only meningeal extrusion.
- Meningomyelocele occur most commonly in the lumbosacral region . The patients manifest with deficits referable to motor & sensory function in lower extremity & disturbance in bladder & bowel control



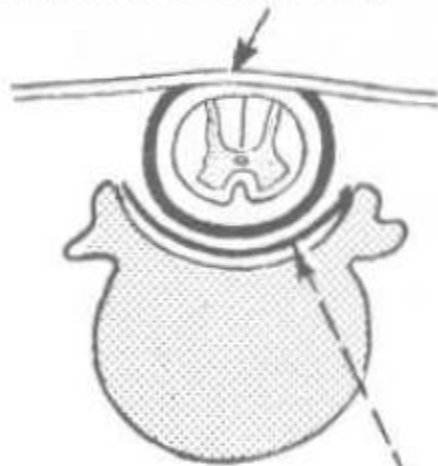
- *Spina bifida occulta* is the mildest form of spinal neural tube defect. It is characterized by defective closure of the posterior vertebral arches, with intact meninges and spinal cord. The site of the defect is sometimes marked by a small skin dimple or tuft of hair. This abnormality, which occurs in about 20% of the general population, is asymptomatic.

SPINA BIFIDA

In the lumbosacral and cervical regions particularly, local defects in the development and closure of the neural tube and vertebral arches result in spina bifida.

(a) Spina bifida occulta

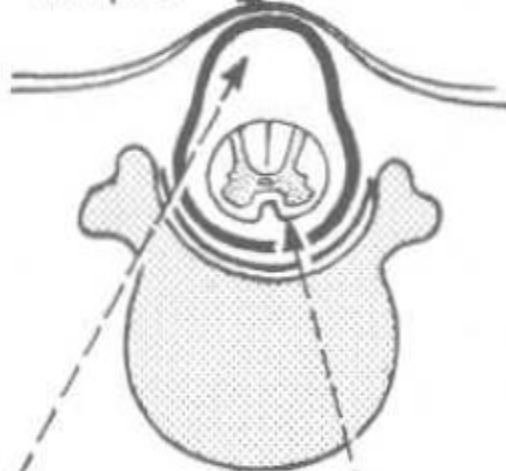
Normal skin covering



Vertebral arches and dura deficient – may be associated with minor nerve defects – particularly affecting bladder function

b) Meningocele

Skin may be thin at apex

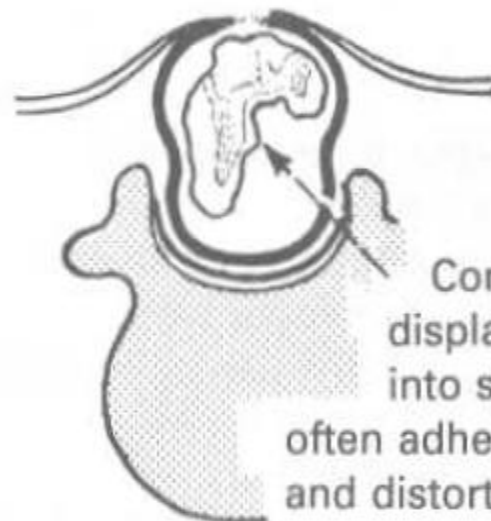


Rounded sac containing CSF

Cord in normal position

(c) Meningomyelocele

Skin usually defective



Cord displaced into sac: often adherent and distorted

May be associated with serious nervous deficit – especially in legs, bladder.

Infection through the deficient skin is a serious complication.

Forebrain anomalies

- The volume of the brain can be abnormally large (megalocephaly) or small (microcephaly).
- Microcephaly is more common & is associated with chromosomal abnormalities, fetal alcohol syndrome & HIV infection acquired in utero.

Trauma

- The anatomic location of the lesion & the limited capacity of the brain for functional repair are two factors of great significance in CNS trauma
- Head injury can be caused by penetrating or blunt trauma . It can be an open or a closed injury

- It can be associated with
 - Skull fractures
 - Parenchymal injury
 - Vascular injury

Skull fractures

- A fracture in which bones displace into cranial cavity by a distance greater than the thickness of the bone is *displaced skull fractures*
- Fractures of skull is related to pattern of fall



- Basal skull fracture typically follow an impact to the occiput or sides of the head. Symptoms referable to lower cranial nerves & the presence of orbital or mastoid hematoma raise clinical suspicion of basal skull fracture. CSF discharge from the nose (rhinorrhea) or ear & infection (meningitis) may follow.



PARENCHYMAL INJURIES

Parenchymal injuries

Concussion

- It is clinical syndrome of alteration of consciousness secondary to head injury typically brought about by a change in the momentum of head (movement of the head arrested by a rigid surface)
- The characteristics neurologic picture include instantaneous onset of neurologic dysfunction including loss of consciousness , temporary respiratory arrest & loss of reflexes. Although neurological recovery is complete , amnesia for the event persists
- Postconcussive neuropsychiatric syndromes may occur

concussion

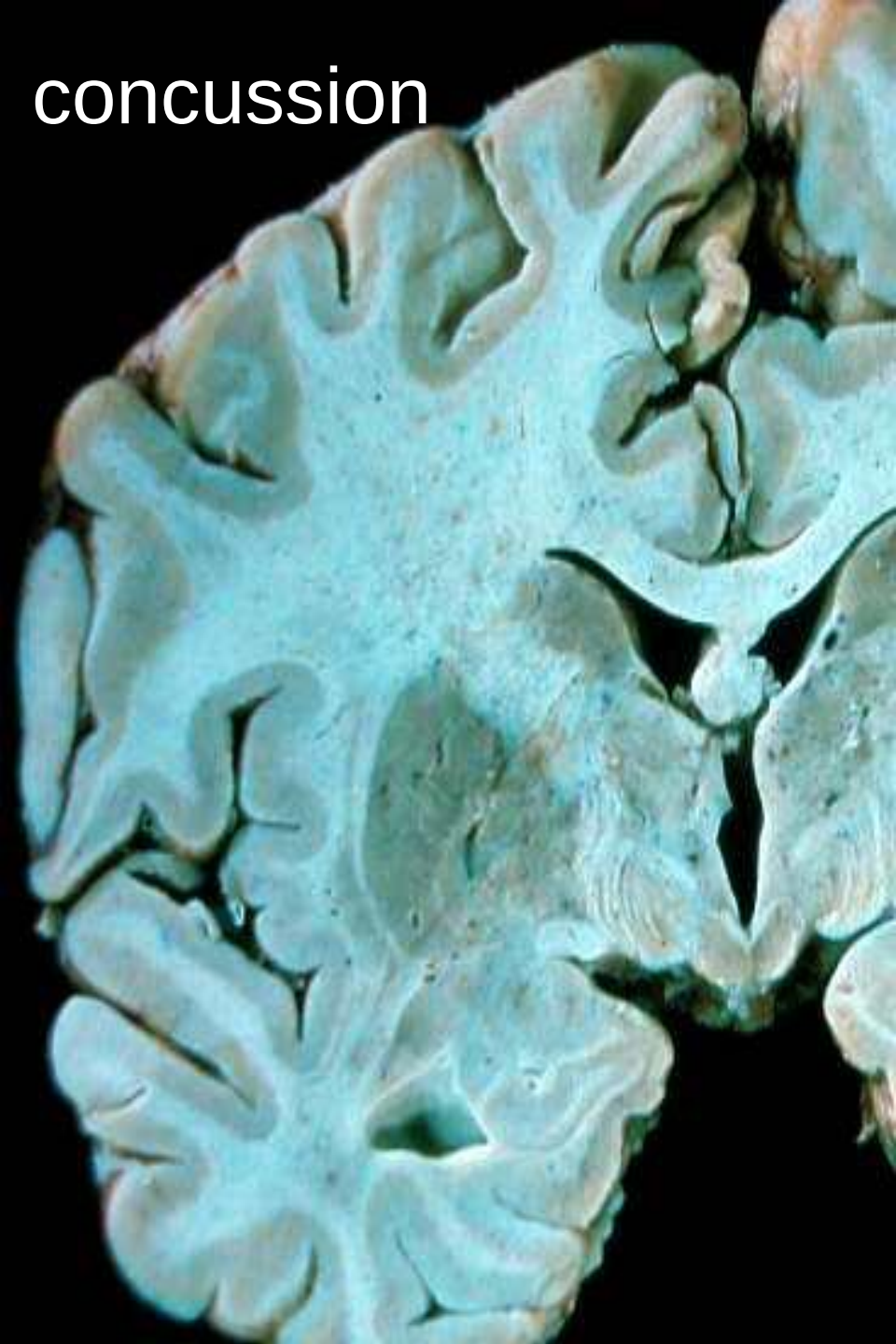
morphology = normal, intact

clinical syndrome:

- onset of transient neurologic dysfunction: loss of consciousness
- amnesia

outcome:

- neurologic recovery complete
- amnesia for the event persists



- Concussions may be associated with minimal morphologic changes; transient injury to the reticular activating system in the brain has been proposed as the mechanism for loss of consciousness

Persistent postconcussion symptoms

- Headache
- Sleep disturbance
- Irritability
- Dizziness
- Imbalance
- Fatigue
- Inattention
- Problems with concentration and memory

Original Article

Detection of Blast-Related Traumatic Brain Injury in U.S. Military Personnel

Christine L. Mac Donald, Ph.D., Ann M. Johnson, Dana Cooper, B.S., Elliot C. Nelson, M.D., Nicole J. Werner, Ph.D., Joshua S. Shimony, M.D., Ph.D., Abraham Z. Snyder, M.D., Ph.D., Marcus E. Raichle, M.D., John R. Witherow, M.D., Raymond Fang, M.D., Stephen F. Flaherty, M.D., and David L. Brody, M.D., Ph.D.

N Engl J Med
Volume 364(22):2091-2100
June 2, 2011



The NEW ENGLAND
JOURNAL of MEDICINE

Study Overview

- In this study of injured U.S. military personnel, an advanced MRI technique found abnormalities consistent with traumatic axonal injury in some patients with mild traumatic brain injury after blasts; these abnormalities were not detected with conventional MRI.

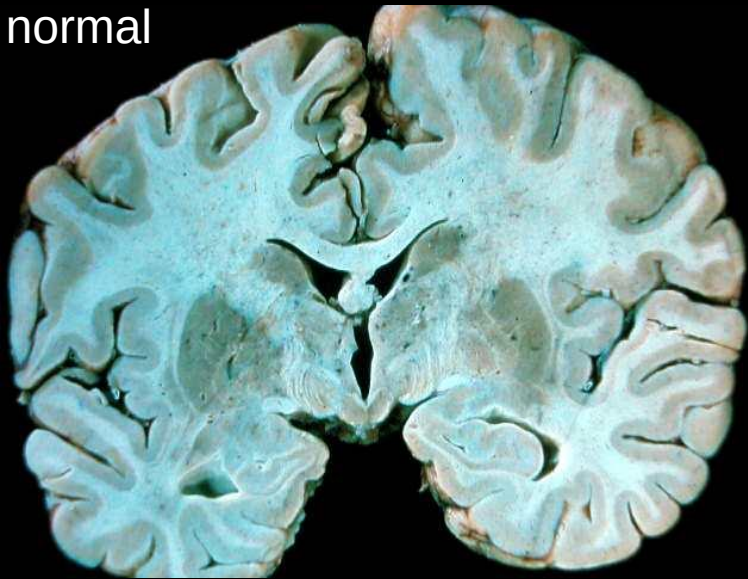
diffusion tensor imaging **DTI**

Conclusions

- DTI findings in U.S. military personnel support the hypothesis that blast-related mild traumatic brain injury can
- **involve axonal injury.**
- Thus, traumatic brain injury remains a clinical diagnosis



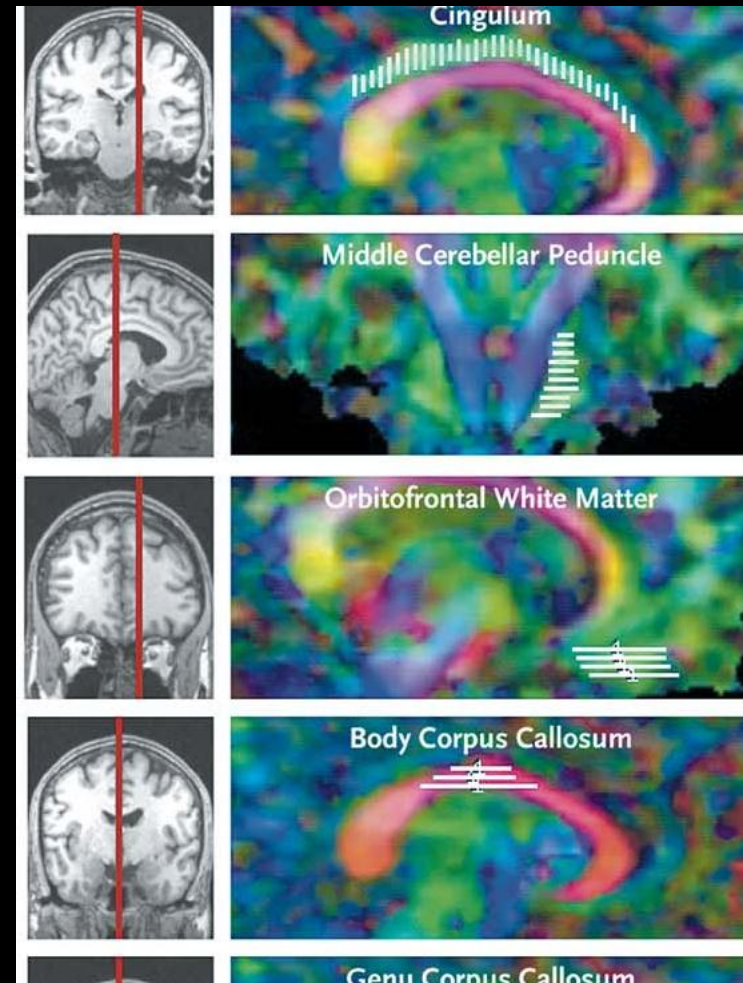
normal



concussion

morphology = normal,
intact

blast-related mild traumatic
brain injury can involve axonal
injury.



Direct parenchymal injury

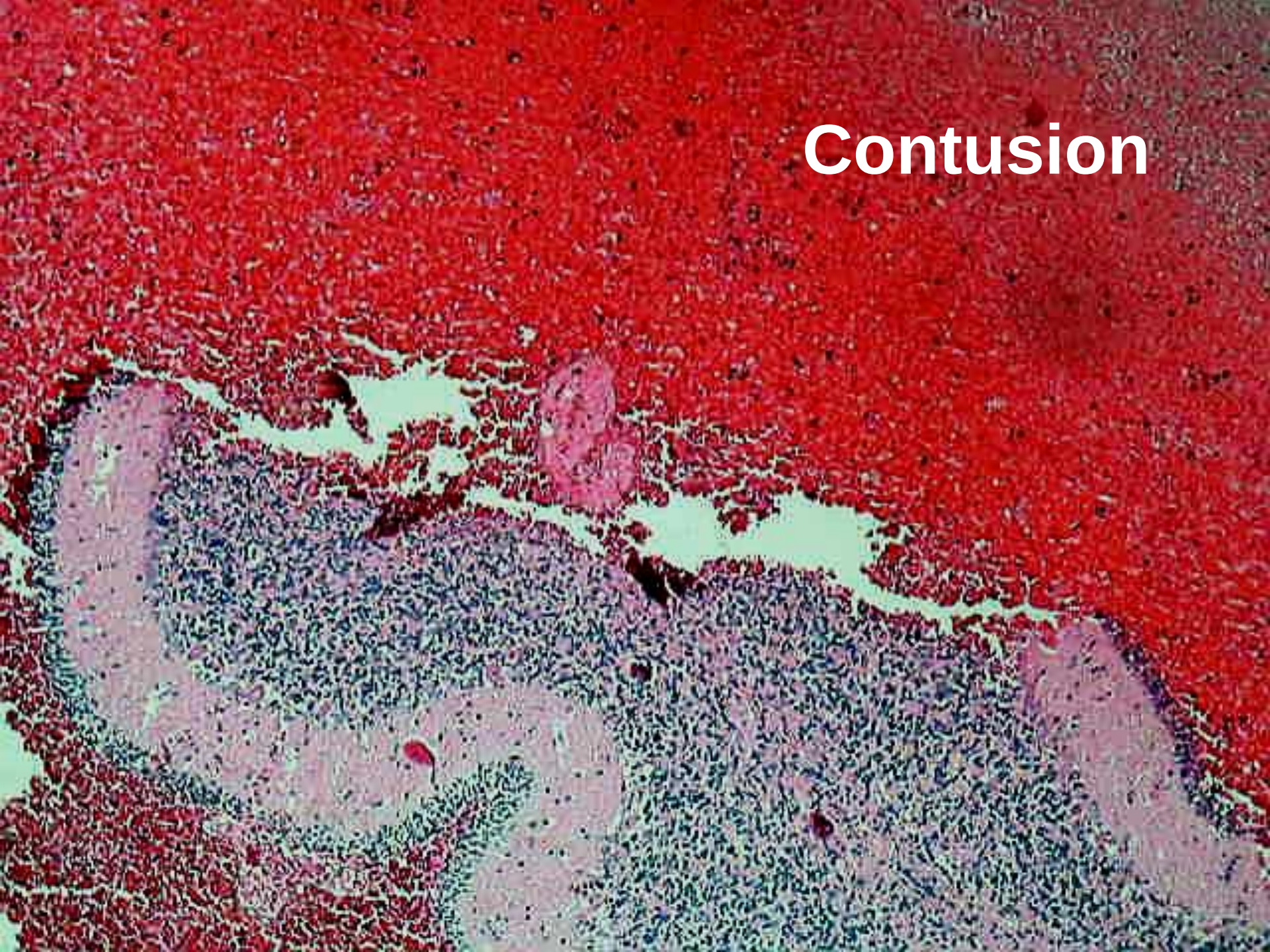
Contusion

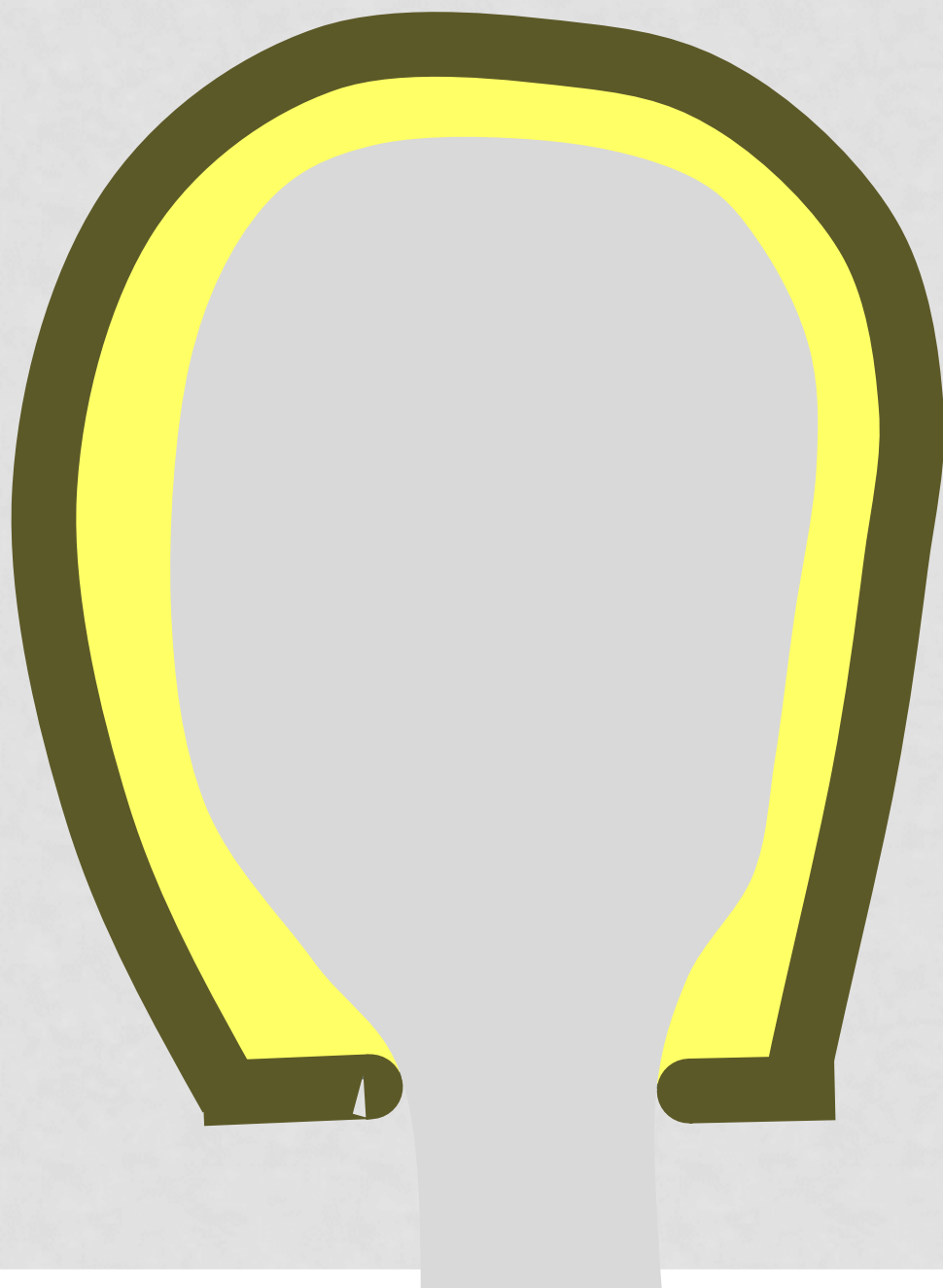
- Direct parenchymal injury brought about by transmission of kinetic energy to brain . It is analogous to bruising

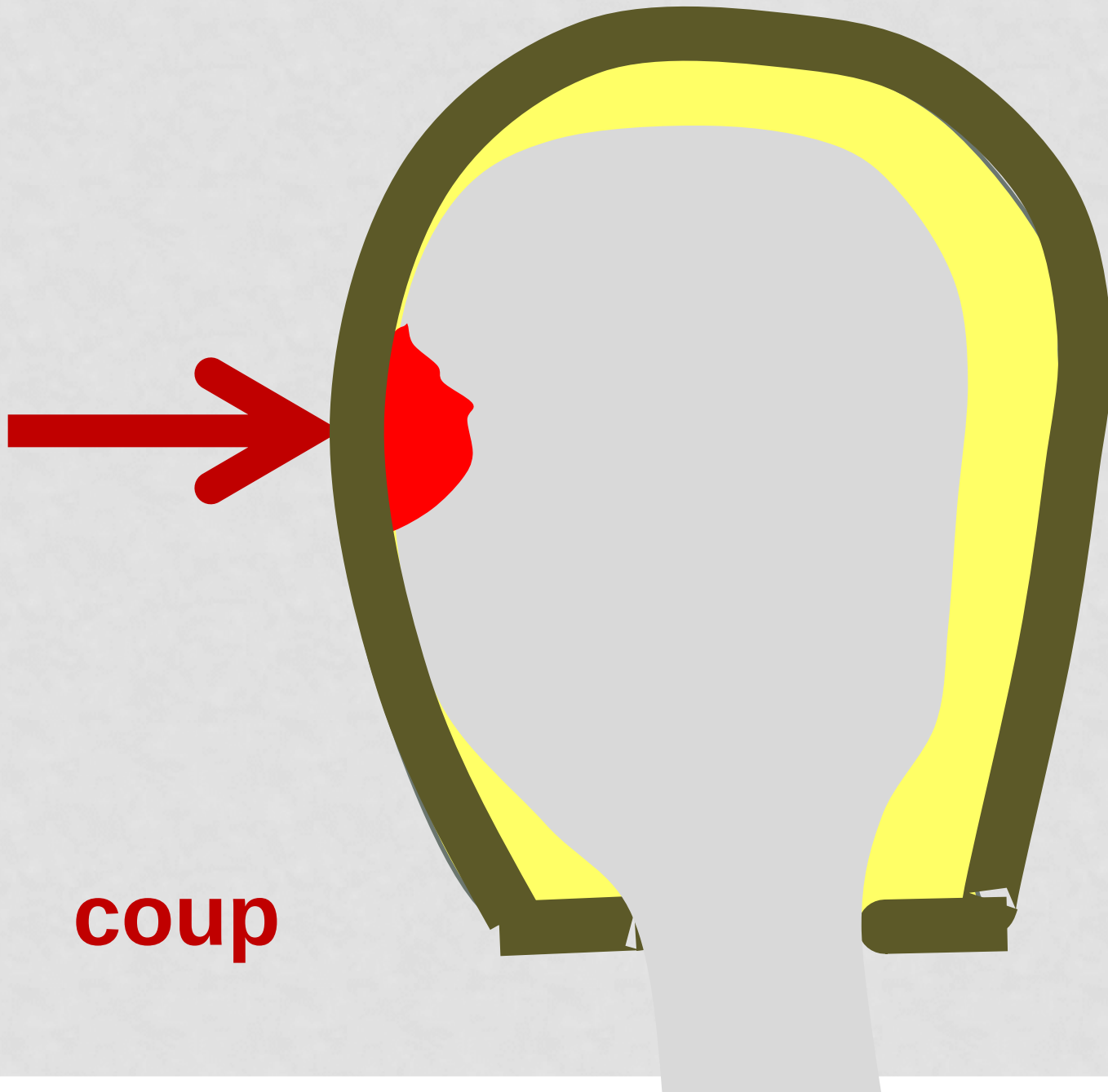
Laceration

- Caused by penetration of an object & tearing of tissue

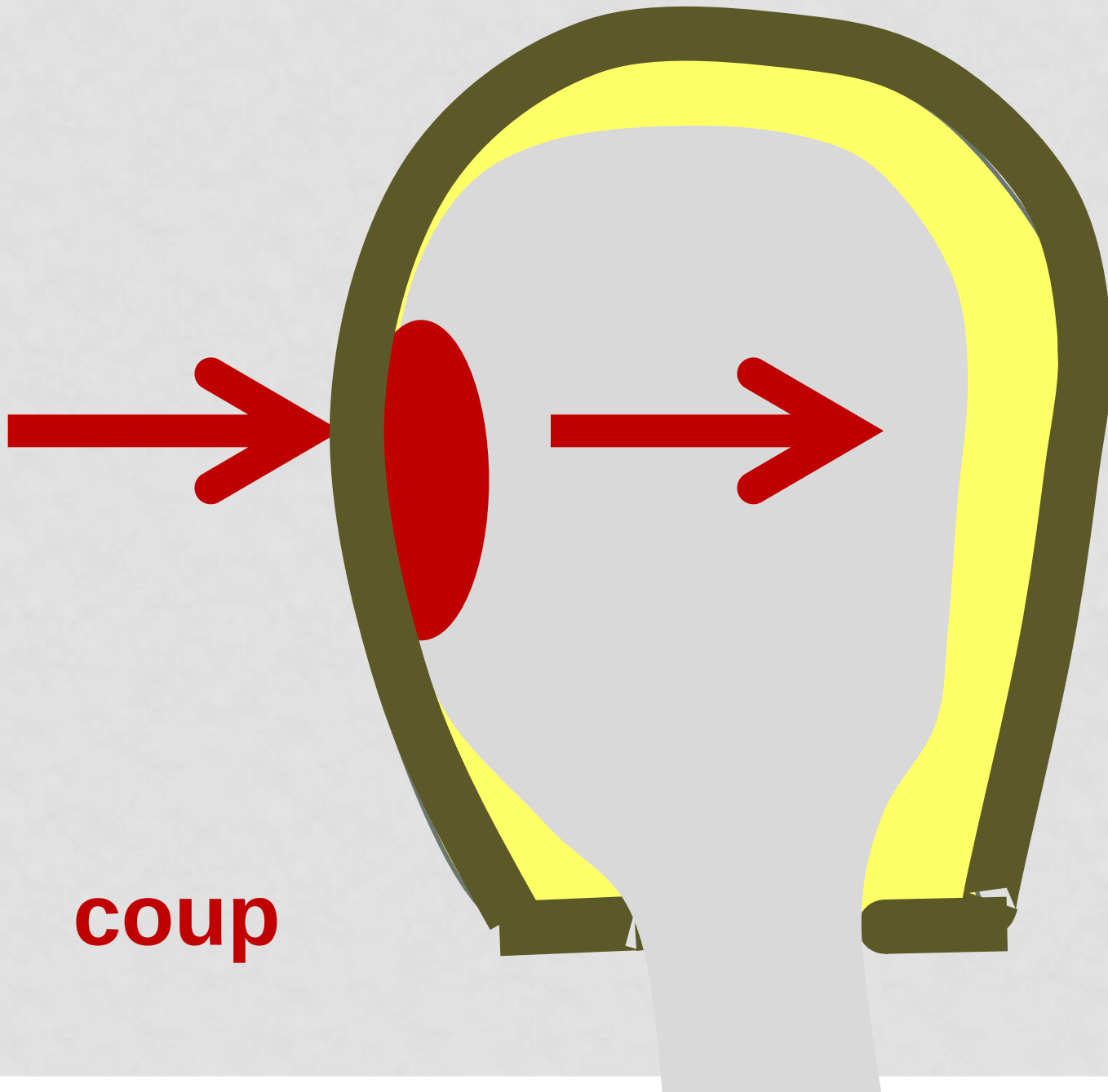
Contusion





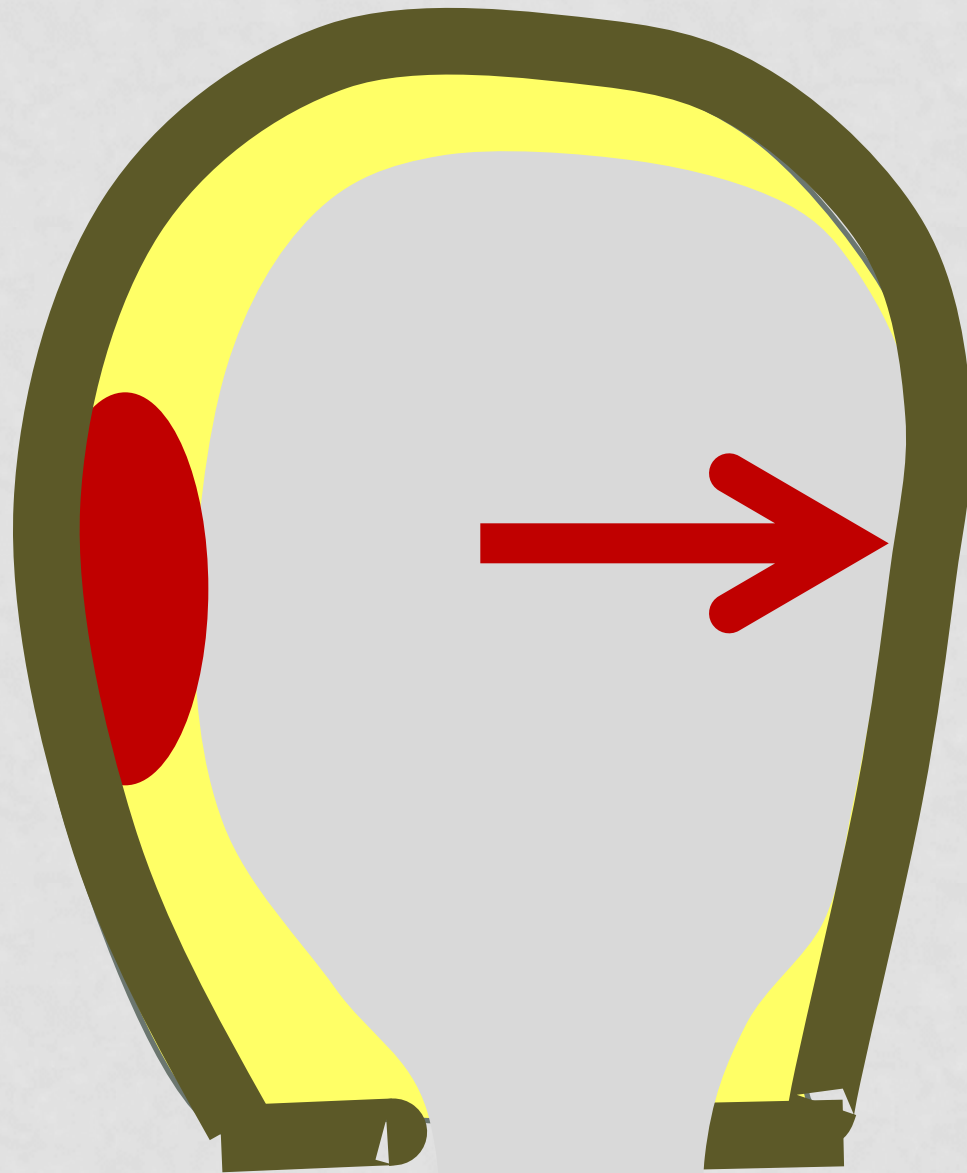


coup

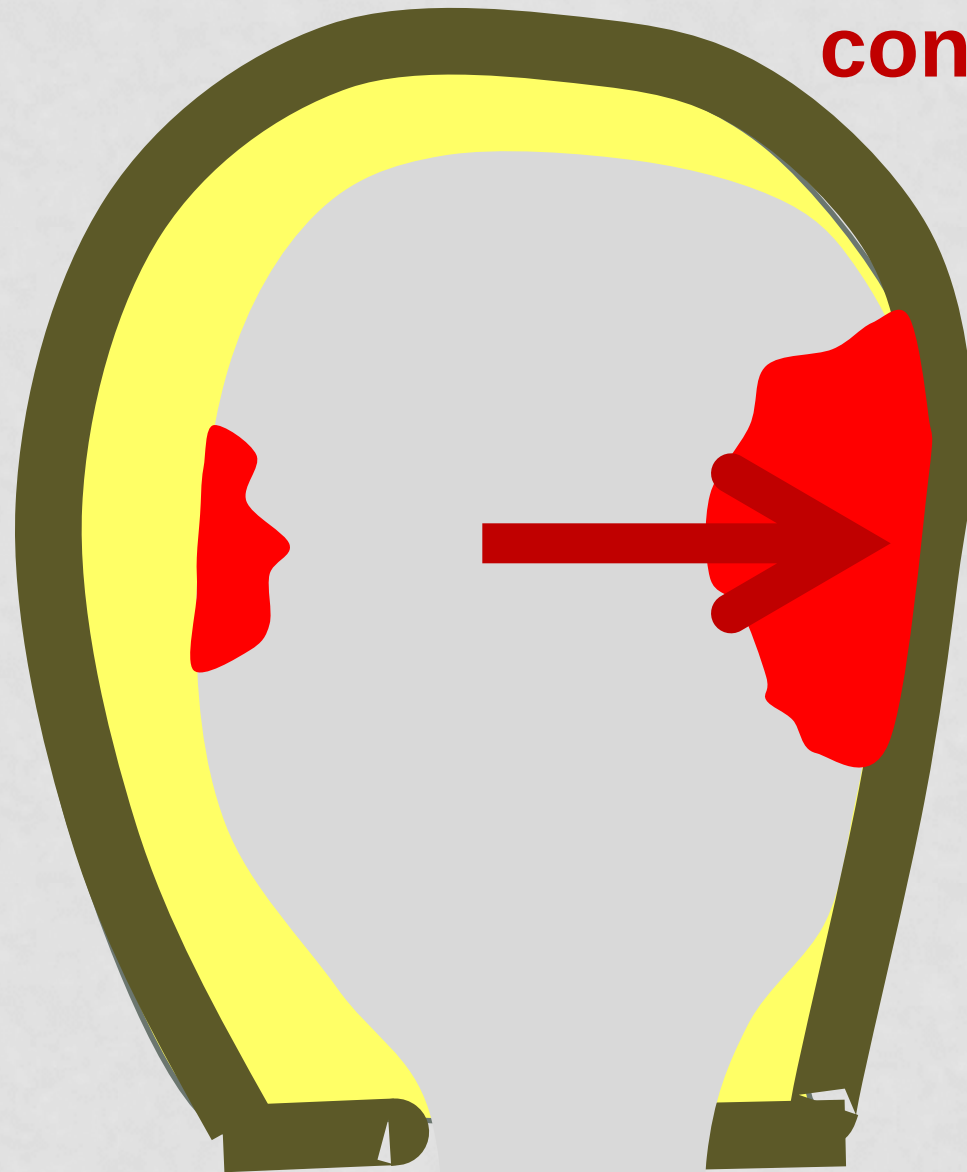


coup

coup



coup

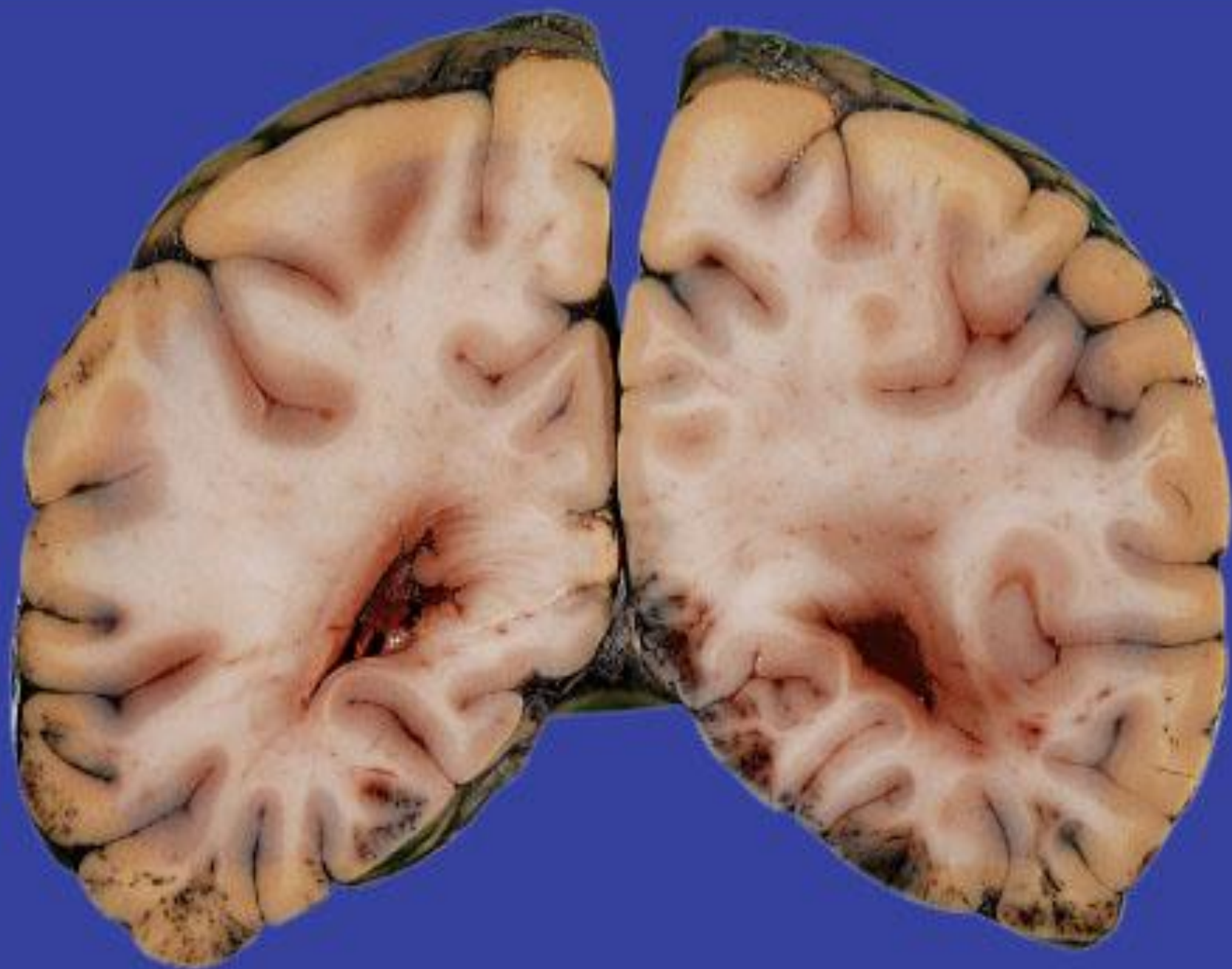


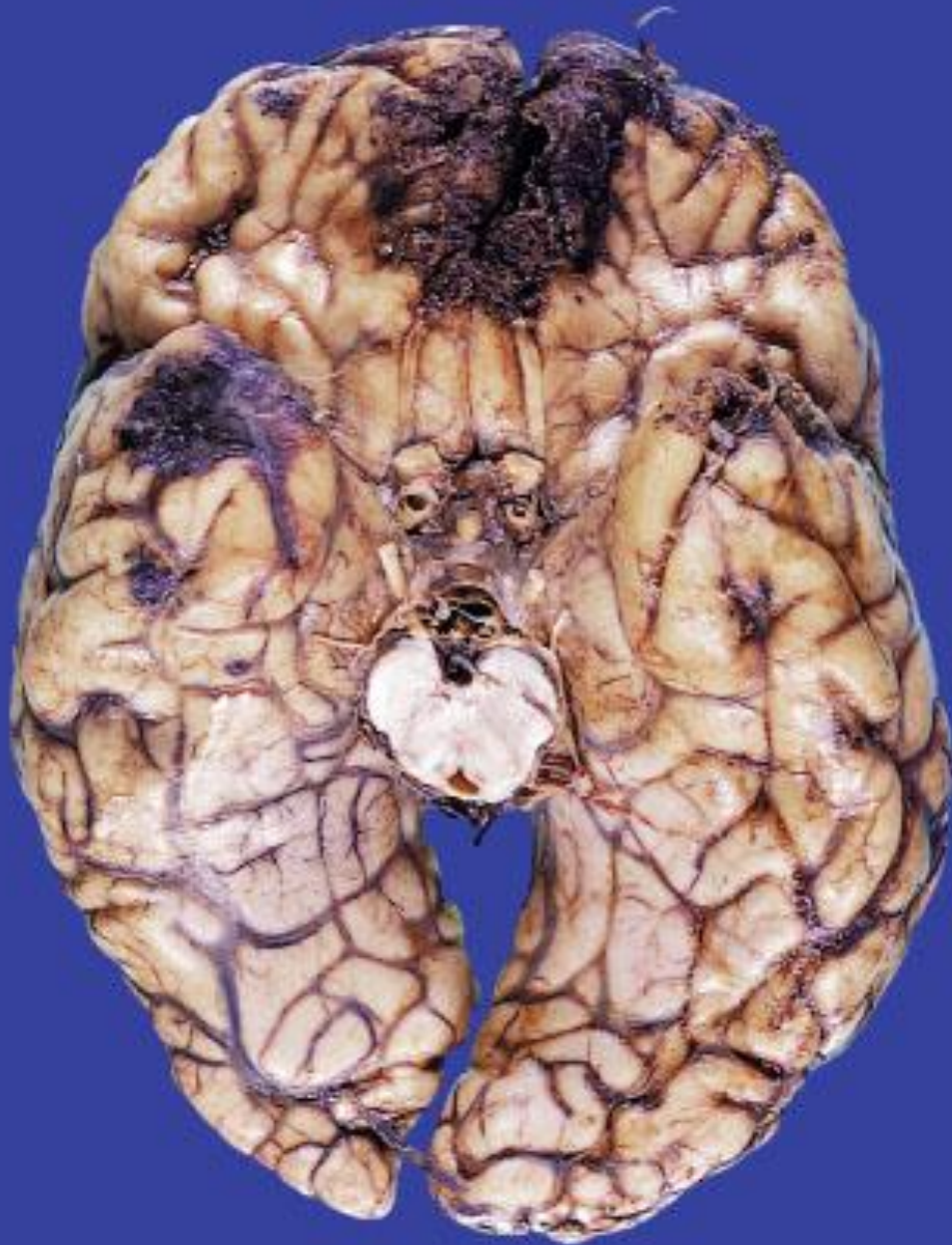
contre - coup

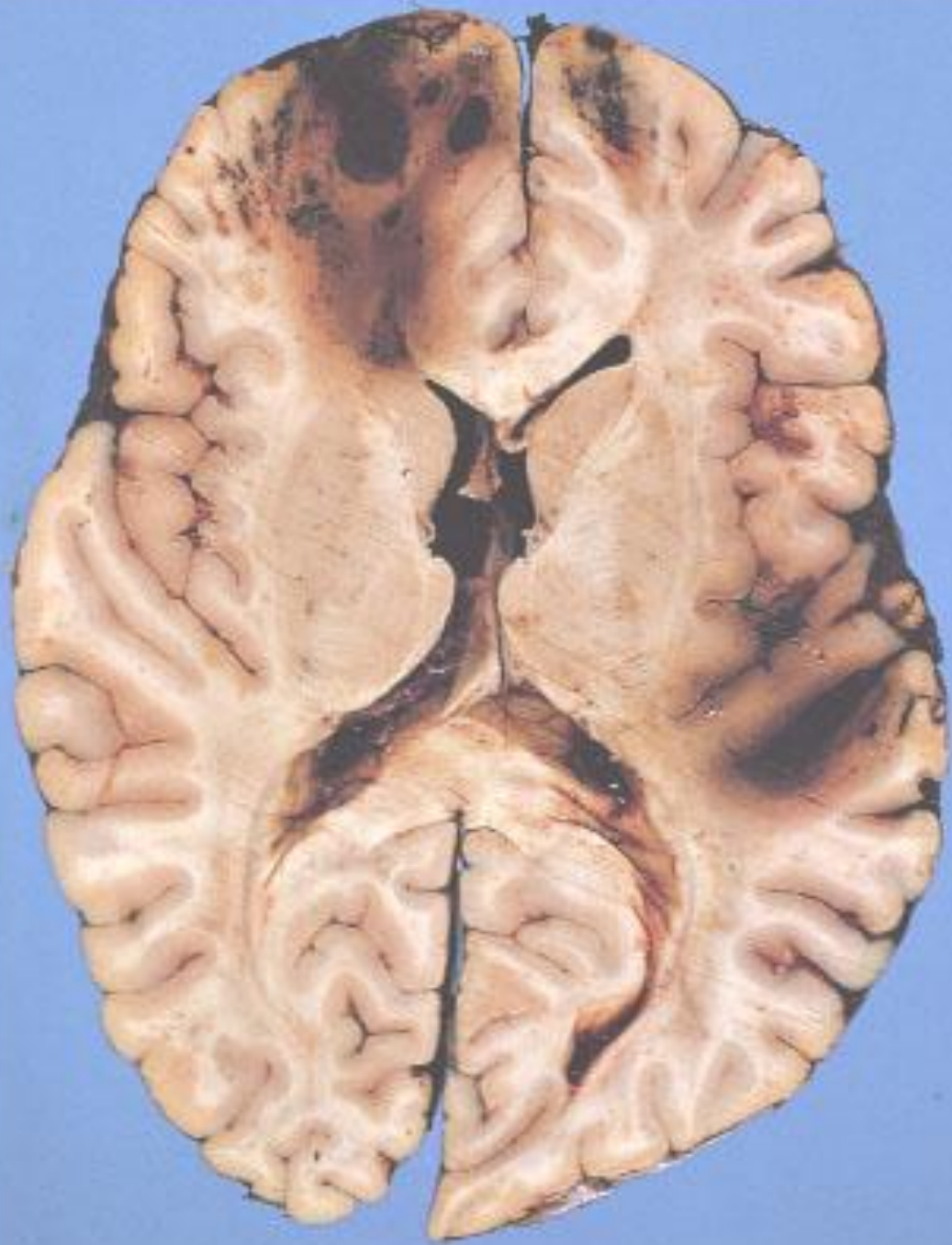
Cerebral injury following a blow may develop at two points

- Point of contact (coup injury)
- Brain surface opposite to it (counter coup)
 - Both are contusions if the head is immobile at the time of trauma, only a coup injury is found. When the head is mobile, there may be coup lesion beneath the impact & countercoup lesions. The coup lesion is caused by the force of direct impact between the brain & skull at the site of impact, the countercoup contusion develops when the brain strikes the opposite inner surface of the skull after sudden deceleration



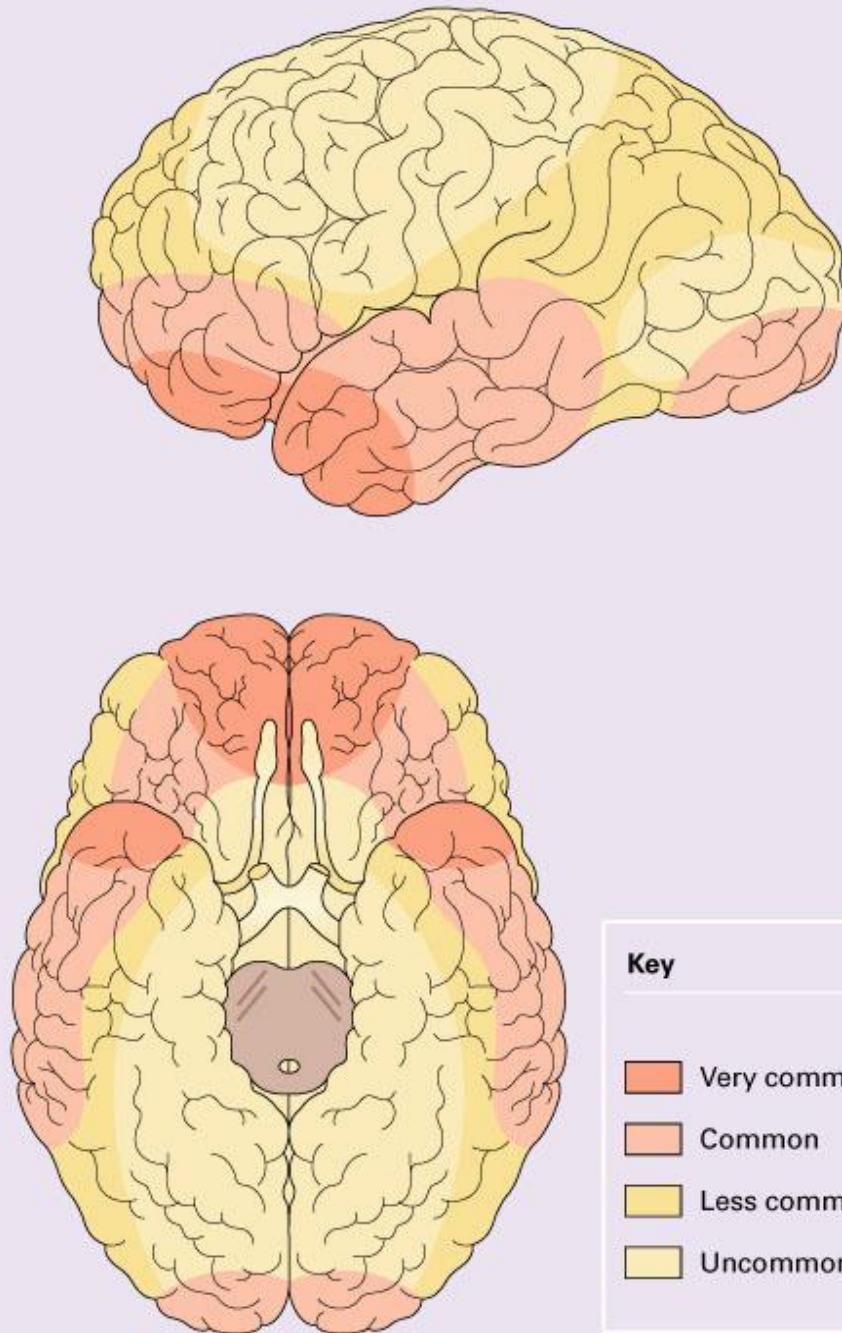








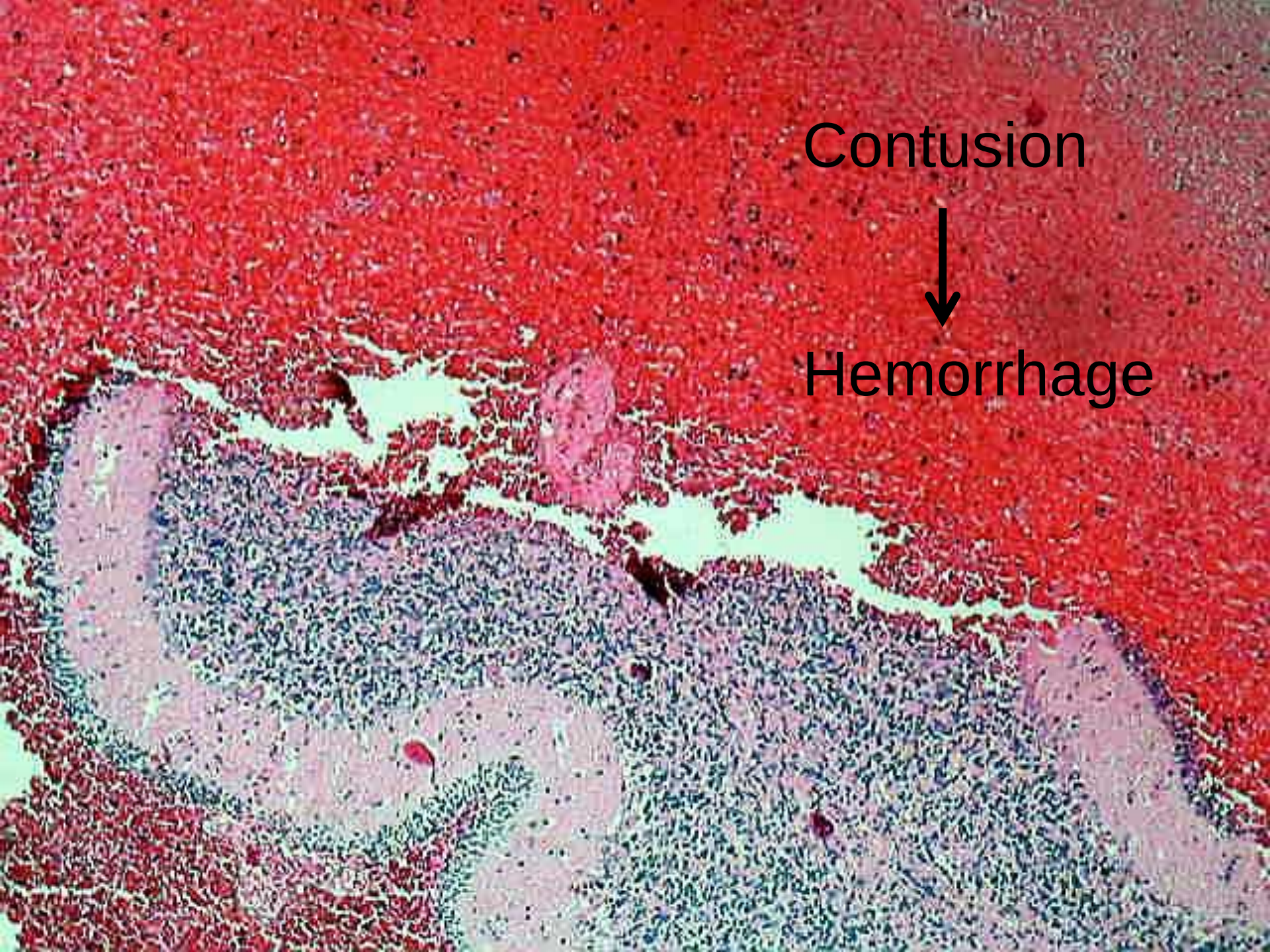
Typical sites of contusions

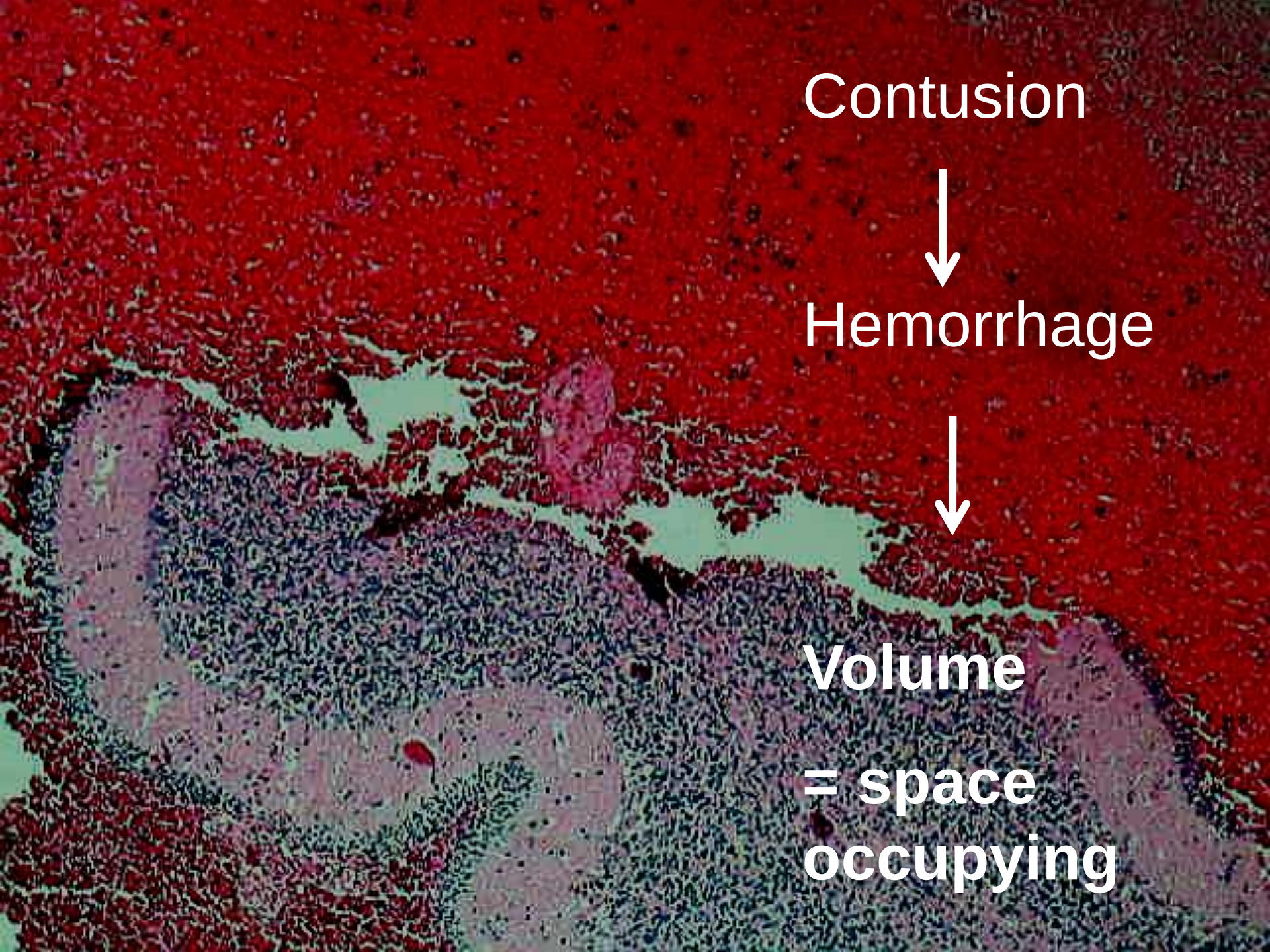


Contusion



Hemorrhage





Contusion



Hemorrhage



Volume

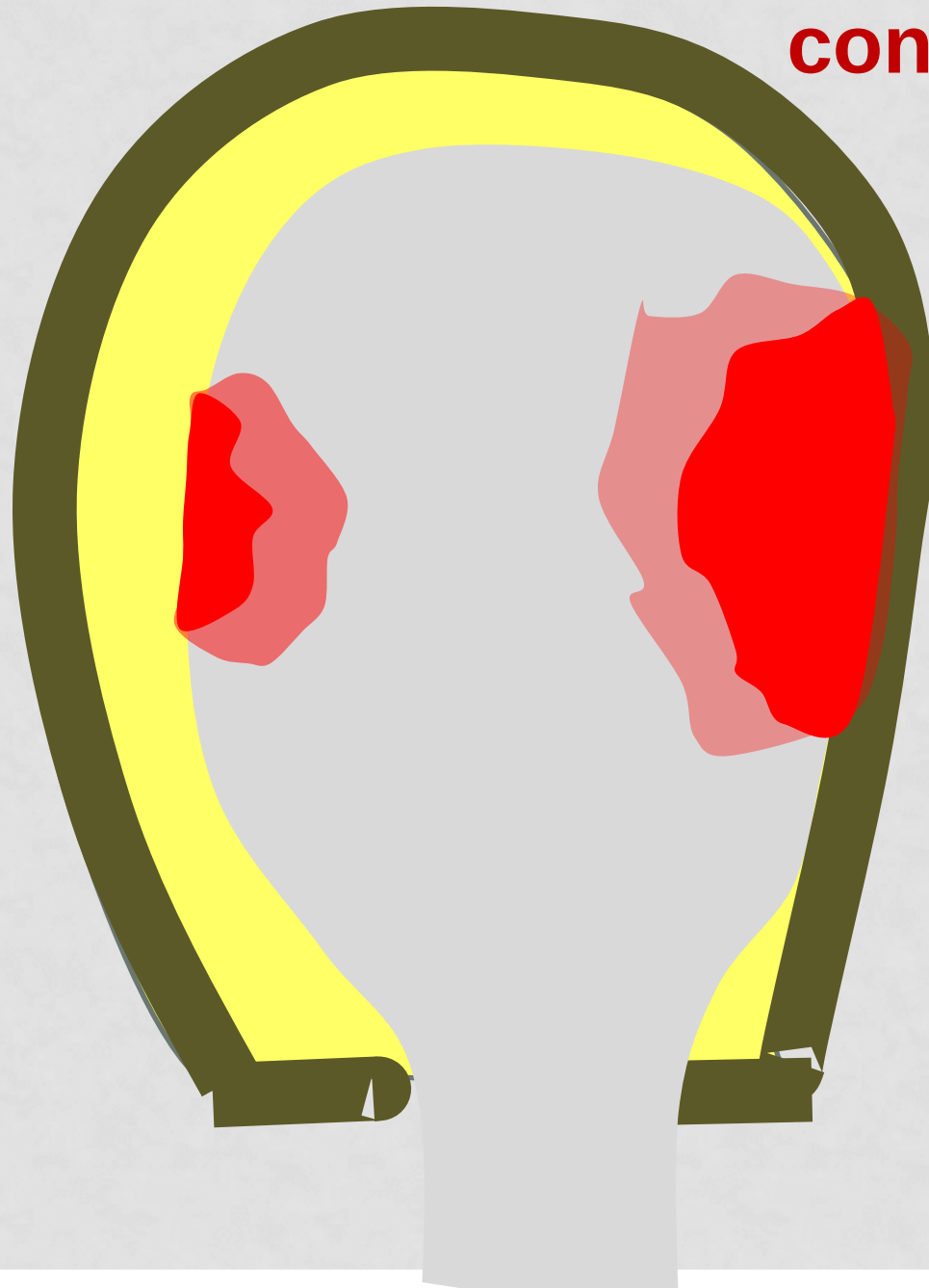
**= space
occupying**

**space
occupying
process**

SOP

contre - coup

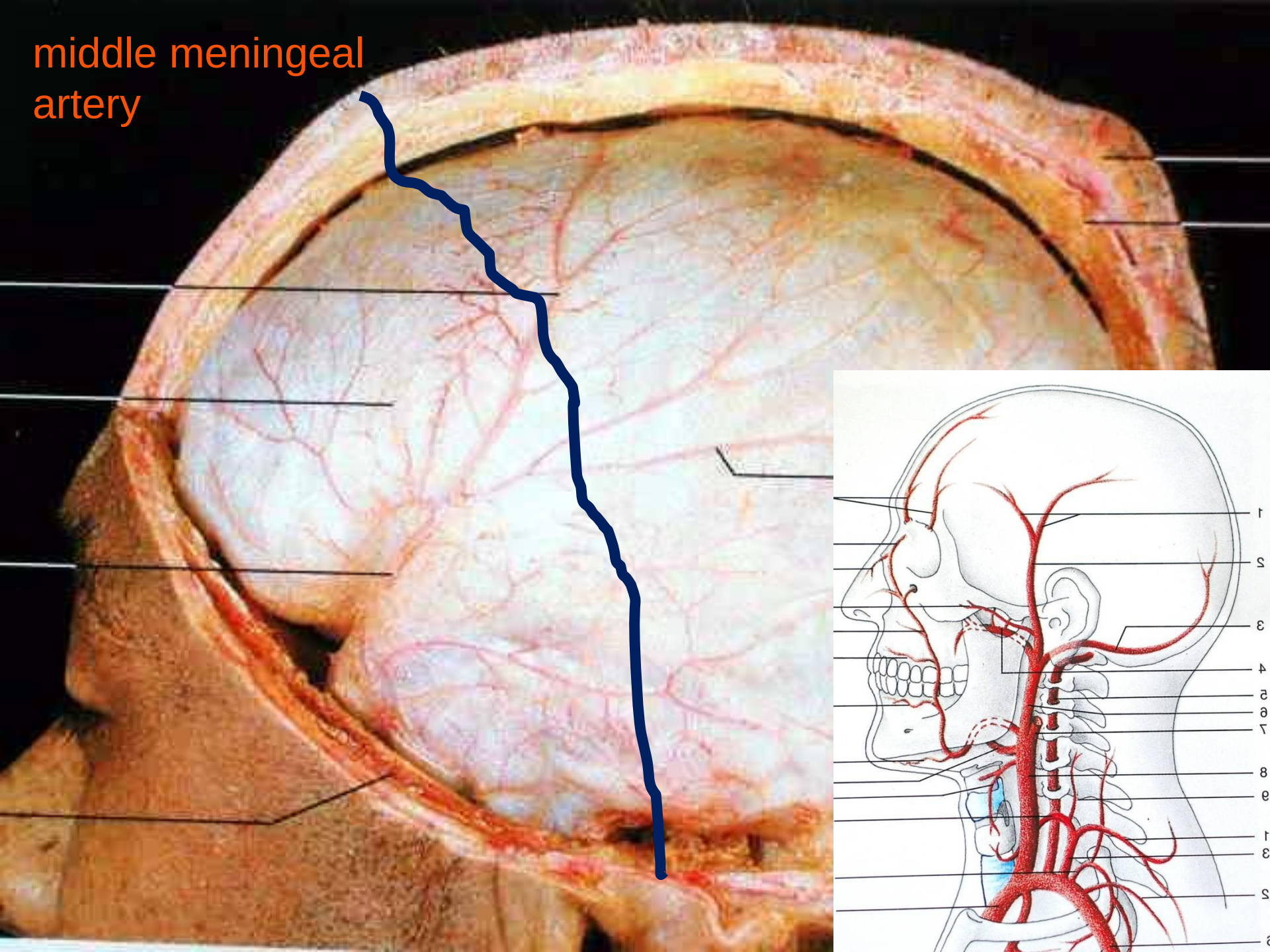
coup



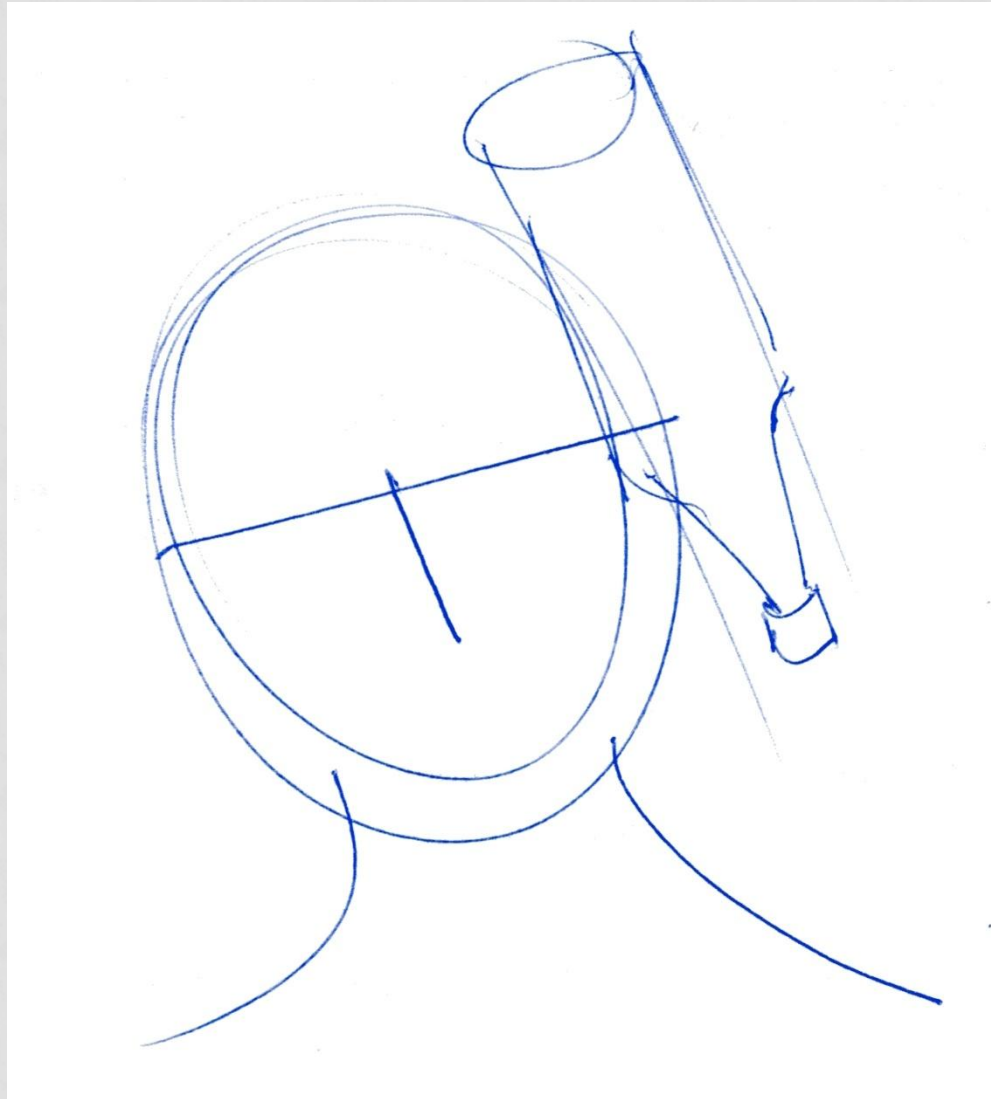
Traumatic vascular injury

- Vascular injury is a frequent component of CNS trauma & results from direct trauma & disruption of the vessel wall leading to hemorrhage.

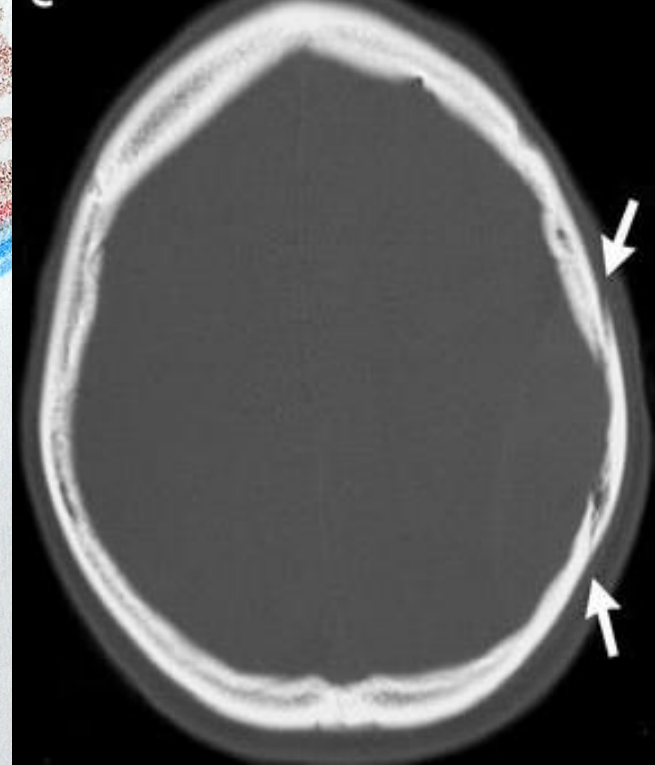
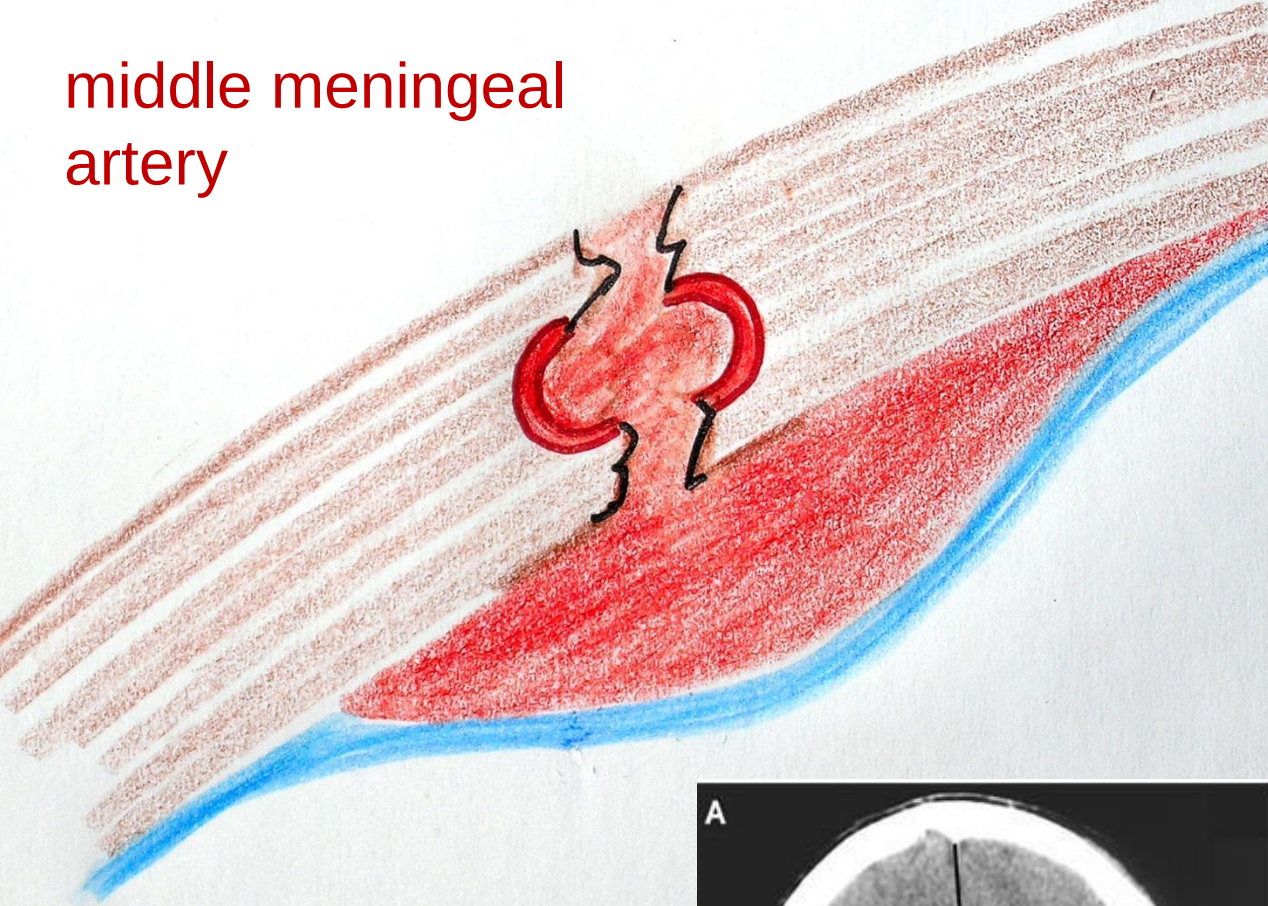
middle meningeal
artery



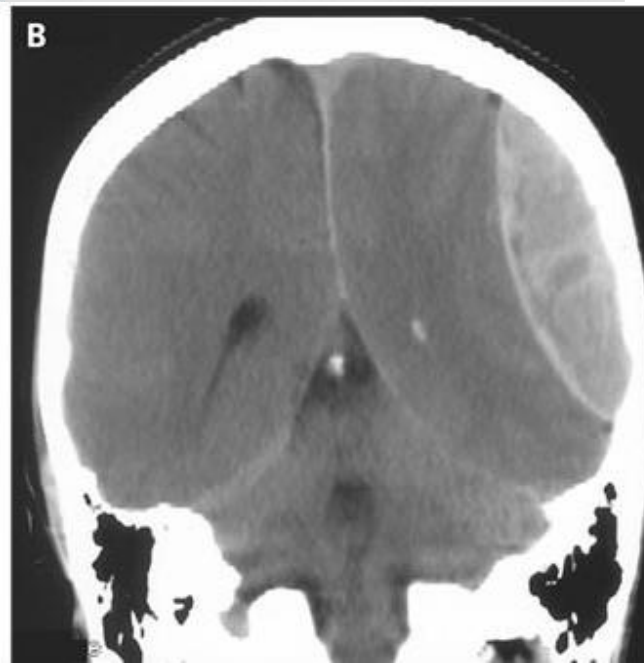
An 18-year-old intoxicated man was assaulted with a glass bottle on the left parietal region of his head and had a 5-minute loss of consciousness



middle meningeal
artery

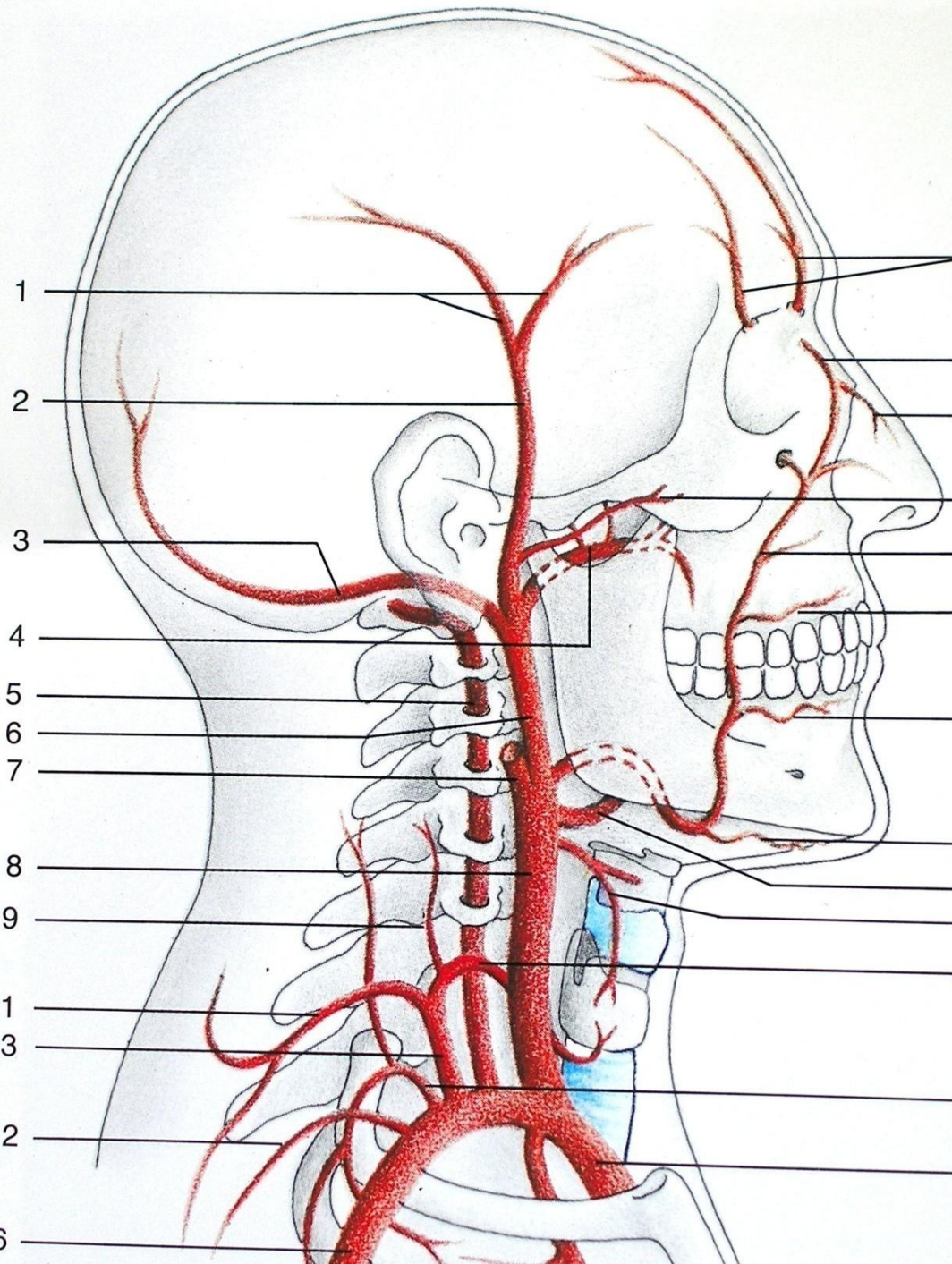


epidural hematoma

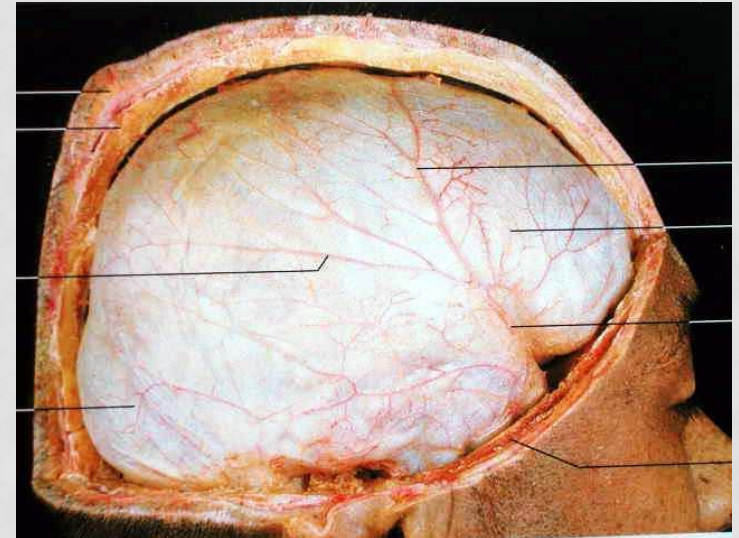


HISTORY

- M 18
- Assaulted with a glass bottle left parietal
- 5 min loss of consciousness
- 2 hours: headache, nausea, vomiting
- Physical examination: small wound on the scalp
- **CT: Epidural hematoma left parietal**
- **After 30 min:**
 - Consciousness dropped, left pupil dilated
- Taken to operating room: epidural hematoma evacuated



Middle meningeal artery

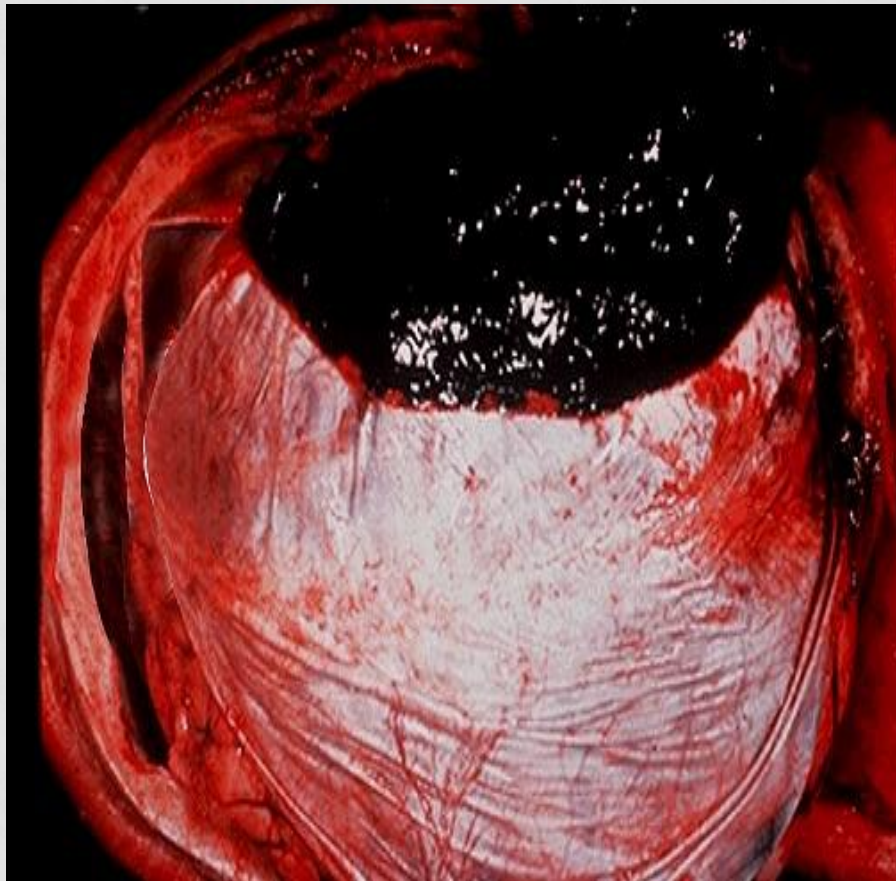


Left pupil dilated

Arterial hemorrhage



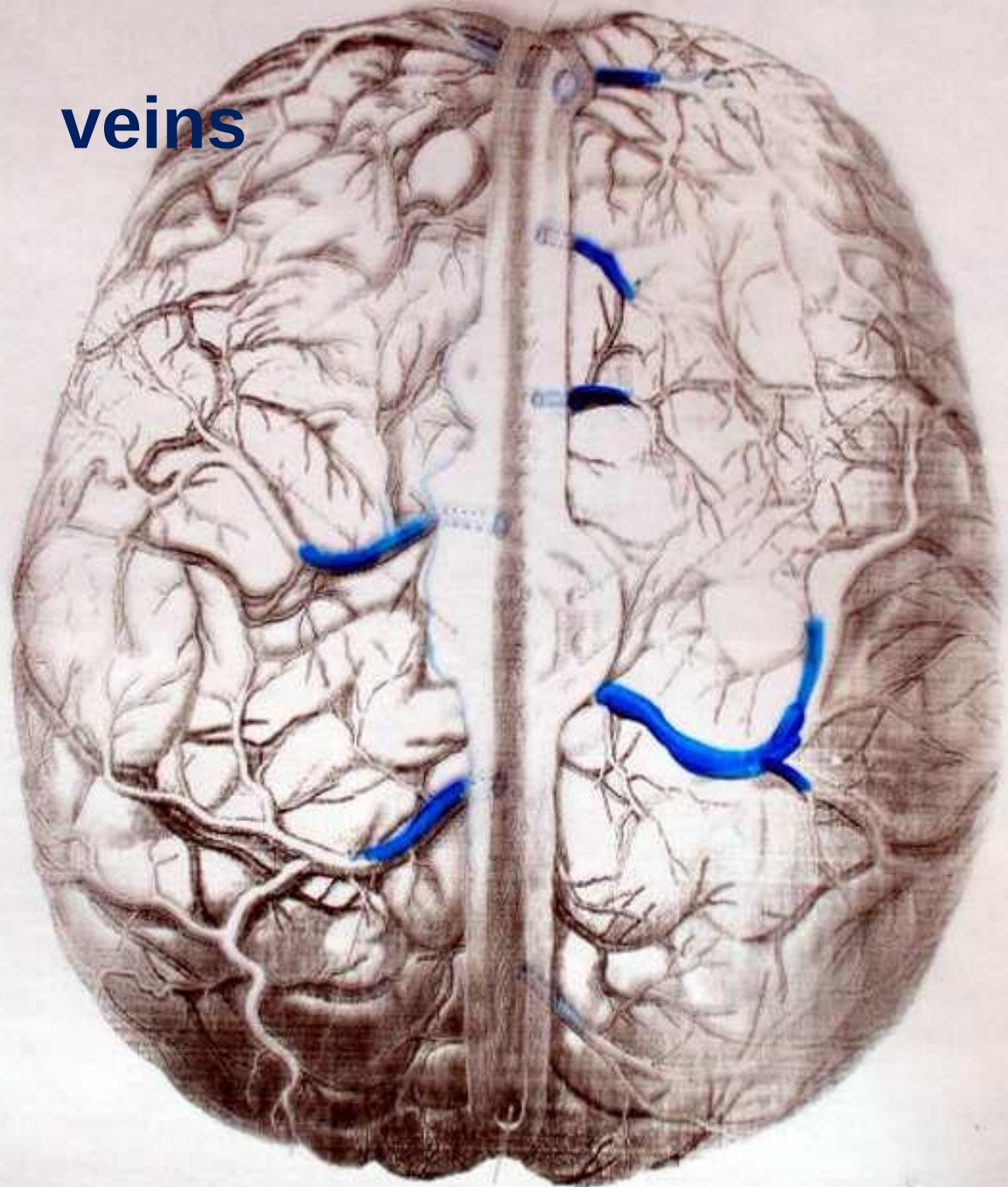
Short interval



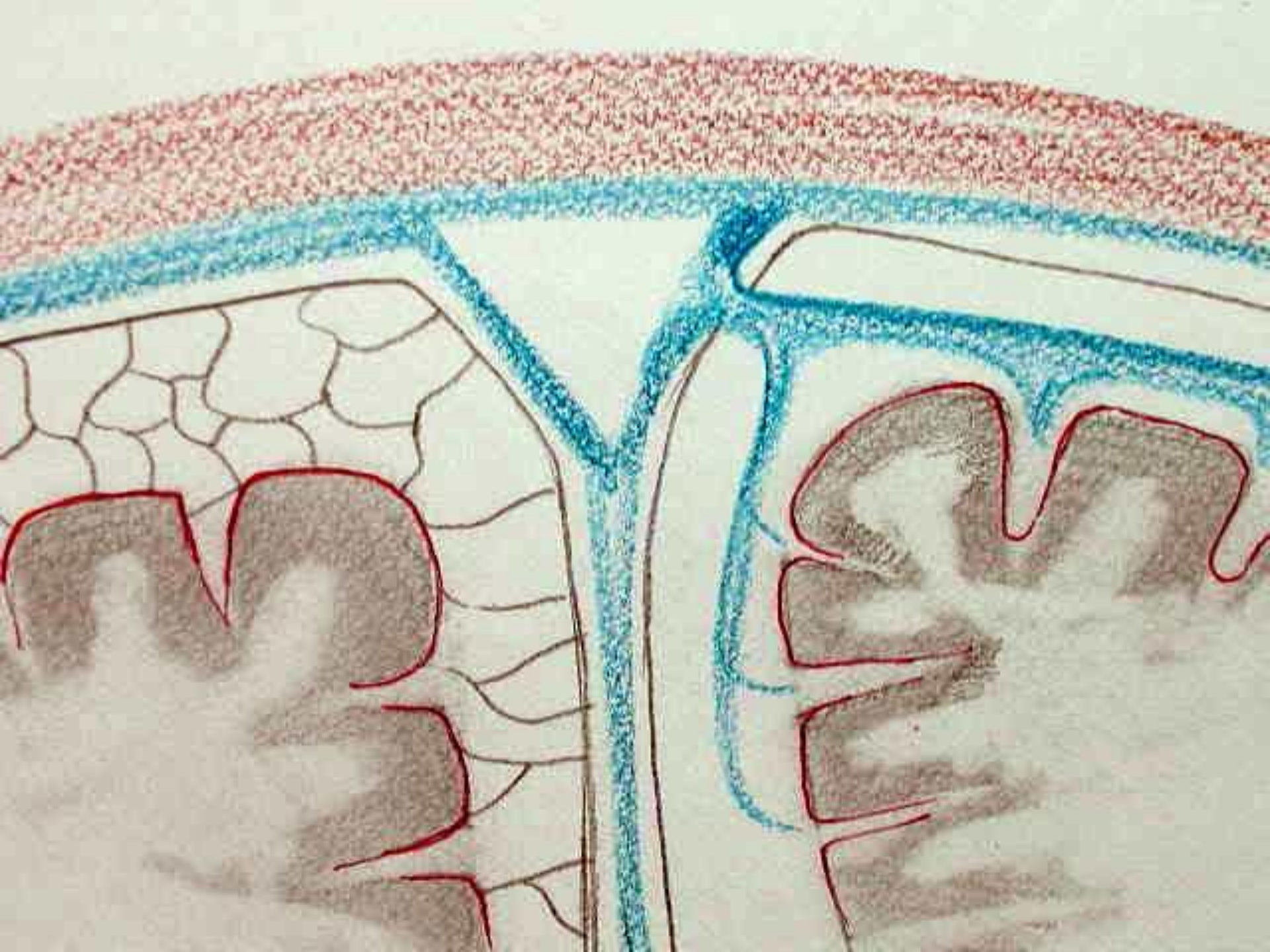
Epidural hematoma

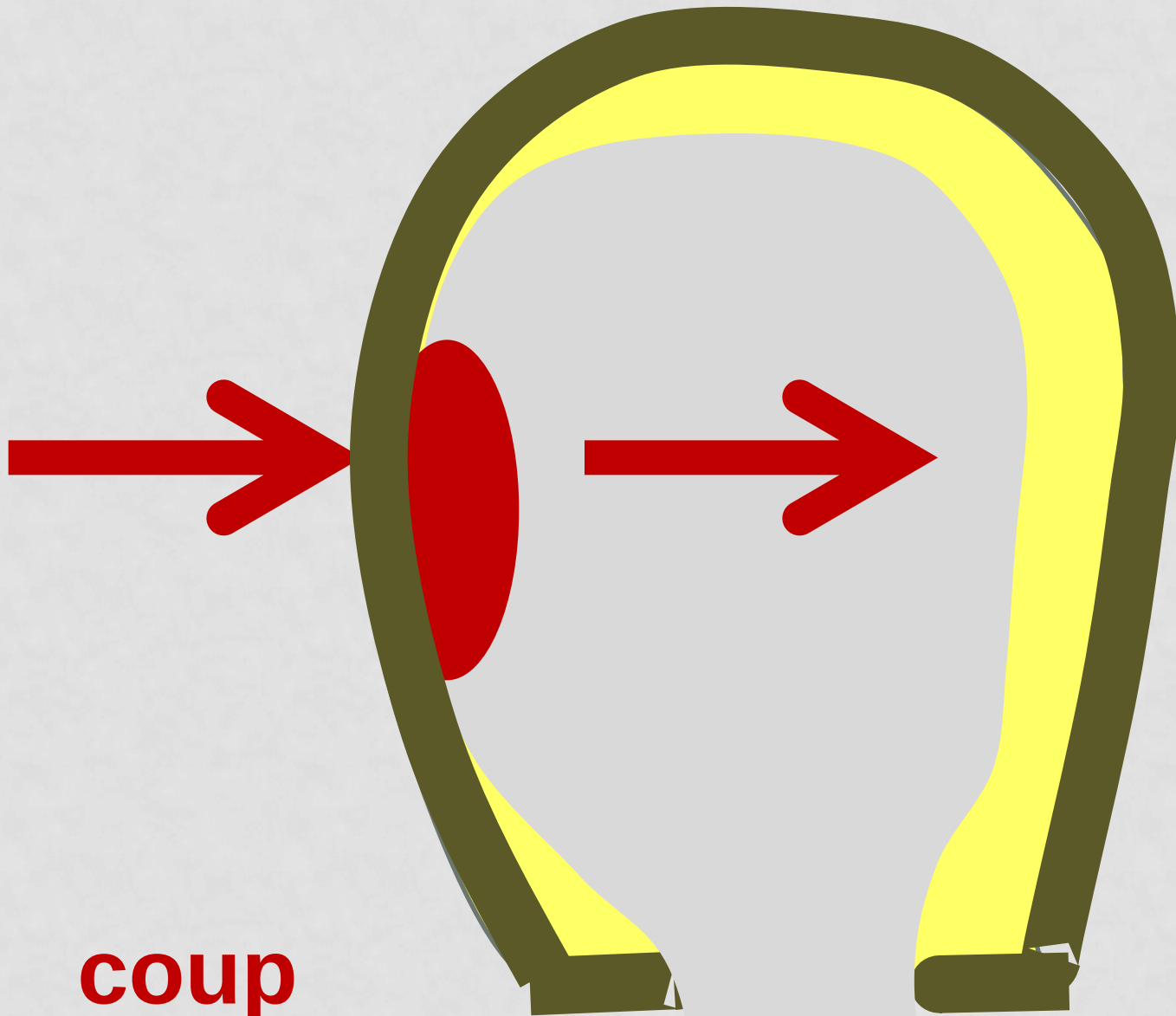
- Epidural space is a potential space between dura & internal surface of skull
 - Trauma to temporal bone can lead to tearing of middle meningeal artery & the accumulation of blood results in separation of the dura from the skull
- c/f – patients can be lucid for several hours b/n the moment of trauma & the development of neurological signs
- An epidural hematoma may expand rapidly & is neurosurgical emergency requiring prompt drainage

bridging veins

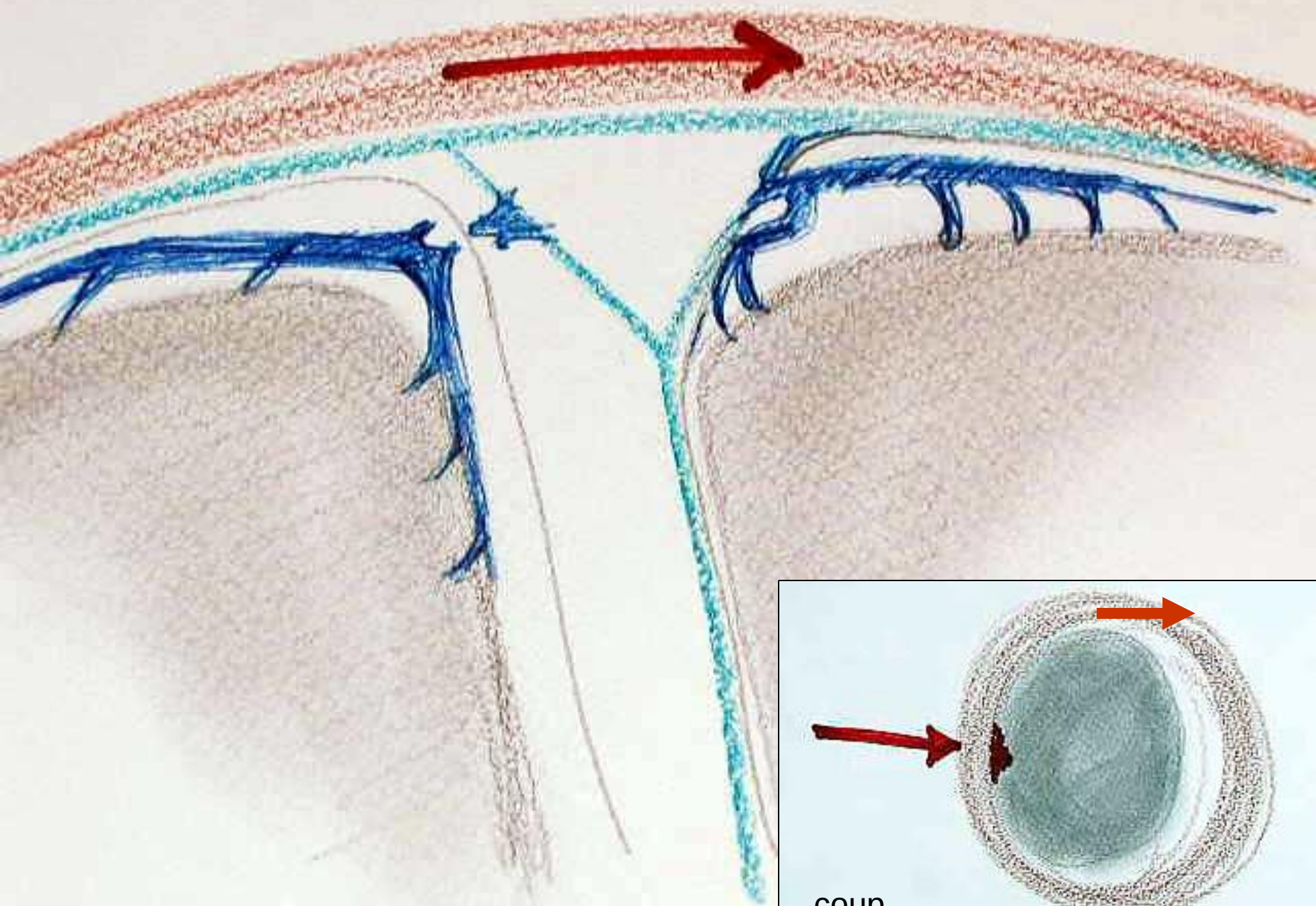




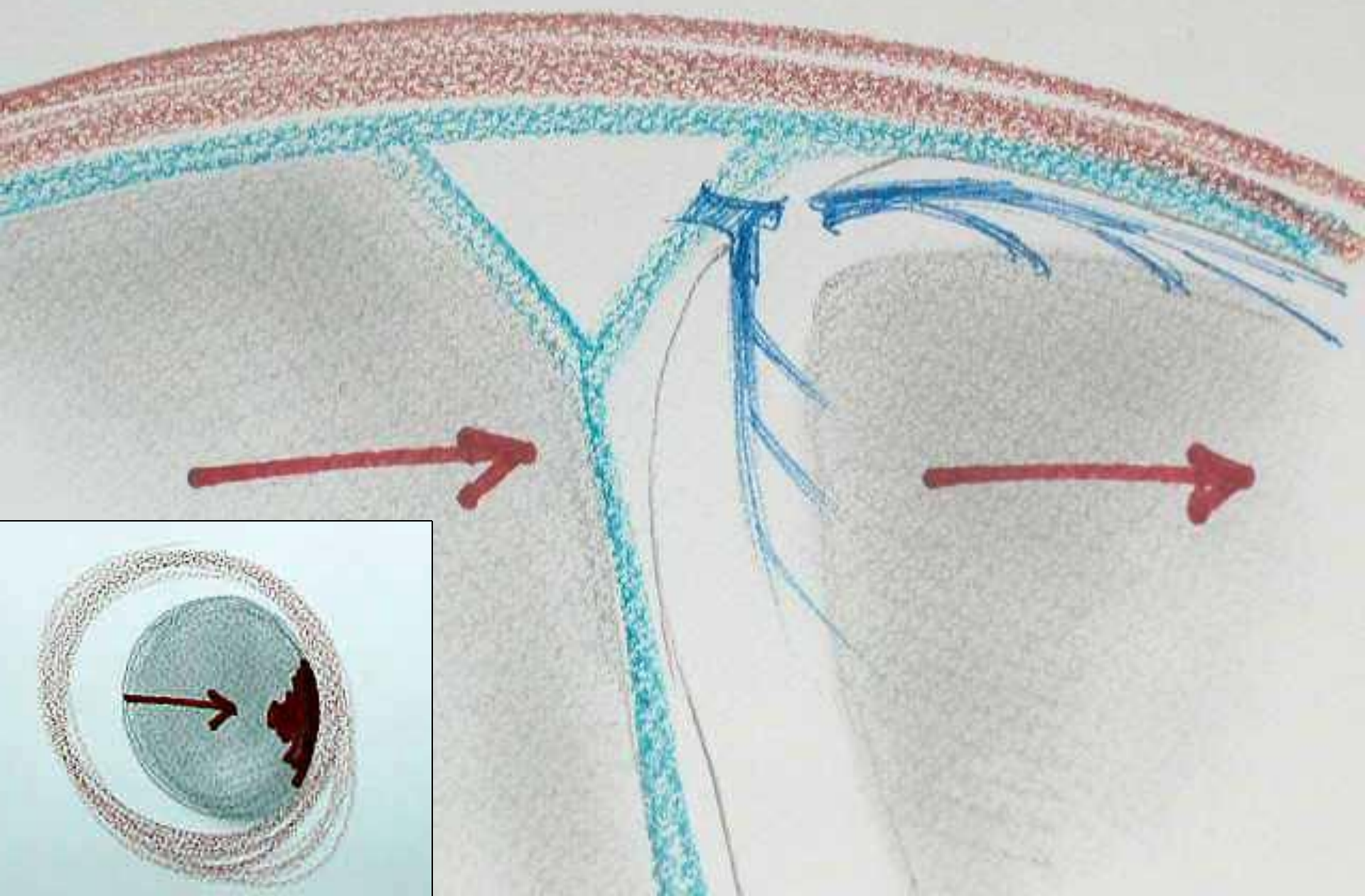




coup



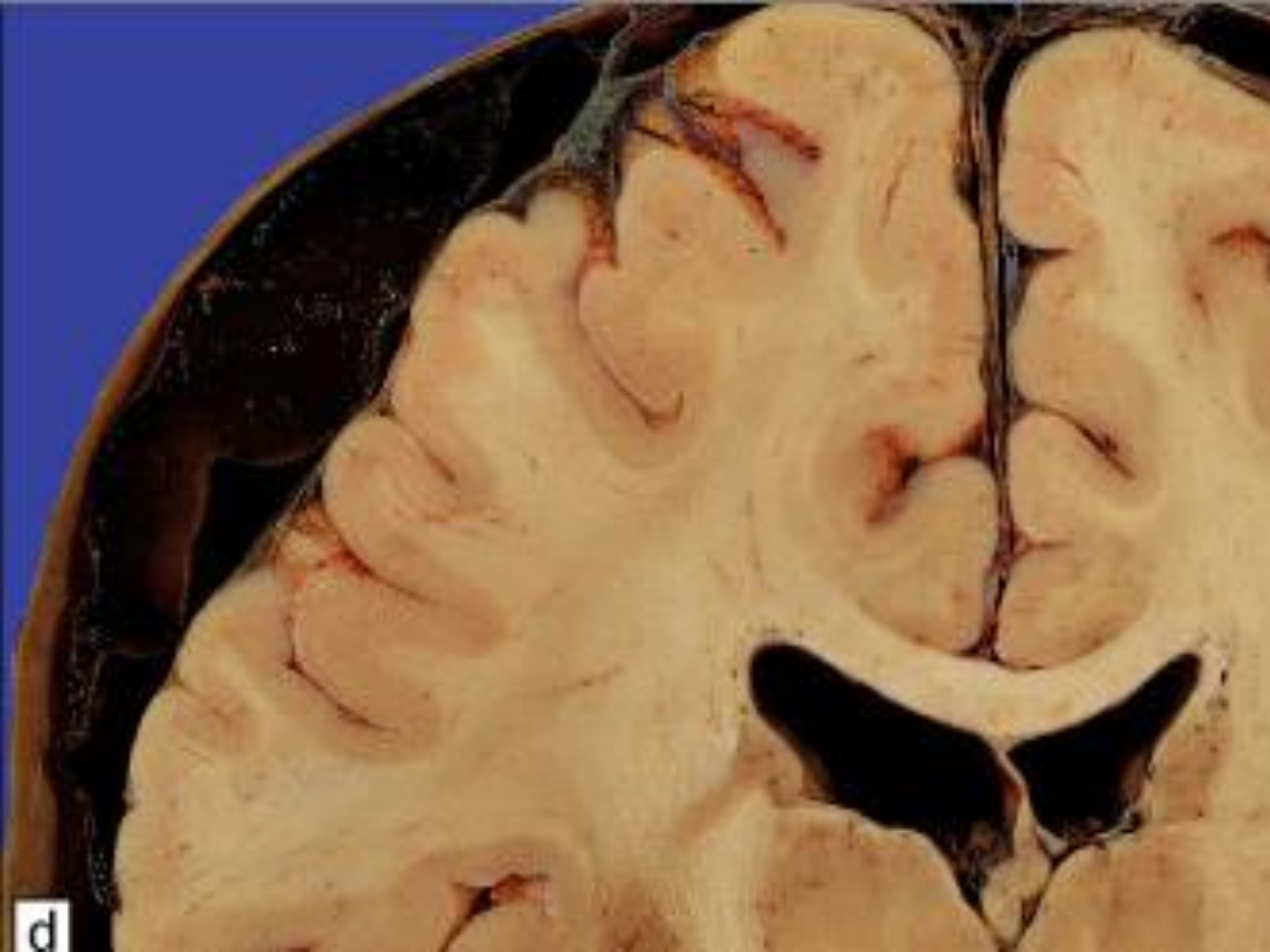
coup





subdural hematoma





d

Subdural hematoma

- Subdural space is a potential space beneath the inner surface of the dura matter & the outer arachnoid layer
- Bridging veins travel from the surface of convexities of the cerebral hemispheres through the subarachnoid & subdural space to empty into the superior sagittal sinus
- These veins are prone to tearing along their course through the subdural space; they are the source of bleeding in most cases of subdural hematoma.

- The most commonly accepted mechanism of damage postulates that the brain floating freely in CSF bath can move within the skull but the veins are fixed.
- The displacement of the brain that occurs in trauma can tear the veins at a point where they penetrate the dura .

- The bridging veins can be teared along their course in dura space even with minor trauma in elderly patients with brain atrophy . Due to this there is increased rate of subdural hematoma in elderly

C/F

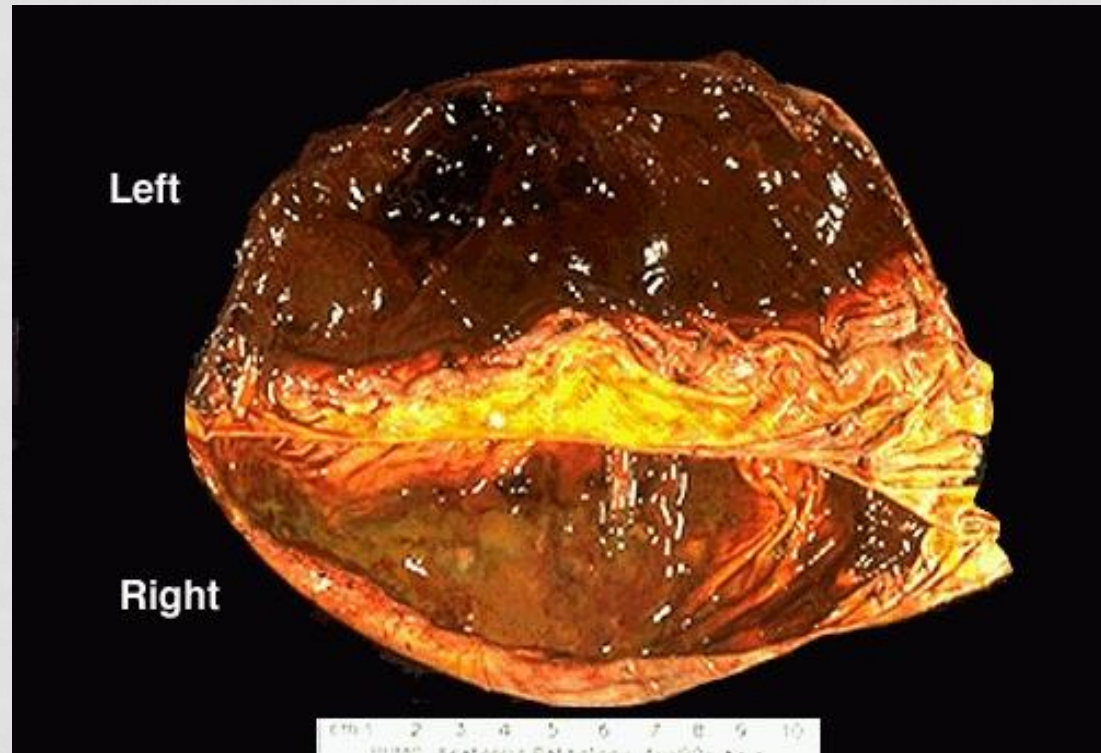
- Manifest with in 48 hrs
- Show progressive neurological deterioration

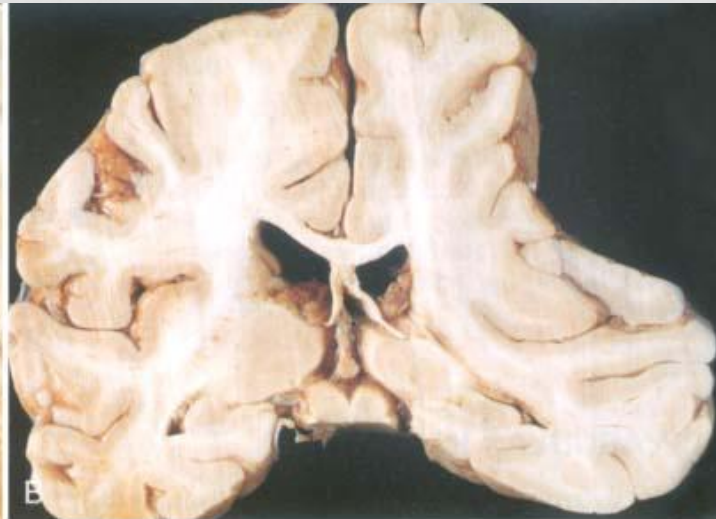
Chronic subdural hematomas

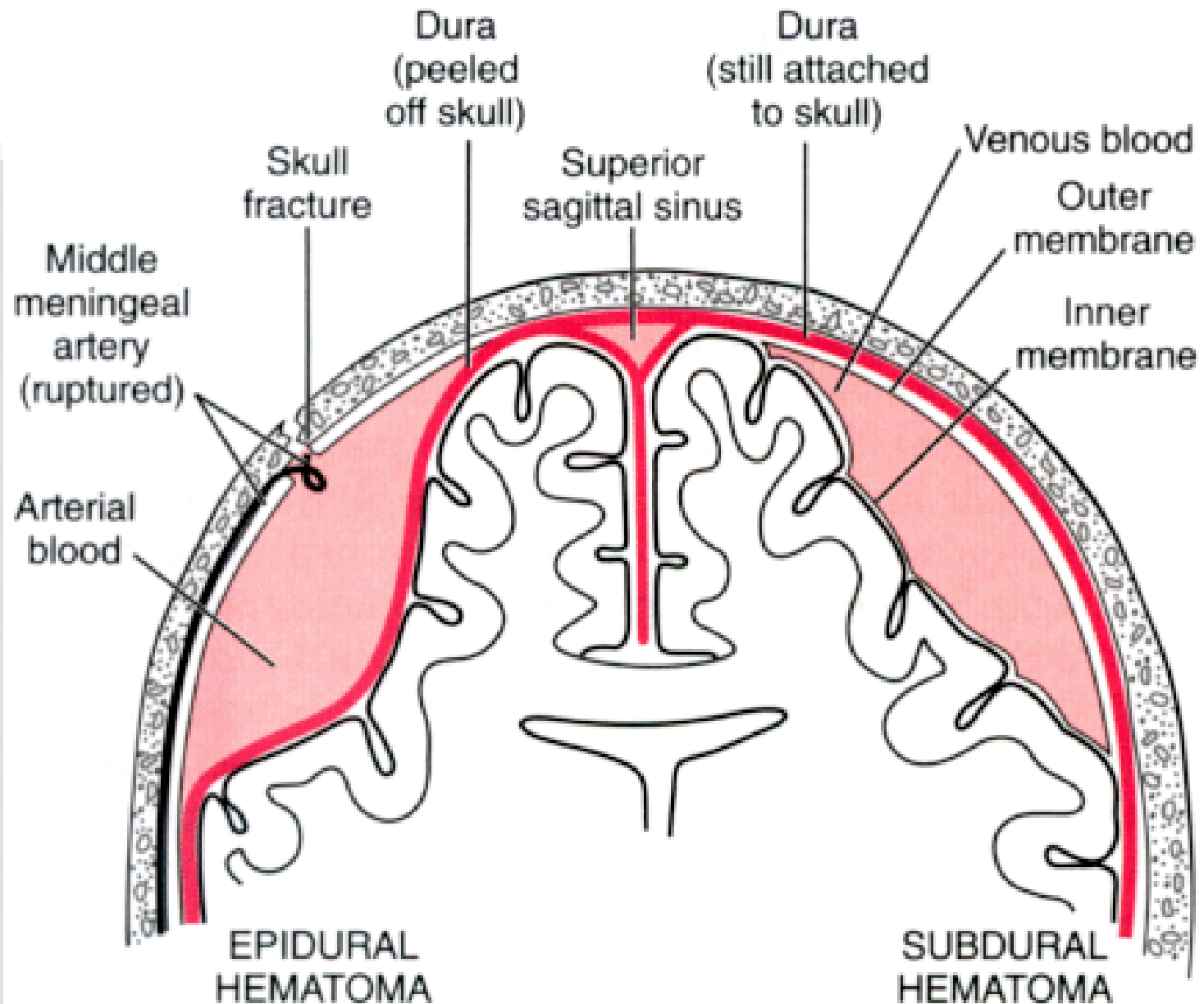
- are less frequently associated with a well-defined history of trauma than are acute lesions. They are often associated with brain atrophy.
- They are often bilateral.

- Chronic subdural hematomas are composed of liquefied blood or yellow-tinged fluid separated from the inner surface of the dura mater and the underlying brain by "neomembranes" composed of granulation tissue and mature collagen, derived from the dura mater. Blood vessels within the neomembranes are abnormally permeable, owing to an incompletely developed endothelial layer, contributing to progressive accumulation of fluid and recurrent hemorrhage within the hematoma.

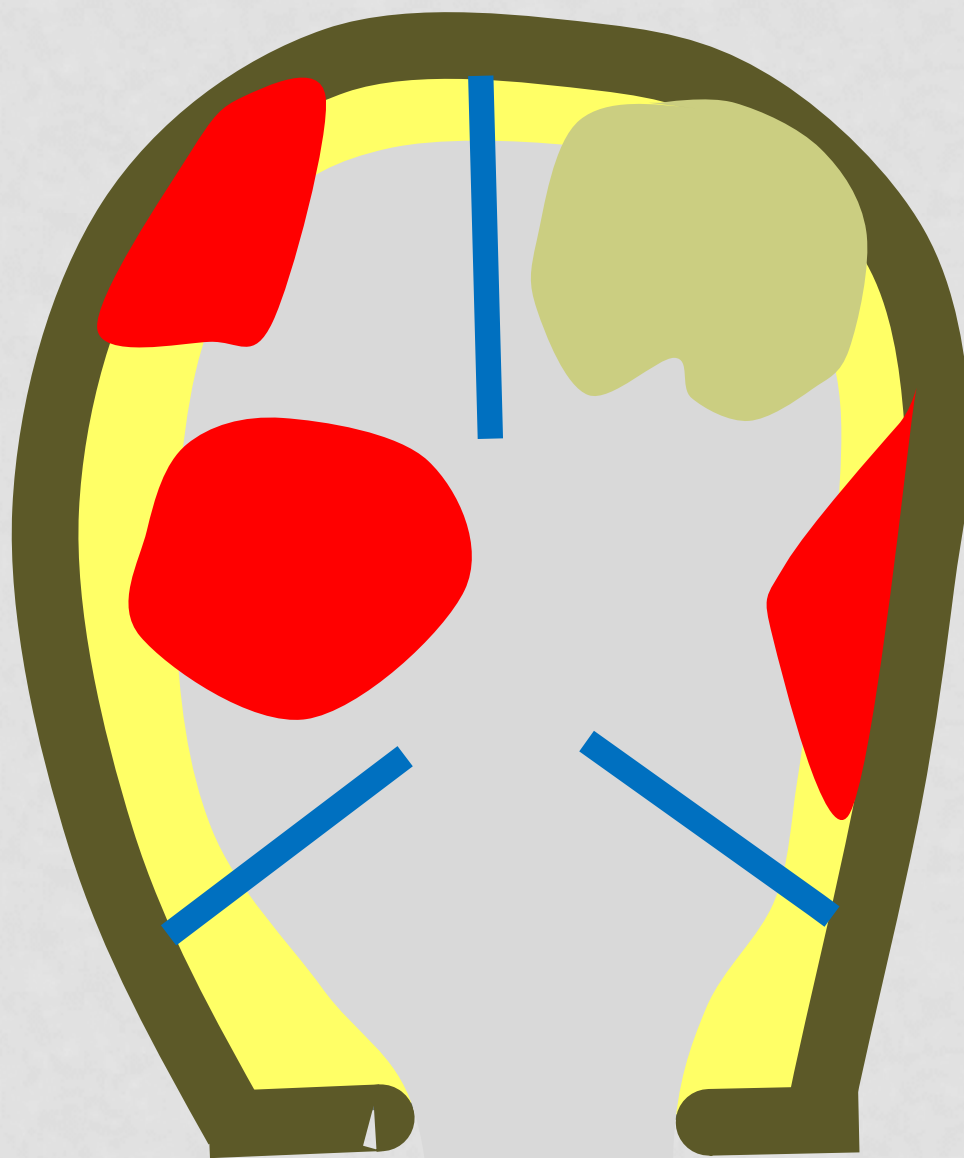
- Clinical symptoms include altered mental status, sometimes accompanied by focal neurologic deficits. Because of the slowly evolving nature of the deficits, chronic subdural hematomas may be confused clinically with dementia caused by neurodegenerative disorders such as Alzheimer disease.







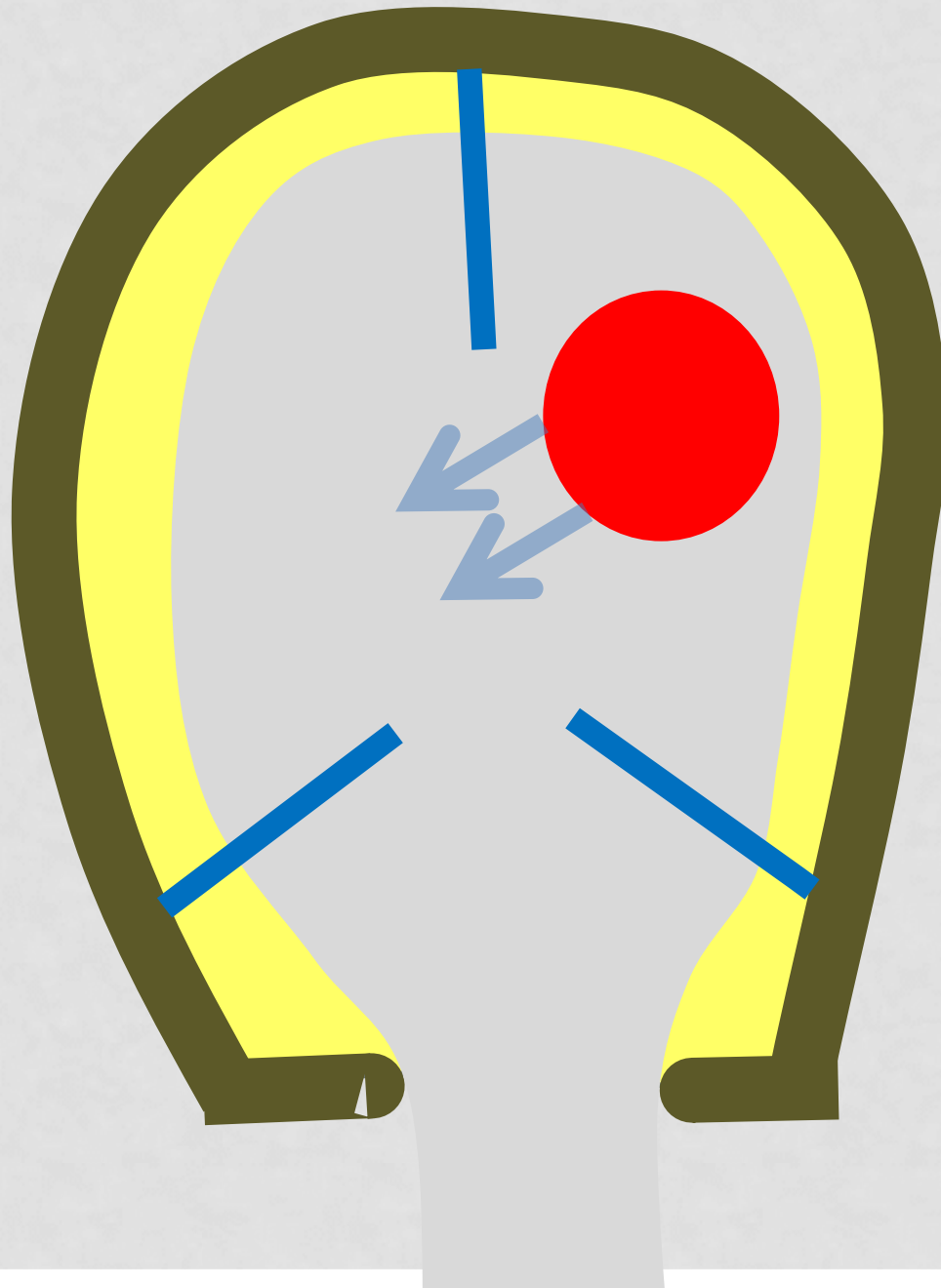
SOP



protective
shield



cage



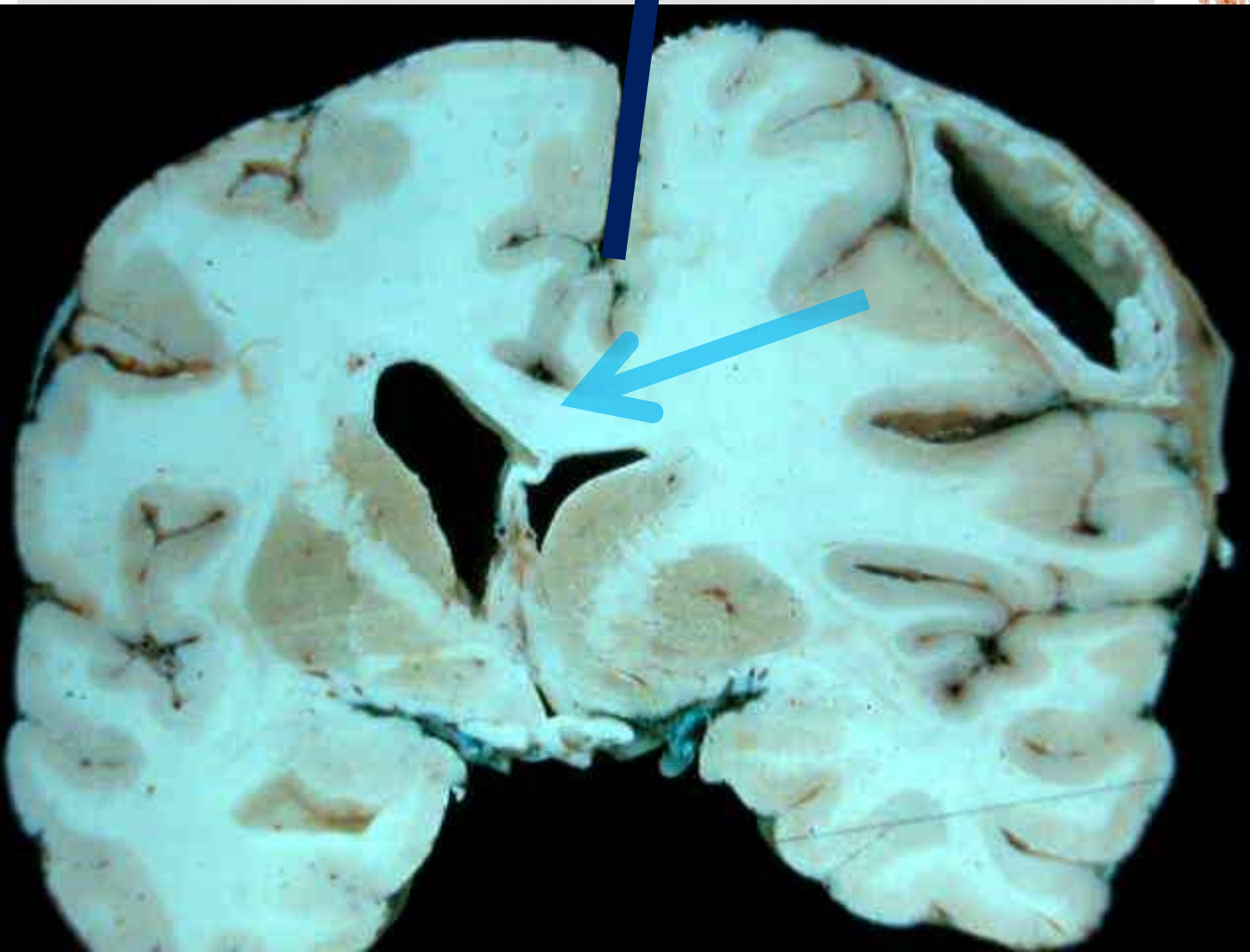
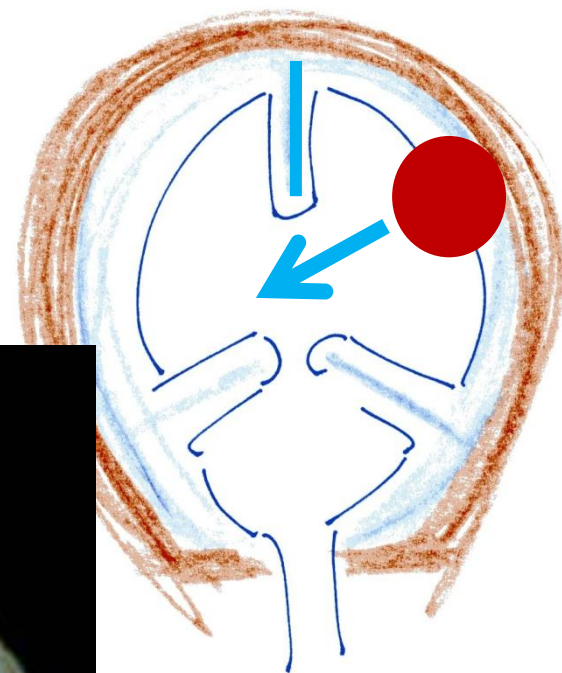
Herniation

volume of brain tissue increases beyond →
intracranial pressure may rise → displacement of
brain through partitions(falx and
tentorium)=herniation

Herniations are named by either the part of the brain
that is displaced or the structure across which it
moves

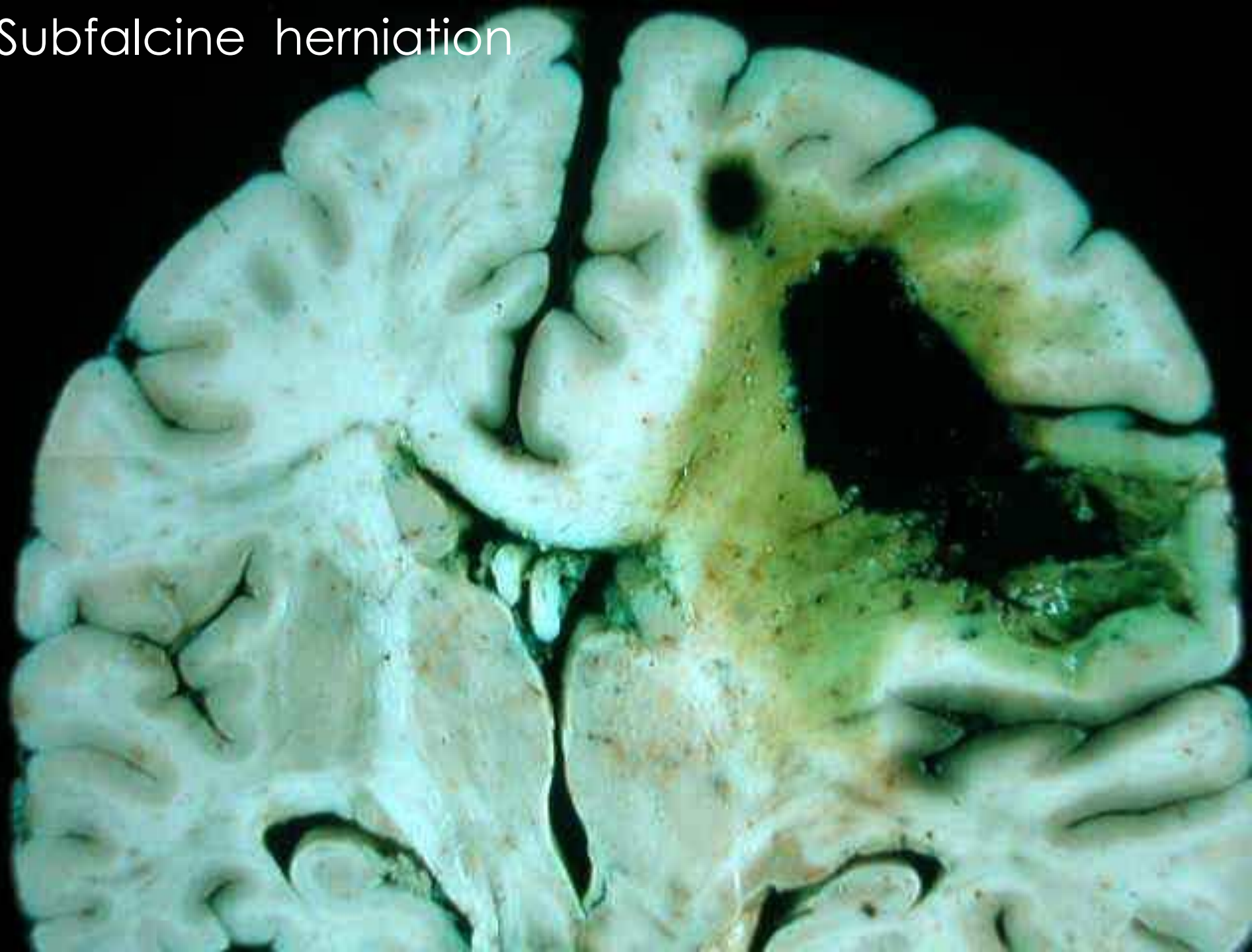
1. Subfalcine (cingulate) herniation occurs when unilateral or asymmetric expansion of a cerebral hemisphere displaces the **cingulate** gyrus under the edge of falx. This may be associated with compression of branches of the anterior cerebral artery.

Herniation

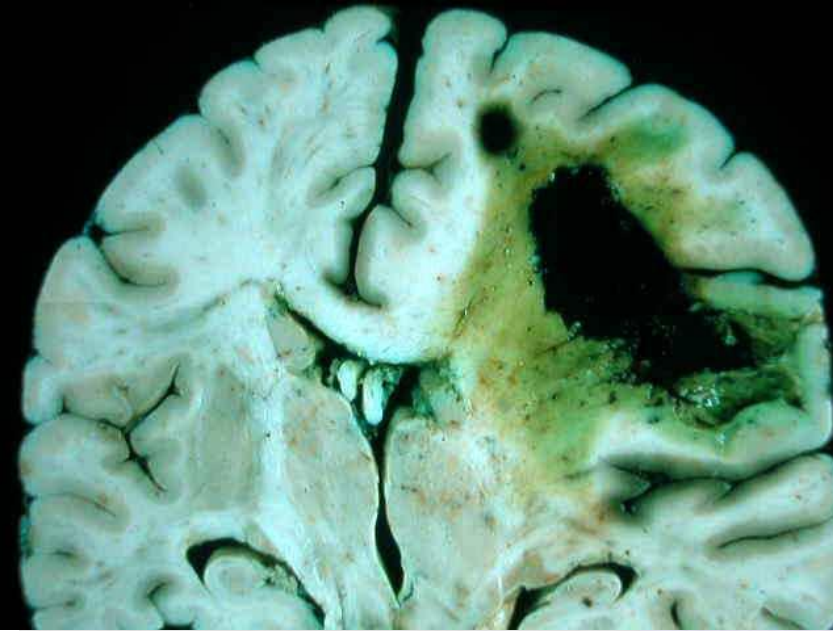


subfalcine

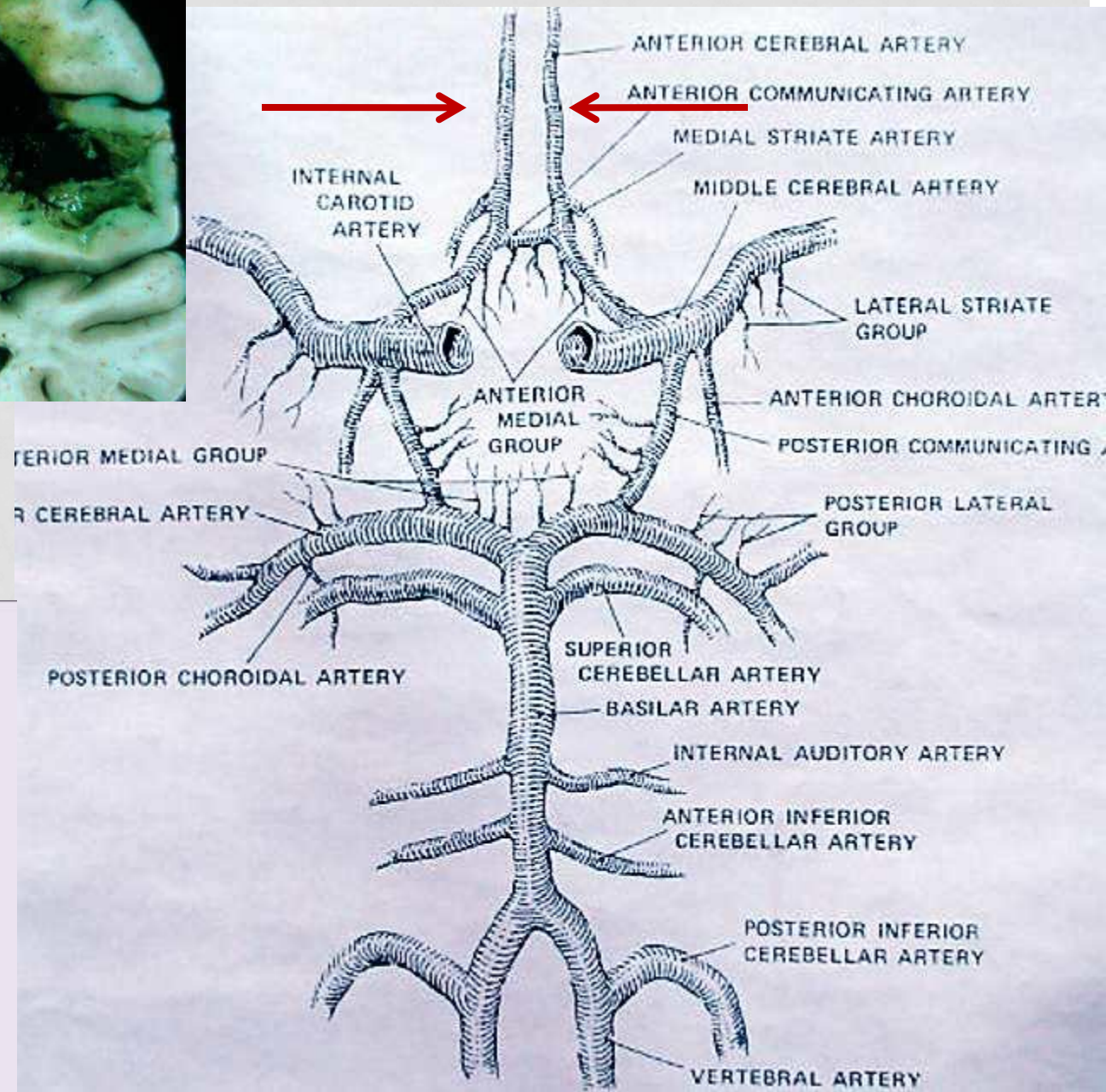
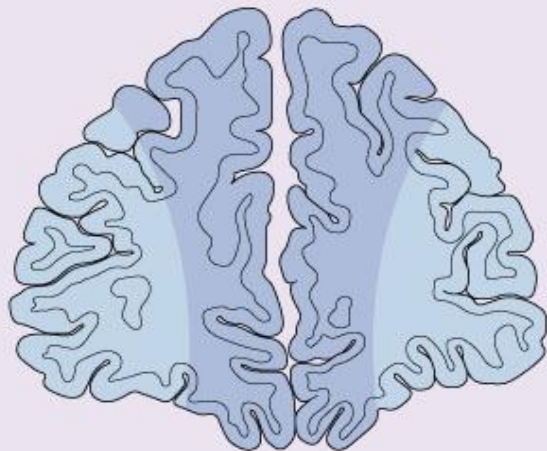
Subfalcine herniation



Subfalcine herniation

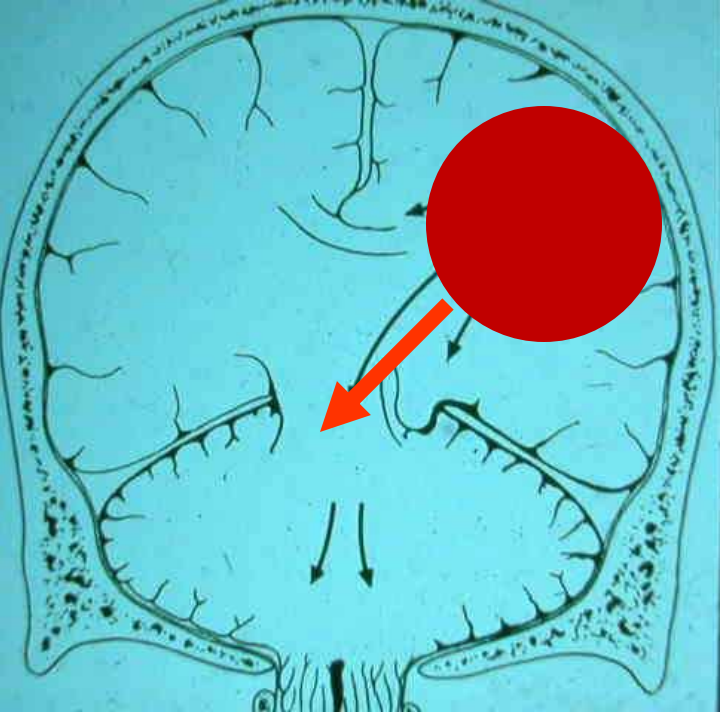


→ may compromise the
anterior cerebral
artery



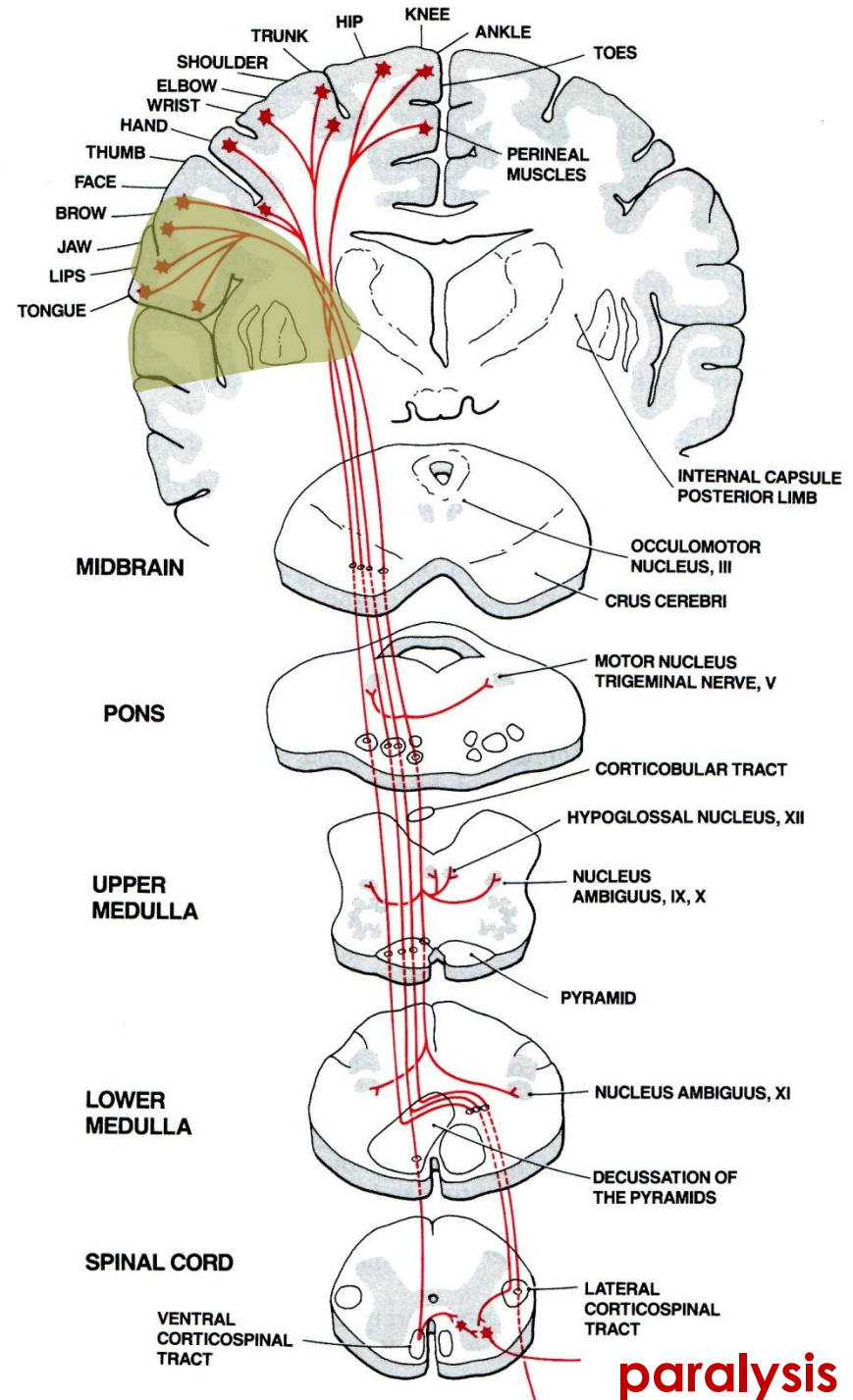
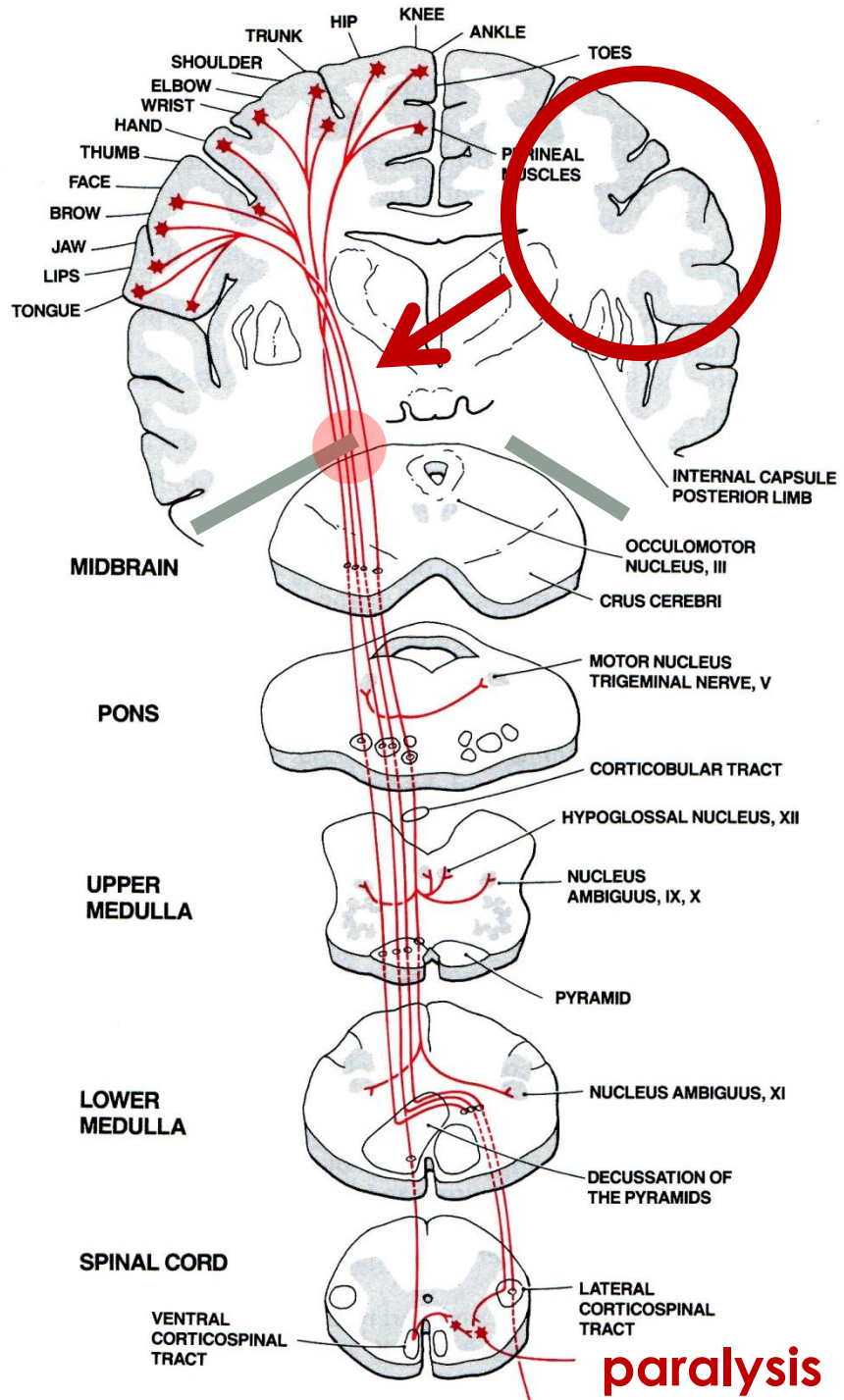
- ***Transtentorial (uncinate) herniation*** – occurs when the medial aspect of temporal lobe is compressed against the free margin of tentorium cerebelli. With increasing displacement of the temporal lobe, the third cranial nerve is compromised, resulting in pupillary dilation & impairment of ocular movements on the side of the lesion.

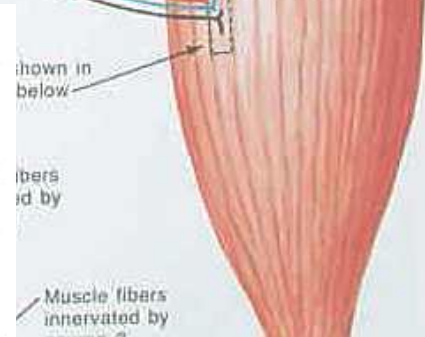
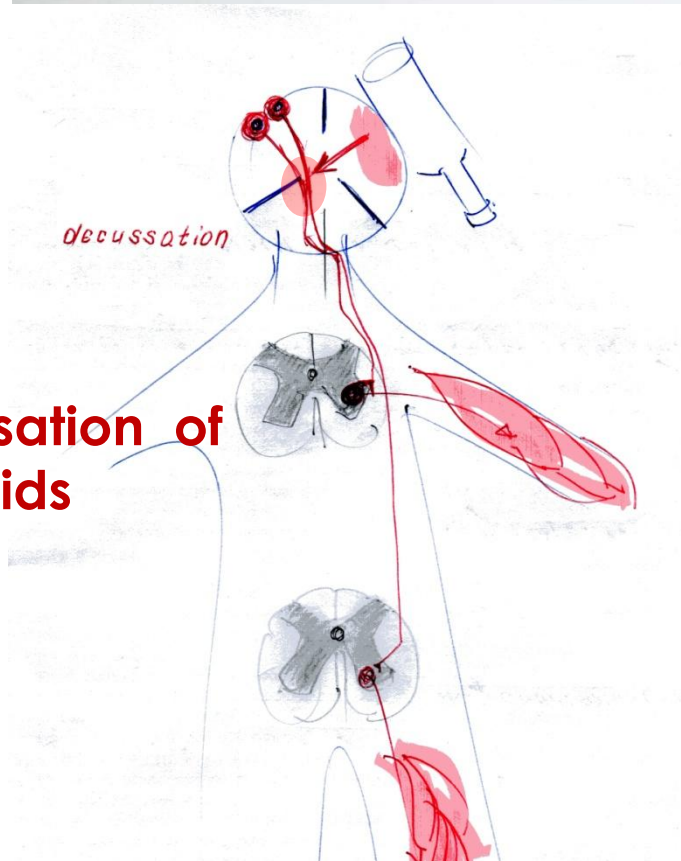
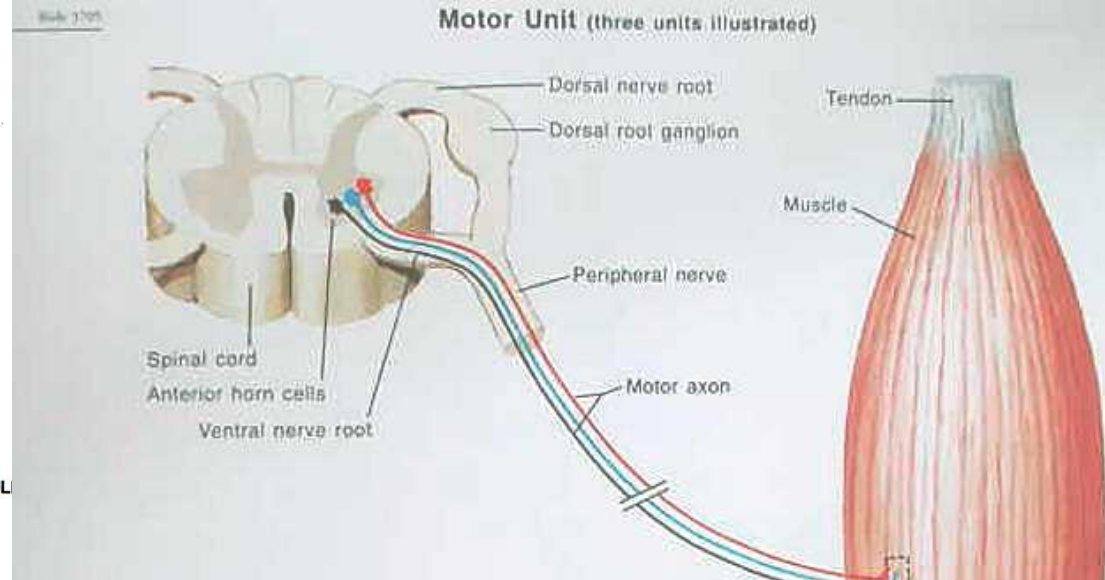
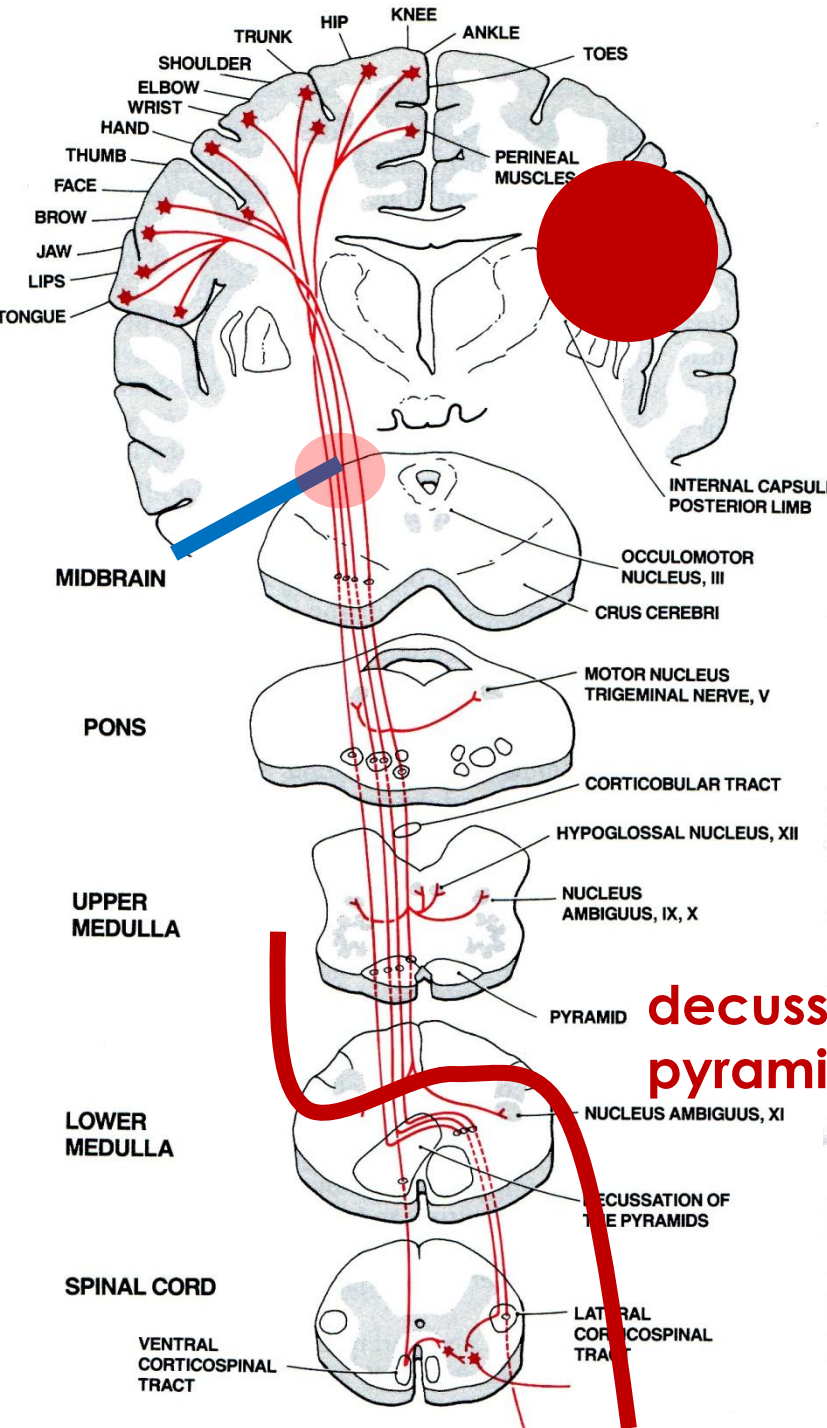
- The posterior cerebral artery may be compressed. When the extent of herniation is large enough , the contralateral cerebral peduncle may be compressed , resulting in hemiparesis ipsilateral to the side of the herniation.



**Tentorial
herniation**



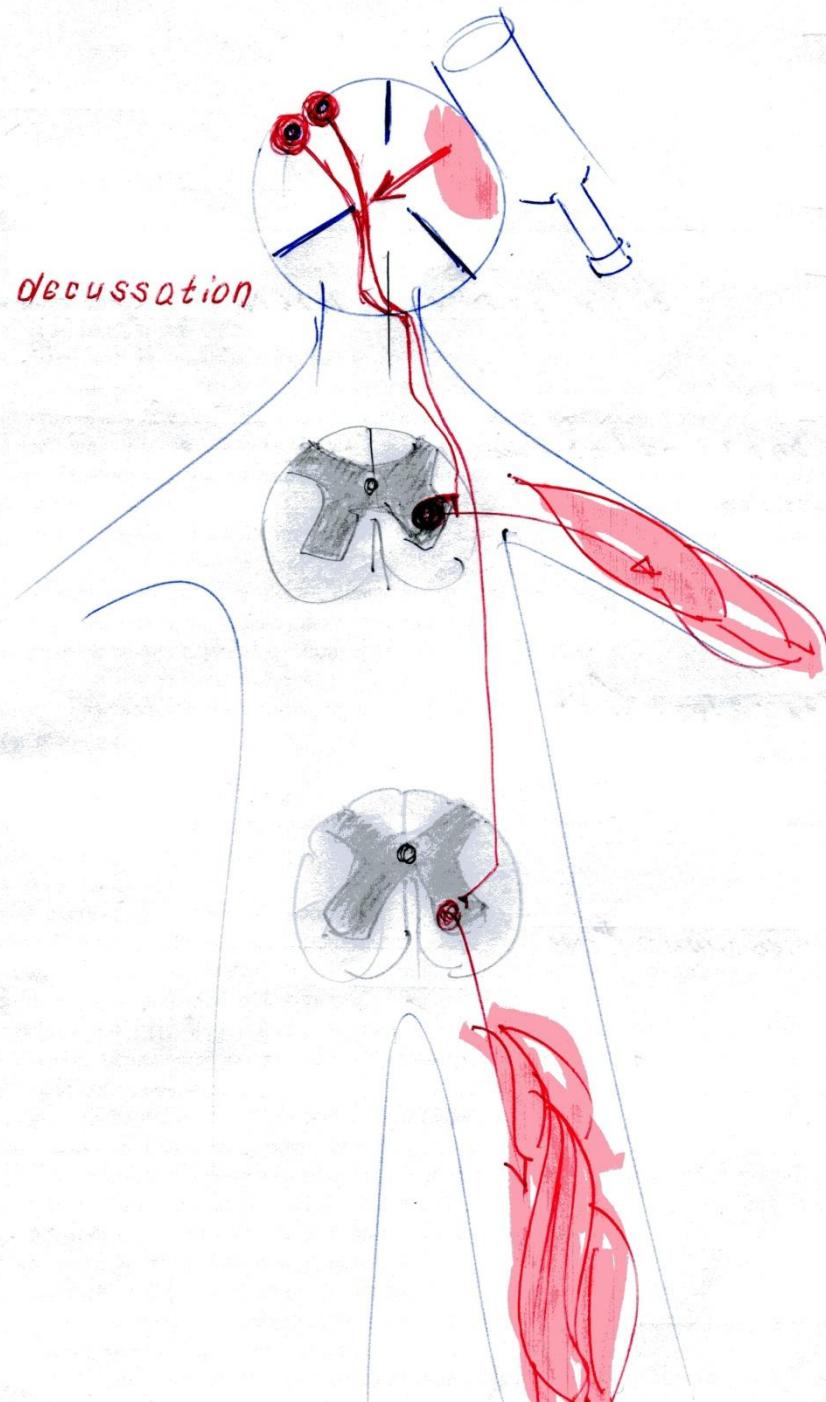


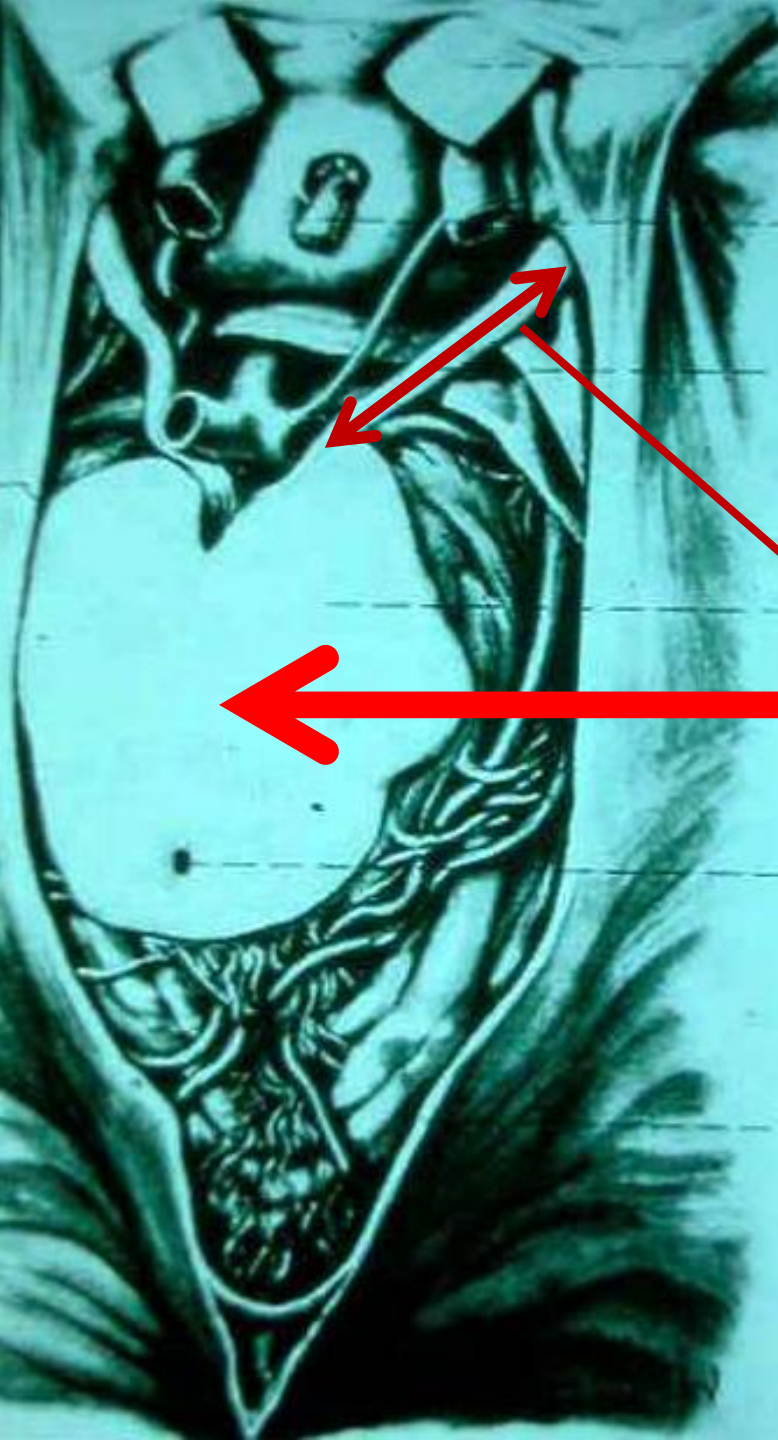


decussation of pyramids

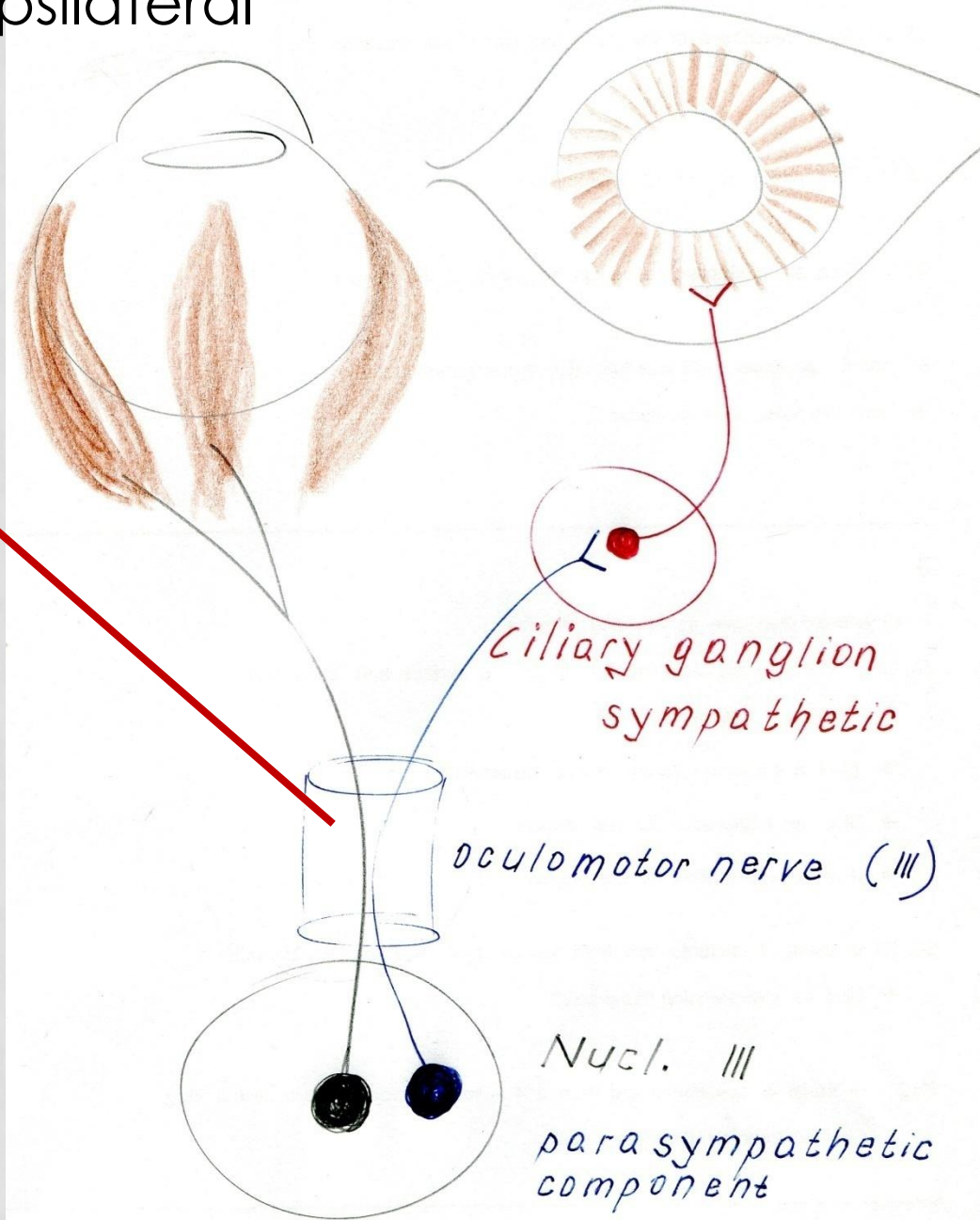
ipsilateral paralysis

decussation

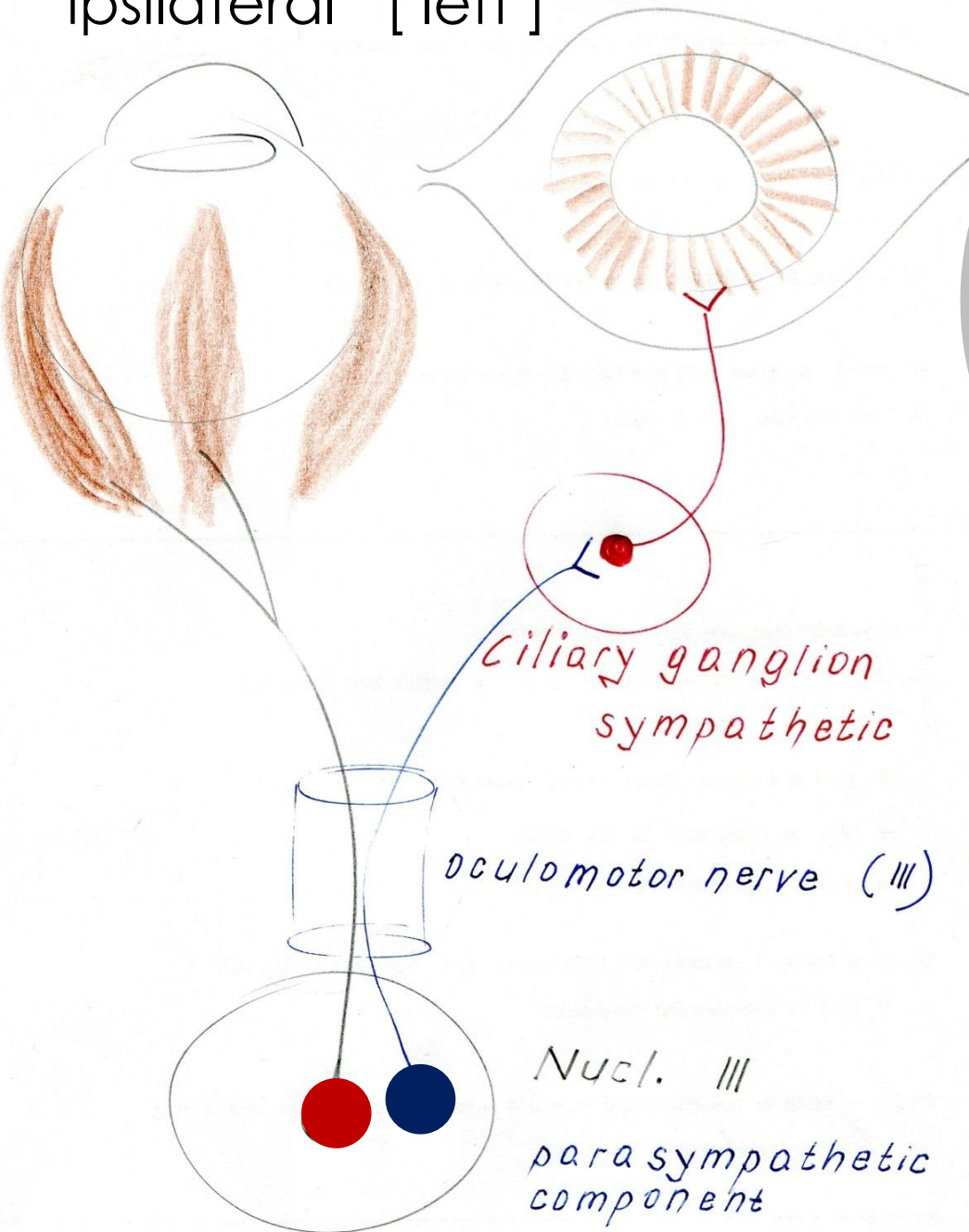




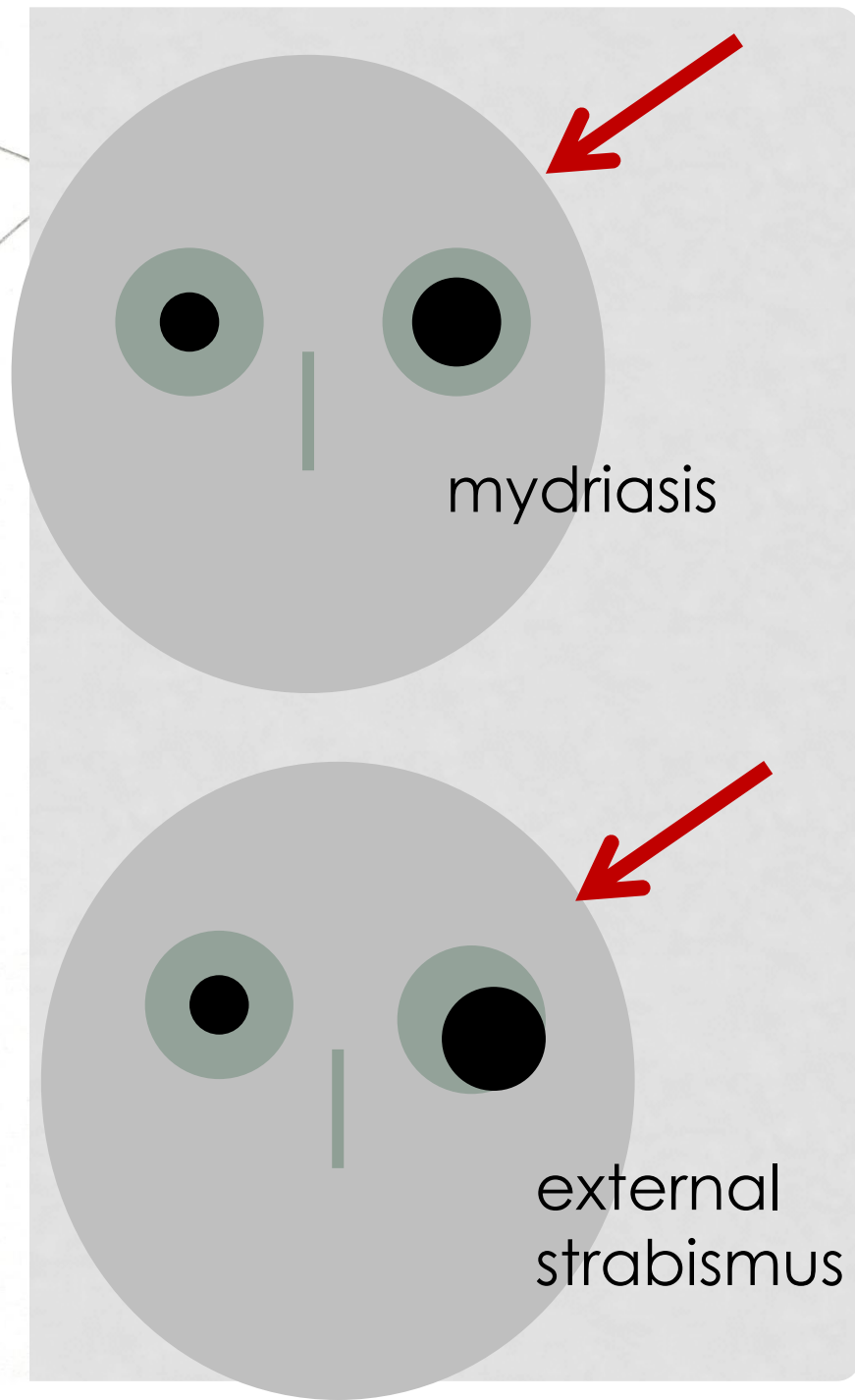
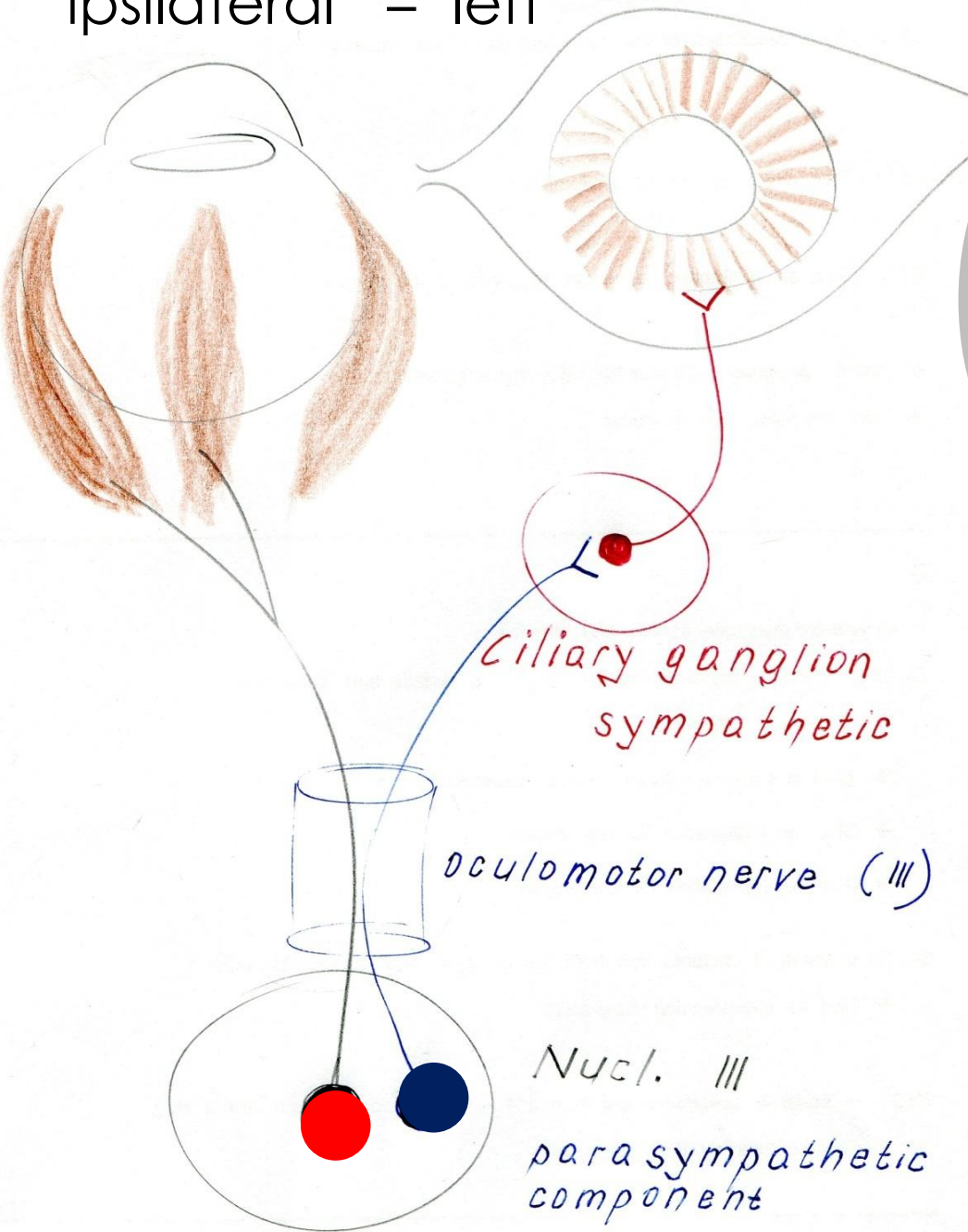
ipsilateral

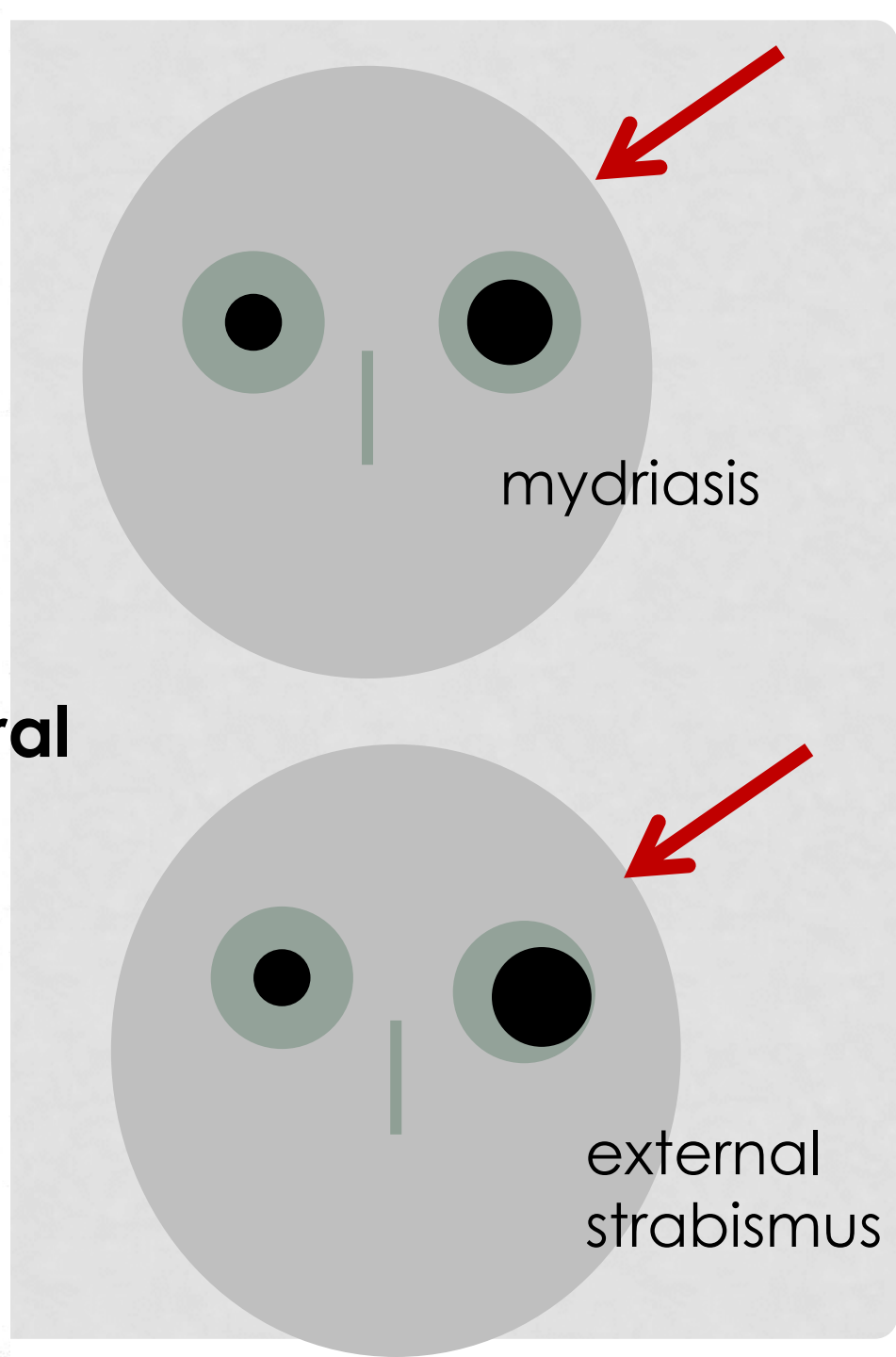
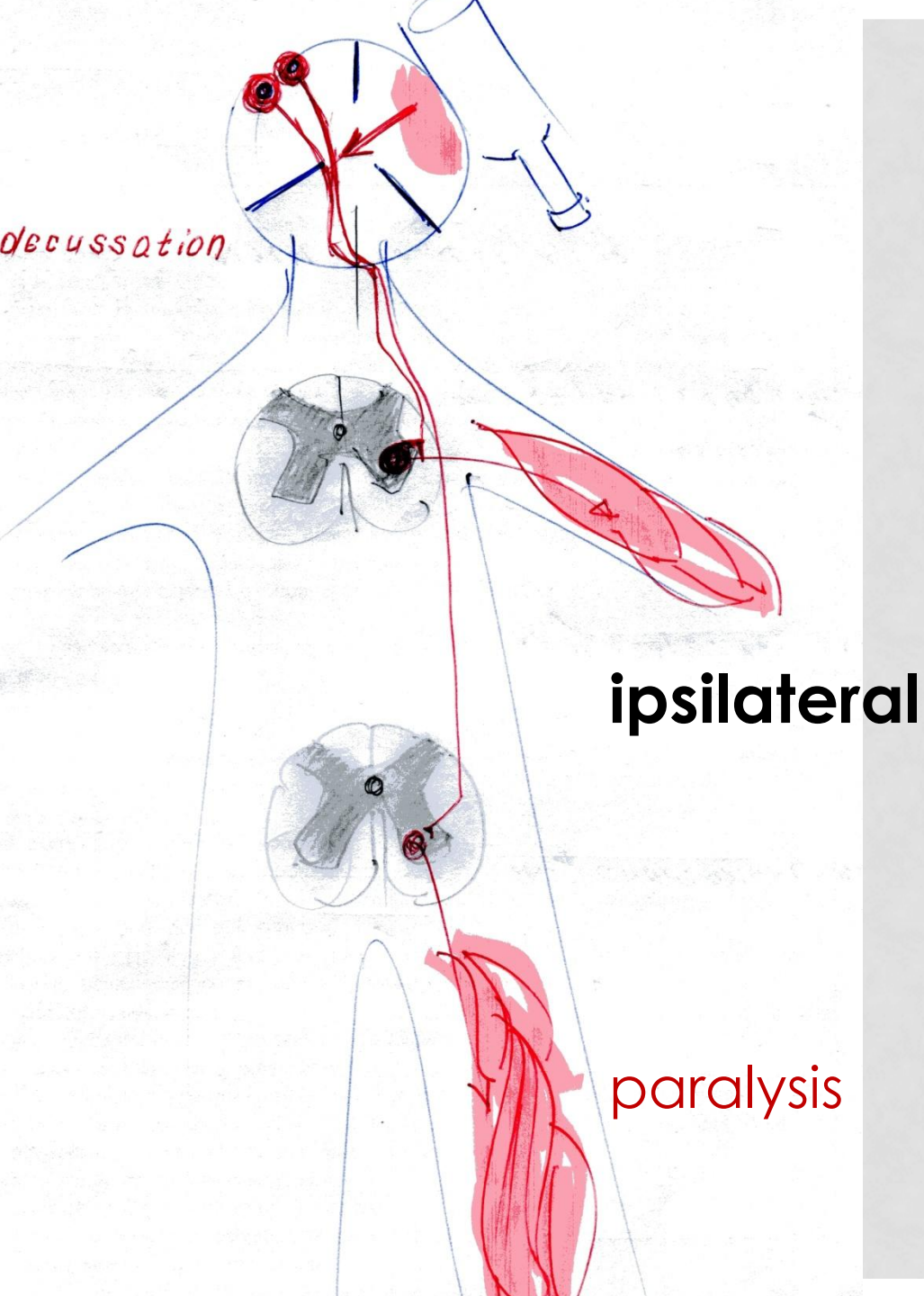


ipsilateral [left]

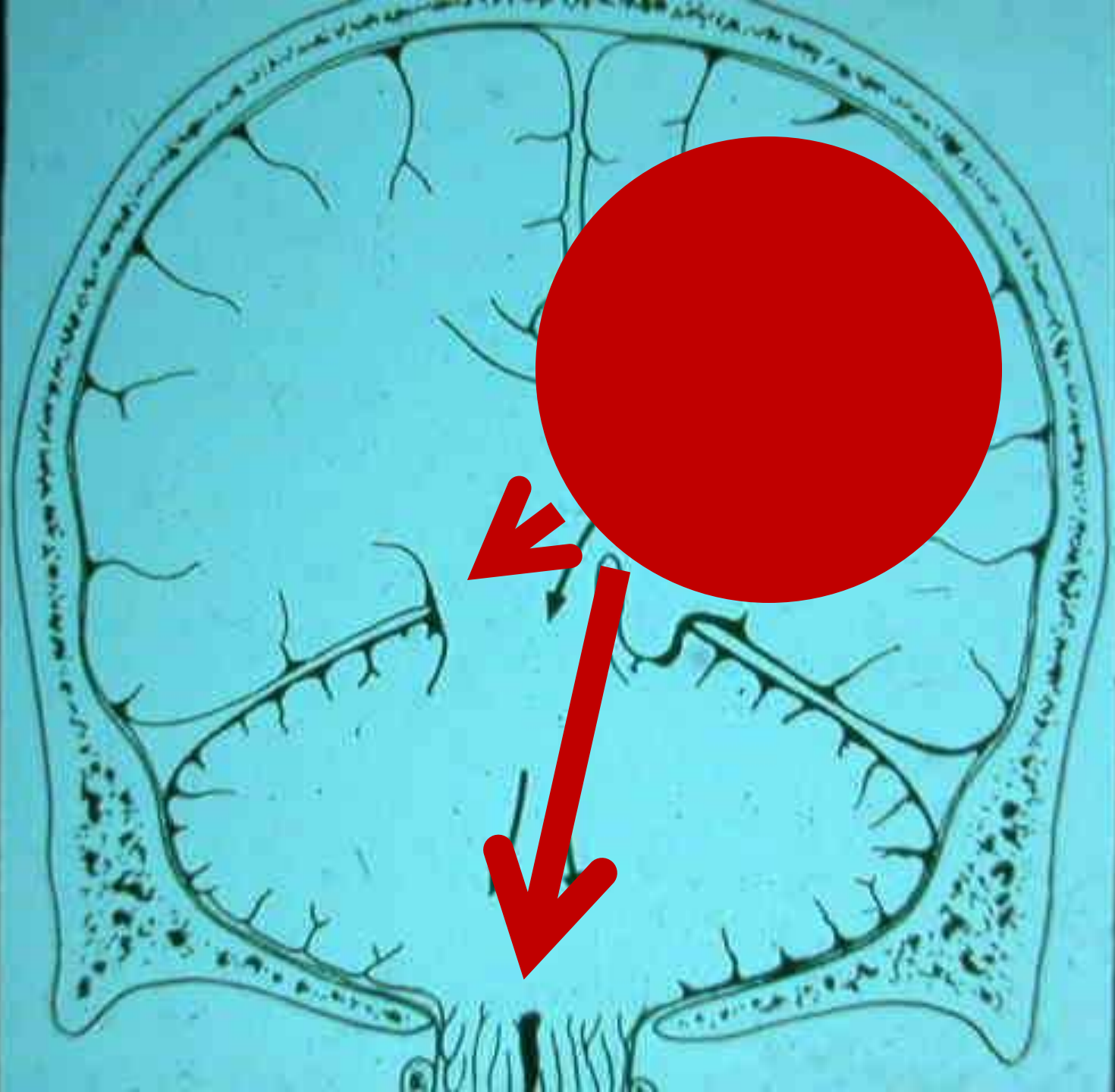


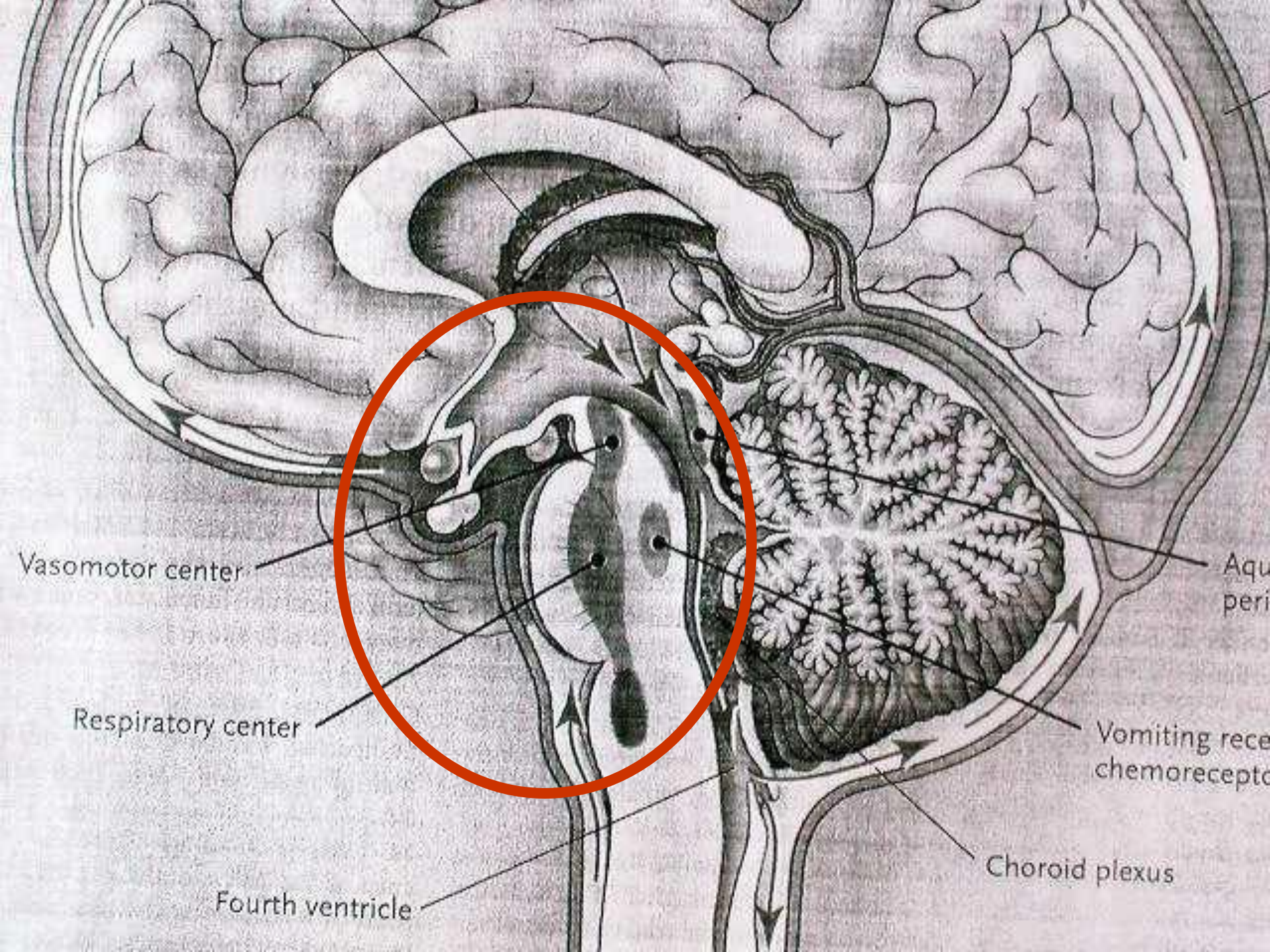
ipsilateral = left





- ***Tonsillar herniation*** – displacement of cerebellar tonsils through foramen magnum which lead to compression of medulla oblongata & result in respiratory distress





Vasomotor center

Respiratory center

Fourth ventricle

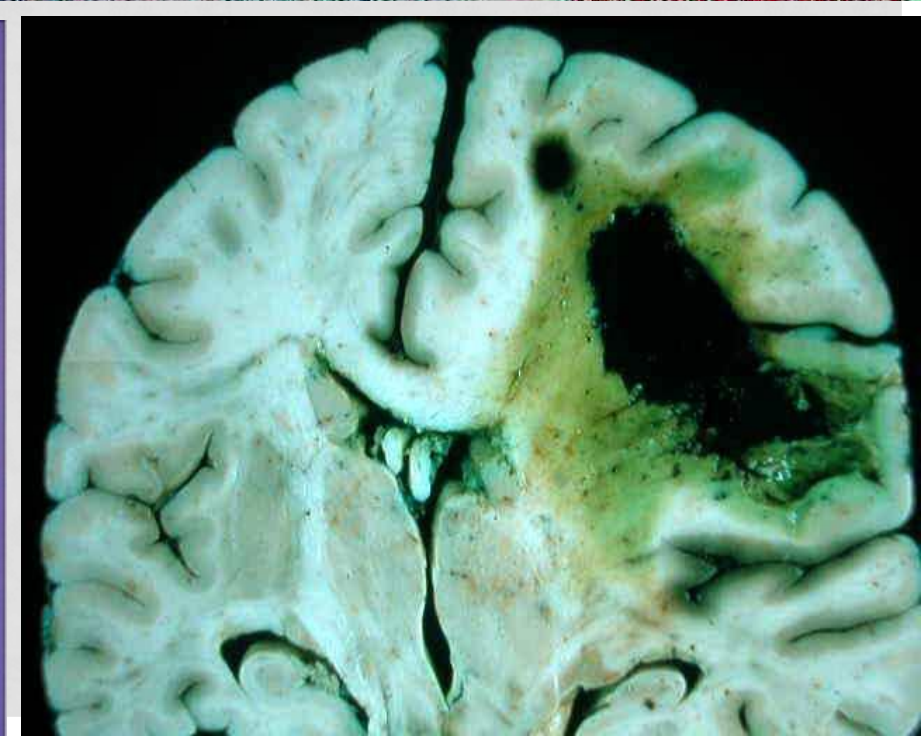
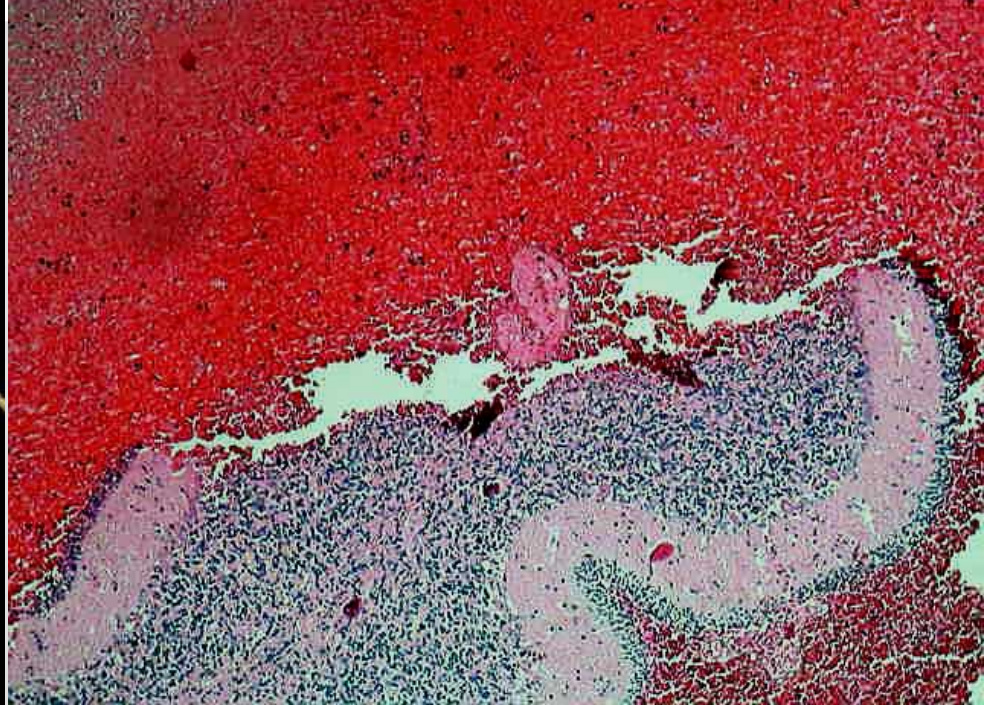
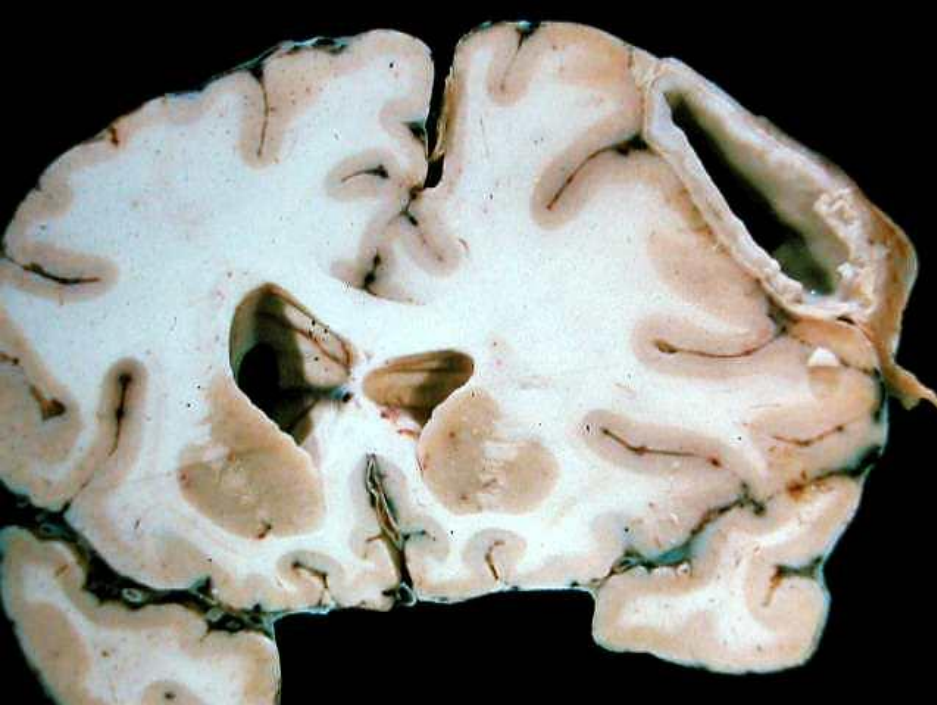
Choroid plexus

Vomiting receptor/chemoreceptor

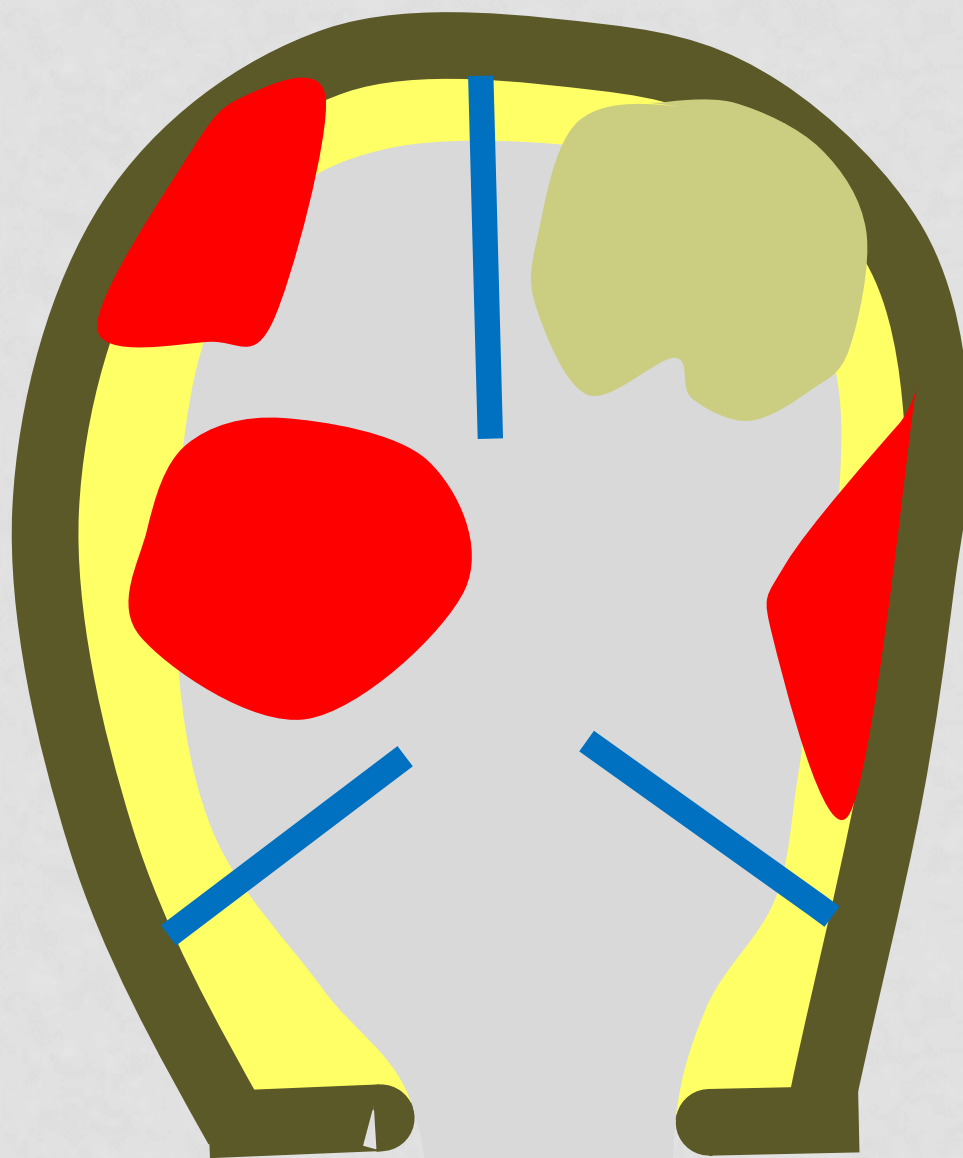
Aqueduct of the midbrain

Sequel of brain trauma

- Post traumatic hydrocephalus
- Posttraumatic dementia
- Posttraumatic epilepsy
- Posttraumatic infections
- Psychiatric disorders



SOP



protective
shield



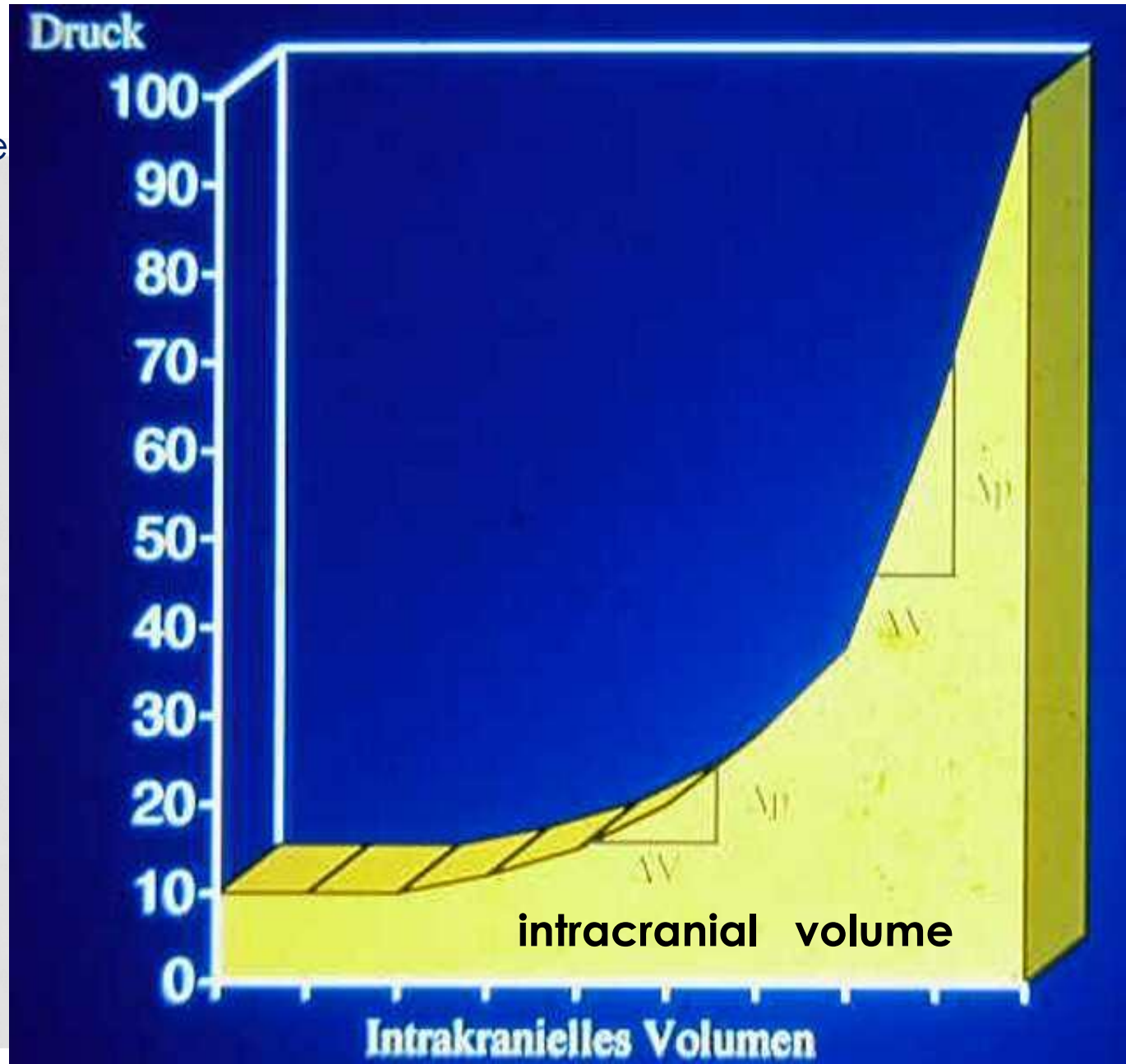
cage

MONROE KELLY DOCTRINE

- A. The cranial compartment is incompressible
- B. Volume inside the cranium is fixed.
- “The cranium and its constituents (blood, CSF, and brain tissue) create a state of volume equilibrium, such that any increase in volume of one of the cranial constituents must be compensated by a decrease in volume of another.”

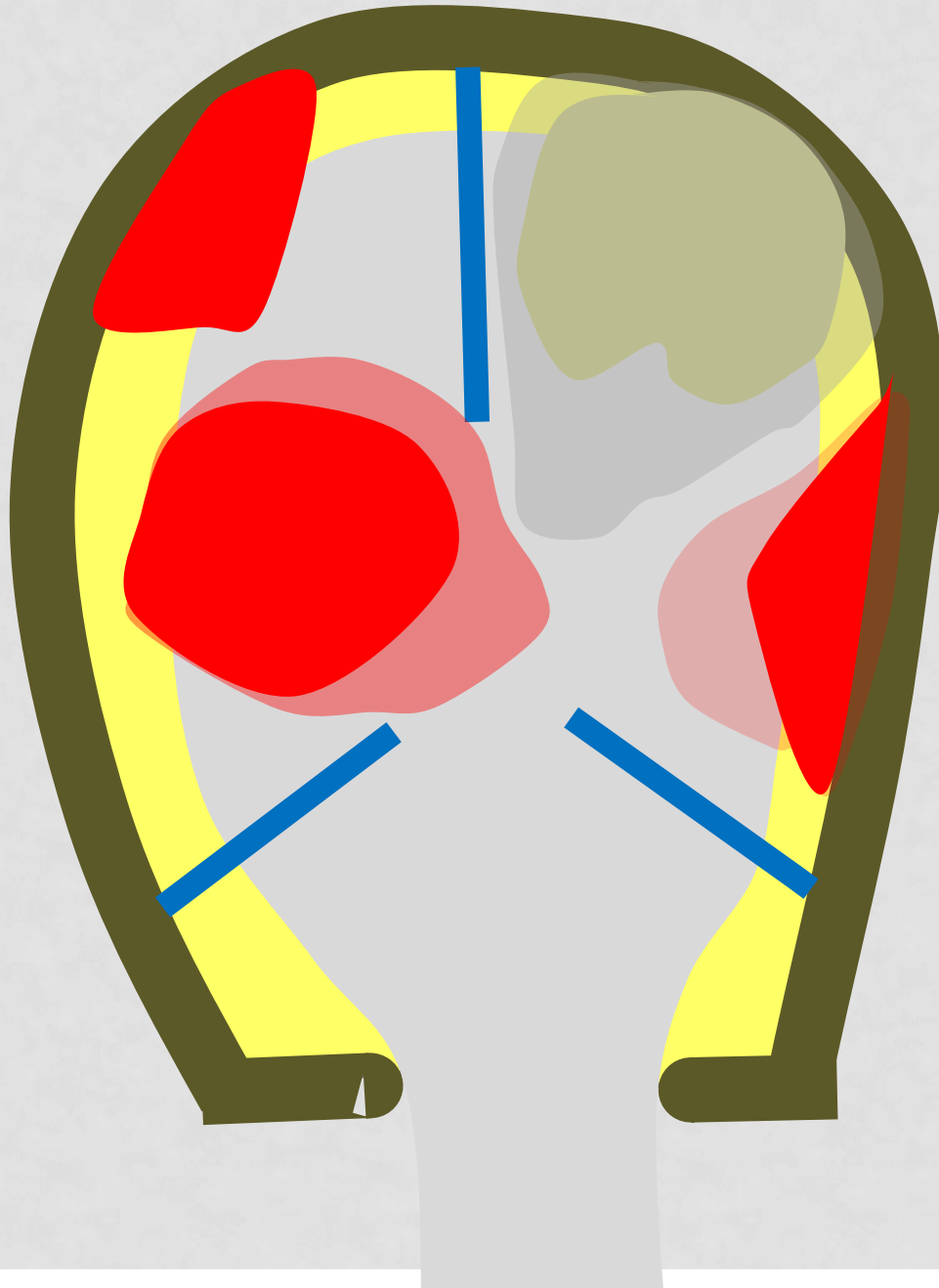
INCREASED INTRACRANIAL PRESSURE

pressure
mmHg



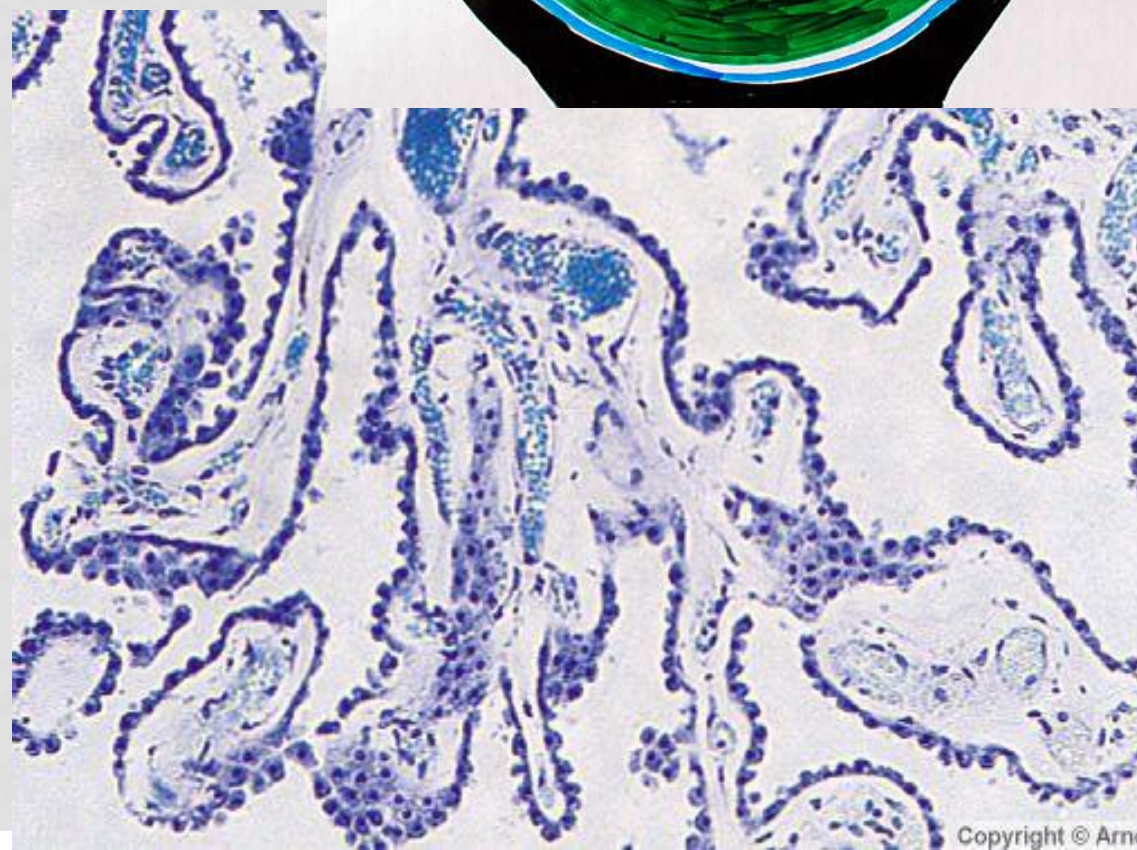
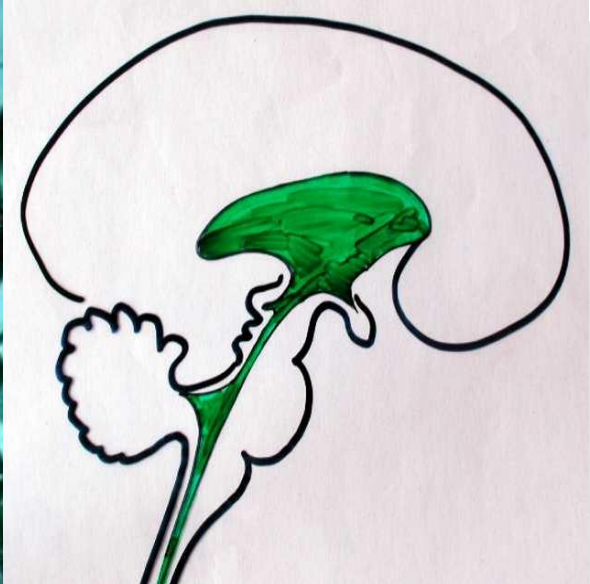
SOP

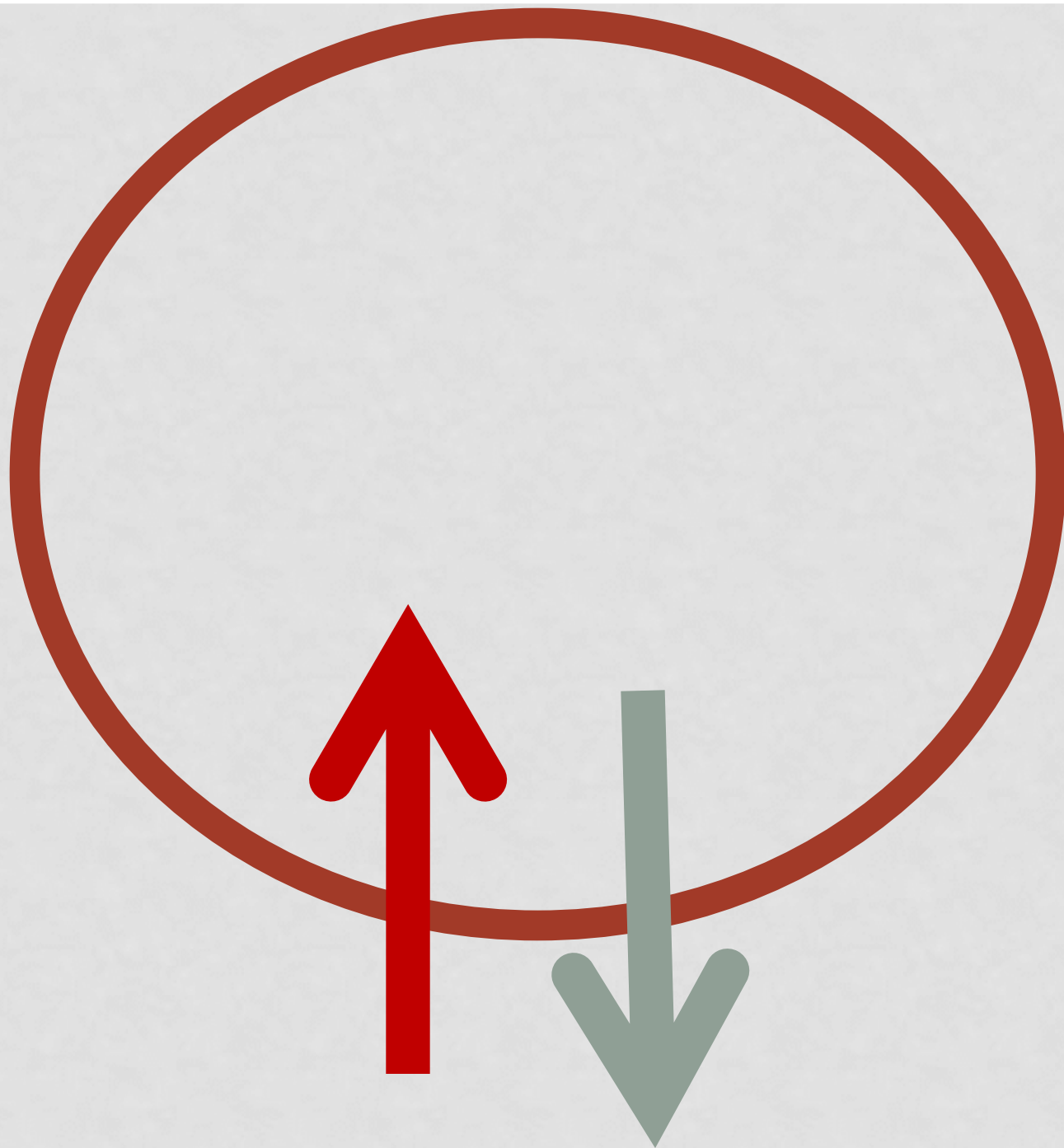
cage

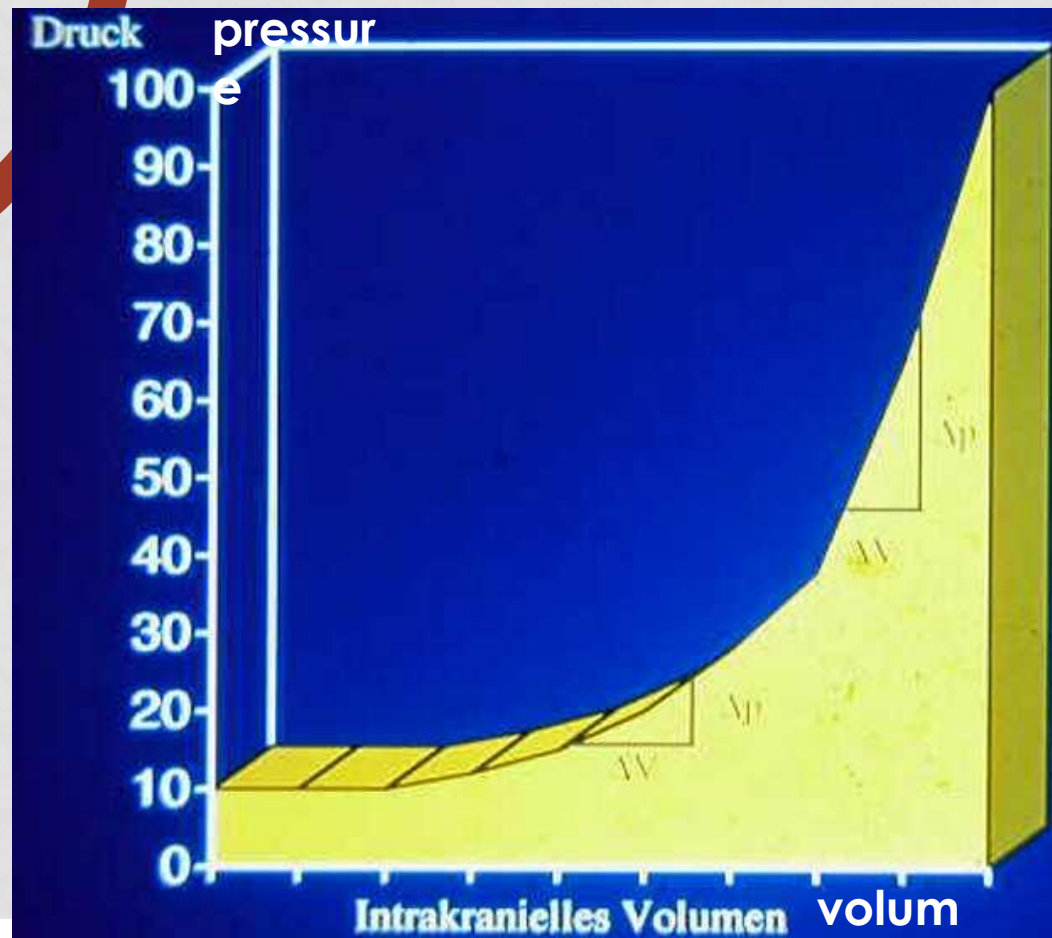
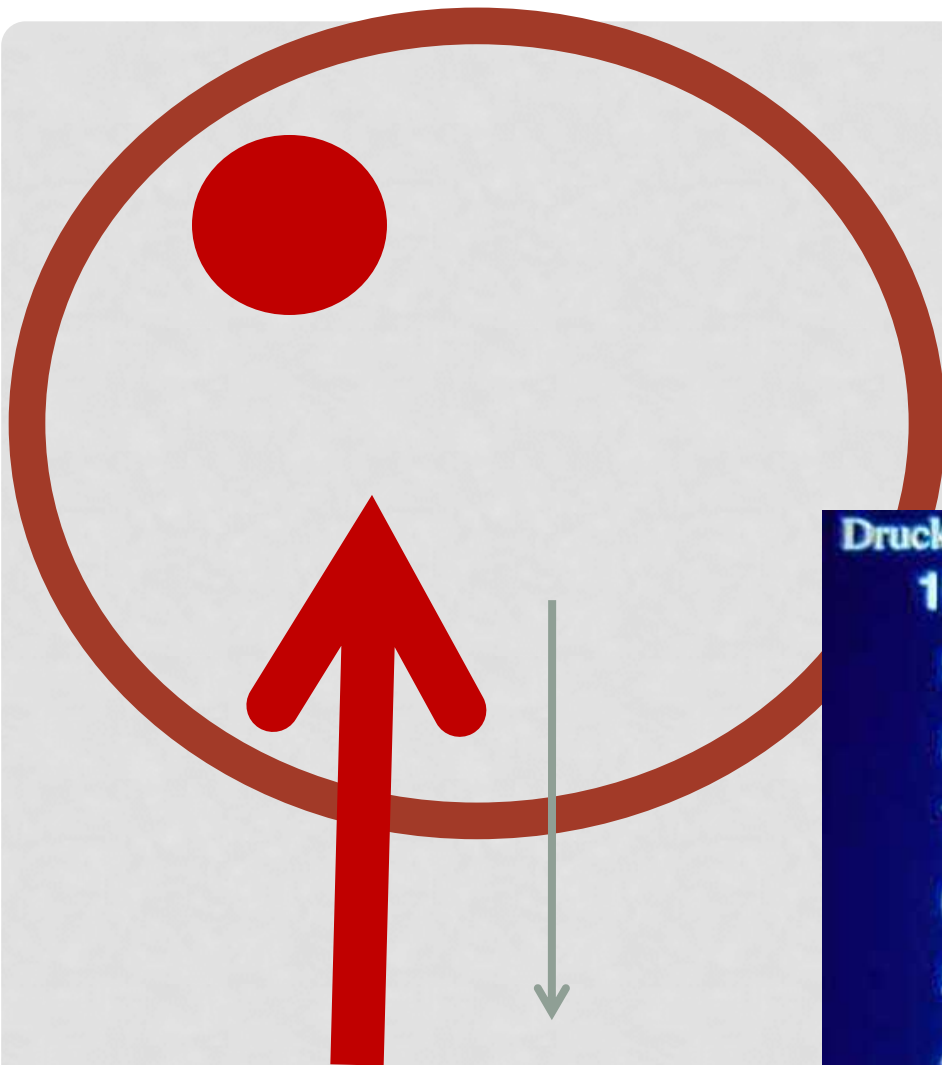


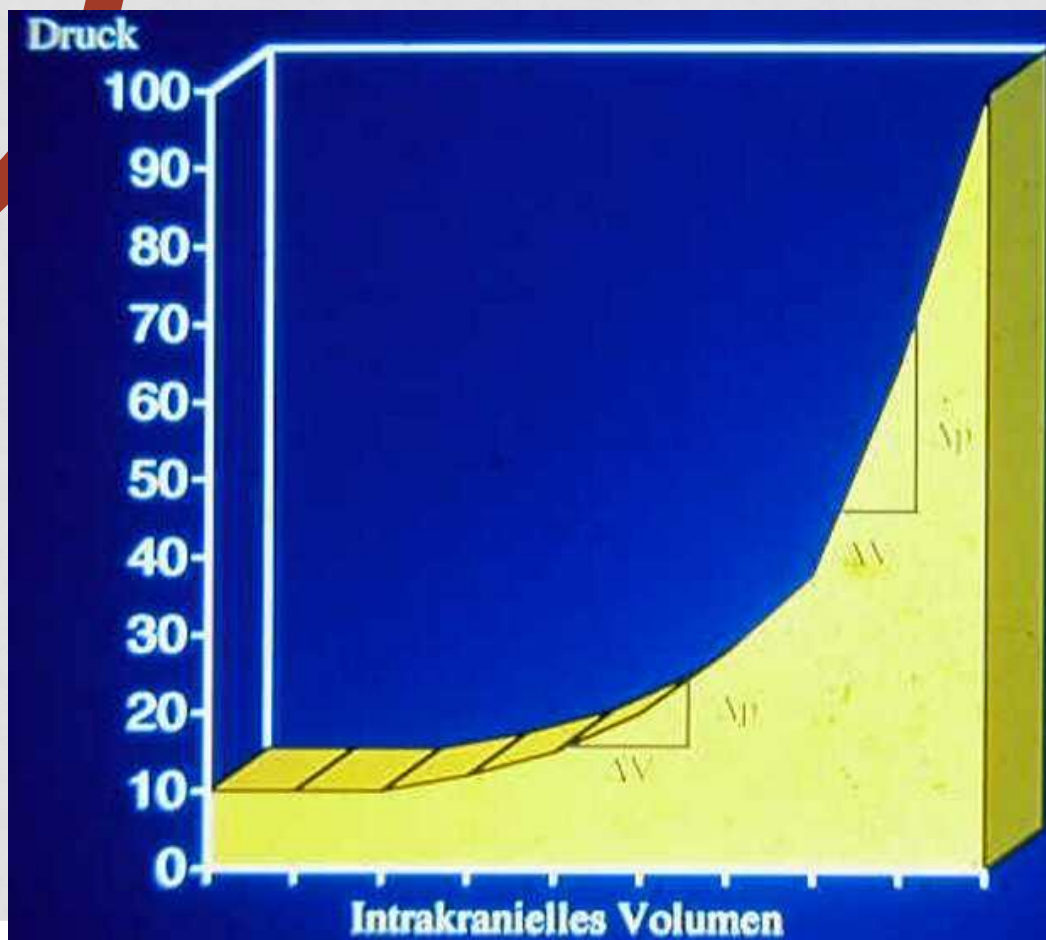
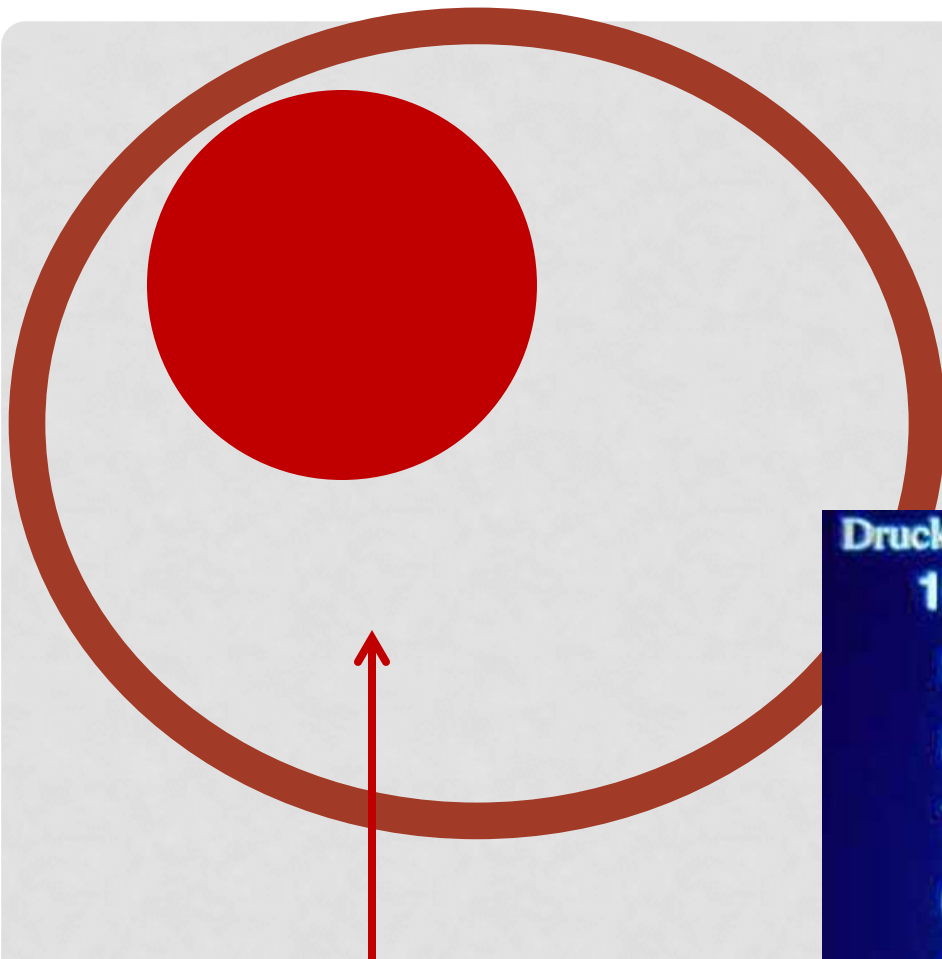


aqueduct

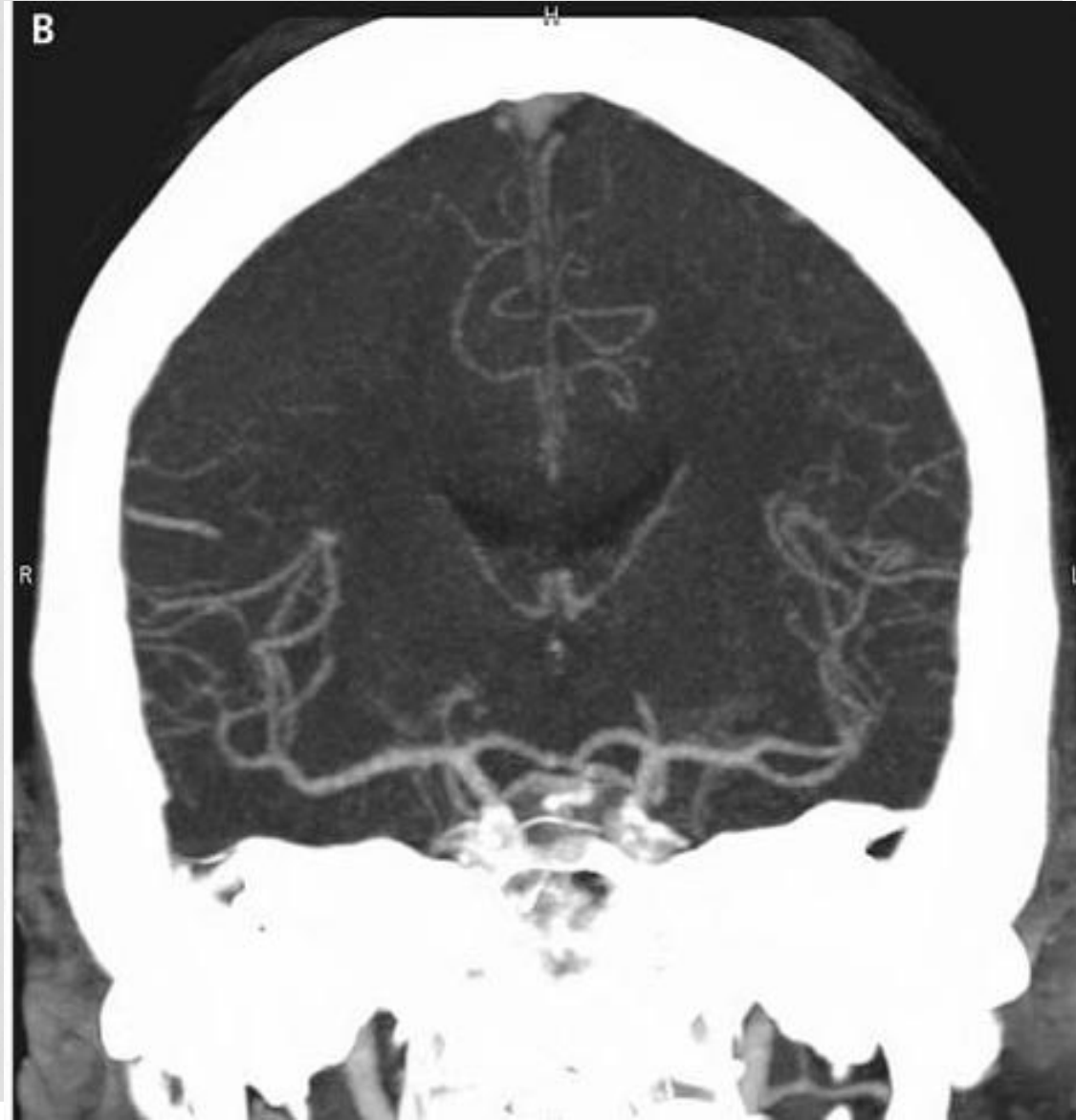
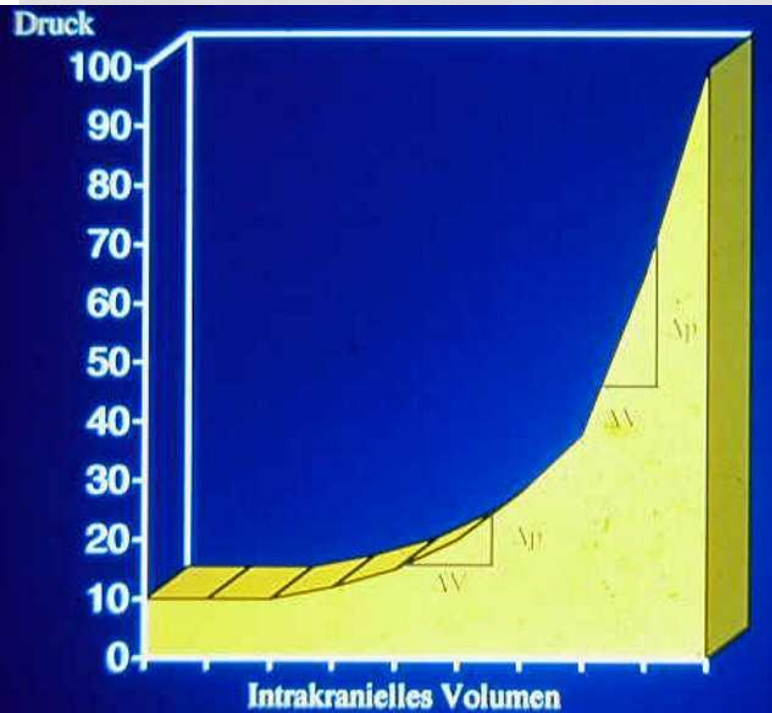
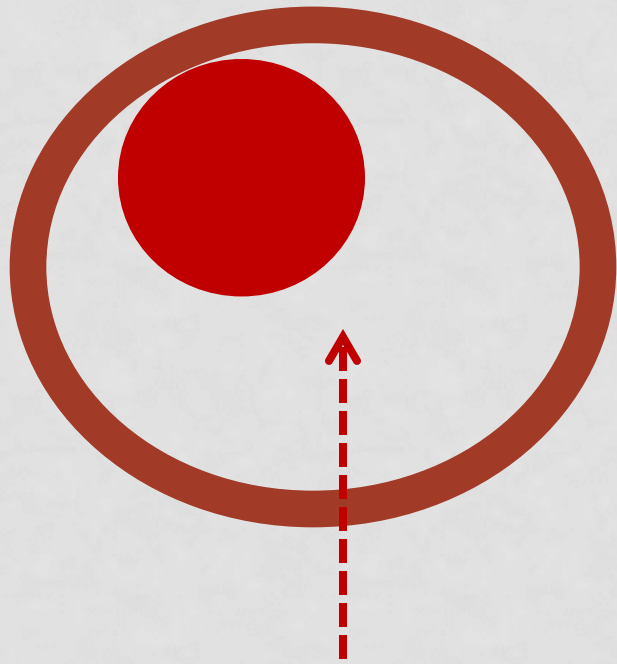








Circulatory standstill =
death



RAISED ICP

The characteristic clinical signs & symptoms of raised intracranial pressure include

Papilloedema due to accumulation of axoplasm in optic papillae when axonal flow is impeded by pressure on the optic nerve

Nausea & vomiting due to pressure on vomiting centers in the pons & medulla

Headache due to compression & distortion of pain & stretch receptors around intracranial blood vessels & within the dura matter

Impairment of consciousness ranging from drowsiness to deep coma

Spinal cord trauma

- Usually associated with vertebral column displacement. The level of cord injury determines the neurological manifestations

Thoracic or lower vertebrae – paraplegia

Cervical lesions – quadriplegia

Those above C4 - respiratory distress

Cerebrovascular disease

- Any abnormality of brain caused by a pathologic process of blood vessels
- 'Stroke' – clinical term that apply when symptoms begin acutely
- Cerebrovascular diseases include: thrombosis, embolism & hemorrhage

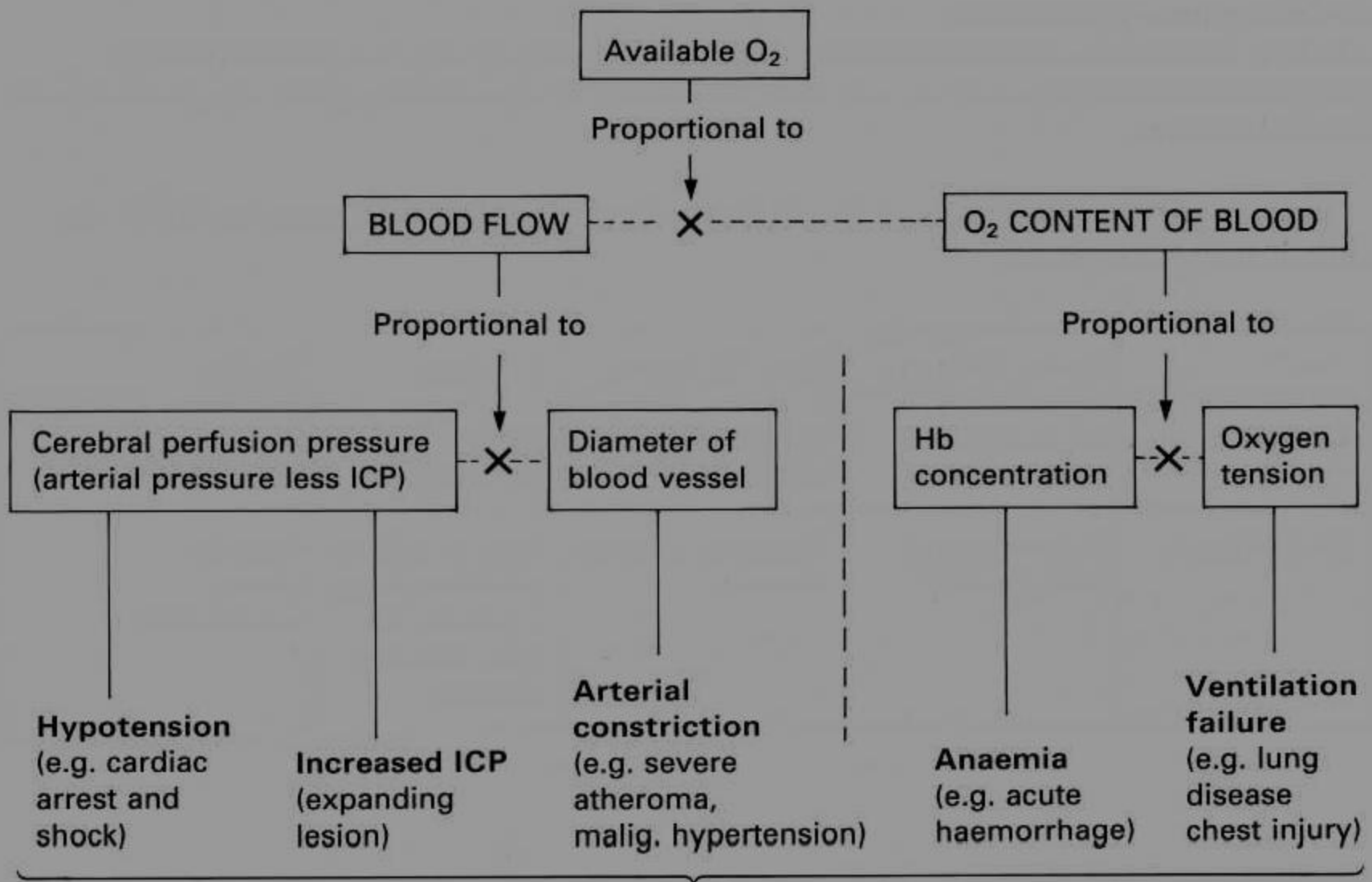
- There are two processes
 - Hypoxia, ischemia & infarction resulting from impairment of blood supply & oxygenation of CNS tissue
 - Hemorrhage resulting from rupture of CNS vessels

- Hypoxia , ischemia & infarction resulting from impairment of blood supply & oxygenation of CNS tissue.
- **Global cerebral ischemia** (ischemic / hypoxic encephalopathy) occurs when there is generalized reduction of cerebral perfusion, such as cardiac arrest or shock

- **Focal cerebral ischemia** follows reduction or cessation of blood flow to a localized area of the brain due to large-vessel disease (such as embolic or thrombotic in setting of atherosclerosis) or to small-vessel disease (such as vasculitis)

Global cerebral ischemia (Hypotension, Hypoperfusion & Low flow state)

- Below a systolic pressure of approximately 50 mm Hg, autoregulatory mechanisms are inadequate to compensate for the reduction in blood flow, and parenchymal injury may occur.



These conditions, singly or together, can be responsible for cerebral hypoxia.

- Within the CNS, certain regions and cell populations are more susceptible than others to hypoxic-ischemic injury. For example, neurons, as a group, are far more vulnerable to ischemic injury than are glial cells. Among the neurons, certain subpopulations, including some of the **pyramidal cells in the hippocampus, the Purkinje cells of the cerebellum are particularly sensitive.**
- **Areas of the brain located at the junctions of arterial territories (i.e., *arterial border zones*, sometimes termed "watershed" areas) are also particularly susceptible**

Clinical Features

- A number of factors may modify the degree of parenchymal injury. These include the *age* of the patient, the *duration* of the circulatory disturbance, and *temperature* (hypothermia increases the resistance of the brain to hypoxic-ischemic injury whereas hyperthermia worsens the effects of hypoxic-ischemic injury).
- The resulting deficits range from transient neurologic disturbances to "brain death," with a global cessation of electrical activity

Focal cerebral ischemia (ischemic stroke)

Reduction or cessation of blood flow to a localized area of the brain due to a large vessel disease (emboli, thrombosis or vasculitis)

- The majority of thrombotic occlusions are due to atherosclerosis. The most common sites are the **carotid bifurcation, the origin of the middle cerebral artery & either end of basilar artery**

EMBOLISM

- Embolism to the brain occurs from wide range origins. Cardiac mural thrombi are among the most common sources. Myocardial infarct, valvular disease & atrial fibrillation are important predisposing factors. Other sources are emboli from atheromatous plaques within the carotid arteries.

Rheumatic Heart Disease 1995-2000

Tikur Anbessa Hospital

Dr. Jilalu Asmera and Dr. Kebede Oli

96 patients

F: 53 %
23 years

M: 47%
19 years (age at death)

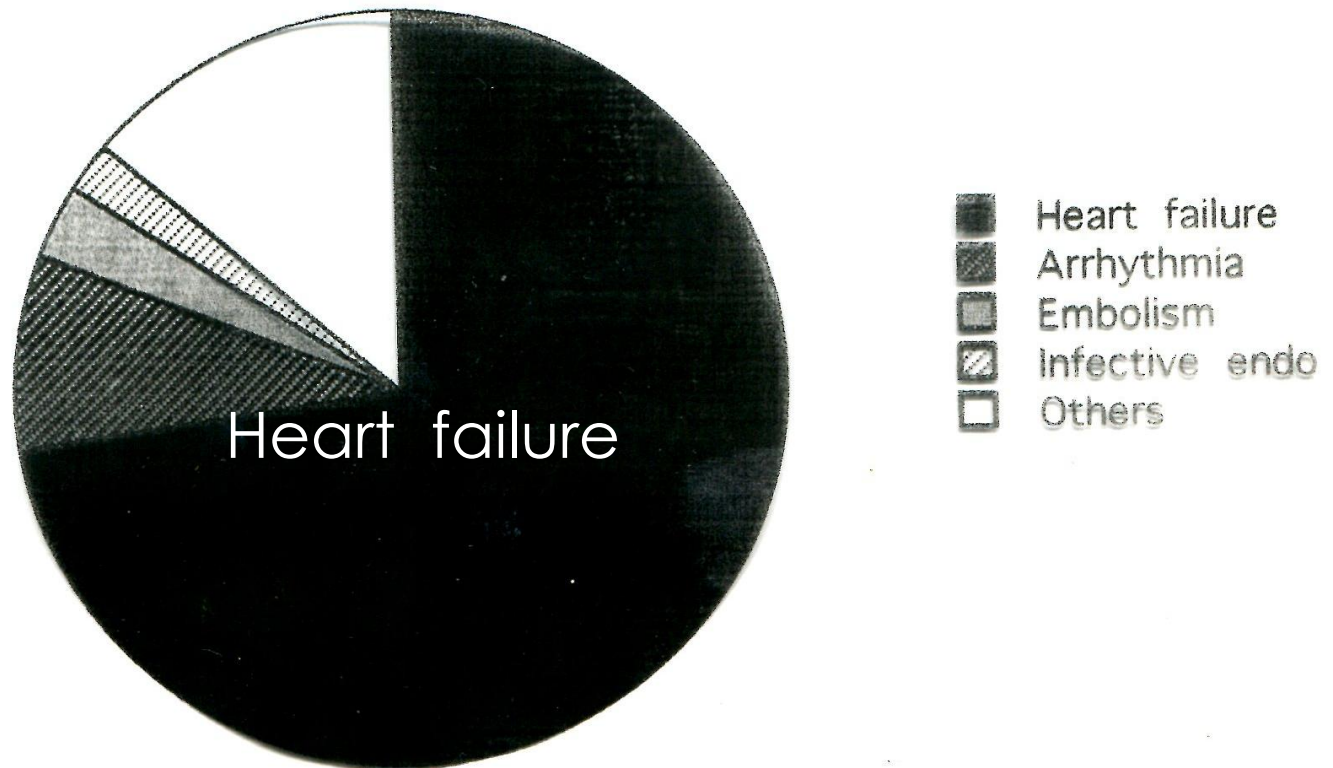
Follow-up: 3 years

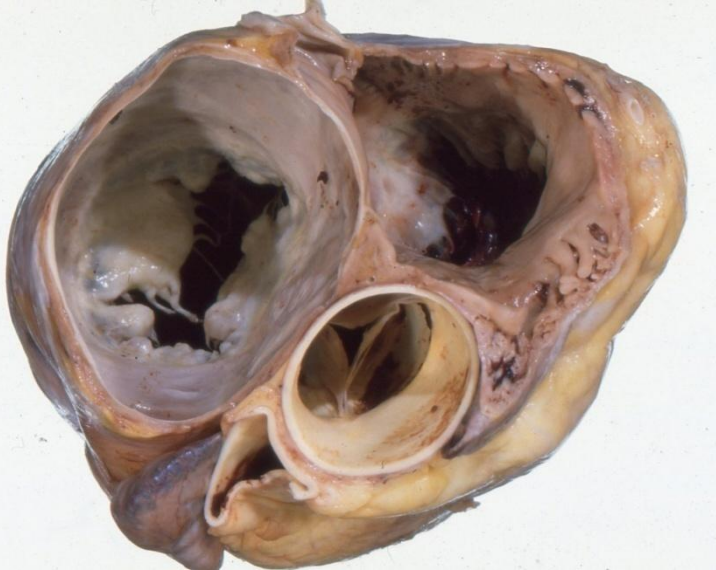
- Complications:

- Infective endocarditis	33%
- Arrhythmia	39%
- Embolism	15%

Dr. Jilalu and Prof.
Kebede

- Cause of death:





Scarring of
valves ↓

**disturbed
function**

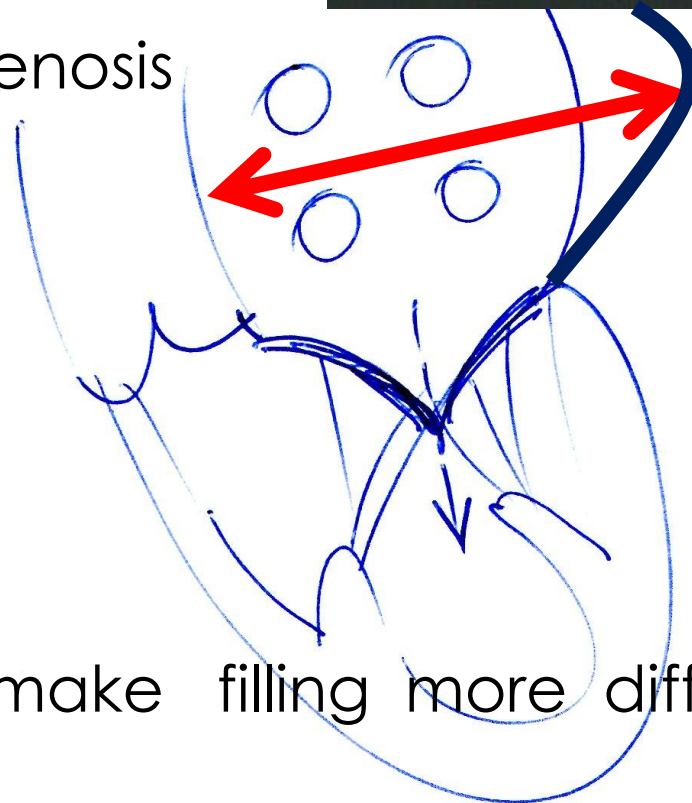


incompetence



regurgitation

stenosis



make filling more difficult

Mitral valve



Insufficiency
Incompetence



Regurgitation



LV: volume overload



eccentric hypertrophy



LA: dilation

Stenosis



makes filling of the LV
more difficult



normal filling



normal LV

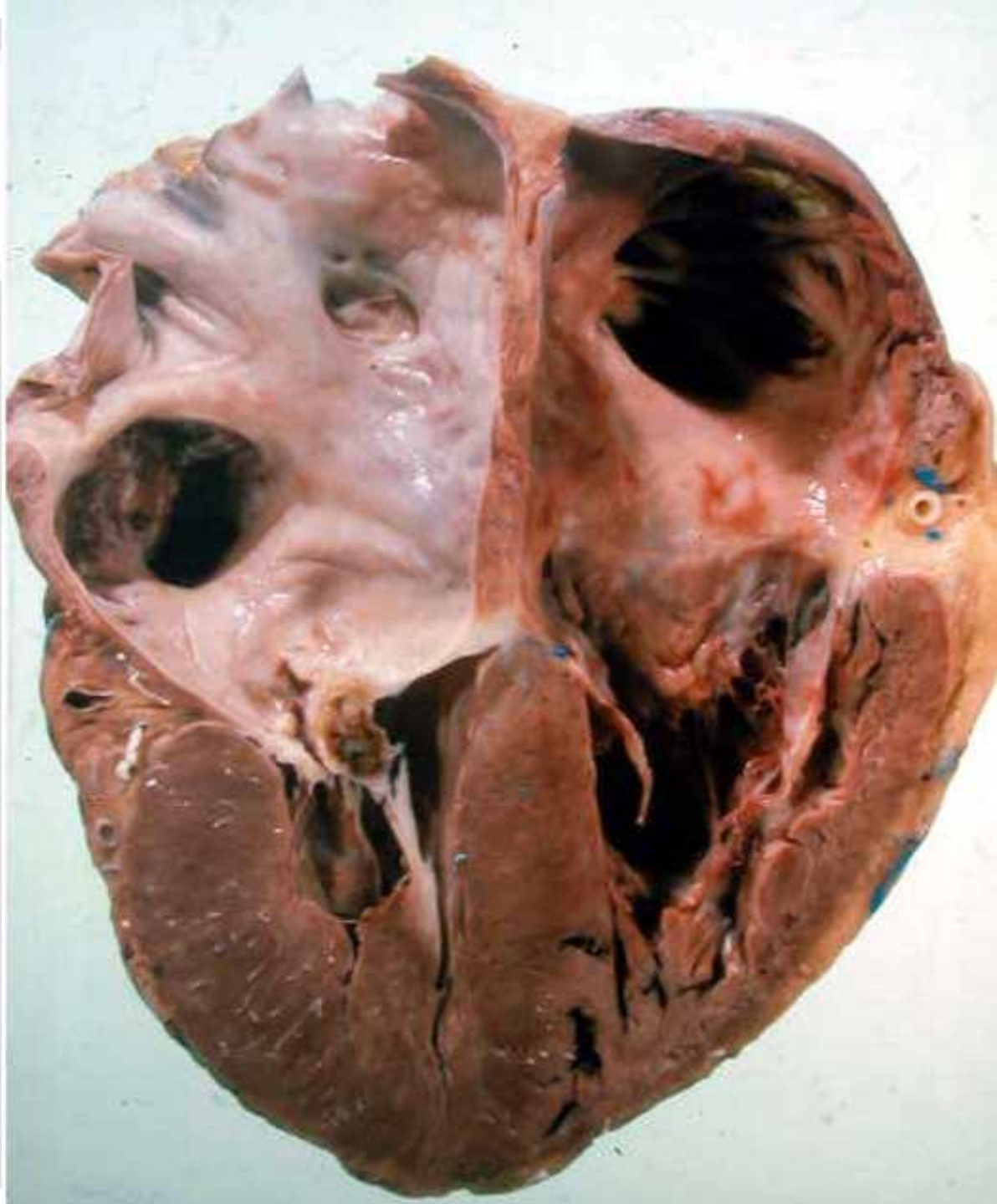


dilation

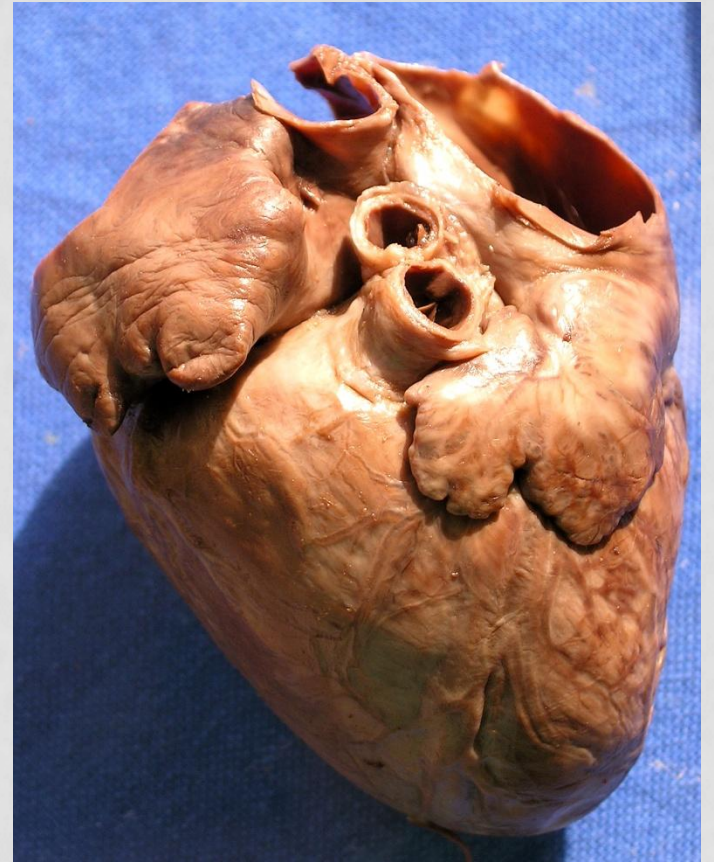
stasis



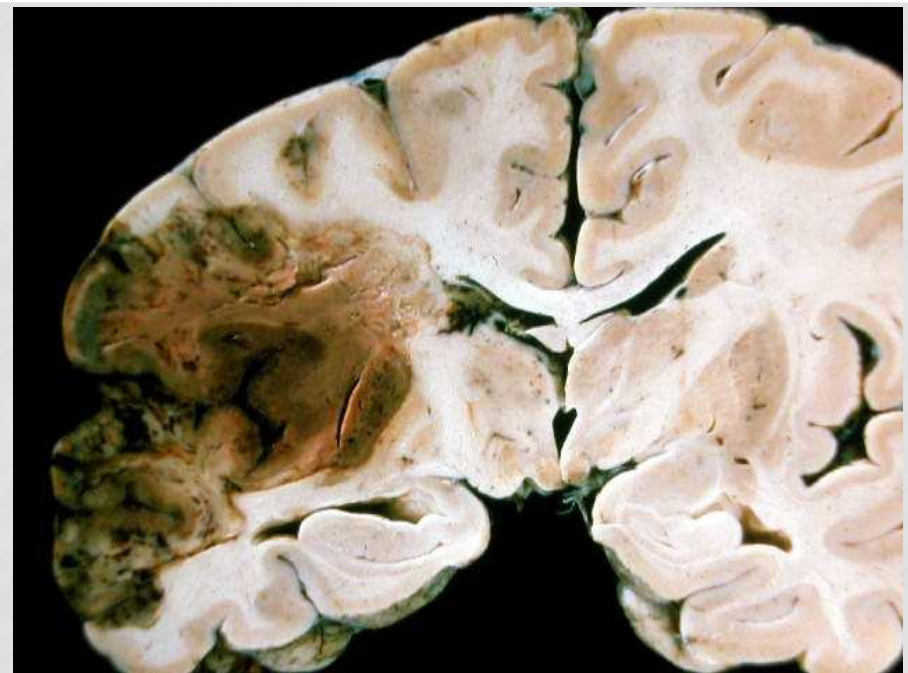
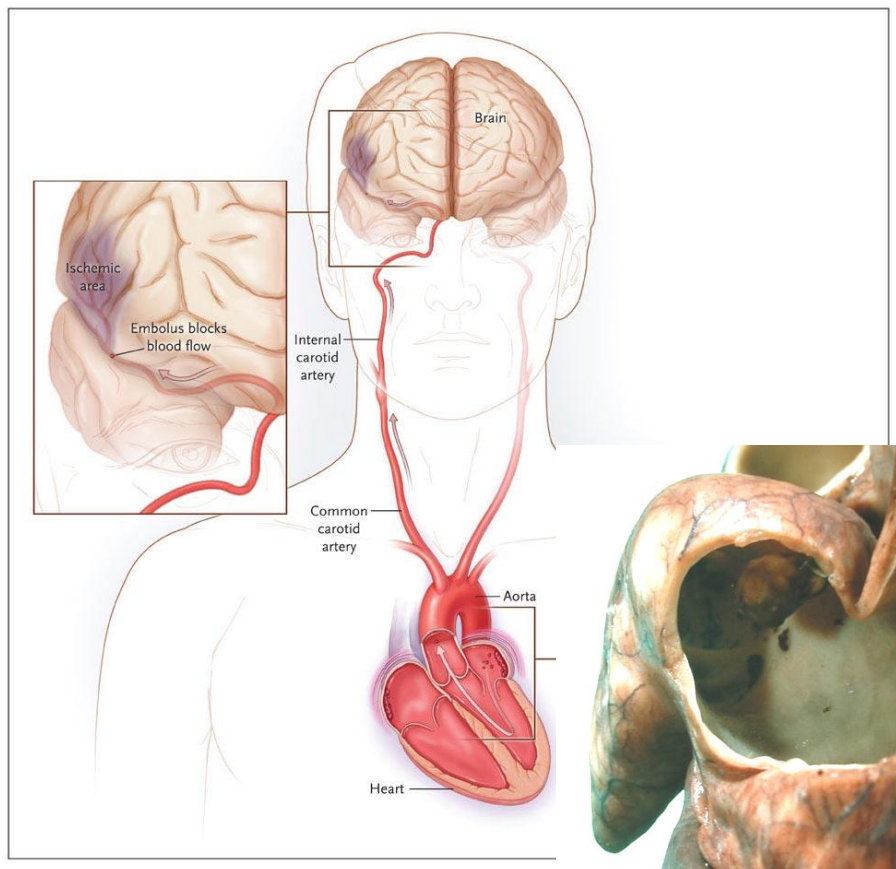
thrombosis







Left auricle

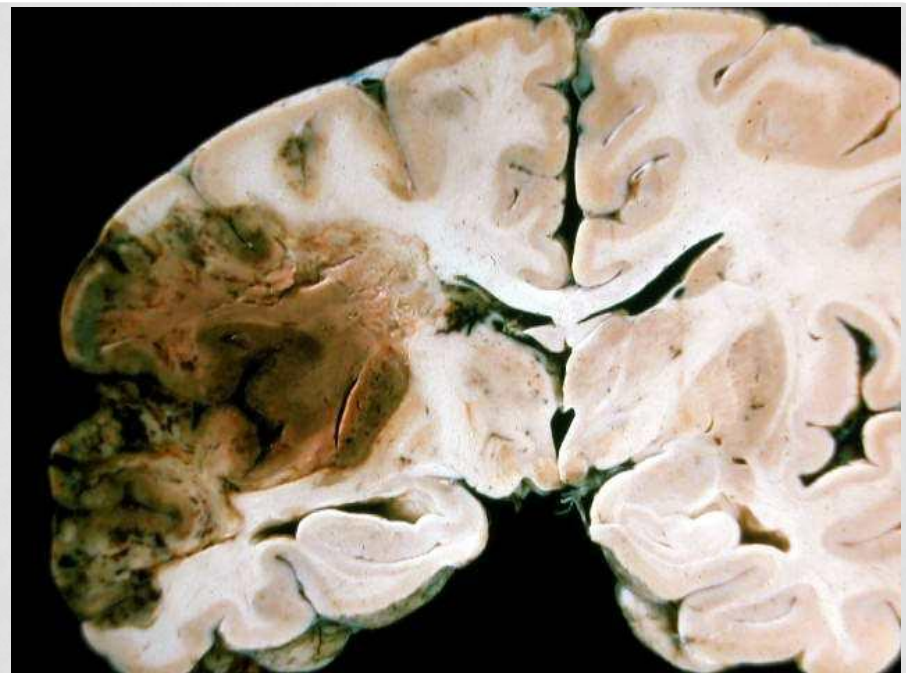
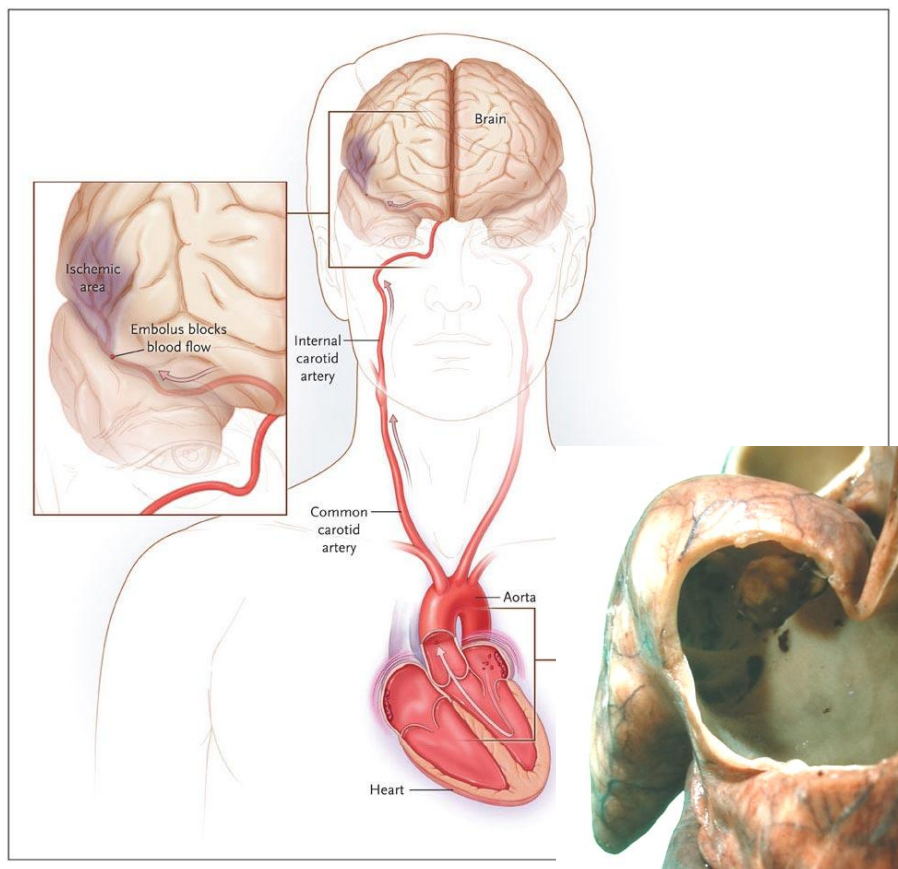


Complications:

Infective endocarditis
33%

Arrhythmia 39%

Embolisms 15%





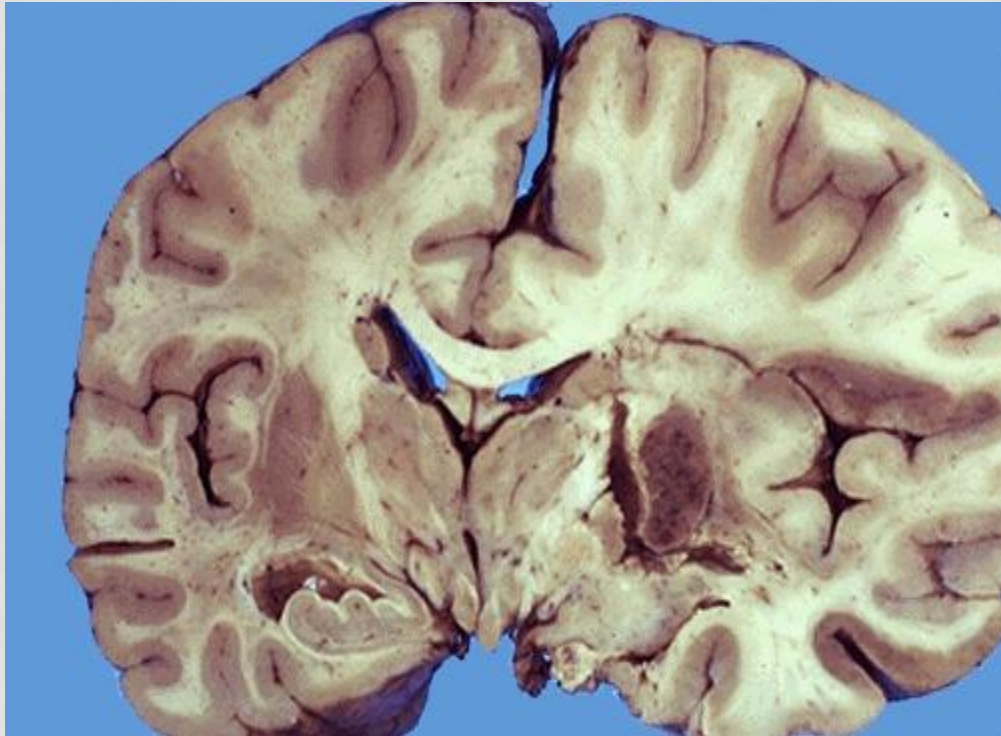
A



- Emboli tend to lodge where blood vessels branch or areas of preexisting luminal stenosis

The distribution of **middle cerebral artery** (the direct extension of the internal carotid artery) is most frequently affected by embolic infarction

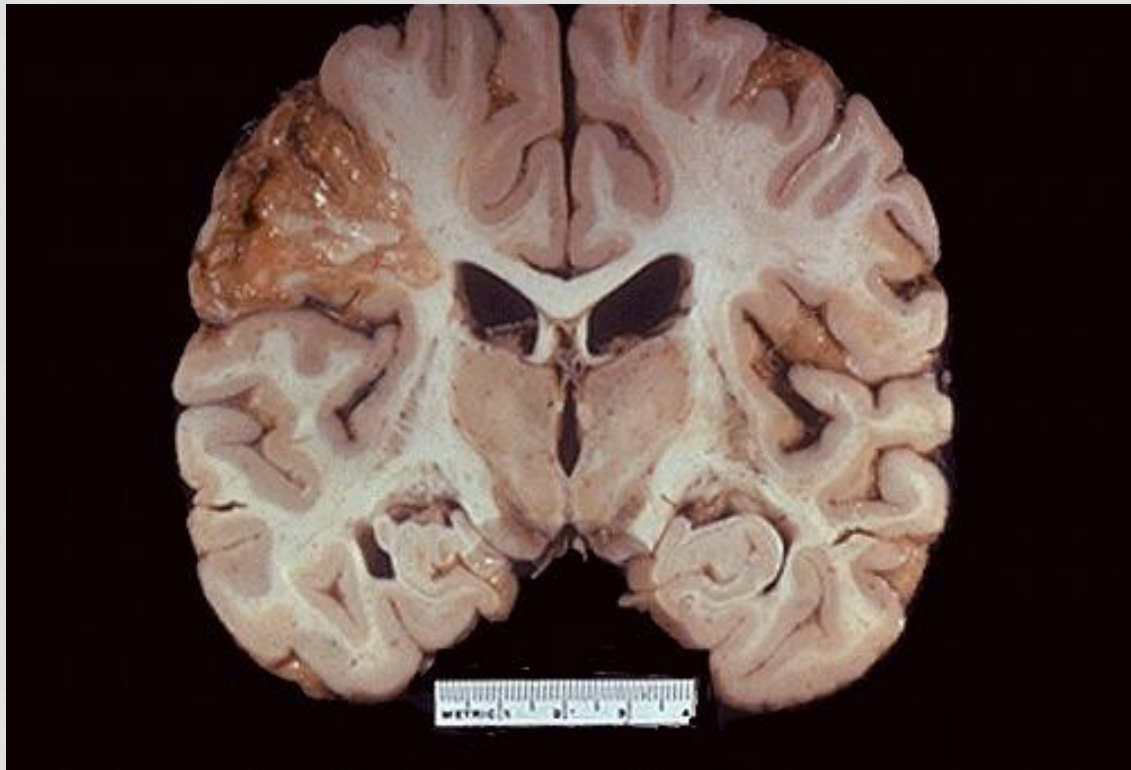
- Infarcts are subdivided into two broad groups based on macroscopic appearance
 - ❖ **Hemorrhagic (red) infarction**, characterized macroscopically by multiple petechial hemorrhages is typically associated embolic events. The hemorrhage is presumed to be secondary to reperfusion of damaged vessels & tissue either through collaterals or after lysis of the occlusive material.



❖ **Nonhemorrhagic (pale, anemic)** infarcts are usually associated with thrombosis.

Morphology

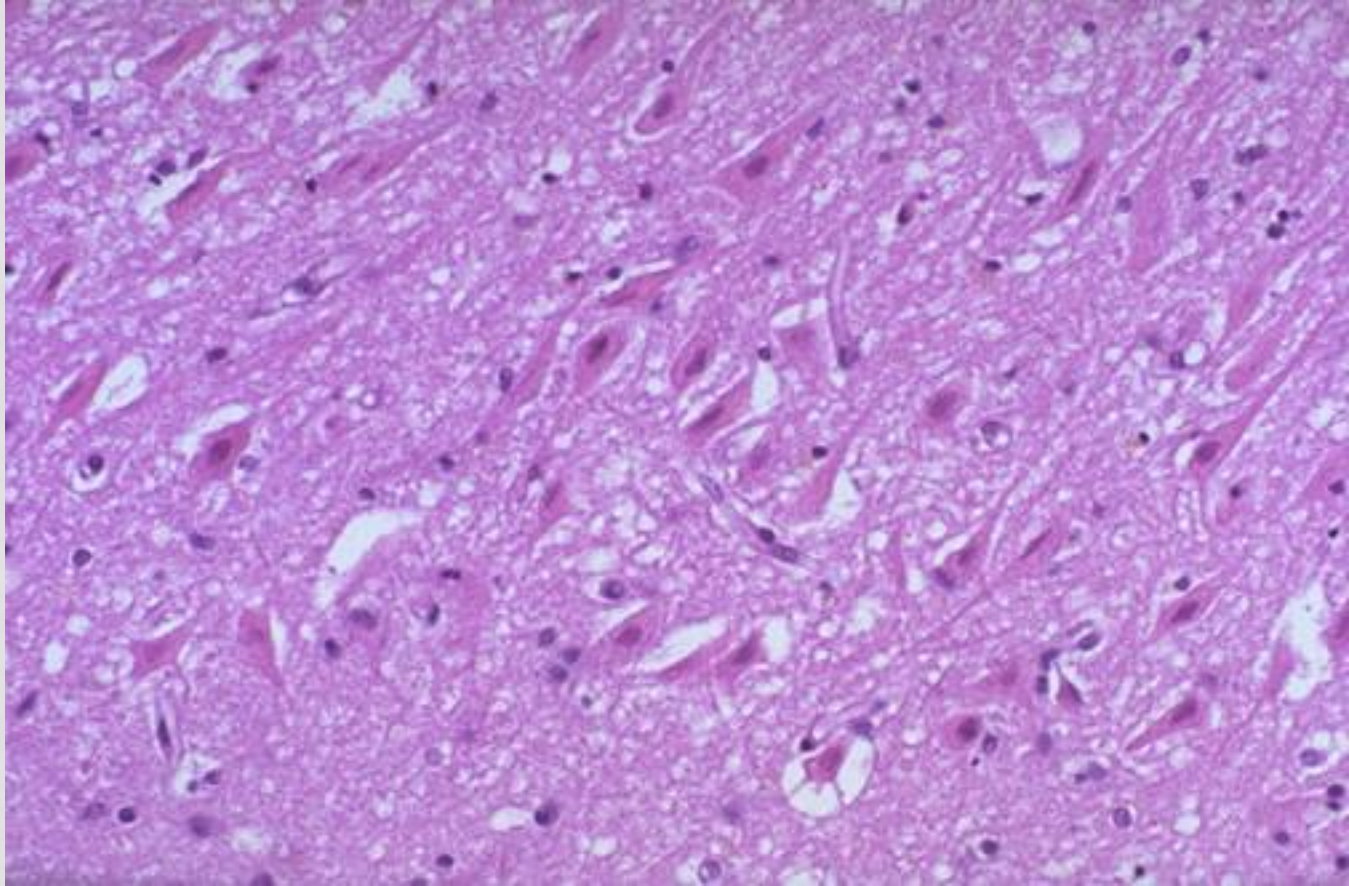
- The macroscopic appearance of nonhemorrhagic infarct changes with time.
- By 48 hrs the brain becomes pale, soft & swollen
- From 2 to 10 days- , the brain becomes gelatinous & friable
- From 10 days to 3 weeks, , the tissue liquifies , eventually leaving fluid filled cavity

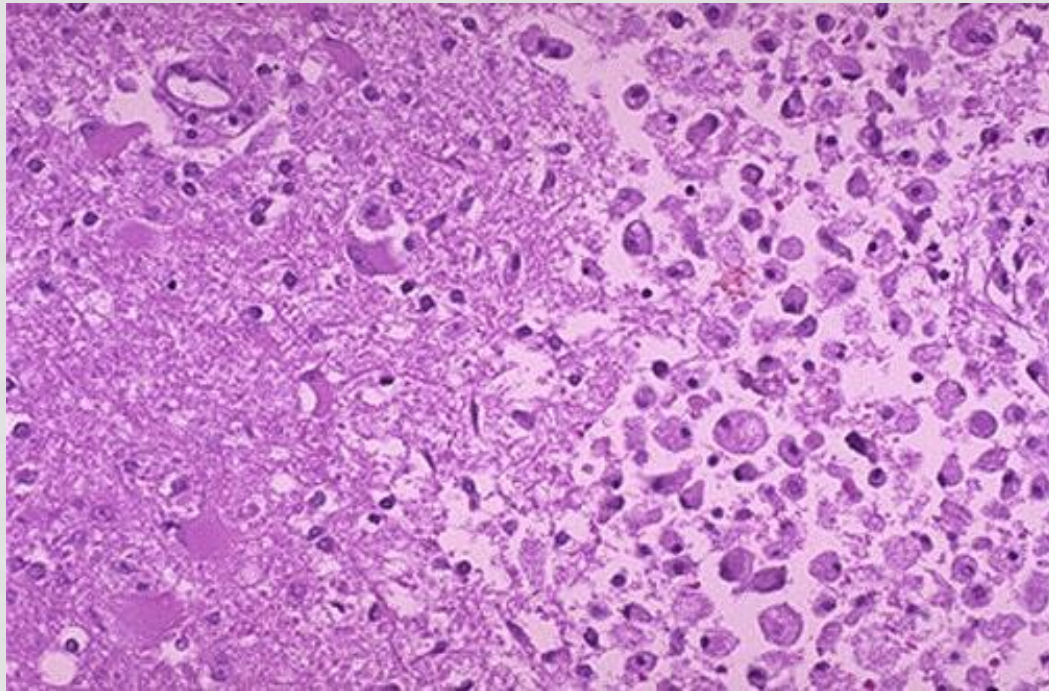


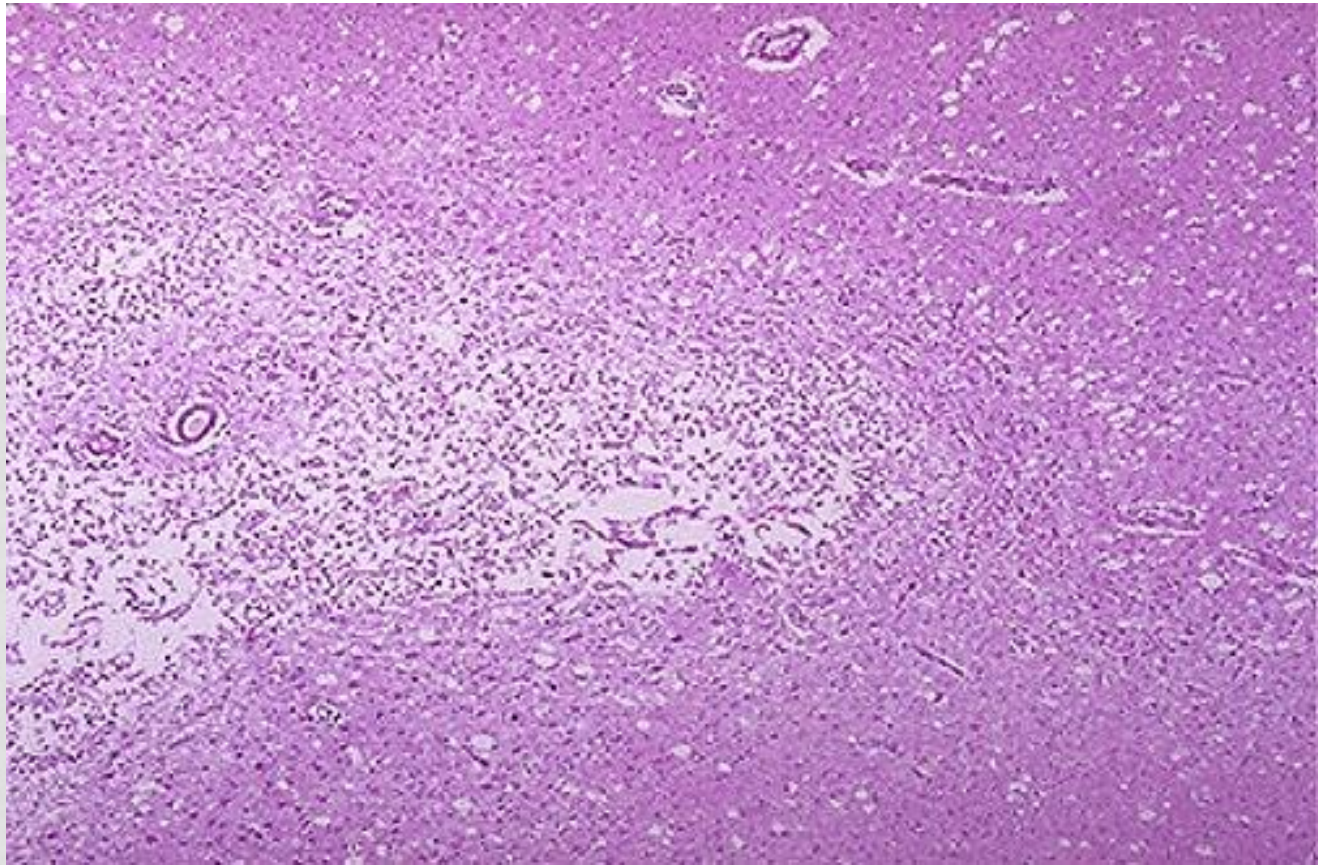


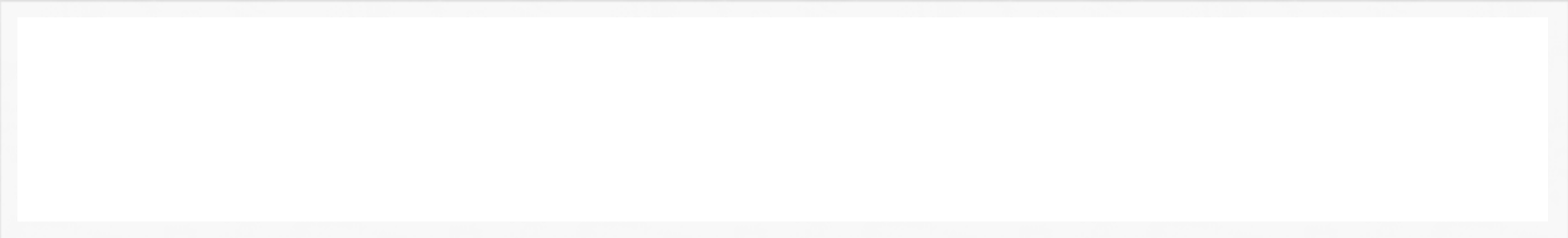


- On microscopic examination
- After the first 12 hrs, ischemic neuronal change (red neurons) & edema predominates.
- Up to 48hrs, neutrophils migration increases & after 48hrs phagocytic cells dominate in the next 2 to 3 wks

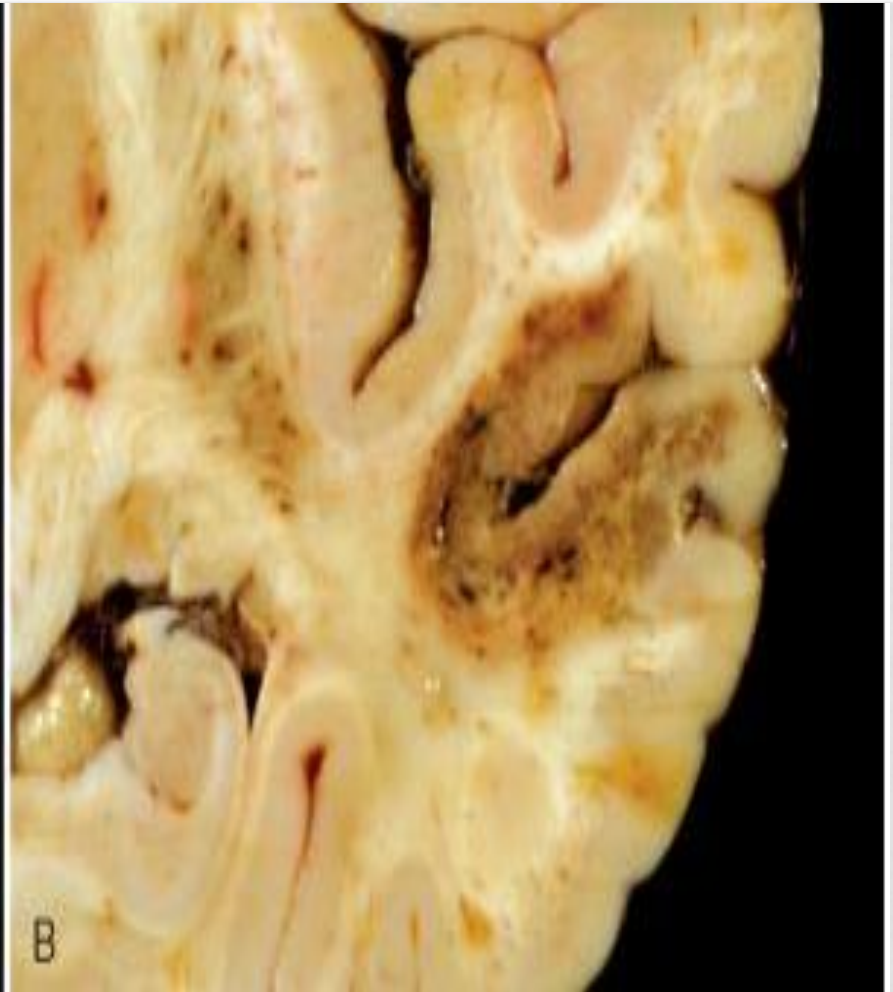
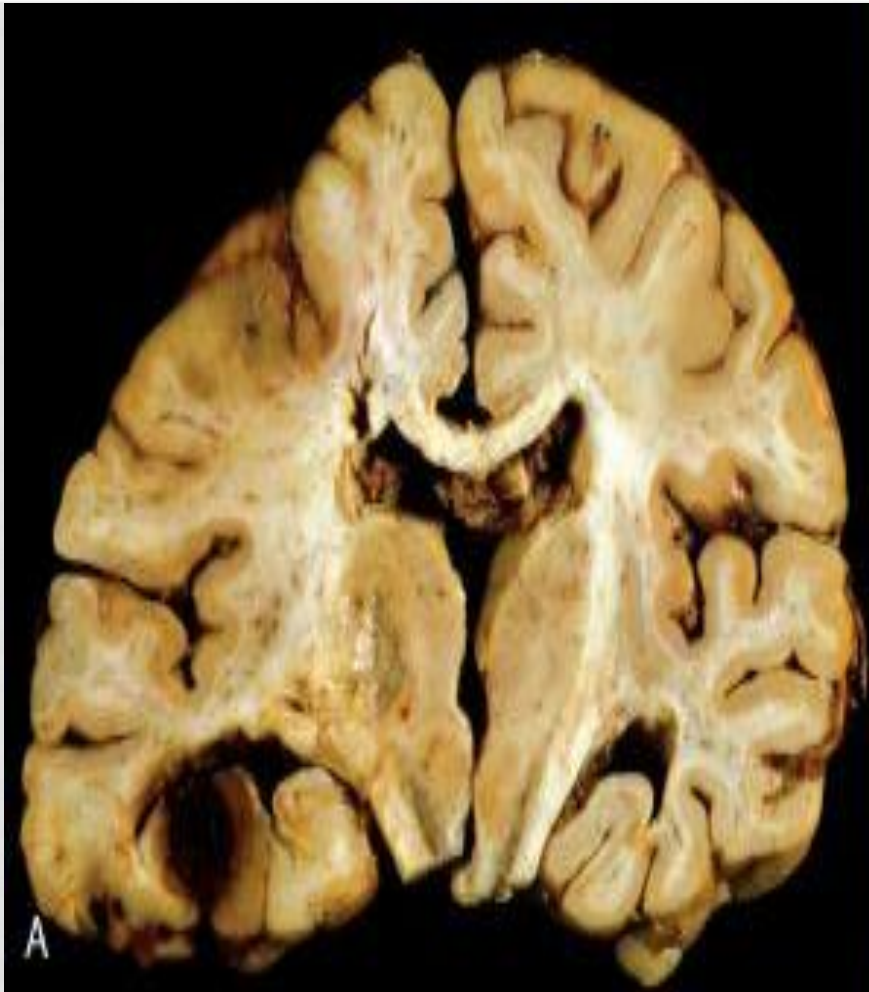


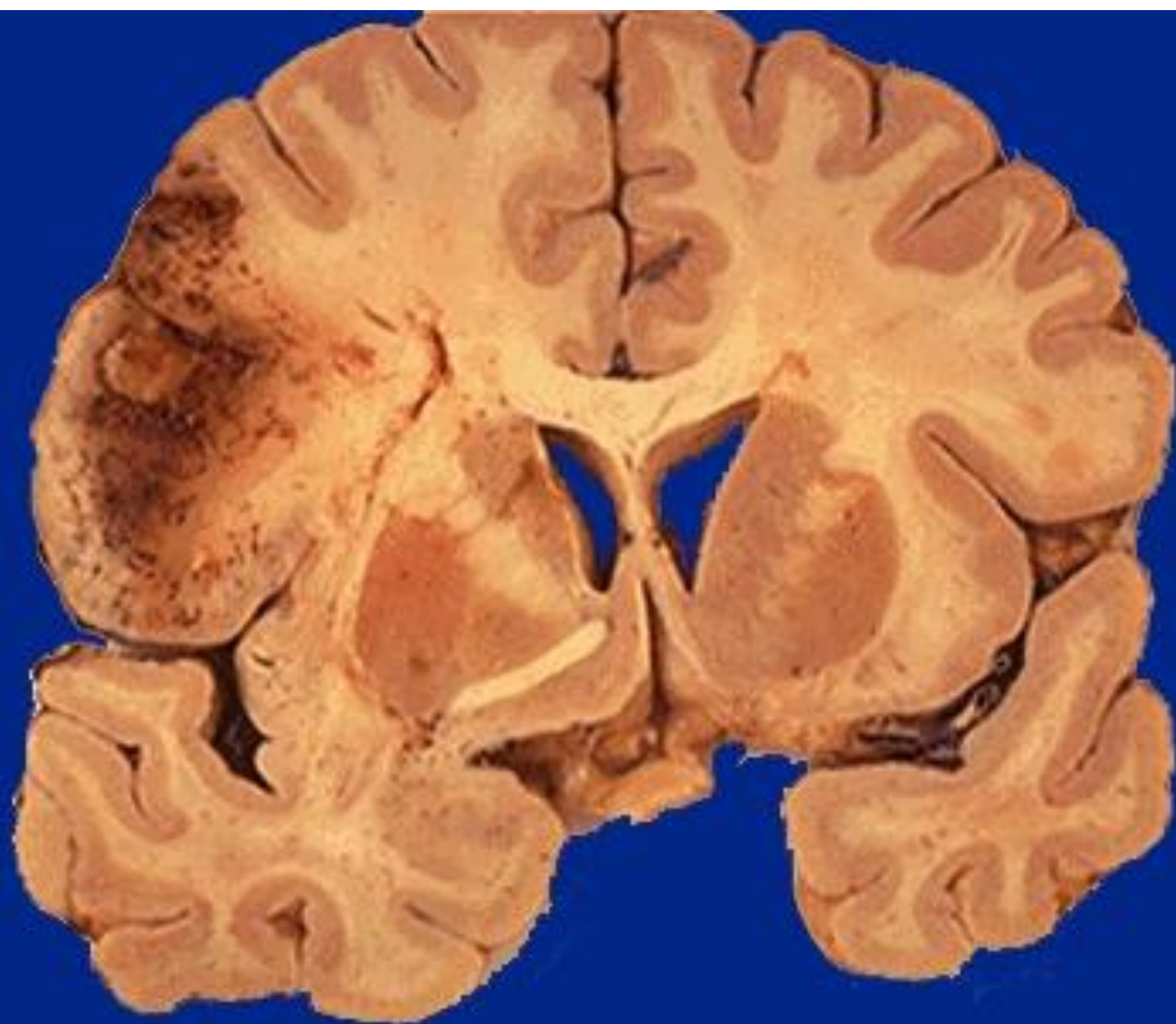




- 
- The microscopic picture of hemorrhagic infarction parallels ischemic infarction with addition of blood extravasation & resorption.

HEMORRHAGIC VS WHITE





Clinical features

- The area of brain that is affected determines whether the patient becomes asymptomatic or develops a hemiplegia , sensory deficit, aphasia or other deficit.

HEMORRHAGE RESULTING FROM RUPTURE OF CNS VESSELS

- Hemorrhages may occur at any site within the CNS. Hemorrhages within the brain parenchyma & subarachnoid space are often manifestation of underlying cerebrovascular disease where as hemorrhages within epidural & subdural space are typically related to trauma

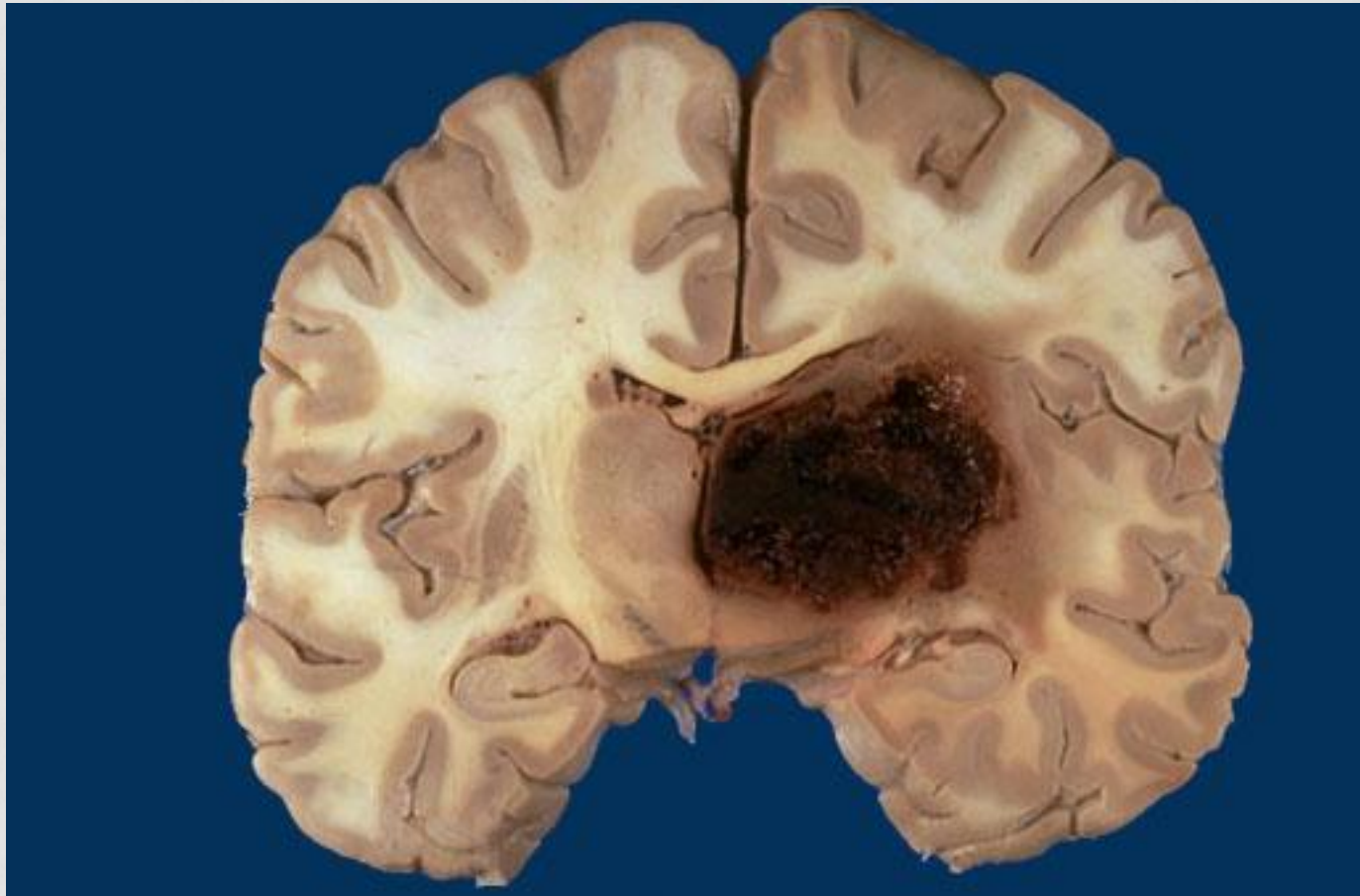
Intracerebral (intraparenchymal) hemorrhage

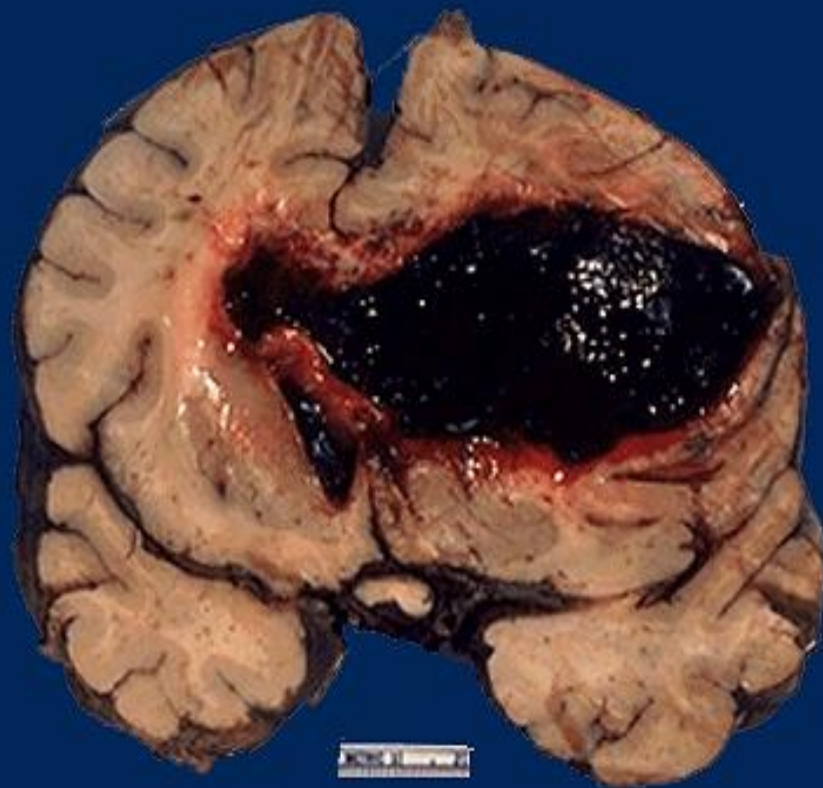
- Spontaneous (non traumatic) intraparenchymal hemorrhages occur most commonly in middle to late adult life with peak incidence at age of 60yrs.
- Most are caused by rupture of a small intraparenchymal vessel.

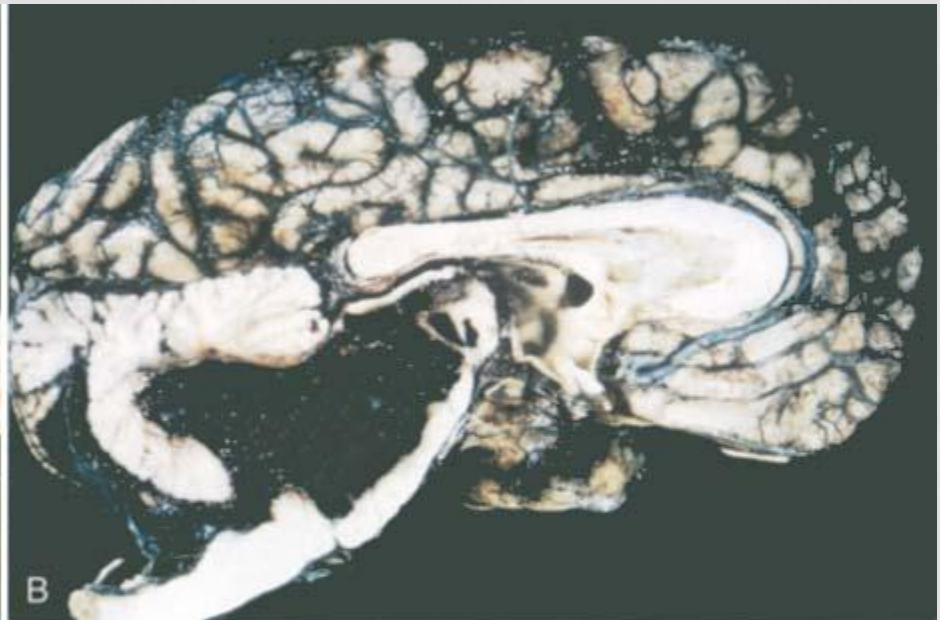
- HTN is the most common underlying cause of primary brain parenchymal hemorrhage (50%).
- HTN causes a number of abnormalities in vessel wall including accelerated atherosclerosis in large arteries, hyaline arteriolosclerosis in small vessels . Arteries with this changes are presumably weaker than are normal vessels & are therefore more vulnerable to rupture

Morphology

- Hypertensive intraparenchymal hemorrhage may originate in the putamen (50-60%, the most common site) , thalamus, pons, cerebellar hemispheres.
- The hemorrhages are characterized by extravasation of blood with compression of the adjacent parenchyma. Old hemorrhages show an area of cavitory destruction of brain with a rim of brownish discoloration.







Clinical features

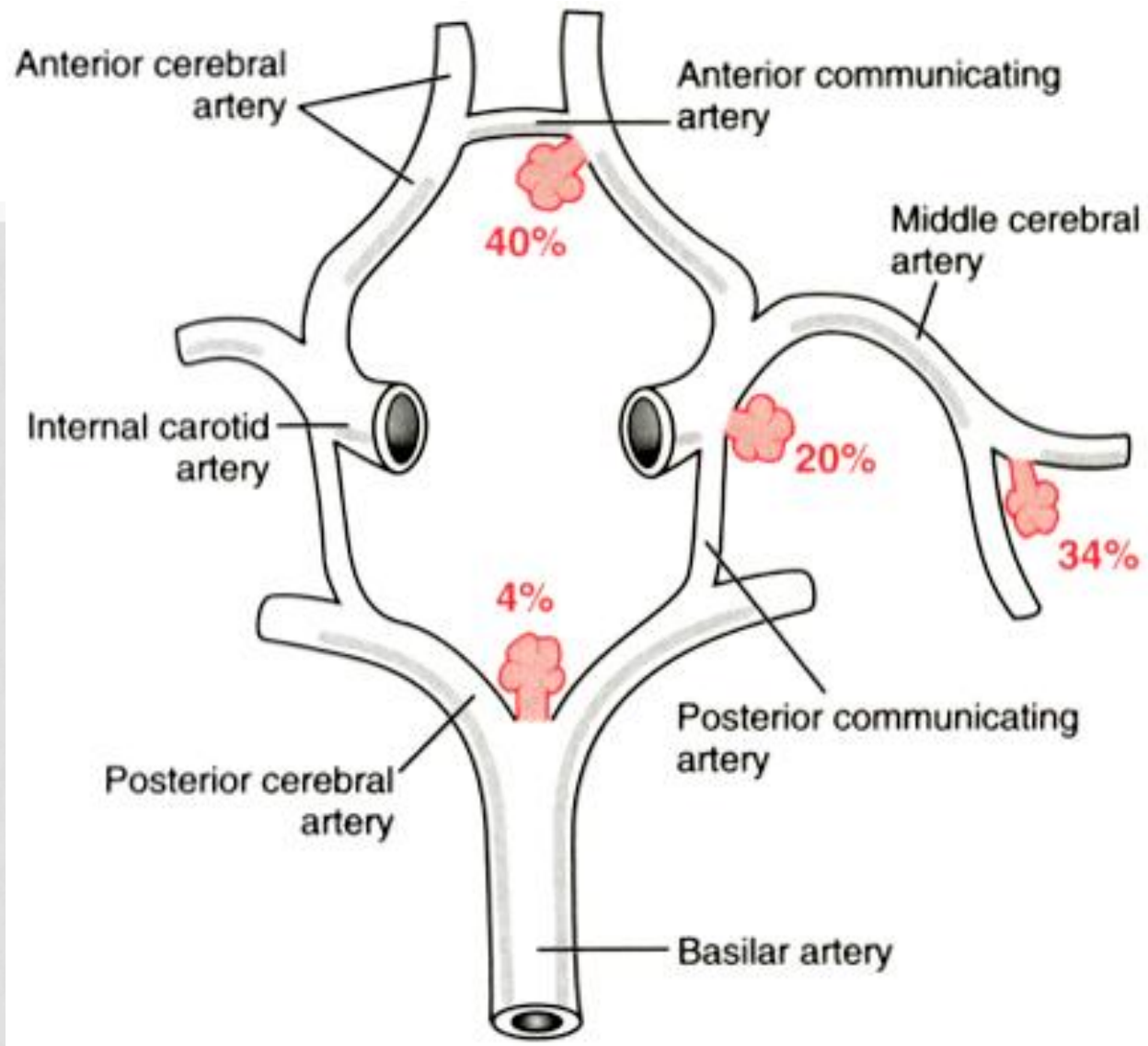
- Intracerebral hemorrhage can be clinically devastating when it affects large portions of the brain & extends into the ventricular system or it can affect small regions & be clinically silent .
- Over weeks or months , there is gradual resolution of the hematoma with clinical improvement.
- The location of the bleed will determine the clinical manifestation.

Subarachnoid hemorrhage

- The most frequent cause is a rupture of a saccular (berry) aneurysm

Berry aneurysm (saccular or congenital aneurysm)

- It is the most common type intracranial aneurysm
- Most occur in anterior circulation of circle of willis



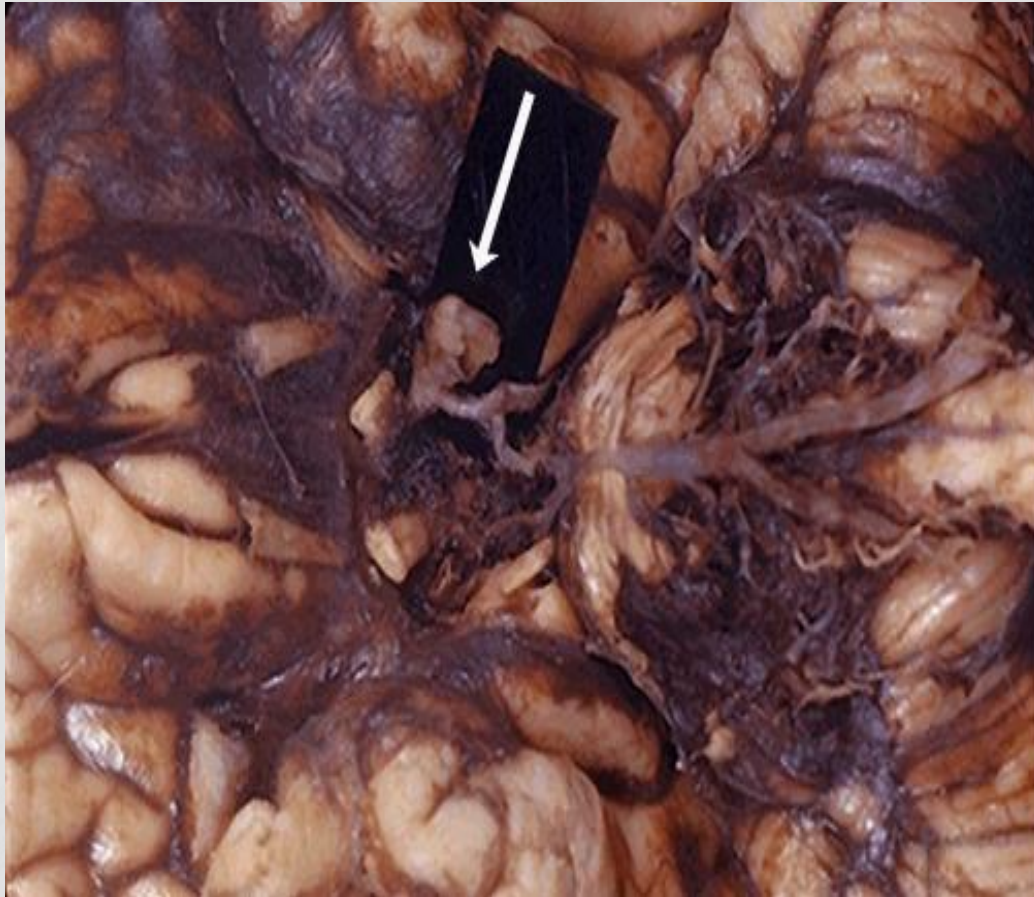
Pathogenesis

- Although majority occur sporadically , genetic factors may play a role.
- There is increased risk of occurrence among patients with certain heritable conditions (autosomal dominant polycystic kidney disease)
- Cigarette smoking & hypertension are predisposing factors.
- The aneurysms are not present at birth but develop over time owing to underlying defect in the media of the vessel.

Morphology

- Unruptured berry aneurysm is a thin walled outpouching at arterial branch point along the circle of willis.
- It measures a few mms to 2 to 3 cm in diameter







Clinical features

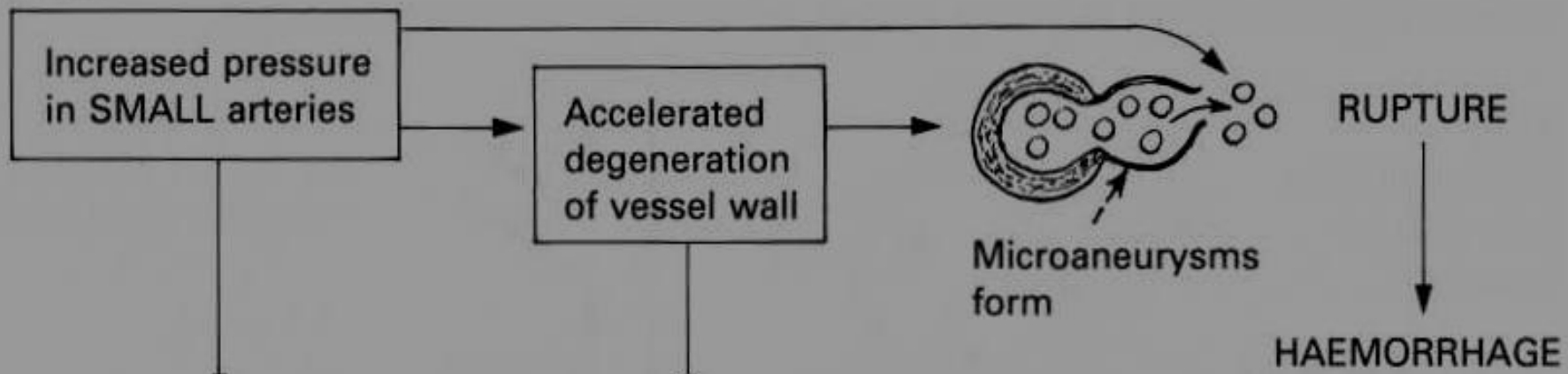
- Rupture of an aneurysm is most frequent in the fifth decade. the probability of rupture increases with the size of the lesion
- In most cases, rupture is associated with acute increases in intracranial pressure such as straining at stool
- Blood under arterial pressure is forced into subarachnoid space & patients are stricken with sudden excruciating headache typically 'worst headache I 've ever had' & rapidly lose consciousness.
- CSF – xanthochromasia

HYPERTENSIVE CEREBROVASCULAR DISEASE

The incidence of strokes in hypertensive subjects is very significantly increased.

1. Intracranial haemorrhage

Mechanism



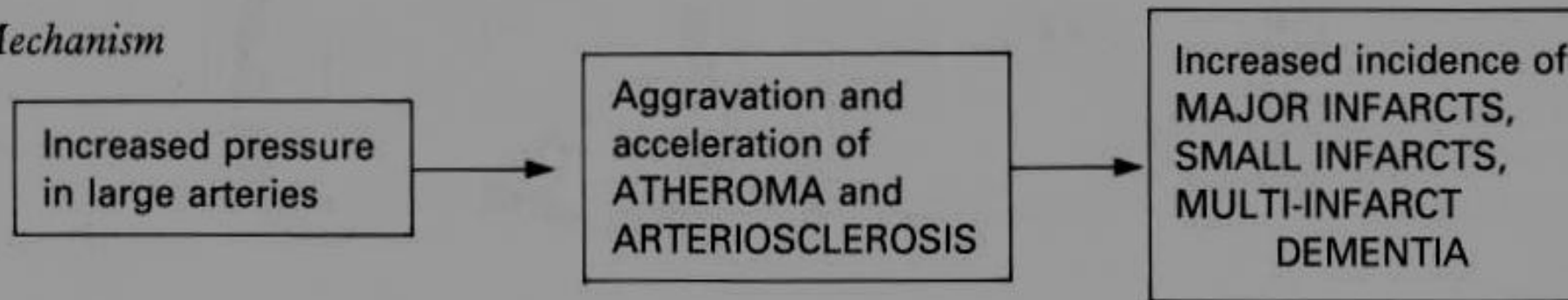
HYPERTENSIVE VASCULAR DISEASE

- Lacunar Infarction- Due to arteriolar sclerosis of the deep penetrating vessels in the basal ganglia and hemispheric white pattern.
 - - Less than 1.5cm
- Slit Hemorrhages-Due to rupture of small caliber penetrating vessels.



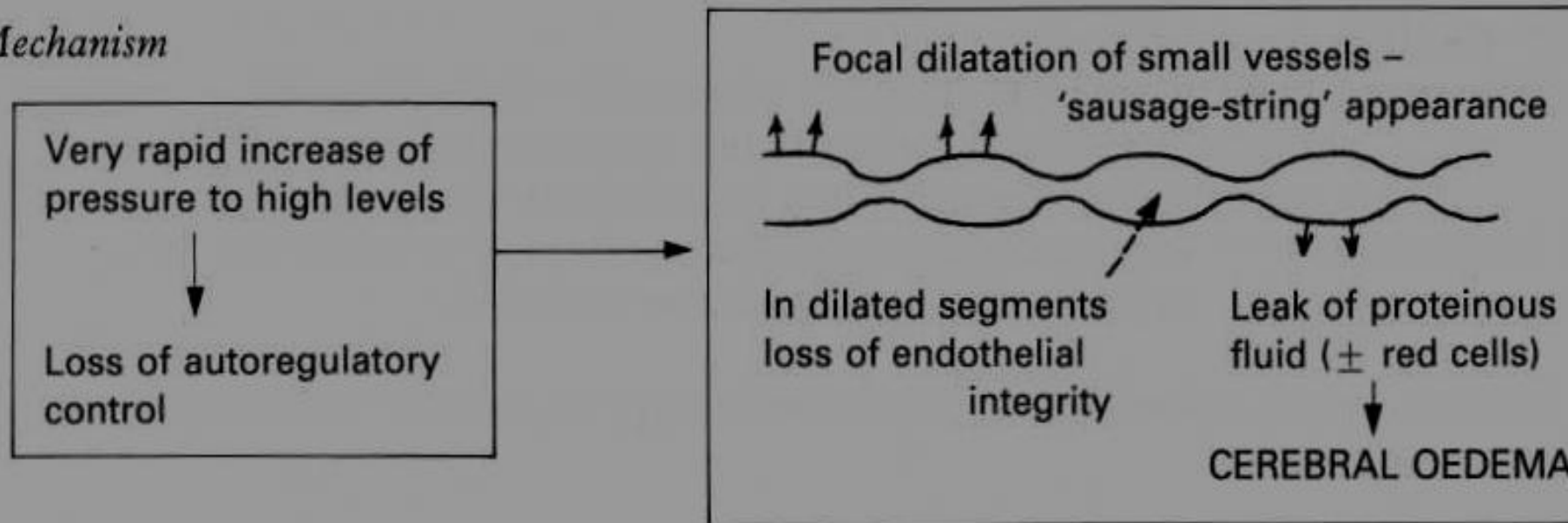
2. Cerebral infarction – risk increased indirectly.

Mechanism



In MALIGNANT HYPERTENSION, an additional dimension, HYPERTENSIVE ENCEPHALOPHATHY, is added.

Mechanism



- Cumulative Effect of the above with repeated insults results in a syndrome called Vascular(Multi-infarct) Dementia.
 - Dementia
 - Gait Abnormality
 - Pseudobulbar signs
 - Focal Neurologic Deficit